

APPENDIX F

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United States Patent and Trademark Office

May 26, 2026

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OF:**

APPLICATION NUMBER: 19/023,905

FILING DATE: *January 16, 2025*

PATENT NUMBER: 12412924

ISSUE DATE: *September 09, 2025*



Certified by

DIRECTOR OF THE UNITED STATES PATENT AND TRADEMARK OFFICE



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PATENT AND TRADEMARK OFFICE

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APPLICATION #
19/023,905

RECEIPT DATE / TIME
01/16/2025 12:06:14 PM Z ET

ATTORNEY DOCKET #
00249/0086

Title of Invention

MULTILAYER NANOPOROUS SEPARATOR

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3614

FILED BY Joy Chapman

PATENT CENTER # 68770157

FILING DATE -

CUSTOMER # 1912

FIRST NAMED
INVENTOR David W. Avison

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY Brian Amos

Documents

TOTAL DOCUMENTS: 8

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
00249-0086_ADS_fillable-pdf.pdf	9	Application Data Sheet	2174 KB
00249-0086_Track-1-Req-for-Prioritized-Exam_fillable-pdf.pdf	2	Track One Request	185 KB
00249-0086_DOCX-Application-APP.TEXT.docx	31	Application body structured text document	69 KB
Warning: Bookmarks were found and have been removed.			
00249-0086_Drawings.pdf	6	Drawings-only black and white line drawings	63 KB
00249-0086_Declarations-for-the-3-Inventors.pdf	3	Oath or Declaration filed	2107 KB

00249-0086_POA-with-Transmittal.pdf	2	Power of Attorney	413 KB
00249-0086_IDS_fillable-pdf.pdf	7	Information Disclosure Statement (IDS) Form (SB08)	1019 KB
00249-0086_AUX-PDF-Application.pdf	31	Auxiliary PDF of Application	262 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
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If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement

Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

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Figure 1

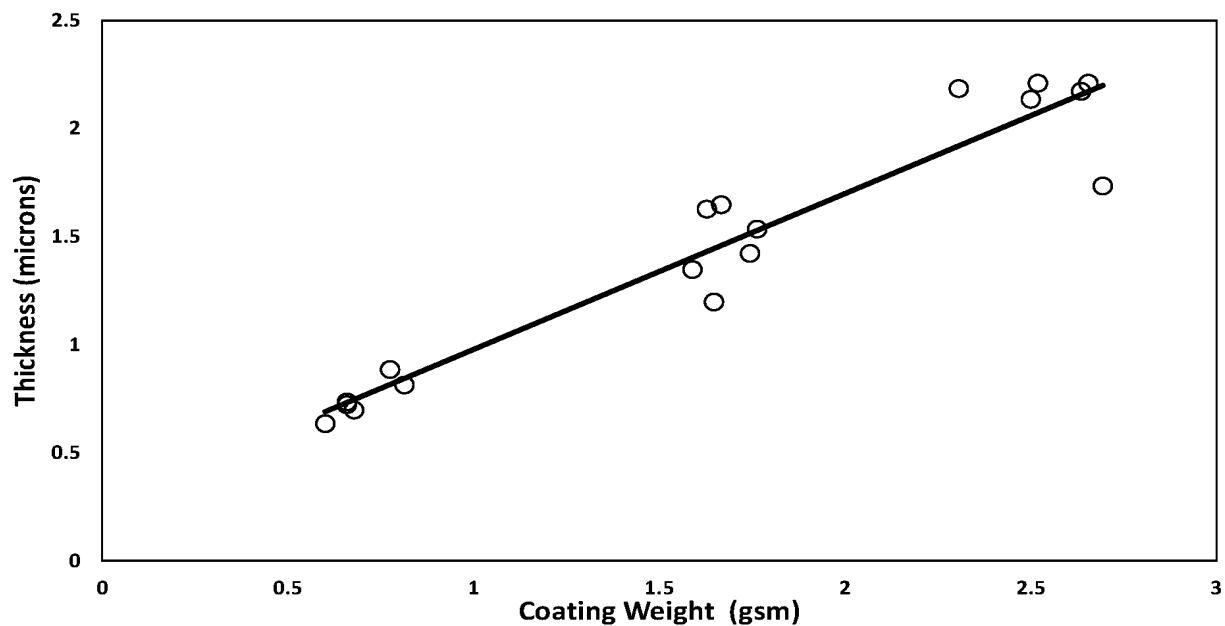
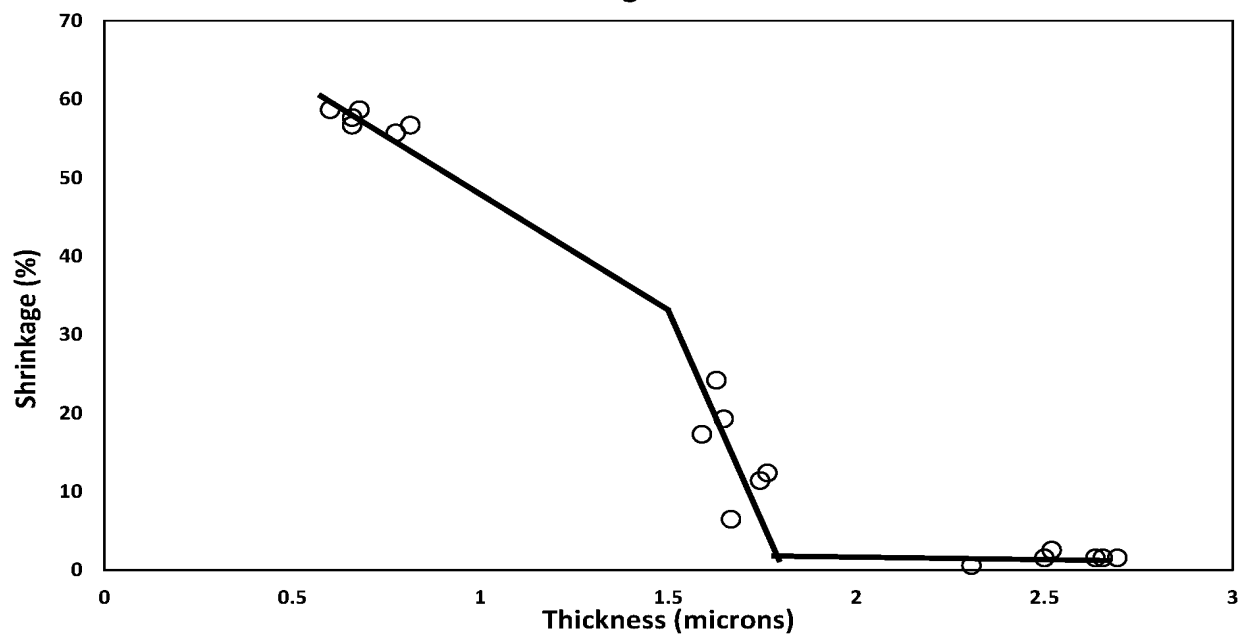
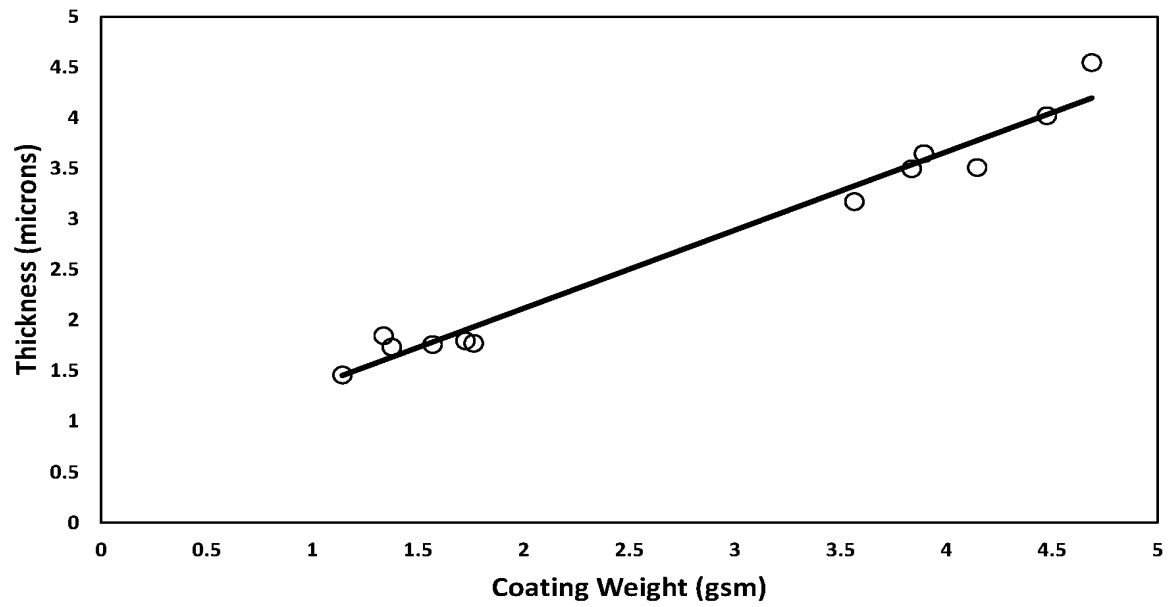
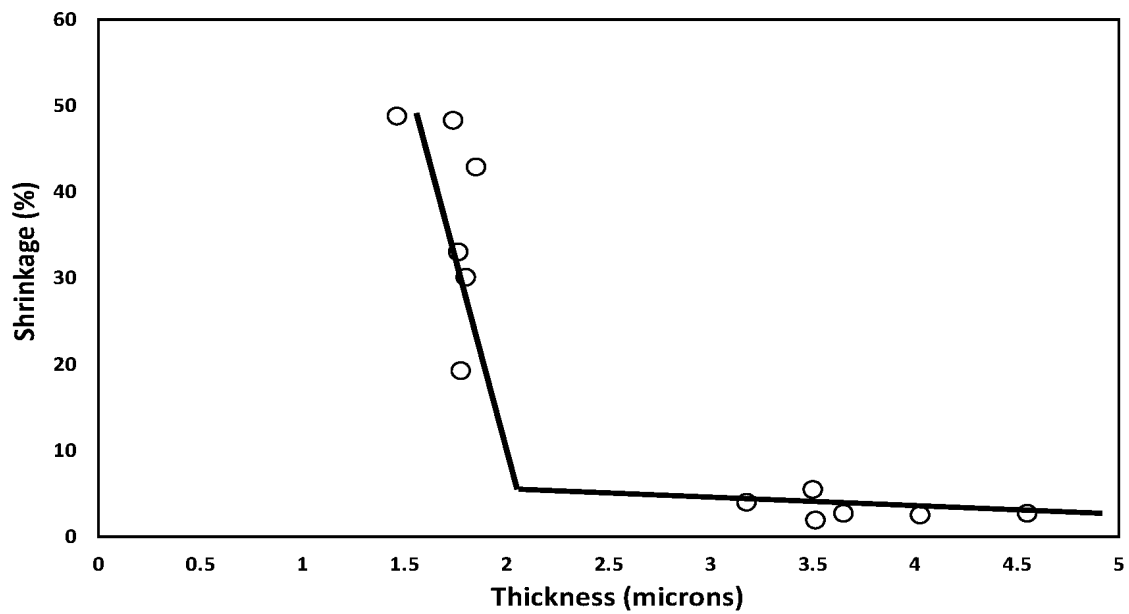


Figure 2



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Figure 3**Figure 4**

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Figure 5

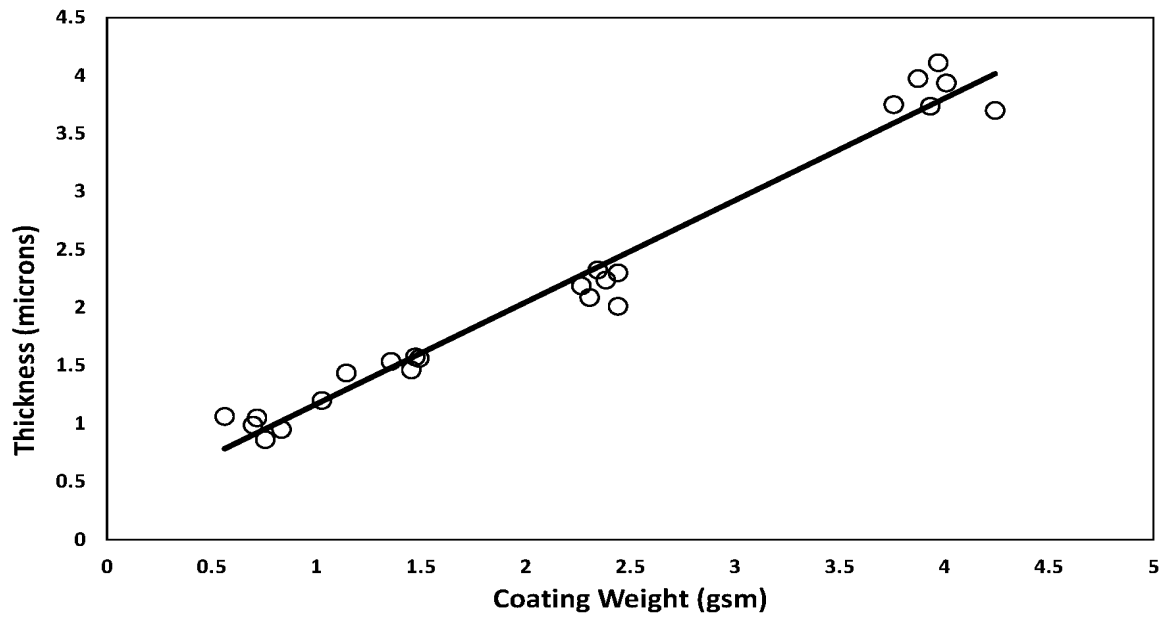
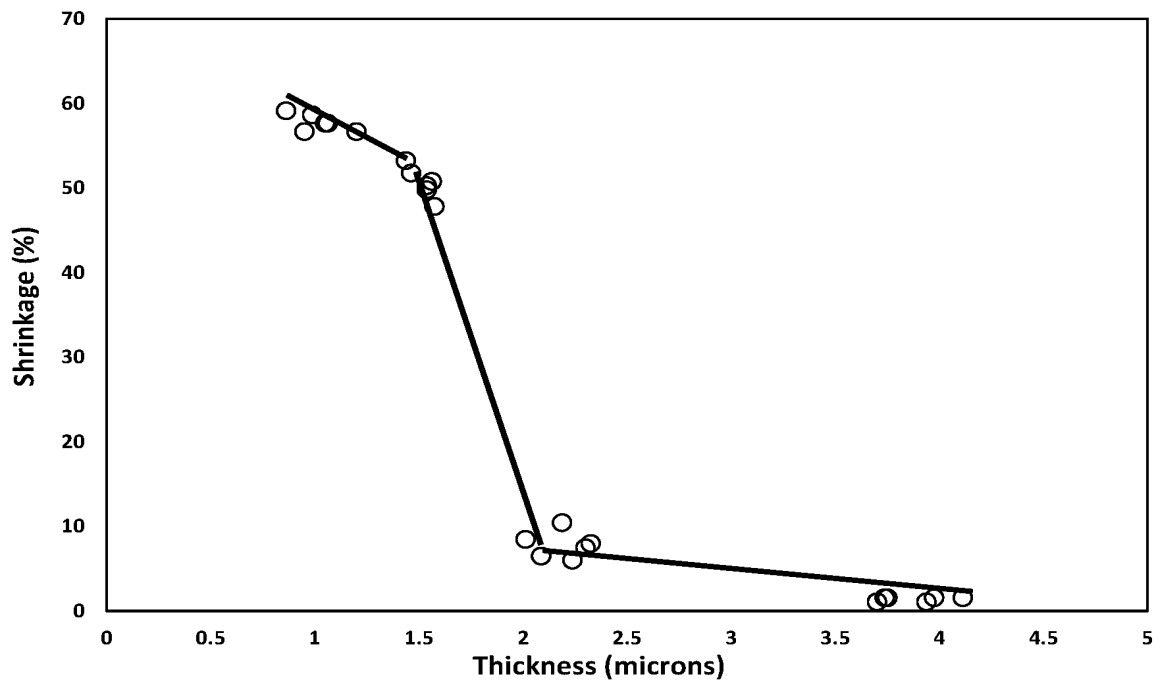
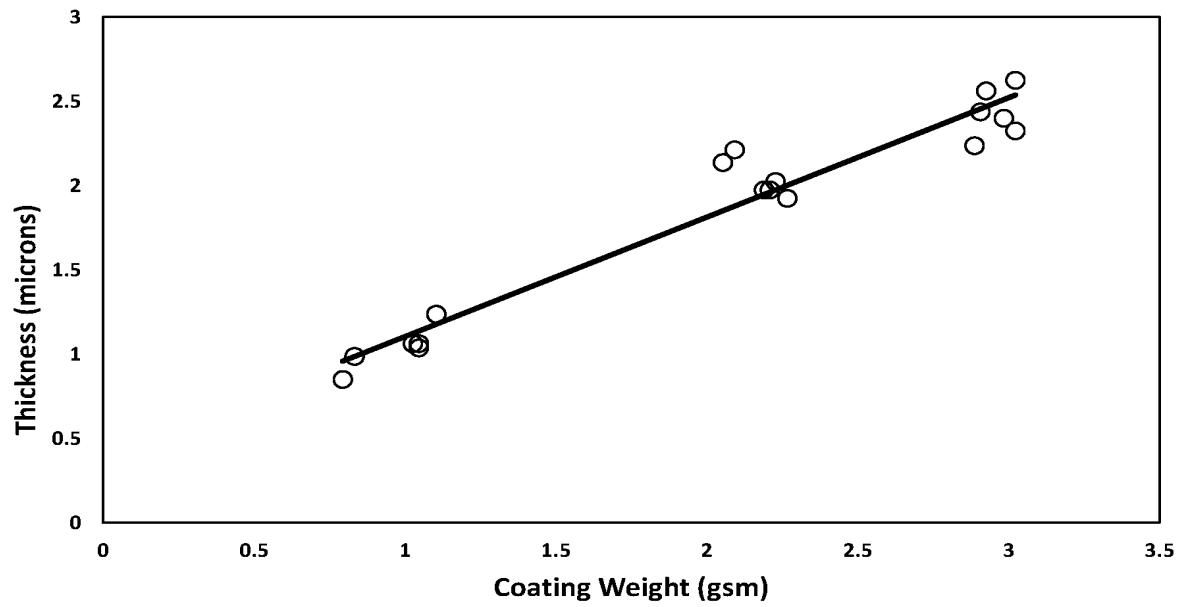
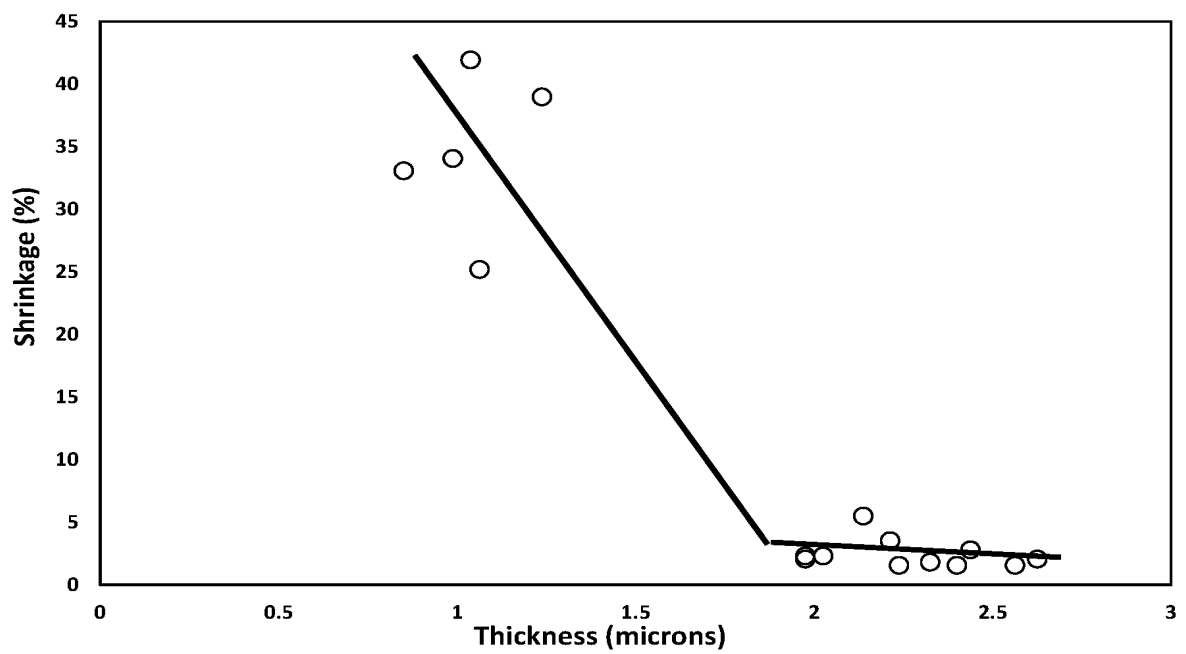


Figure 6



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Figure 7**Figure 8**

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Figure 9

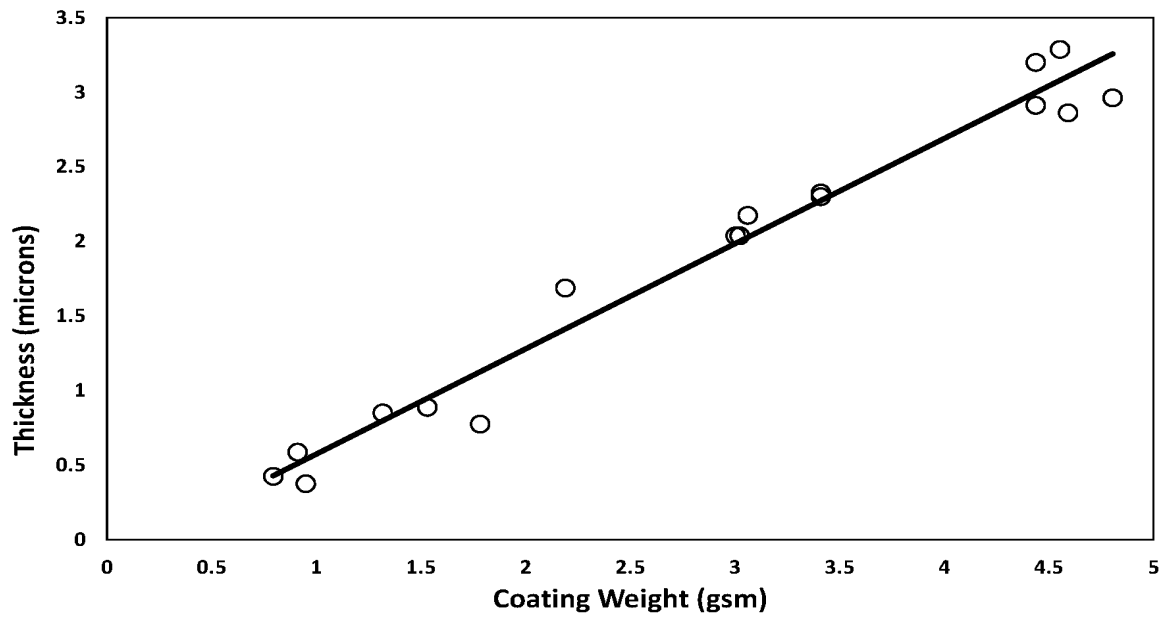
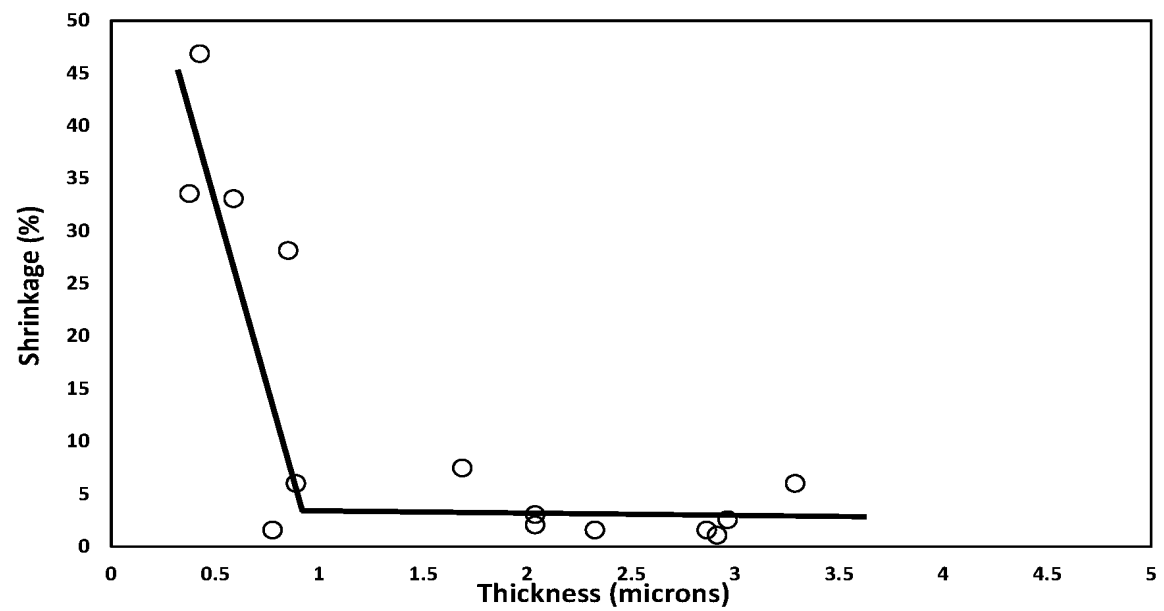


Figure 10



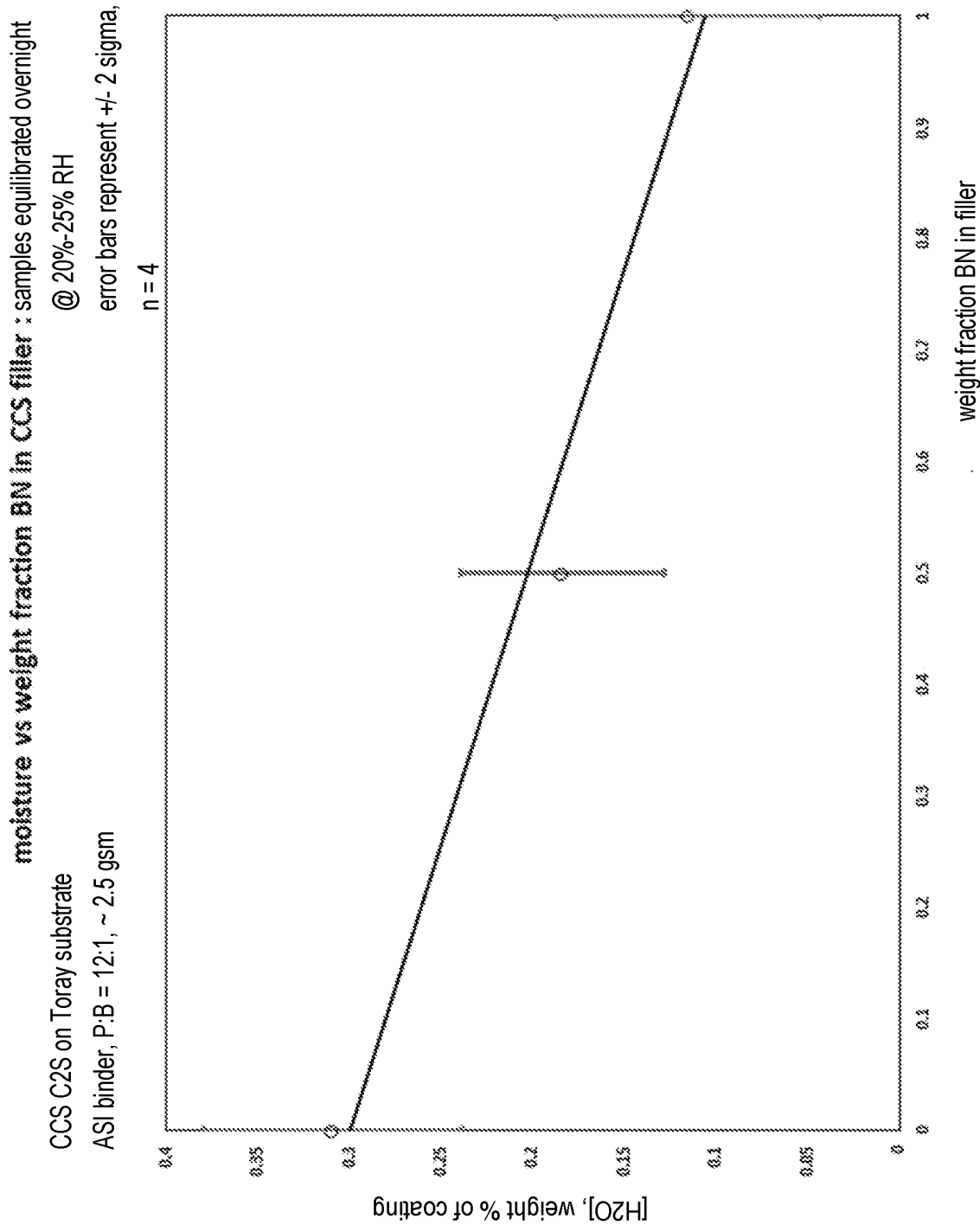


Fig. 11

**CERTIFICATION AND REQUEST FOR PRIORITIZED EXAMINATION
UNDER 37 CFR 1.102(e) (Page 1 of 1)**

First Named Inventor:	David W. Avison	Nonprovisional Application Number (if known):	
Title of Invention:	MULTILAYER NANOPOROUS SEPARATOR		

APPLICANT HEREBY CERTIFIES THE FOLLOWING AND REQUESTS PRIORITIZED EXAMINATION FOR THE ABOVE-IDENTIFIED APPLICATION.

1. The processing fee set forth in 37 CFR 1.17(i)(1) and the prioritized examination fee set forth in 37 CFR 1.17(c) have been filed with the request. The publication fee requirement is met because that fee, set forth in 37 CFR 1.18(d), is currently \$0. The basic filing fee, search fee, and examination fee are filed with the request or have been already been paid. I understand that any required excess claims fees or application size fee must be paid for the application.
2. I understand that the application may not contain, or be amended to contain, more than four independent claims, more than thirty total claims, or any multiple dependent claims, and that any request for an extension of time will cause an outstanding Track I request to be dismissed.
3. The applicable box is checked below:

I. ☒ Original Application (Track One) - Prioritized Examination under § 1.102(e)(1)

i. (a) The application is an original nonprovisional utility application filed under 35 U.S.C. 111(a). This certification and request is being filed with the utility application via USPTO patent electronic filing system.

---OR---

(b) The application is an original nonprovisional plant application filed under 35 U.S.C. 111(a). This certification and request is being filed with the plant application in paper.

ii. An executed inventor's oath or declaration under 37 CFR 1.63 or 37 CFR 1.64 for each inventor, or the application data sheet meeting the conditions specified in 37 CFR 1.53(f)(3)(i) is filed with the application.

II. ☐ Request for Continued Examination - Prioritized Examination under § 1.102(e)(2)

- A request for continued examination has been filed with, or prior to, this form.
- If the application is a utility application, this certification and request is being filed via USPTO patent electronic filing system.
- The application is an original nonprovisional utility application filed under 35 U.S.C. 111(a), or is a national stage entry under 35 U.S.C. 371.
- This certification and request is being filed prior to the mailing of a first Office action responsive to the request for continued examination.
- No prior request for continued examination has been granted prioritized examination status under 37 CFR 1.102(e)(2).

Signature <u>/Brian J. Amos/</u>	Date <u>January 16, 2025</u>
Name (Print/Typed) <u>Brian J. Amos</u>	Practitioner Registration Number <u>63,679</u>

Note: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. Submit multiple forms if more than one signature is required.*

☒ *Total of 1 forms are submitted.

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- 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
- 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
- 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
- 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
- 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
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DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

Title of Invention	MULTILAYER NANOPOROUS SEPARATOR
--------------------	---------------------------------

As the below named inventor, I hereby declare that:

This declaration is directed to:

☐

The attached application, or

☒United States application or PCT international application number 16/755,294filed on April 10, 2020

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

WARNING:

Petitioner/applicant is cautioned to avoid submitting personal information in documents filed in a patent application that may contribute to identity theft. Personal information such as social security numbers, bank account numbers, or credit card numbers (other than a check or credit card authorization form PTO-2038 submitted for payment purposes) is never required by the USPTO to support a petition or an application. If this type of personal information is included in documents submitted to the USPTO, petitioners/applicants should consider redacting such personal information from the documents before submitting them to the USPTO. Petitioner/applicant is advised that the record of a patent application is available to the public after publication of the application (unless a non-publication request in compliance with 37 CFR 1.213(a) is made in the application) or issuance of a patent. Furthermore, the record from an abandoned application may also be available to the public if the application is referenced in a published application or an issued patent (see 37 CFR 1.14). Checks and credit card authorization forms PTO-2038 submitted for payment purposes are not retained in the application file and therefore are not publicly available.

LEGAL NAME OF INVENTOR

Inventor: David W. Avison

Date (Optional): _____

Signature: David W. Avison

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity, must accompany this form or must have been previously filed. Use an additional PTO/AIA/01 form for each additional inventor.

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 1 minute to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

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DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

Title of Invention	MULTILAYER NANOPOROUS SEPARATOR
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This declaration
is directed to

☐

The attached application, or

☒

United States application or PCT international application number 16/755,294

filed on April 10, 2020

The above-identified application was made or authorized to be made by me.

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LEGAL NAME OF INVENTOR

Inventor: Steven A. Carlson

Date (Optional): _____

Signature: Steven A. Carlson

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DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

Title of Invention	MULTILAYER NANOPOROUS SEPARATOR
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As the below named inventor, I hereby declare that:

This declaration is directed to: ☐ The attached application, or ☒ United States application or PCT international application number 16/755,294 filed on April 10, 2020

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

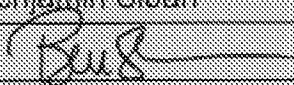
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LEGAL NAME OF INVENTOR

Inventor: Benjamin Sloan Date (Optional): _____

Signature: 

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00249/0086
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

Inventor Information:

Inventor 1					Remove	
Legal Name						
Prefix	Given Name	Middle Name	Family Name	Suffix		
	David	W.	Avison			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service						
City	Boxborough	State/Province	MA	Country of Residenceⁱ	US	
Mailing Address of Inventor:						
Address 1	31D Liberty Road					
Address 2						
City	Boxborough	State/Province	MA			
Postal Code	01719	Countryⁱ	US			
Inventor 2					Remove	
Legal Name						
Prefix	Given Name	Middle Name	Family Name	Suffix		
	Steven	A.	Carlson			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service						
City	Cambridge	State/Province	MA	Country of Residenceⁱ	US	
Mailing Address of Inventor:						
Address 1	993 Memorial Drive, Apt. 101					
Address 2						
City	Cambridge	State/Province	MA			
Postal Code	02138	Countryⁱ	US			
Inventor 3					Remove	
Legal Name						
Prefix	Given Name	Middle Name	Family Name	Suffix		
	Benjamin		Sloan			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service						

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00249/0086
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

City	Exeter	State/Province	NH	Country of Residence	US
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Mailing Address of Inventor:

Address 1	11 Gardner Street				
Address 2					
City	Exeter	State/Province	NH		
Postal Code	03833	Country	US		
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.					

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence Information of this application.			
Customer Number	1912		
Email Address	ptodocket@arelaw.com	Add Email	Remove Email

Application Information:

Title of the Invention	MULTILAYER NANOPOROUS SEPARATOR		
Attorney Docket Number	00249/0086	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	6	Suggested Figure for Publication (if any)	

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐ Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00249/0086
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.			
Please Select One: ☒ Customer Number ☐ US Patent Practitioner ☐ Limited Recognition (37 CFR 11.9)			
Customer Number	1912		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status		Pending		Remove	
Application Number		Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
		Continuation of		18/631783	2024-04-10
Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
18/631783	Continuation of	17/543259	2021-12-06	11973181	2024-04-30
Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
17/543259	Continuation of	17/223845	2021-04-06	11769903	2023-09-26
Prior Application Status		Patented		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
17/223845	Continuation of	16/755294	2020-04-10	10998536	2021-05-04
Prior Application Status		Expired		Remove	
Application Number		Continuity Type		Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
16/755294		a 371 of international		PCT/US2018/055862	2018-10-15

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00249/0086
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		
Prior Application Status	Expired	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
PCT/US2018/055862	Claims benefit of provisional	62/572083	2017-10-13
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			Add

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This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

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Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00249/0086
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

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The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00249/0086
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.			
Applicant 1			Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.			
Clear			
☒ Assignee	☐ Legal Representative under 35 U.S.C. 117	☐ Joint Inventor	
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If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:			
Name of the Deceased or Legally Incapacitated Inventor:			
If the Applicant is an Organization check here. ☒			
Organization Name	LG Energy Solution, Ltd.		
Mailing Address Information For Applicant:			
Address 1	Parc.1, 108 Yeoui-daero, Yeongdeungpo-gu		
Address 2			
City	Seoul	State/Province	
Country	KR	Postal Code	07335
Phone Number		Fax Number	
Email Address			
Additional Applicant Data may be generated within this form by selecting the Add button. Add			

Assignee Information including Non-Applicant Assignee Information:

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00249/0086
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

Assignee 1

Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.

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If the Assignee or Non-Applicant Assignee is an Organization check here. ☐

Prefix	Given Name	Middle Name	Family Name	Suffix

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This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Brian J. Amos/		Date (YYYY-MM-DD)	2025-01-16
First Name	Brian J.	Last Name	Amos	Registration Number
				63,679

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	00249/0086
		Application Number	
Title of Invention	MULTILAYER NANOPOROUS SEPARATOR		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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- 3 A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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- 6 A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
- 7 A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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- 9 A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

MULTILAYER NANOPOROUS SEPARATOR

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. Patent Application No. 18/631,783, filed April 10, 2024, which is a continuation of U.S. Patent Application No. 17/543,259, filed December 6, 2021, now U.S. Patent No. 11,973,181, which is a continuation of U.S. Patent Application No. 17/223,845, filed April 6, 2021, now U.S. Patent No. 11,769,903, which is a continuation of U.S. Patent Application No. 16/755,294, filed April 10, 2020, now U.S. Patent No. 10,998,536, which is a U.S. National Stage of PCT International Application No. PCT/US2018/055862, filed October 15, 2018, which claims the benefit of and priority to U.S. Provisional Patent Application No. 62/572,083, filed October 13, 2017, the entire contents of each of which are incorporated by reference herein.

FIELD OF THE INVENTION

[0002] The present invention relates generally to porous separators for batteries and other electric current producing cells. In particular, it pertains to multilayer porous separators comprising inorganic oxide/polymer layers on a polymeric layer, and to lithium batteries comprising such separators.

BACKGROUND OF THE INVENTION

[0003] Lithium batteries are widely used in portable electronics, such as smartphones and portable computers. Among the new applications for lithium batteries are high power batteries for hybrid, plug-in hybrid, and fully electric vehicles. Existing lithium batteries, including rechargeable and non-rechargeable lithium batteries, often utilize multilayer separators. Such separators may include an extruded polyethylene or polypropylene porous film that is coated with an inorganic oxide/polymer coating layer. As the demand for lower cost lithium batteries with increased size and energy density has grown, manufacturers have turned to less safe battery active materials in order to meet these goals. This has resulted in a need for battery separators with increased dimensional stability at temperatures such as 150°C, or higher.

[0004] Thus, it would be advantageous if the inorganic oxide/polymer coating layers could provide even greater dimensional stability at elevated temperatures while, at the same time, requiring lower coating weights and coating thicknesses, and providing lower manufacturing costs.

SUMMARY OF THE INVENTION

[0005] It is an object of the present invention to provide lighter weight, lower cost, and thinner inorganic oxide/polymer coating layers for polymer-based separators for lithium batteries and other electric current producing cells.

[0006] The present invention achieves these goals by providing a separator for a lithium battery comprising: (a) a porous polymeric layer, and (b) nanoporous inorganic oxide/polymer composite layers coated on both sides of the polymeric layer. In one embodiment, the nanoporous layer comprises an inorganic oxide (e.g., boehmite, SiO_2) and one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to about 45%, and wherein the crystallite size of the inorganic oxide particles is 5 nm to 90 nm. This combination of inorganic oxide particles (e.g., boehmite particles) having an extremely small crystallite size, and a high volume of organic polymer, produces a porous heat stable layer that provides increased dimensional stability at high temperature and a high level of porosity for excellent ionic conductivity of the electrolyte in the pores of the separator. This combination was further found to enable a significantly lower coating weight and coating thickness of the heat stable layer.

[0007] In another embodiment the nanoporous layer comprises: (a) a blend of inorganic oxide particles (e.g., boehmite particles) and inorganic nitride particles (e.g., boron nitride or aluminum nitride particles), and (b) one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to about 45%, and wherein the crystallite size of the inorganic particles is 5 nm to 90 nm. This combination of: (1) inorganic oxide and nitride particles (e.g., boehmite and boron nitride or aluminum nitride particles) having extremely small crystallite sizes, and (2) a high volume of organic polymer, produces a porous heat stable layer that provides increased dimensional stability at high temperature and a high level of porosity for excellent ionic conductivity of the electrolyte in the pores of the separator.

[0008] Preferably, the heat stable inorganic oxide/polymer layer (or inorganic oxide and nitride/polymer layer) is coated on both sides of the porous polymeric layer, such as a porous polyethylene layer or a porous polypropylene layer, as opposed to being on only one side of the polymeric layer. It has been found that this approach provides improved thermal dimensional stability in a thin and low cost separator. However, having the heat stable layer on only one side of the polymeric layer is also useful.

[0009] Another aspect of the separators for a lithium battery of this invention relates to a separator comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic oxide/polymer composite layer on one or both sides of the polymeric layer, wherein the nanoporous layer comprises an inorganic oxide (e.g., boehmite, SiO_2) and a polymer, and wherein the crystallite size of the inorganic oxide is 5 nm to 25 nm. It has been found that this smaller crystallite size results in a thinner coating layer with more cohesive strength and with more adhesion to the polymeric layer at an even lower coating weight than comparable layers where the crystallite size for the inorganic oxide is larger such as, for example, about 80 nm.

[0010] In another embodiment, the present invention includes lithium batteries that comprise the improved separators of this invention.

[0011] A further aspect of the present invention are methods for making separators that comprise coating a solution comprising an inorganic oxide and/or inorganic nitride, and one or more polymers, on one or both surfaces of a porous polymeric layer to form the separators of this invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The features and advantages of the present disclosure will be more fully understood with reference to the following, detailed description when taken in conjunction with the accompanying figures, wherein:

[0013] FIG. 1 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 1.

[0014] FIG. 2 is a graph of the shrinkage as a function of coating thickness of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 1.

[0015] FIG. 3 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0016] FIG. 4 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0017] FIG. 5 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0018] FIG. 6 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0019] FIG. 7 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0020] FIG. 8 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0021] FIG. 9 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0022] FIG. 10 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0023] FIG. 11 is a graph of the moisture content as function of the weight fraction of boron nitride in a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 4.

DETAILED DESCRIPTION

[0024] The separators of the present invention provide superior safety, lower cost, lower thickness, and other key performance properties for use in lithium batteries and in other electric current producing cells.

[0025] One aspect of this invention pertains to a separator for a lithium battery comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic oxide/polymer composite layer on both sides of the polymeric layer, wherein the nanoporous layer comprises boehmite and one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to 45%, and wherein the crystallite size of the boehmite is 5 nm to 90 nm. Alternatively, the nanoporous layer may further comprise other inorganic oxide (e.g., SiO₂) or nitride (e.g., BN or AlN) particles, or combinations thereof.

[0026] The term “lithium battery” may include, for example, batteries in which the electroactive anode material comprises lithium. This includes, but is not limited to, secondary or rechargeable lithium ion batteries, secondary lithium metal batteries, and primary or non-rechargeable lithium metal batteries.

[0027] The term “separator” may include, for example, porous materials in a battery that are interposed between the cathode and the anode of the battery to provide electrical insulation against short circuiting and to provide pores for filling with the electrolyte of the battery.

[0028] The term “porous polymeric layer” may include, for example, a layer of a polymer material that is porous such that it has permeability to the flow of air through the polymeric layer. The battery industry typically measures this air permeability using a porosimeter that provides a value for the time in seconds for 100 cubic centimeters (cc) of air to pass through the porous layer. This value is commonly called the Gurley number. The battery industry prefers that the Gurley number for a separator be as low as possible without sacrificing mechanical integrity. Typical Gurley numbers for the air permeability of porous polyethylene or porous polypropylene separators, as used in the battery industry, are 100 seconds/100 cc to 300 seconds/100 cc. For this invention, porous polymer layer does not include any of the non-woven polymeric layers, such as with polyester fibers, that are used as separators in the battery industry.

[0029] The term “nanoporous” may include layers having pores up to about 100 nm in diameter, preferably, with at least 90% of the pores being less than 100 nm in diameter. The size and relative number of pores can be measured by mercury porosimetry, or by scanning electron microscope (SEM) or transmission electron microscope (TEM) cross-sections of the layer.

[0030] As discussed herein, the nanoporous layer may be present on both sides of a substrate, support, layer, or coating. For example, the nanoporous layer may be present on both sides of the porous polymeric layer, such as a porous polyethylene layer. The general benefits of the nanoporous layers of this invention are also found when the nanoporous layer is present on only one side of the porous polymeric layer, but the overall performance is typically superior when the nanoporous layer is present on both sides of the porous polymeric layer.

[0031] The superior performance features of two-side coating of the polymeric layer, compared to one-side coating at the same total coating weight, includes lower shrinkage at

high temperatures and improved battery safety. Improved safety is typically the most important performance feature desired in separators for lithium batteries, especially as the batteries become larger and higher in energy density and use less safe battery active materials in order to achieve higher energy densities and lower costs.

[0032] Another advantage of two-side coating of the porous polymeric layer with the nanoporous coating layer is that it provides the opportunity to make the coatings on each side different in composition. This enables the nanoporous layer to be optimized for the best performance when it is facing the anode and when it is facing the cathode. In general, by tailoring the formulations and coating thicknesses of the ceramic coatings on either side of the separator, it is possible to optimize: (a) adhesion between separator and a specific electrode, either anode or cathode; (b) interfacial impedance between separator and a specific electrode, either anode or cathode; (c) complexation of migratory species (e.g., solubilized transition metal ions from cathode).

[0033] For example, when the coating is facing the cathode, it is particularly useful that the coating be resistant to oxidation, especially at higher charge voltages, e.g., above 4.25 volts. This can be done, for example, by selecting the organic polymer of the nanoporous layer to be resistant to oxidation up to the higher voltage desired, e.g., up to 5.0 volts. In addition, the nanoporous layer on one of the sides can be designed with particularly small pore sizes, such as under 10 nm diameter pores, for example, by using boehmite with a primary crystallite size of less than 10 nm in diameter. It should be understood that the present invention is not limited to the examples provided herein. For example, surface-modified alumina particles, including boehmite particles, as described in PCT/IB2015/000272 to Loscutova *et al.* of Sasol, may be utilized in the separator layers described herein. This patent application discloses the dispersion of these surface-modified aluminas into nano-sized single particle crystallites. This reference also describes that the average crystallite size can be measured by X-ray diffraction on a 120 plane. As an example, the crystallite size (or an average crystallite size) of the inorganic particles in a separator comprising a nanoporous layer according to an embodiment of the invention may be measured by: (i) taking a cross-section sample of the nanoporous layer; (ii) using a TEM or very high power SEM on this sample to measure the diameter/width, in their widest dimension, of 100 single primary (inorganic) particles selected at random in the TEM or SEM image; (iii) excluding the 25 highest and 25 lowest diameter numbers in the 100 diameter/width numbers; and (iv) recording the crystallite size as the average of the middle

50 diameter/width numbers, i.e., an average crystallite size of the measured particles. Since the pore size diameters of ceramic particles in a nanoporous layer usually approximate the crystallite size of the ceramic particles, such as boehmite particles, used to make the nanoporous layer, the pore size diameters may indicate the crystallite size of the ceramic particles for nanoporous layers containing ceramic particles of a narrow crystallite size. However, for blends of ceramic particles of different crystallite sizes in a nanoporous coating, the pore size diameters of the coating layer typically are intermediate, or a blend, among the crystallite size diameters of the different ceramic particles and not individual peaks reflecting the different crystallite sizes. The TEM or SEM method described above for measuring the crystallite size does not have this limitation since it is analyzing single particles, and not blends, and thus is a preferred method to use for this invention. These separator configurations using very small pore sizes are useful when placed against a lithium metal anode, where the very small size of the pores assists in preventing the growth of lithium metal dendrites into and through the separator, which reduces cycle life, capacity, and safety for the cell. Where very small boehmite particles are used and some of these boehmite particles penetrate into the pores of the polymeric separator substrate, this is particularly useful in preventing the growth of lithium metal dendrites in and through the separator.

[0034] In another example, boehmite could be modified by reaction with a fluorinated ethylene carbonate for improved cycle life when such nanoporous layer is positioned against an anode comprising silicon or lithium metal. This nanoporous layer with the boehmite covalently reacted with a fluorinated ethylene carbonate could also be used against the cathode to help achieve excellent stability under high voltage operation of the cell, such as at up to 5.0 volts.

[0035] In a further example, the nanoporous layer with pores that are less than 10 nm in diameter could be positioned against the cathode to inhibit the diffusion of transition metals, such as nickel ions, from the cathode to the anode, which degrades the cycle life and other performance properties of the cell. Optionally, the very small boehmite crystallite size of less than 10 nm could be combined with reaction of this boehmite with fluorinated ethylene carbonate or another organic carbonate to make a nanoporous layer positioned against the cathode to lessen the diffusion of transition metal ions from the cathode.

[0036] The term “Boehmite” is understood in the art. It, for example, may include hydrated aluminum oxide of the empirical chemical formula of $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ where x is 1.0 to 2.0. Another term used interchangeably herein for boehmite is AlOOH . Boehmite may be

characterized by its distinctive Fourier-transform infrared spectroscopy (“FTIR”) fingerprint. Boehmite may also be characterized by its distinctive X-ray diffraction pattern and by its distinctive infrared spectrum of double peaks of hydroxyl groups around 3200 cm^{-1} . Boehmite may have values of x for molecules of water that are above 1.5 and even up to about 2. However, the commercial grades of boehmite of all particle sizes typically have a value of x that is in the range of 1.0 to 1.5. The value of x can be measured by first drying the boehmite at about 120°C for 1 hour to remove the surface adsorbed water and then measuring the loss of water that occurs when the boehmite sample is heated to 1000°C for 1 hour and is thereby dehydrated to anhydrous aluminum oxide, Al_2O_3 .

[0037] Examples of polymers that may be utilized in the present invention include, but are not limited to, cellulosic polymers such as hydroxyethyl cellulose, vinyl pyrrolidone polymers and copolymers, polysaccharides such as chitosan, polyethylene oxides, and polyvinyl alcohols. Preferably, the polymers, alone or in combination, provide cohesive and adhesive strength to the nanoporous layer, are insoluble in the electrolyte of the lithium battery, and are electrochemically stable over the voltage range of use of the lithium battery. If needed to achieve these properties, one or more of the polymers may be crosslinked.

[0038] By the term “volume fraction of the one or more polymers in the solid portion of the nanoporous layer” is meant the fraction expressed as a percentage by volume of the polymer content of the solid portion of the inorganic particle/polymer composite layer. As described hereinafter, the nanoporous layer has a % porosity of between about 30% to about 70%. In other words, much of the volume of the nanoporous layer is air, and not a solid material. The air volume (e.g., pores) is not included in the solid portion for purposes of determining the volume fraction of the one or more polymers in the solid portion of the nanoporous layer. The volume fraction of the polymer and inorganic particle content of the nanoporous layer can be calculated based on the relative densities of the materials. For example, when the inorganic particles are only boehmite with a density of 3.03 g/cc , the organic polymer has a density of 1.3 g/cc , and the weight ratio of the boehmite to the organic polymer is 12:1, the volume fraction of boehmite would be 83.7% and the volume fraction of organic polymer would be 16.3%. Changing the weight ratio of the boehmite to the organic polymer to 4:1 results in a volume fraction of boehmite of 63.2% and of organic polymer of 36.8%. If the density of the organic polymer is lower, such as 1.1 g/cc , then its volume fraction at the same weight ratio of inorganic particle/polymer will be higher than at the higher polymer density of 1.3 g/cc . To determine the volume fraction of one or more organic

polymers in the solid portion of the nanoporous layer, for example, the nanoporous layer is removed from the separator, such as by careful scraping of the layer and not removing any of the polymer substrate. Next, a weighed nanoporous layer sample in a crucible is combusted at a sufficient temperature to combust the organic polymer to leave the residue of the ceramic particles. This provides the weight of combusted ceramic particles. When the particles are only boehmite, this weight of the combusted residue is divided by 0.85 in the case of the equivalent of one water molecule per boehmite molecule to obtain the weight of boehmite in the sample prior to the combustion. Although most boehmite particles have a 15% weight loss upon combustion, it should be noted that boehmite particles may have a different weight loss factor, such as 0.84 or 0.82. This same information can be obtained by using a thermogravimetric analyzer to record the weight of the nanoporous layer sample as it is heated to 900 °C or higher, until a constant weight is achieved. As noted above, boehmite has a density of 3.03 g/cc. Most organic polymers have a density in the range of 1.1 to 1.4 g/cc, although some polymers, such as polyvinylidene difluoride (PVdF), have a density of about 1.8 g/cc. After the density of the organic polymers is known by analyzing for the identity and weight ratios of the organic polymers, for boehmite-only nanoporous layers, the volume % of the organic polymers in the solid portion of the nanoporous layer can be calculated. For example, if the weight percent of boehmite is 88.9%, and the weight percent of the organic polymers is 11.1% and its density is 1.3 g/cc--with a pigment:binder ratio of 8:1 by weight--dividing the weight percents of boehmite and the organic polymers by their densities gives relative volume percents of 29.34% for boehmite and 8.54% for the organic polymers. Normalizing these volume percents to a total volume percent of 100% for the solid portion of the nanoporous layer gives a volume percent of 22.54% for the organic polymers. If the nanoporous layer is a blend of boehmite and other ceramic particles or if the density of the organic polymers is not known, further analysis, as known in the art, may be undertaken to measure the volume percent of the organic polymers.

[0039] The term “crystallite size” of the particles refers to the average size of the primary or single crystals of the inorganic particles. This is typically measured using X-ray diffraction, a transmission electron microscope (TEM), or a very high resolution scanning electron microscope (SEM). PCT/IB2015/000272 to Loscutova *et al.* of Sasol cited above provides examples of using X-ray diffraction to determine an average size of the crystals of the inorganic particles, such as alumina and boehmite. As described earlier, a TEM or SEM method may be used to determine the crystallite size. Many inorganic oxide particles, such

as boehmite particles, are nearly cubic in shape, although they can have other shapes such as platelets and needles. In the dry solid state and in solution, boehmite particles are typically agglomerated into larger particles comprised of many primary boehmite particles, such as, for example, 10 to 20 primary particles. The inorganic particles in the nanoporous layers of this invention result in features, such as pore size diameter, that are related to the crystallite size of the particles.

[0040] In one embodiment of the separators of the present invention, the ratio of the total thickness of the nanoporous layer on both sides of the polymeric layer to the total thickness of the separator is 10% to 80%. The thickness of these layers may be measured by SEM analysis of cross-sections of the separator. Because of the nanoporous pore sizes and the high volume fraction of organic polymer in the nanoporous layer, it has been found that a lower limit for the proportional thickness of the nanoporous layer of about 10% of the total thickness of the separator, rather than 15% or more typically 25%, is able to provide an excellent improvement in the dimensional stability of the separator at high temperatures, which is needed for improved safety. It has been found that the combination of nanoporous pore sizes and very high volume fractions of organic polymer in the nanoporous layer (e.g., from about 10% to about 50%) provides acceptable % porosity (e.g., from about 30% to about 70%) with good ionic conductivity of the electrolyte. It has also been found that the separator of this invention functions well with the total thickness of the nanoporous layer as high as 80% of the total thickness of the separator.

[0041] In one embodiment, the heat shrinkage of the separator at 150°C for 1 hour is less than 5%. This heat shrinkage may be measured by the change in the area of the separator after heating in an oven for 1 hour. This measurement of the heat shrinkage may be done by hanging a large sheet of the separator in the oven after marking and measuring the dimensions and area of a square or rectangle such as a 10 cm by 10 cm square, on a smaller section of the large sheet. After the heating in the oven, the marked area is measured for the % change or % shrinkage of the dimensions and the area.

[0042] In one embodiment, the heat shrinkage of the separator at 200°C for 1 hour is less than 5%. The nanoporous separators of the present invention have been found to retain their dimensional stability even when heated well above the melting point of the polymeric layer, such as a porous polyethylene layer.

[0043] In one embodiment of the separators of this invention, the total coating weight of the nanoporous layer on both sides is less than 4.5 grams per square meter (gsm), while

providing a heat shrinkage of the separator at 150°C for 1 hour of less than 5%. It should be noted that the use of boehmite with its lower density of about 3.03 g/cc, as compared to the higher density anhydrous aluminum oxide or alumina (which has a density of about 4 g/cc), helps to further lower the coating weight of the heat stabilizing layer. The lower density of the inorganic oxide particle proportionately lowers the weight of coating needed for a specific thickness and % porosity.

[0044] For example, a 4 micron thick alumina/polymer layer with a 45% porosity for polyethylene-based separators typically weighs about 8 grams per square meter (gsm). However, switching to boehmite (or a blend of boehmite and other inorganic particles of a similar or lower density) at the same thickness and % porosity lowers the coating weight to about 6 gsm or less. The high volume fraction of polymer in the nanoporous layers of the separators of this invention, combined with the use of boehmite with a crystallite size of 5 nm to 90 nm, provides increased dimensional stability for a specific thickness of the separator layer. Although it is possible to still make the total thickness of the inorganic particle/polymer layer in the separator 4 microns or higher, it is desirable for lower cost and for lower total thickness of the separator to make the thickness of the inorganic particle/polymer layer as thin as possible, such as about 3 microns total, while still achieving an excellent dimensional stability at high temperature. In one embodiment, the heat shrinkage of the separator at 200°C for 1 hour is less than 5%.

[0045] In one embodiment, other inorganic oxides and/or inorganic nitrides with a density of about 3.0 g/cc or lower, such as silicon dioxide and boron nitride, may be blended with the boehmite pigment. As discussed further below, this provides lower total coating weight of the nanoporous layer and other desirable properties, such as lower water content.

[0046] In another embodiment of the separators of this invention, the total coating weight of the nanoporous layer of both sides is less than 3.5 grams per square meter (gsm), while providing a heat shrinkage of the separator at 150°C for 1 hour of less than 5%. In one embodiment, the heat shrinkage of the separator at 200°C for 1 hour is less than 5%.

[0047] In one embodiment of the separators of the present invention, the crystallite size of the inorganic particles may be about 30 nm to about 50 nm, preferably about 5 nm to about 25 nm, and more preferably about 5 nm to about 10 nm. In one embodiment, the inorganic particles comprise boehmite particles having a crystallite size of about 30 nm to about 50 nm. In one embodiment, the crystallite size of the boehmite may be about 5 nm to about 10 nm, and preferably about 5 nm to about 8 nm.

[0048] In another embodiment of this invention, the % porosity of the nanoporous layer is between 30% and 55%. The % porosity of the nanoporous layer may be calculated from the density in g/cc of the materials in the dry coating assuming it is a 100% solid coating with 0% porosity and comparing it to the actual density of the coating. The actual density of the nanoporous layer may be calculated by dividing its coat weight in gsm by its coating thickness to provide the density in g/cc of the nanoporous layer. Dividing the density of the nanoporous layer by the calculated density of the same materials in a non-porous layer and multiplying by 100% provides the % solid fraction by volume in the nanoporous layer. Subtracting this value of the % solid fraction from 100% gives the % pore volume or % porosity of the nanoporous layer. The % porosity can also be measured by mercury porosimetry analysis.

[0049] In one embodiment, the pore volume of the nanoporous layer is between 0.8 cc/g and 1.0 cc/g. The pore volume can be calculated by measuring the coat weight of the nanoporous layer in gsm and measuring the thickness of the nanoporous layer to determine the total cc in a square meter. The pore volume in cc/g is then calculated by dividing the total cc in a square meter by the grams of the gsm measurement. In one embodiment, the % porosity of the nanoporous layer is 55% to 70%. Above this level of % porosity, it is difficult to obtain the cohesive and adhesive strength of the nanoporous layer that are preferred for long term, safer lithium battery separator applications.

[0050] In one embodiment of the separators of the present invention, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 15% to 45%. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 20% to 45%. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 25% to 45%.

[0051] In one embodiment of the separators of this invention, the nanoporous layer comprises pores having an average pore diameter of 80 nm or less and at least 70% of the pores have a diameter less than 80 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of 50 nm or less and at least 70% of the pores have a diameter less than 50 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of 25 nm or less and at least 70% of the pores have a diameter less than 25 nm.

[0052] In one embodiment of the separators of this invention, the ratio of the air permeability of the separator to the air permeability of the polymeric layer without any

nanoporous layer present is 1.0 to 1.4. As described hereinabove, this air permeability is typically measured on an instrument which provides a Gurley number of the number of seconds for 100 cc of air to pass through the separator. A low level of increase of the air permeability is an important indicator of an expected low decrease in the ionic conductivity of electrolyte in the separator due to adding the inorganic oxide/polymer layer. If the ionic conductivity decreases too much, this can interfere with the cycling rate capability of the lithium battery particularly its power rate properties, and also interfere with the cycling lifetime of the lithium battery.

[0053] In one embodiment of the separators of the present invention, the polymeric layer comprises a polyolefin. Suitable polyolefins include, but are not limited to, polyethylenes and polypropylenes. Other examples of the polyolefin polymers include, but are not limited to, homopolymers of 1-butene, 1-pentene, 1-hexene, and 1-octene, as well as copolymers and terpolymers thereof, used alone or in combination with one or more types of polymers. In one embodiment, the polymeric layer comprises polyethylene. The polymeric layer may be of the “wet process” type which is produced by extrusion followed by solvent extraction to provide the porosity or of the “dry process” type which is produced by extrusion followed by stretching to provide the porosity. Various molecular weights, densities, and stereochemistries of polyethylene and the other polymers, with various melt flow and mechanical properties, may be utilized in the porous polymeric layers.

[0054] For the purposes of better uniform wetting of the coating fluid of the nanoporous layer and of better adhesion of the nanoporous layer to the substrate, the porous polymeric layer may have a coating layer on the surface to which the nanoporous layer is applied.

[0055] In one embodiment of the separators comprising polyethylene in the polymeric layer, the heat shrinkage of the separator at 120°C for 1 hour is less than 1.0%. Without the nanoporous layer of the separators of this invention, the heat shrinkage of a polyethylene separator of comparable total thickness is typically greater than 1.0% at 105°C. In one embodiment, the ratio of the air permeability of the separator after the heat shrinkage at 120°C for 1 hour to the air permeability of the separator before the heat shrinkage at 120°C is 0.8 to 1.2. This is an important feature for allowing vacuum drying of the polyethylene-based separators of this invention at a higher temperature of 120°C without any negative effect on the air permeability and ionic conductivity and, in some cases, with a positive effect on ionic conductivity due to the heat treatment. Vacuum drying is beneficial for lowering the moisture

content of separators for increased lithium battery cycling lifetimes and capacities and for higher lithium battery cycling rate capabilities. Vacuum drying is typically done for 4 to 24 hours at about 80°C for polyethylene separators and at about 90°C for polyethylene separators with one or more inorganic oxide/polymer layers. The capability of using a higher temperature of 120°C and a shorter drying time of 1 to 3 hours for the polyethylene-based separators of this invention is useful.

[0056] In another embodiment of the separators of the present invention, the nanoporous layer is a laminated layer on the polymeric layer. By the term “laminated layer” is meant that the nanoporous layer is not initially formed on the polymeric layer, such as by direct coating, but is instead laminated to the polymeric layer in a subsequent step. For example, the nanoporous layer is coated on a release substrate, and then this nanoporous layer is transferred by lamination onto the polymeric layer. This lamination may optionally include the use of heat and/or the use of some organic solvent to adhere the nanoporous layer to the polymeric layer and to aid in the subsequent delamination of the release substrate without interfering with the quality of the laminated multilayer separator.

[0057] Another aspect of the separators for a lithium battery of this invention relates to a separator comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic particle/polymer composite layer on one or both sides of the polymeric layer, wherein the nanoporous layer comprises an inorganic oxide (e.g., boehmite), or a blend of an inorganic oxide and an inorganic nitride (e.g., BN or AlN), and a polymer, and wherein the crystallite size of the inorganic particles is about 5 nm to about 25 nm, and preferably about 5 nm to about 8 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of about 25 nm or less and at least about 70% of the pores have a diameter of about 25 nm or less.

[0058] A further aspect of the separators for a lithium battery of the present invention pertains to a separator comprising (a) a porous polyethylene layer, and (b) a nanoporous inorganic particles/polymer composite layer on both sides of the polyethylene layer, wherein the nanoporous layer comprises boehmite, or a blend of boehmite and boron nitride, and one or more polymers, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is about 10% to about 50%, the total coating weight of the nanoporous layer on both sides is 5.5 grams per square meter or less, and the heat shrinkage of the separator at 150°C for 1 hour is less than 5%. In one embodiment, the total coating weight of the

nanoporous layer on both sides is 3.5 grams per square meter or less and the thickness of the polyethylene layer is between 5 microns and 16 microns.

[0059] In another embodiment, this invention is lithium batteries comprising a separator of the present invention.

[0060] Another aspect of the present invention are methods for making the separator of this invention, which method comprises coating a solution comprising inorganic particles and one or more polymers on one or both surfaces of a porous polymeric layer. In one embodiment, the inorganic particles comprise boehmite particles having a crystallite size of 5 nm to 90 nm. In one embodiment, the boehmite particles have a crystallite size of about 5 nm to about 25 nm, and preferably about 5 nm to about 8 nm. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the coating after drying is about 10% to about 50%, preferably about 12% to about 45 %, and more preferably about 15% to about 45%.

[0061] EXAMPLES

[0062] Several embodiments of the present invention are described in the following examples, which are offered by way of illustration and not by way of limitation.

[0063] Example 1

[0064] A boehmite slurry was prepared by adding 12 parts of boehmite (DISPAL 10F4 supplied by Sasol Inc., Houston, Texas) and 1 part of chitosan (deacetylated chitin, poly(D-glucosamine), medium molecular weight, of a polysaccharide supplied by Sigma-Aldrich, St. Louis, MO) to 73 parts of 9:1 water:isopropyl alcohol and stirring the slurry. The primary particle size of the boehmite was about 40 nm. The boehmite slurry was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of coating weights for both sides from about 1.2 grams per square meter (gsm) to about 5.2 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. The boehmite coating was then applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0065] The 12:1 by weight ratio of boehmite pigment to chitosan binder is calculated to have 15.3% by volume of chitosan binder in the solid portion of the porous boehmite

coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.4 g/cc for chitosan.

[0066] The coating thickness was measured using a Dorsey gauge on the polyethylene separator before and after the coating of the boehmite coating on both sides. The difference in the thickness values before and after the coating is the coating thickness in microns. An alternative method for measuring the coating thickness is using the Dorsey gauge on the coated separator before and after removing the boehmite coating on both sides, either by pulling off the coating with an adhesive tape or by rubbing off the coating with an alcohol-soaked cloth.

[0067] The coating weight was measured by weighing a 10 cm x 10 cm sample of the polyethylene separator before and after the coating of the boehmite coating on both sides. The difference in the weights before and after the coating multiplied by 1,000 is the coat weight in grams per square meter (gsm). An alternative method for measuring the coating weight was weighing a 10 cm x 10 cm sample of the coated separator before and after removing the boehmite coating on both sides, either by pulling off the coating with an adhesive tape or by rubbing off the coating with an alcohol-soaked cloth.

[0068] Figure 1 shows a plot of the average boehmite coating thickness and coating weight per side of about 0.6 gsm to about 2.6 gsm. For two sided coatings, these numbers are doubled to give a total coating weight range of about 1.2 gsm to about 5.2 gsm. The total boehmite coating thickness for both sides varies from about 1.2 microns to about 4.4 microns. If a more accurate measurement of the coating thickness is desired, scanning electron microscope (SEM) analysis of cross-sections of the boehmite-coated plastic separator can be utilized.

[0069] Using a SEM analysis for the total coating thickness showed a density of about 1.06 gsm per micron of coating thickness. From this, the % porosity for this 12:1 pigment to binder ratio is estimated as 62%.

[0070] One objective for minimizing the cost of the boehmite coating is to apply the total weight of coating that is needed to reduce the heat shrinkage to the target value, such as less than 5% shrinkage at 150°C. Figure 2 shows that this level of heat shrinkage is obtained with an average coating thickness per side of about 1.9 microns or thicker on the 12 micron polyethylene substrate of this example. For coatings on two sides, this number is doubled to give a total coating thickness of about 3.8 microns or thicker. Based on the density of the

boehmite coating determined by SEM analysis, a total coating thickness of about 3.8 microns on both sides equates to about 4.0 gsm of coating weight.

[0071] For better wetting and coating adhesion of the boehmite slurry, a corona treatment of the polyethylene substrate is preferred. Besides wire wound coating rods, a variety of other coating application methods may be used, such as, but not limited to, gravure coating, screen printer coating, reverse roll coating, blade coating, and slot die coating.

[0072] Example 2

[0073] A boehmite slurry was prepared by adding 12 parts of boehmite (DISPERAL D60 supplied by Sasol Inc., Hamburg, Germany) and 1 part of a vinyl pyrrolidone copolymer (Soteras CCS grade supplied by Ashland Inc., Wilmington, DE) to 73 parts of 9:1 water:isopropyl alcohol and stirring the slurry. The primary particle size of the boehmite was about 75 nm. The boehmite slurry was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of coating weights for both sides from about 2.4 grams per square meter (gsm) to about 9.2 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. Then, the boehmite coating was applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0074] The 12:1 by weight ratio of boehmite pigment to vinyl pyrrolidone copolymer binder is calculated to have 18.7% by volume of polymer binder in the solid portion of the porous boehmite coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.1 g/cc for the vinyl pyrrolidone copolymer.

[0075] Figure 3 shows a plot of the average boehmite coating thickness and coating weight per side of about 1.2 gsm to about 4.6 gsm. For two sided coating, these numbers are doubled to give a total coating weight range of about 2.4 gsm to about 9.2 gsm. The total boehmite coating thickness for both sides varies from about 2.8 microns to about 9.0 microns.

[0076] As discussed, one objective for minimizing the cost of the boehmite coating is to apply the total weight of coating that is needed to reduce the heat shrinkage to the target value, such as less than 5% shrinkage at 150°C for 1 hour. Figure 4 shows that this level of heat shrinkage is obtained with an average coating thickness per side of about 2.0 microns or thicker on the 12 micron polyethylene substrate of this example. For the coatings on two sides, this number is doubled to give a total coating thickness of about 4.0 microns or thicker.

Based on Figure 3, a total coating thickness of about 4.0 microns on both sides equates to about 3.9 gsm of coating weight.

[0077] A boehmite slurry of an 8 to 1 ratio by weight of the DISPERAL D60 to vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol at 15% solids was also coated using wire wound rods on the 12 micron polyethylene substrate and dried, as for the 12:1 boehmite D60:polymer coating in this example. It was calculated to have a volume percent of polymer binder of 25.6%. Figure 5 shows a plot of the average boehmite coating weight and coating thickness per side of about 0.8 gsm to about 4.2 gsm. For the coatings on two sides for this boehmite coating with a 8:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a range of about 1.6 gsm to 8.4 gsm. The corresponding total boehmite coating thicknesses for both sides varies from about 1.6 microns to 8.2 microns. Figure 6 shows that a less than 5% shrinkage at 150°C for 1 hour is obtained with an average coating thickness per side of about 2.2 microns or thicker. For the coatings on two sides, this number is doubled to give a total coating thickness on both sides of about 4.4 microns or thicker. Based on Figure 5, a total coating thickness of about 4.4 microns equates to about 4.8 gsm of coating weight.

[0078] Example 3

[0079] Using the procedure of Example 2, a boehmite slurry of a 12:1 by weight of DISPAL 25F4 boehmite and the vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol at 15% solids was prepared. The DISPAL 25F4 is supplied by Sasol, Lake Charles, LA, and has a primary crystallite particle size of 8 nm. This boehmite slurry was coated on the EP12 12 micron polyethylene substrate from ENTEK and dried, similarly to the process used in Examples 1 and 2. It was calculated to have a volume percent of polymer binder of 18.7%. Figure 7 shows a plot of the average boehmite coating thickness and the coating weight per side of about 0.8 gsm to about 3.0 gsm. For the coatings on two sides for this boehmite coating with a 12:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a total coating weight range of about 1.6 gsm to about 6.0 gsm. Figure 8 shows that a less than 5% shrinkage at 150°C for 1 hour is obtained with an average coating thickness per side of about 1.8 microns or thicker. For two sided coatings, this number is doubled to give a total coating thickness on both sides of about 3.6 microns or thicker. Based on Figure 7, a total coating thickness of about 3.6 microns equates to about 3.6 gsm.

[0080] A boehmite slurry of an 8 to 1 ratio by weight of the DISPAL 25F4 to vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol was also coated on the 12 micron polyethylene substrate and dried, similarly to the process used in Examples 1 and 2. It was calculated to have a volume percent of polymer binder of 25.6%. Figure 9 shows a plot of the average boehmite coating thickness and coating weight per side of about 0.8 gsm to about 4.7 gsm. For the coatings on two sides for this boehmite coating with a 8:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a total coating weight range of about 1.6 gsm to 9.4 gsm. Figure 10 shows that a less than 5% shrinkage at 150°C for 1 hour is obtained with an average coating thickness per side of about 0.9 microns or thicker. For the coatings on two sides, this number is doubled to give a total coating thickness on both sides of about 1.8 microns or thicker. Based on Figure 9, a total coating thickness of about 1.8 microns equates to about 2.9 gsm of coating weight. For a thickness of about 1.8 microns of the boehmite coating, the coating weight should have been about 2.0 gsm based on the density of the boehmite coating of about 1.1 g/cc shown in Example 1, while factoring in the lower boehmite weight % of the 8:1 boehmite pigment:vinyl pyrrolidone copolymer and the higher vinyl pyrrolidone copolymer weight % and lower density of the vinyl pyrrolidone copolymer compared to chitosan. This shows that about 0.9 gsm of the boehmite coating penetrated into and dried in the pores of the polyethylene substrate to produce a boehmite/polyethylene substrate with a boehmite coating on its surface. At the 8:1 pigment:binder (P:B) ratio of this coating, the about 0.9 gsm of boehmite coating in the pores equates to about 0.8 gsm of boehmite particles in the pores of the boehmite/polyethylene substrate or layer. Since this boehmite coating was applied on both sides of the polyethylene substrate, about one-half, or about 0.4 gsm of boehmite, would have penetrated and dried in the pores on each side of the polyethylene substrate to form the boehmite/polyethylene substrate or layer that resulted in a boehmite-coated boehmite/polyethylene substrate.

[0081] A similar 8:1 boehmite coating with Dispall 25F4, with chitosan substituted for the vinyl pyrrolidone copolymer and a 12 micron thick polyethylene substrate from Shenzhen Senior, Shenzhen, China, used instead of the polyethylene substrate from Entek, was made using the procedure of this Example to give a coat weight of 1.8 gsm. This coating had a heat shrinkage of less than 5% when heated at 150°C for 1 hour. When the procedure of removing the boehmite coating with an adhesive tape was used, even with very careful tape pulls, it was not possible to remove the boehmite coating from the 10 x 10 cm samples without pulling off

some of the polyethylene substrate. This was in contrast to the tape pull results of easy and complete removal of the boehmite coatings from the polyethylene substrate for Examples 1 and 2 and Comparative Examples 1 and 2. This result was consistent with some of the boehmite coating penetrating into the pores of the polyethylene layer and, upon drying, providing increased adhesive and cohesive strength to the boehmite coating such that the point of failure in the tape pull was below the surface of the polyethylene layer at most areas, rather than at the interface of the polyethylene surface and the boehmite coating on the polyethylene surface. After the tape pull, some areas were found where the polyethylene layer had not been disturbed. From these areas, the added thickness of the boehmite layer was found to be about 1.8 microns.

[0082] Since the tape pulls were not able to cleanly remove the boehmite coating from the polyethylene layer, a methanol-saturated soft cloth was used to wash and rub the coating off. The 10 x 10 cm sample after washing had a weight of 0.0757 grams (g). Two more methanol rubs did not change this weight of the washed and rubbed sample. The polyethylene substrate used in this Example contained no inorganic pigment, such as alumina or boehmite. A furnace combustion of the methanol-washed sample in a crucible, at a temperature of 1000 °C for 70 minutes showed 0.0045 g of anhydrous aluminum oxide after dehydrating the boehmite pigment with a weight loss of 15% of the boehmite pigment weight. Using this boehmite to aluminum oxide conversion factor by dividing the aluminum oxide weight by 0.85, this gives a boehmite content of about 0.0053, or at least about 0.53 gsm of boehmite pigment in the pores of the polyethylene layer. Since the wet rubbing to remove the boehmite coating on the surface may have removed a small amount of the boehmite coating in the upper pores of the polyethylene layer, it is possible that the weight of boehmite pigment in the pores was somewhat greater than 0.53 gsm. Since there were two coatings on the polyethylene layer, the amount of boehmite in the pores from a single coating is about one-half of this total, or about 0.27 gsm. As further evidence for penetration of the boehmite coating into the pores of the polyethylene substrate, the dimensional stability of this “washed” polyethylene with about 0.53 gsm of boehmite in the pores after heating at 150°C for 1 hour was 54% and 36.5% in the two directions(--machine direction (MD) and cross direction (CD)). By contrast, the polyethylene substrate before coating showed heat shrinkage of 70% and 75.5%. This increased heat stability with boehmite pigment in the pores of the polyethylene substrate is one explanation for the greater dimensional stability at 150°C with a much lower coat weight and coating thickness added to the surface of the

polyethylene layer. For analyzing the presence and amount of boehmite particles in the pores of the polymeric layer, the removal of the surface coating of the nanoporous layer and subsequent combustion of all of the organic polymer, as described above, in a combustion and/or thermogravimetric method may be used for the case of only boehmite particles in the nanoporous layer and a polymer layer that contained no inorganic or boehmite particles before the application of the boehmite coating. Another method is to use the TEM or SEM method used for measuring the average crystallite size to make images of the presence of boehmite particles in the pores of the polymeric layer and to determine their crystallite size at the same time. Combining this method with the combustion and/or thermogravimetric method above to quantify the weight of the boehmite, or other ceramic particles in the pores, if any, would provide the weight in gsm of boehmite particles in the separator. The known analytical techniques for boehmite particles, such as FTIR, X-ray diffraction, thermogravimetric analysis (TGA), high resolution photoelectron spectroscopy (ESCA), and ^{27}Al nuclear magnetic resonance (nmr) can be utilized to further confirm the presence and quantity of boehmite particles in the pores of the polymeric layer, if needed. An alternative approach for collecting the ceramic particles, such as boehmite particles, in the pores of the polymeric layer, after removing any surface coating from the polymeric layer, would be to do a Soxhlet extraction of the polymeric layer with a hot solvent that dissolves the polymer from the polymeric layer to isolate the insoluble inorganic particles. Besides providing the weight of inorganic particles, this could be combined with the above analytical methods, combustion and/or thermogravimetric analysis, and TEM or SEM method, to confirm the presence and quantity of boehmite particles. One benefit of this penetration of nano-sized boehmite particles would be the ability to make and sell boehmite/polyethylene substrates, with no or minimal surface boehmite or other coating, for various applications, including for manufacturing CCS, where the boehmite/polyethylene substrate would have multiple advantages over polyethylene substrates, including, but not limited to, a much greater dimensional stability at high heats, a modified and smaller pore size distribution, and other benefits, such as ease of wetting by subsequent coatings. One example of a particularly useful application for boehmite/polymeric substrates would be filtration applications, where the very small nano-sized boehmite particles, with and without various surface treatments, in the pores of the polymer could provide the desired nanofiltration and ultrafiltration properties not available from the polymeric substrate without boehmite particles in its pores, and could be adapted for desalination of sea water. This benefit would also be available for other

boehmite/polymeric substrates besides polyethylene, such as for boehmite/polypropylene and boehmite/other polyolefin substrates.

[0083] Example 4

[0084] The procedure of Example 1 was used to make a nanoporous separator comprising a 50:50 blend by weight of boehmite and boron nitride pigments, except that a 12 micron thick porous polyethylene separator from Toray Industries, Tokyo, Japan, was substituted for the Entek polyethylene separator, and the vinyl pyrrolidone polymer, Soteras CCS, was substituted for the chitosan. The boron nitride pigment was NX1, supplied by Momentive Performance Materials. The total coating weight on both sides was 2.5 gsm. Figure 11 shows the reduced water content of this separator compared to the control with only boehmite pigment and no boron nitride. Specifically, as shown in Figure 11, at a 50:50 by weight ratio of boehmite and boron nitride, the water content at 20% RH was reduced by one-third from its value with boehmite pigment only.

[0085] Comparative Example 1

[0086] An alumina mix was prepared in water by adding 20 parts of alumina pigment (HJA-0719 supplied by AnHui Estone Materials Technology Co., Ltd., Anhui Province, China) and 1 part of a hydroxypropyl cellulose polymer (Klucel H grade supplied by Ashland Inc., Wilmington, DE) to 63 parts of water and stirring the slurry. This 20:1 weight ratio of pigment to binder provided a similar Gurley air permeability and similar ionic conductivity in electrolyte as the 8:1 weight ratio of pigment to binder in Example 2. Reducing the 20:1 weight ratio had a progressively negative effect with higher Gurley air permeability and lower ionic conductivity. The primary particle size of the alumina was greater than 400 nm. The alumina mix was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of total coating weights for both sides from about 0.8 grams per square meter (gsm) to about 4.4 gsm. The alumina coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. Then, the alumina coating was applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0087] The 20:1 by weight ratio of alumina pigment to hydroxypropyl cellulose polymer binder is calculated to have 13.6% by volume of polymer binder in the solid portion of the porous alumina coating layer. This calculation is based on a specific gravity of 4.0 g/cc for alumina and of 1.27 g/cc for the hydroxypropyl cellulose polymer. At a total coating

weight of about 4.4 gsm, this alumina coating had a shrinkage of about 60% at 150°C for 1 hour.

[0088] Comparative Example 2

[0089] A boehmite mix was prepared in water by adding 20 parts of boehmite pigment (BG-613 supplied by AnHui Estone Materials Technology Co., Ltd., Anhui Province, China) and 1 part of a hydroxypropyl cellulose polymer (Klucel H grade supplied by Ashland Inc., Wilmington, DE) to 63 parts of water and stirring the slurry. This 20:1 weight ratio of pigment to binder provided a similar Gurley air permeability and similar ionic conductivity in electrolyte as the 8:1 weight ratio of pigment to binder in Example 2. Reducing the 20:1 weight ratio had a progressively negative effect with higher Gurley air permeability and lower ionic conductivity. The primary particle size of the boehmite was greater than 400 nm. The boehmite mix was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of total coating weights for both sides from about 0.8 grams per square meter (gsm) to about 7.0 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. Then, the alumina coating was applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0090] The 20:1 by weight ratio of boehmite pigment to hydroxypropyl cellulose polymer binder is calculated to have 10.7% by volume of polymer binder in the solid portion of the porous boehmite coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.27 g/cc for the hydroxypropyl cellulose polymer. At a total coating weight of about 7.0 gsm, this boehmite coating had a shrinkage of about 60% at 150°C for 1 hour.

[0091] Table 1 shows a summary of the data from Examples 1 to 3.

Table 1

Example	Boehmite Type	Primary Particle Size (nm)	Pigment:Polymer Binder Weight Ratio	Volume % Polymer	Total Coat Weight for <5% Shrinkage at 150°C for 1 hour
1	10F4	40	12:1	15.3%	≥4.0 gsm
2	D60	75	12:1	18.7%	≥3.9 gsm
2	D60	75	8:1	25.6%	≥4.8 gsm

3	25F4	8	12:1	18.7%	≥ 3.6 gsm
3	25F4	8	8:1	25.6%	≥ 2.9 gsm

[0092] One challenge in using nano-sized ceramic particles, such as nano-sized boehmite particles, is their increased tendency to absorb water because of their much larger surface areas compared to micro-sized ceramic particles, such as micro-sized alumina and micro-sized boehmite particles. For lithium ion cells, lithium metal cells, and other electrochemical cells with sensitivity to water content, the separator needs to have a low water content, such as under 600 ppm of water in the separator, preferably less than 300 ppm of water, and more preferably under 100 ppm of water as typically found in porous polymer only separators with no ceramic materials in them. Although vacuum drying of the separator at up to 80 °C can help to lower this moisture content in separators with nano-sized ceramic particles, and the separators of this invention have increased thermal stability that permits even higher vacuum drying temperatures of up to 90 °C to about 110 °C, some users of separators may prefer not to perform this extra step of vacuum drying. It would be preferable if the greater heat stability, lower cost from the use of a lower amount of ceramic coating materials, and lower ceramic coating thickness, including from the penetration of the ceramic coating and ceramic particles into the porous polymeric separator layer, of the separators of this invention would include a lower moisture content in the separators from the use of more hydrophobic ceramic particles and organic polymers. Having a lower amount of ceramic particles from the lower coating weights of the invention is a significant benefit for lowering the moisture content of the separator, but it would be further useful if the ceramic particles were more hydrophobic and less prone to moisture pickup.

[0093] One approach which provides this lower moisture absorption is to treat the surface of the ceramic particles with a hydrophobic material, such as by using hydrophobically-treated boehmite particles, particularly if the hydrophobic material is chemically reacted with the boehmite particles and thus is permanently bonded to the boehmite particles. Suitable hydrophobic materials for chemical modification of boehmite pigments include, but are not limited to, organic sulfonic acids, organic carbonates, and polymers with hydrophobic groups and with reactive groups, such as carboxylic acid and hydroxyl groups, that can be made to react with hydroxyl groups of the boehmite or other ceramic particles.

[0094] Examples of the organic sulfonic acid-treated boehmite particles include Dispal® 25SR, available from Sasol in Lake Charles, LA, a boehmite pigment chemically reacted with

p-toluene sulfonic acid through the sulfonic acid groups. The reaction leaves the hydrophobic p-toluene group on the outer surface of the boehmite particle to make it less prone to pick up water. An even more hydrophobic example from Sasol is p-dodecylbenzene sulfonic acid surface reacted to boehmite particles through the sulfonic acid group and having the very hydrophobic dodecyl group on the outer surface of the boehmite pigment. This makes the boehmite particles so hydrophobic that this treated boehmite can be dispersed in hydrocarbon solvents. The use of boehmite particles surface treated with organic sulfonic acids in boehmite coatings for separators is disclosed in U.S. Patent Application Publication No. 2013/0171500 by Xu et al.

[0095] Examples of the organic carbonate-treated boehmite particles include, but are not limited to, those with ethylene carbonate and fluoroethylene carbonate as the organic carbonates. Besides the option of choosing an organic carbonate that is very hydrophobic in the part of the molecule that will face away from the surface of the boehmite, the choice can also be made with a view to picking an organic carbonate which will provide increased ionic conductivity to the separator. The use of boehmite particles surface treated with ethylene carbonate to provide separators with increased conductivity is described in U.S. Pat. No. 8,883,354 to Carlson et al.

[0096] The use of boehmite particles surface treated with organic polymers to provide separators is described in U.S. Pat. No. 9,871,239, to Carlson et al. As the primary crystallite size and resulting pore size become smaller in the boehmite separators and the ionic conductivity of the separator becomes lower because of the smaller pore sizes, it is useful to have surface treatments, such as the three types described above, that can be utilized to increase the ionic conductivity of the separator, as well as to decrease the water content of the separator.

[0097] The preparation of a ceramic mix or slurry with the surface-treated hydrophobic materials can be done conveniently by adding a precursor, such as an organic carbonate, to the slurry containing the untreated boehmite particles and heating the slurry to a temperature, such as 80 °C, for a sufficient time, such as 1 hour, to react the surface treatment material, such as an organic carbonate, with the boehmite particles to the desired level of surface treatment. There is no need to isolate the surface-treated boehmite particles by a drying process, but rather the slurry can be used as is, usually after some dilution with water and/or solvents to the desired % solids level for the coating of the porous polymer substrate. If it is desired to further reduce the particle size of the ceramic particles, such as boehmite particles,

and/or to homogenize the slurry for more uniformity and perhaps a lower viscosity at the desired % solids level for coating, the slurry, before and/or after any surface treatment, can be further mixed in an impingement mill or similar milling device. Examples of the use of an impingement mill for reducing and homogenizing coating mixes containing pigments is described in U.S. Pat. Nos. 5,210,114 and 5,292,588, to Katsen.

[0098] While the invention has been described in detail and with reference to specific and general embodiments thereof, it will be apparent to one skilled in the art that various changes and modifications can be made therein without departing from the spirit and scope thereof.

CLAIMS

What is claimed:

1. A separator for a lithium battery comprising:
a porous polymeric layer having a first side and a second side;
a first layer adjacent the first side of the porous polymeric layer, said first layer comprising a porous portion and a solid portion, wherein
 - (a) the solid portion comprises inorganic oxide particles and one or more polymers;
 - (b) a plurality of the inorganic oxide particles in the first layer having a crystallite size of about 5 nm to about 90 nm;
 - (c) the thickness of the first layer is about 3 microns or less and the total coating weight of the first layer is about 0.8 gsm to about 4.7 gsm; and
 - (d) the ratio of the total thickness of the first layer to the total thickness of the separator is 10% to 80%.
2. The separator of claim 1, wherein a plurality of pores in the first layer have a diameter of about 100 nm or less when measured by an SEM or TEM cross section of the porous layer.
3. The separator of claim 1, wherein the crystallite size of the inorganic oxide particle comprises about 5-25 nm, 5-80 nm, 30 nm-50 nm, 5-8 nm, or 5-75 nm.
4. The separator of claim 1, wherein the inorganic oxide particles of the first layer comprise a hydrated Al_2O_3 .

5. The separator of claim 1, wherein the inorganic oxide particles of the first layer comprise one or more of boehmite, SiO_2 , boron nitride pigments, AlN , hydrated aluminum oxide, AlOOH , and $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$.

6. The separator of claim 5, wherein when the inorganic oxide particles of the first layer comprise $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$, x is about 1.0 to 2.0.

7. The separator of claim 1, wherein the at least one polymer in the first layer comprises one or more of the following: cellulosic polymers, hydroxyethyl cellulose, vinyl pyrrolidone polymers and copolymers, polysaccharides, chitosan, polyethylene oxides, polyvinyl alcohols, polyolefin (e.g., polyethylene, polypropylene, homopolymers of 1-butene, 1-pentene, 1-hexene, and 1-octene, as well as copolymers and terpolymers thereof.

8. The separator of claim 1, wherein the volume fraction is about 12% to about 45%, 15% to about 45%, 20% to about 45%, about 25% to about 45%, or about 18% to about 45%.

9. The separator of claim 1, wherein the porous portion of the first layer has porosity of between about 30% to about 70%.

10. The separator of claim 1, further comprising a second layer adjacent to the second side of the porous polymeric layer, said second layer comprising inorganic oxide particles and one or more polymers.

11. The separator of claim 10, wherein the combined total coating weight of the first layer and second layer is less than about 9.4 gsm or less than about 8.4 gsm.

12. The separator of claim 1, wherein the total coating weight of the first layer is about 0.8 gsm to about 3.0 gsm.

13. The separator of Claim 10, wherein the composition of the first layer is different in composition from the composition of the second layer.

14. The separator of Claim 1, wherein the ratio of inorganic oxide to polymer in the first layer is 12:1 or 8:1.

15. The separator of Claim 10, wherein the combined total thickness of the first layer and second layer is less than about 3.8 microns.

16. The separator of Claim 1, wherein the pore volume of the first layer is between 0.8 cc/g and 1.0 cc/g.

17. The separator of Claim 1, wherein the porous polymeric layer comprises a polyolefin.

18. The separator of Claim 1, wherein the polyolefin is selected from the group consisting of polyethylenes, polypropylenes, homopolymers of 1-butene,

homopolymers of 1-pentene, homopolymers of 1-hexene, homopolymers of 1-octene, and copolymers and terpolymers thereof.

19. The separator of claim 9, wherein the porous portion of the first layer has porosity of between about 30% - 55%.

ABSTRACT

A separator for a lithium battery having (a) a porous polymeric layer, such as a polyethylene layer; and (b) a nanoporous inorganic particle/polymer layer on both sides of the polymeric layer, the nanoporous layer having an inorganic oxide and one or more polymers; the volume fraction of the polymers in the nanoporous layer is about 15% to about 50%, and the crystallite size of the inorganic oxide is 5 nm to 90 nm.

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Filing Date	Herewith
First Named Inventor	David W. Avison
Title	MULTILAYER NANOPOROUS SEPARATOR
Art Unit	
Examiner Name	
Attorney Docket Number	00249/0086

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Signature	/Brian J. Amos/	Date (Optional)	January 16, 2025
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	9	8318360	B2	2012-11-27	Seo et al.	
	10	9484565	B2	2016-11-01	Miyazaki et al.	
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	3	20180019457	A1	2018-01-18	Alharizah et al.	(U.S. counterpart of WO 2016/123404 A1 listed below)

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	2	10-2016-0072009	KR	A	2016-06-22	Samsung SDI Co., Ltd.		☒
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	1	F. Sola et al., "A new scanning electron microscopy approach to image aerogels at the nanoscale; A new scanning electron microscopy approach to image aerogels at the nanoscale," Nanotechnology, IOP, BRISTOL, GB, Vol. 22, No. 17, March 16, 2011, page 175704, XP020188888, ISSN: 0957-4484, DOI: 10.1088/0957-4484/22/17/175704.	☐
	2	Chongwen Yang et al., "Boehmite particle coating modified microporous polyethylene membrane: A promising separator for lithium ion batteries," Journal of Power Sources, Vol. 348, March 3, 2017, pp. 80-86, XP029949881, ISSN: 0378-7753, DOI: 10.1016/J.POWSOUR.2017.02.078.	☐
	3	International Search Report issued in PCT Intl. Patent Application No. PCT/US2018/055862, mailed on April 11, 2019, 6 pages.	☐
	4	Written Opinion issued in PCT Intl. Patent Application No. PCT/US2018/055862, mailed on April 11, 2019, 13 pages.	☐
	5	Chinese Office Action dated April 12, 2022 issued in Chinese Patent Application No. 201880079213.4.	☐

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	David W. Avison	
Art Unit		
Examiner Name		
Attorney Docket Number	00249/0086	

	6	Communication Pursuant to Article 94(3) EPC dated September 8, 2023 issued in European Patent Application No. 18 799 922.2.	☐
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Signature	/Brian J. Amos/	Date (YYYY-MM-DD)	2025-01-16
Name/Print	Brian J. Amos	Registration Number	63,679

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MULTILAYER NANOPOROUS SEPARATOR

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. Patent Application No. 18/631,783, filed April 10, 2024, which is a continuation of U.S. Patent Application No. 17/543,259, filed December 6, 2021, now U.S. Patent No. 11,973,181, which is a continuation of U.S. Patent Application No. 17/223,845, filed April 6, 2021, now U.S. Patent No. 11,769,903, which is a continuation of U.S. Patent Application No. 16/755,294, filed April 10, 2020, now U.S. Patent No. 10,998,536, which is a U.S. National Stage of PCT International Application No. PCT/US2018/055862, filed October 15, 2018, which claims the benefit of and priority to U.S. Provisional Patent Application No. 62/572,083, filed October 13, 2017, the entire contents of each of which are incorporated by reference herein.

FIELD OF THE INVENTION

[0002] The present invention relates generally to porous separators for batteries and other electric current producing cells. In particular, it pertains to multilayer porous separators comprising inorganic oxide/polymer layers on a polymeric layer, and to lithium batteries comprising such separators.

BACKGROUND OF THE INVENTION

[0003] Lithium batteries are widely used in portable electronics, such as smartphones and portable computers. Among the new applications for lithium batteries are high power batteries for hybrid, plug-in hybrid, and fully electric vehicles. Existing lithium batteries, including rechargeable and non-rechargeable lithium batteries, often utilize multilayer separators. Such separators may include an extruded polyethylene or polypropylene porous film that is coated with an inorganic oxide/polymer coating layer. As the demand for lower cost lithium batteries with increased size and energy density has grown, manufacturers have turned to less safe battery active materials in order to meet these goals. This has resulted in a need for battery separators with increased dimensional stability at temperatures such as 150°C, or higher.

[0004] Thus, it would be advantageous if the inorganic oxide/polymer coating layers could provide even greater dimensional stability at elevated temperatures while, at the same time, requiring lower coating weights and coating thicknesses, and providing lower manufacturing costs.

SUMMARY OF THE INVENTION

[0005] It is an object of the present invention to provide lighter weight, lower cost, and thinner inorganic oxide/polymer coating layers for polymer-based separators for lithium batteries and other electric current producing cells.

[0006] The present invention achieves these goals by providing a separator for a lithium battery comprising: (a) a porous polymeric layer, and (b) nanoporous inorganic oxide/polymer composite layers coated on both sides of the polymeric layer. In one embodiment, the nanoporous layer comprises an inorganic oxide (e.g., boehmite, SiO_2) and one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to about 45%, and wherein the crystallite size of the inorganic oxide particles is 5 nm to 90 nm. This combination of inorganic oxide particles (e.g., boehmite particles) having an extremely small crystallite size, and a high volume of organic polymer, produces a porous heat stable layer that provides increased dimensional stability at high temperature and a high level of porosity for excellent ionic conductivity of the electrolyte in the pores of the separator. This combination was further found to enable a significantly lower coating weight and coating thickness of the heat stable layer.

[0007] In another embodiment the nanoporous layer comprises: (a) a blend of inorganic oxide particles (e.g., boehmite particles) and inorganic nitride particles (e.g., boron nitride or aluminum nitride particles), and (b) one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to about 45%, and wherein the crystallite size of the inorganic particles is 5 nm to 90 nm. This combination of: (1) inorganic oxide and nitride particles (e.g., boehmite and boron nitride or aluminum nitride particles) having extremely small crystallite sizes, and (2) a high volume of organic polymer, produces a porous heat stable layer that provides increased dimensional stability at high temperature and a high level of porosity for excellent ionic conductivity of the electrolyte in the pores of the separator.

[0008] Preferably, the heat stable inorganic oxide/polymer layer (or inorganic oxide and nitride/polymer layer) is coated on both sides of the porous polymeric layer, such as a porous polyethylene layer or a porous polypropylene layer, as opposed to being on only one side of the polymeric layer. It has been found that this approach provides improved thermal dimensional stability in a thin and low cost separator. However, having the heat stable layer on only one side of the polymeric layer is also useful.

[0009] Another aspect of the separators for a lithium battery of this invention relates to a separator comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic oxide/polymer composite layer on one or both sides of the polymeric layer, wherein the nanoporous layer comprises an inorganic oxide (e.g., boehmite, SiO_2) and a polymer, and wherein the crystallite size of the inorganic oxide is 5 nm to 25 nm. It has been found that this smaller crystallite size results in a thinner coating layer with more cohesive strength and with more adhesion to the polymeric layer at an even lower coating weight than comparable layers where the crystallite size for the inorganic oxide is larger such as, for example, about 80 nm.

[0010] In another embodiment, the present invention includes lithium batteries that comprise the improved separators of this invention.

[0011] A further aspect of the present invention are methods for making separators that comprise coating a solution comprising an inorganic oxide and/or inorganic nitride, and one or more polymers, on one or both surfaces of a porous polymeric layer to form the separators of this invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The features and advantages of the present disclosure will be more fully understood with reference to the following, detailed description when taken in conjunction with the accompanying figures, wherein:

[0013] FIG. 1 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 1.

[0014] FIG. 2 is a graph of the shrinkage as a function of coating thickness of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 1.

[0015] FIG. 3 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0016] FIG. 4 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0017] FIG. 5 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0018] FIG. 6 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 2.

[0019] FIG. 7 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0020] FIG. 8 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0021] FIG. 9 is a graph of the coating thickness as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0022] FIG. 10 is a graph of the shrinkage as a function of coating weight of a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 3.

[0023] FIG. 11 is a graph of the moisture content as function of the weight fraction of boron nitride in a nanoporous separator, according to an embodiment of the present invention made in accordance with Example 4.

DETAILED DESCRIPTION

[0024] The separators of the present invention provide superior safety, lower cost, lower thickness, and other key performance properties for use in lithium batteries and in other electric current producing cells.

[0025] One aspect of this invention pertains to a separator for a lithium battery comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic oxide/polymer composite layer on both sides of the polymeric layer, wherein the nanoporous layer comprises boehmite and one or more polymers, wherein the volume fraction of the one or more polymers in the solid portion of the nanoporous layer may be about 10% to about 50%, and preferably about 12% to 45%, and wherein the crystallite size of the boehmite is 5 nm to 90 nm. Alternatively, the nanoporous layer may further comprise other inorganic oxide (e.g., SiO₂) or nitride (e.g., BN or AlN) particles, or combinations thereof.

[0026] The term “lithium battery” may include, for example, batteries in which the electroactive anode material comprises lithium. This includes, but is not limited to, secondary or rechargeable lithium ion batteries, secondary lithium metal batteries, and primary or non-rechargeable lithium metal batteries.

[0027] The term “separator” may include, for example, porous materials in a battery that are interposed between the cathode and the anode of the battery to provide electrical insulation against short circuiting and to provide pores for filling with the electrolyte of the battery.

[0028] The term “porous polymeric layer” may include, for example, a layer of a polymer material that is porous such that it has permeability to the flow of air through the polymeric layer. The battery industry typically measures this air permeability using a porosimeter that provides a value for the time in seconds for 100 cubic centimeters (cc) of air to pass through the porous layer. This value is commonly called the Gurley number. The battery industry prefers that the Gurley number for a separator be as low as possible without sacrificing mechanical integrity. Typical Gurley numbers for the air permeability of porous polyethylene or porous polypropylene separators, as used in the battery industry, are 100 seconds/100 cc to 300 seconds/100 cc. For this invention, porous polymer layer does not include any of the non-woven polymeric layers, such as with polyester fibers, that are used as separators in the battery industry.

[0029] The term “nanoporous” may include layers having pores up to about 100 nm in diameter, preferably, with at least 90% of the pores being less than 100 nm in diameter. The size and relative number of pores can be measured by mercury porosimetry, or by scanning electron microscope (SEM) or transmission electron microscope (TEM) cross-sections of the layer.

[0030] As discussed herein, the nanoporous layer may be present on both sides of a substrate, support, layer, or coating. For example, the nanoporous layer may be present on both sides of the porous polymeric layer, such as a porous polyethylene layer. The general benefits of the nanoporous layers of this invention are also found when the nanoporous layer is present on only one side of the porous polymeric layer, but the overall performance is typically superior when the nanoporous layer is present on both sides of the porous polymeric layer.

[0031] The superior performance features of two-side coating of the polymeric layer, compared to one-side coating at the same total coating weight, includes lower shrinkage at

high temperatures and improved battery safety. Improved safety is typically the most important performance feature desired in separators for lithium batteries, especially as the batteries become larger and higher in energy density and use less safe battery active materials in order to achieve higher energy densities and lower costs.

[0032] Another advantage of two-side coating of the porous polymeric layer with the nanoporous coating layer is that it provides the opportunity to make the coatings on each side different in composition. This enables the nanoporous layer to be optimized for the best performance when it is facing the anode and when it is facing the cathode. In general, by tailoring the formulations and coating thicknesses of the ceramic coatings on either side of the separator, it is possible to optimize: (a) adhesion between separator and a specific electrode, either anode or cathode; (b) interfacial impedance between separator and a specific electrode, either anode or cathode; (c) complexation of migratory species (e.g., solubilized transition metal ions from cathode).

[0033] For example, when the coating is facing the cathode, it is particularly useful that the coating be resistant to oxidation, especially at higher charge voltages, e.g., above 4.25 volts. This can be done, for example, by selecting the organic polymer of the nanoporous layer to be resistant to oxidation up to the higher voltage desired, e.g., up to 5.0 volts. In addition, the nanoporous layer on one of the sides can be designed with particularly small pore sizes, such as under 10 nm diameter pores, for example, by using boehmite with a primary crystallite size of less than 10 nm in diameter. It should be understood that the present invention is not limited to the examples provided herein. For example, surface-modified alumina particles, including boehmite particles, as described in PCT/IB2015/000272 to Loscutova *et al.* of Sasol, may be utilized in the separator layers described herein. This patent application discloses the dispersion of these surface-modified aluminas into nano-sized single particle crystallites. This reference also describes that the average crystallite size can be measured by X-ray diffraction on a 120 plane. As an example, the crystallite size (or an average crystallite size) of the inorganic particles in a separator comprising a nanoporous layer according to an embodiment of the invention may be measured by: (i) taking a cross-section sample of the nanoporous layer; (ii) using a TEM or very high power SEM on this sample to measure the diameter/width, in their widest dimension, of 100 single primary (inorganic) particles selected at random in the TEM or SEM image; (iii) excluding the 25 highest and 25 lowest diameter numbers in the 100 diameter/width numbers; and (iv) recording the crystallite size as the average of the middle

50 diameter/width numbers, i.e., an average crystallite size of the measured particles. Since the pore size diameters of ceramic particles in a nanoporous layer usually approximate the crystallite size of the ceramic particles, such as boehmite particles, used to make the nanoporous layer, the pore size diameters may indicate the crystallite size of the ceramic particles for nanoporous layers containing ceramic particles of a narrow crystallite size. However, for blends of ceramic particles of different crystallite sizes in a nanoporous coating, the pore size diameters of the coating layer typically are intermediate, or a blend, among the crystallite size diameters of the different ceramic particles and not individual peaks reflecting the different crystallite sizes. The TEM or SEM method described above for measuring the crystallite size does not have this limitation since it is analyzing single particles, and not blends, and thus is a preferred method to use for this invention. These separator configurations using very small pore sizes are useful when placed against a lithium metal anode, where the very small size of the pores assists in preventing the growth of lithium metal dendrites into and through the separator, which reduces cycle life, capacity, and safety for the cell. Where very small boehmite particles are used and some of these boehmite particles penetrate into the pores of the polymeric separator substrate, this is particularly useful in preventing the growth of lithium metal dendrites in and through the separator.

[0034] In another example, boehmite could be modified by reaction with a fluorinated ethylene carbonate for improved cycle life when such nanoporous layer is positioned against an anode comprising silicon or lithium metal. This nanoporous layer with the boehmite covalently reacted with a fluorinated ethylene carbonate could also be used against the cathode to help achieve excellent stability under high voltage operation of the cell, such as at up to 5.0 volts.

[0035] In a further example, the nanoporous layer with pores that are less than 10 nm in diameter could be positioned against the cathode to inhibit the diffusion of transition metals, such as nickel ions, from the cathode to the anode, which degrades the cycle life and other performance properties of the cell. Optionally, the very small boehmite crystallite size of less than 10 nm could be combined with reaction of this boehmite with fluorinated ethylene carbonate or another organic carbonate to make a nanoporous layer positioned against the cathode to lessen the diffusion of transition metal ions from the cathode.

[0036] The term “Boehmite” is understood in the art. It, for example, may include hydrated aluminum oxide of the empirical chemical formula of $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ where x is 1.0 to 2.0. Another term used interchangeably herein for boehmite is AlOOH . Boehmite may be

characterized by its distinctive Fourier-transform infrared spectroscopy (“FTIR”) fingerprint. Boehmite may also be characterized by its distinctive X-ray diffraction pattern and by its distinctive infrared spectrum of double peaks of hydroxyl groups around 3200 cm^{-1} . Boehmite may have values of x for molecules of water that are above 1.5 and even up to about 2. However, the commercial grades of boehmite of all particle sizes typically have a value of x that is in the range of 1.0 to 1.5. The value of x can be measured by first drying the boehmite at about 120°C for 1 hour to remove the surface adsorbed water and then measuring the loss of water that occurs when the boehmite sample is heated to 1000°C for 1 hour and is thereby dehydrated to anhydrous aluminum oxide, Al_2O_3 .

[0037] Examples of polymers that may be utilized in the present invention include, but are not limited to, cellulosic polymers such as hydroxyethyl cellulose, vinyl pyrrolidone polymers and copolymers, polysaccharides such as chitosan, polyethylene oxides, and polyvinyl alcohols. Preferably, the polymers, alone or in combination, provide cohesive and adhesive strength to the nanoporous layer, are insoluble in the electrolyte of the lithium battery, and are electrochemically stable over the voltage range of use of the lithium battery. If needed to achieve these properties, one or more of the polymers may be crosslinked.

[0038] By the term “volume fraction of the one or more polymers in the solid portion of the nanoporous layer” is meant the fraction expressed as a percentage by volume of the polymer content of the solid portion of the inorganic particle/polymer composite layer. As described hereinafter, the nanoporous layer has a % porosity of between about 30% to about 70%. In other words, much of the volume of the nanoporous layer is air, and not a solid material. The air volume (e.g., pores) is not included in the solid portion for purposes of determining the volume fraction of the one or more polymers in the solid portion of the nanoporous layer. The volume fraction of the polymer and inorganic particle content of the nanoporous layer can be calculated based on the relative densities of the materials. For example, when the inorganic particles are only boehmite with a density of 3.03 g/cc , the organic polymer has a density of 1.3 g/cc , and the weight ratio of the boehmite to the organic polymer is 12:1, the volume fraction of boehmite would be 83.7% and the volume fraction of organic polymer would be 16.3%. Changing the weight ratio of the boehmite to the organic polymer to 4:1 results in a volume fraction of boehmite of 63.2% and of organic polymer of 36.8%. If the density of the organic polymer is lower, such as 1.1 g/cc , then its volume fraction at the same weight ratio of inorganic particle/polymer will be higher than at the higher polymer density of 1.3 g/cc . To determine the volume fraction of one or more organic

polymers in the solid portion of the nanoporous layer, for example, the nanoporous layer is removed from the separator, such as by careful scraping of the layer and not removing any of the polymer substrate. Next, a weighed nanoporous layer sample in a crucible is combusted at a sufficient temperature to combust the organic polymer to leave the residue of the ceramic particles. This provides the weight of combusted ceramic particles. When the particles are only boehmite, this weight of the combusted residue is divided by 0.85 in the case of the equivalent of one water molecule per boehmite molecule to obtain the weight of boehmite in the sample prior to the combustion. Although most boehmite particles have a 15% weight loss upon combustion, it should be noted that boehmite particles may have a different weight loss factor, such as 0.84 or 0.82. This same information can be obtained by using a thermogravimetric analyzer to record the weight of the nanoporous layer sample as it is heated to 900 °C or higher, until a constant weight is achieved. As noted above, boehmite has a density of 3.03 g/cc. Most organic polymers have a density in the range of 1.1 to 1.4 g/cc, although some polymers, such as polyvinylidene difluoride (PVdF), have a density of about 1.8 g/cc. After the density of the organic polymers is known by analyzing for the identity and weight ratios of the organic polymers, for boehmite-only nanoporous layers, the volume % of the organic polymers in the solid portion of the nanoporous layer can be calculated. For example, if the weight percent of boehmite is 88.9%, and the weight percent of the organic polymers is 11.1% and its density is 1.3 g/cc--with a pigment:binder ratio of 8:1 by weight--dividing the weight percents of boehmite and the organic polymers by their densities gives relative volume percents of 29.34% for boehmite and 8.54% for the organic polymers. Normalizing these volume percents to a total volume percent of 100% for the solid portion of the nanoporous layer gives a volume percent of 22.54% for the organic polymers. If the nanoporous layer is a blend of boehmite and other ceramic particles or if the density of the organic polymers is not known, further analysis, as known in the art, may be undertaken to measure the volume percent of the organic polymers.

[0039] The term “crystallite size” of the particles refers to the average size of the primary or single crystals of the inorganic particles. This is typically measured using X-ray diffraction, a transmission electron microscope (TEM), or a very high resolution scanning electron microscope (SEM). PCT/IB2015/000272 to Loscutova *et al.* of Sasol cited above provides examples of using X-ray diffraction to determine an average size of the crystals of the inorganic particles, such as alumina and boehmite. As described earlier, a TEM or SEM method may be used to determine the crystallite size. Many inorganic oxide particles, such

as boehmite particles, are nearly cubic in shape, although they can have other shapes such as platelets and needles. In the dry solid state and in solution, boehmite particles are typically agglomerated into larger particles comprised of many primary boehmite particles, such as, for example, 10 to 20 primary particles. The inorganic particles in the nanoporous layers of this invention result in features, such as pore size diameter, that are related to the crystallite size of the particles.

[0040] In one embodiment of the separators of the present invention, the ratio of the total thickness of the nanoporous layer on both sides of the polymeric layer to the total thickness of the separator is 10% to 80%. The thickness of these layers may be measured by SEM analysis of cross-sections of the separator. Because of the nanoporous pore sizes and the high volume fraction of organic polymer in the nanoporous layer, it has been found that a lower limit for the proportional thickness of the nanoporous layer of about 10% of the total thickness of the separator, rather than 15% or more typically 25%, is able to provide an excellent improvement in the dimensional stability of the separator at high temperatures, which is needed for improved safety. It has been found that the combination of nanoporous pore sizes and very high volume fractions of organic polymer in the nanoporous layer (e.g., from about 10% to about 50%) provides acceptable % porosity (e.g., from about 30% to about 70%) with good ionic conductivity of the electrolyte. It has also been found that the separator of this invention functions well with the total thickness of the nanoporous layer as high as 80% of the total thickness of the separator.

[0041] In one embodiment, the heat shrinkage of the separator at 150°C for 1 hour is less than 5%. This heat shrinkage may be measured by the change in the area of the separator after heating in an oven for 1 hour. This measurement of the heat shrinkage may be done by hanging a large sheet of the separator in the oven after marking and measuring the dimensions and area of a square or rectangle such as a 10 cm by 10 cm square, on a smaller section of the large sheet. After the heating in the oven, the marked area is measured for the % change or % shrinkage of the dimensions and the area.

[0042] In one embodiment, the heat shrinkage of the separator at 200°C for 1 hour is less than 5%. The nanoporous separators of the present invention have been found to retain their dimensional stability even when heated well above the melting point of the polymeric layer, such as a porous polyethylene layer.

[0043] In one embodiment of the separators of this invention, the total coating weight of the nanoporous layer on both sides is less than 4.5 grams per square meter (gsm), while

providing a heat shrinkage of the separator at 150°C for 1 hour of less than 5%. It should be noted that the use of boehmite with its lower density of about 3.03 g/cc, as compared to the higher density anhydrous aluminum oxide or alumina (which has a density of about 4 g/cc), helps to further lower the coating weight of the heat stabilizing layer. The lower density of the inorganic oxide particle proportionately lowers the weight of coating needed for a specific thickness and % porosity.

[0044] For example, a 4 micron thick alumina/polymer layer with a 45% porosity for polyethylene-based separators typically weighs about 8 grams per square meter (gsm). However, switching to boehmite (or a blend of boehmite and other inorganic particles of a similar or lower density) at the same thickness and % porosity lowers the coating weight to about 6 gsm or less. The high volume fraction of polymer in the nanoporous layers of the separators of this invention, combined with the use of boehmite with a crystallite size of 5 nm to 90 nm, provides increased dimensional stability for a specific thickness of the separator layer. Although it is possible to still make the total thickness of the inorganic particle/polymer layer in the separator 4 microns or higher, it is desirable for lower cost and for lower total thickness of the separator to make the thickness of the inorganic particle/polymer layer as thin as possible, such as about 3 microns total, while still achieving an excellent dimensional stability at high temperature. In one embodiment, the heat shrinkage of the separator at 200°C for 1 hour is less than 5%.

[0045] In one embodiment, other inorganic oxides and/or inorganic nitrides with a density of about 3.0 g/cc or lower, such as silicon dioxide and boron nitride, may be blended with the boehmite pigment. As discussed further below, this provides lower total coating weight of the nanoporous layer and other desirable properties, such as lower water content.

[0046] In another embodiment of the separators of this invention, the total coating weight of the nanoporous layer of both sides is less than 3.5 grams per square meter (gsm), while providing a heat shrinkage of the separator at 150°C for 1 hour of less than 5%. In one embodiment, the heat shrinkage of the separator at 200°C for 1 hour is less than 5%.

[0047] In one embodiment of the separators of the present invention, the crystallite size of the inorganic particles may be about 30 nm to about 50 nm, preferably about 5 nm to about 25 nm, and more preferably about 5 nm to about 10 nm. In one embodiment, the inorganic particles comprise boehmite particles having a crystallite size of about 30 nm to about 50 nm. In one embodiment, the crystallite size of the boehmite may be about 5 nm to about 10 nm, and preferably about 5 nm to about 8 nm.

[0048] In another embodiment of this invention, the % porosity of the nanoporous layer is between 30% and 55%. The % porosity of the nanoporous layer may be calculated from the density in g/cc of the materials in the dry coating assuming it is a 100% solid coating with 0% porosity and comparing it to the actual density of the coating. The actual density of the nanoporous layer may be calculated by dividing its coat weight in gsm by its coating thickness to provide the density in g/cc of the nanoporous layer. Dividing the density of the nanoporous layer by the calculated density of the same materials in a non-porous layer and multiplying by 100% provides the % solid fraction by volume in the nanoporous layer. Subtracting this value of the % solid fraction from 100% gives the % pore volume or % porosity of the nanoporous layer. The % porosity can also be measured by mercury porosimetry analysis.

[0049] In one embodiment, the pore volume of the nanoporous layer is between 0.8 cc/g and 1.0 cc/g. The pore volume can be calculated by measuring the coat weight of the nanoporous layer in gsm and measuring the thickness of the nanoporous layer to determine the total cc in a square meter. The pore volume in cc/g is then calculated by dividing the total cc in a square meter by the grams of the gsm measurement. In one embodiment, the % porosity of the nanoporous layer is 55% to 70%. Above this level of % porosity, it is difficult to obtain the cohesive and adhesive strength of the nanoporous layer that are preferred for long term, safer lithium battery separator applications.

[0050] In one embodiment of the separators of the present invention, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 15% to 45%. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 20% to 45%. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is 25% to 45%.

[0051] In one embodiment of the separators of this invention, the nanoporous layer comprises pores having an average pore diameter of 80 nm or less and at least 70% of the pores have a diameter less than 80 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of 50 nm or less and at least 70% of the pores have a diameter less than 50 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of 25 nm or less and at least 70% of the pores have a diameter less than 25 nm.

[0052] In one embodiment of the separators of this invention, the ratio of the air permeability of the separator to the air permeability of the polymeric layer without any

nanoporous layer present is 1.0 to 1.4. As described hereinabove, this air permeability is typically measured on an instrument which provides a Gurley number of the number of seconds for 100 cc of air to pass through the separator. A low level of increase of the air permeability is an important indicator of an expected low decrease in the ionic conductivity of electrolyte in the separator due to adding the inorganic oxide/polymer layer. If the ionic conductivity decreases too much, this can interfere with the cycling rate capability of the lithium battery particularly its power rate properties, and also interfere with the cycling lifetime of the lithium battery.

[0053] In one embodiment of the separators of the present invention, the polymeric layer comprises a polyolefin. Suitable polyolefins include, but are not limited to, polyethylenes and polypropylenes. Other examples of the polyolefin polymers include, but are not limited to, homopolymers of 1-butene, 1-pentene, 1-hexene, and 1-octene, as well as copolymers and terpolymers thereof, used alone or in combination with one or more types of polymers. In one embodiment, the polymeric layer comprises polyethylene. The polymeric layer may be of the “wet process” type which is produced by extrusion followed by solvent extraction to provide the porosity or of the “dry process” type which is produced by extrusion followed by stretching to provide the porosity. Various molecular weights, densities, and stereochemistries of polyethylene and the other polymers, with various melt flow and mechanical properties, may be utilized in the porous polymeric layers.

[0054] For the purposes of better uniform wetting of the coating fluid of the nanoporous layer and of better adhesion of the nanoporous layer to the substrate, the porous polymeric layer may have a coating layer on the surface to which the nanoporous layer is applied.

[0055] In one embodiment of the separators comprising polyethylene in the polymeric layer, the heat shrinkage of the separator at 120°C for 1 hour is less than 1.0%. Without the nanoporous layer of the separators of this invention, the heat shrinkage of a polyethylene separator of comparable total thickness is typically greater than 1.0% at 105°C. In one embodiment, the ratio of the air permeability of the separator after the heat shrinkage at 120°C for 1 hour to the air permeability of the separator before the heat shrinkage at 120°C is 0.8 to 1.2. This is an important feature for allowing vacuum drying of the polyethylene-based separators of this invention at a higher temperature of 120°C without any negative effect on the air permeability and ionic conductivity and, in some cases, with a positive effect on ionic conductivity due to the heat treatment. Vacuum drying is beneficial for lowering the moisture

content of separators for increased lithium battery cycling lifetimes and capacities and for higher lithium battery cycling rate capabilities. Vacuum drying is typically done for 4 to 24 hours at about 80°C for polyethylene separators and at about 90°C for polyethylene separators with one or more inorganic oxide/polymer layers. The capability of using a higher temperature of 120°C and a shorter drying time of 1 to 3 hours for the polyethylene-based separators of this invention is useful.

[0056] In another embodiment of the separators of the present invention, the nanoporous layer is a laminated layer on the polymeric layer. By the term “laminated layer” is meant that the nanoporous layer is not initially formed on the polymeric layer, such as by direct coating, but is instead laminated to the polymeric layer in a subsequent step. For example, the nanoporous layer is coated on a release substrate, and then this nanoporous layer is transferred by lamination onto the polymeric layer. This lamination may optionally include the use of heat and/or the use of some organic solvent to adhere the nanoporous layer to the polymeric layer and to aid in the subsequent delamination of the release substrate without interfering with the quality of the laminated multilayer separator.

[0057] Another aspect of the separators for a lithium battery of this invention relates to a separator comprising (a) a porous polymeric layer, and (b) a nanoporous inorganic particle/polymer composite layer on one or both sides of the polymeric layer, wherein the nanoporous layer comprises an inorganic oxide (e.g., boehmite), or a blend of an inorganic oxide and an inorganic nitride (e.g., BN or AlN), and a polymer, and wherein the crystallite size of the inorganic particles is about 5 nm to about 25 nm, and preferably about 5 nm to about 8 nm. In one embodiment, the nanoporous layer comprises pores having an average pore diameter of about 25 nm or less and at least about 70% of the pores have a diameter of about 25 nm or less.

[0058] A further aspect of the separators for a lithium battery of the present invention pertains to a separator comprising (a) a porous polyethylene layer, and (b) a nanoporous inorganic particles/polymer composite layer on both sides of the polyethylene layer, wherein the nanoporous layer comprises boehmite, or a blend of boehmite and boron nitride, and one or more polymers, the volume fraction of the one or more polymers in the solid portion of the nanoporous layer is about 10% to about 50%, the total coating weight of the nanoporous layer on both sides is 5.5 grams per square meter or less, and the heat shrinkage of the separator at 150°C for 1 hour is less than 5%. In one embodiment, the total coating weight of the

nanoporous layer on both sides is 3.5 grams per square meter or less and the thickness of the polyethylene layer is between 5 microns and 16 microns.

[0059] In another embodiment, this invention is lithium batteries comprising a separator of the present invention.

[0060] Another aspect of the present invention are methods for making the separator of this invention, which method comprises coating a solution comprising inorganic particles and one or more polymers on one or both surfaces of a porous polymeric layer. In one embodiment, the inorganic particles comprise boehmite particles having a crystallite size of 5 nm to 90 nm. In one embodiment, the boehmite particles have a crystallite size of about 5 nm to about 25 nm, and preferably about 5 nm to about 8 nm. In one embodiment, the volume fraction of the one or more polymers in the solid portion of the coating after drying is about 10% to about 50%, preferably about 12% to about 45 %, and more preferably about 15% to about 45%.

[0061] EXAMPLES

[0062] Several embodiments of the present invention are described in the following examples, which are offered by way of illustration and not by way of limitation.

[0063] Example 1

[0064] A boehmite slurry was prepared by adding 12 parts of boehmite (DISPAL 10F4 supplied by Sasol Inc., Houston, Texas) and 1 part of chitosan (deacetylated chitin, poly(D-glucosamine), medium molecular weight, of a polysaccharide supplied by Sigma-Aldrich, St. Louis, MO) to 73 parts of 9:1 water:isopropyl alcohol and stirring the slurry. The primary particle size of the boehmite was about 40 nm. The boehmite slurry was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of coating weights for both sides from about 1.2 grams per square meter (gsm) to about 5.2 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. The boehmite coating was then applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0065] The 12:1 by weight ratio of boehmite pigment to chitosan binder is calculated to have 15.3% by volume of chitosan binder in the solid portion of the porous boehmite

coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.4 g/cc for chitosan.

[0066] The coating thickness was measured using a Dorsey gauge on the polyethylene separator before and after the coating of the boehmite coating on both sides. The difference in the thickness values before and after the coating is the coating thickness in microns. An alternative method for measuring the coating thickness is using the Dorsey gauge on the coated separator before and after removing the boehmite coating on both sides, either by pulling off the coating with an adhesive tape or by rubbing off the coating with an alcohol-soaked cloth.

[0067] The coating weight was measured by weighing a 10 cm x 10 cm sample of the polyethylene separator before and after the coating of the boehmite coating on both sides. The difference in the weights before and after the coating multiplied by 1,000 is the coat weight in grams per square meter (gsm). An alternative method for measuring the coating weight was weighing a 10 cm x 10 cm sample of the coated separator before and after removing the boehmite coating on both sides, either by pulling off the coating with an adhesive tape or by rubbing off the coating with an alcohol-soaked cloth.

[0068] Figure 1 shows a plot of the average boehmite coating thickness and coating weight per side of about 0.6 gsm to about 2.6 gsm. For two sided coatings, these numbers are doubled to give a total coating weight range of about 1.2 gsm to about 5.2 gsm. The total boehmite coating thickness for both sides varies from about 1.2 microns to about 4.4 microns. If a more accurate measurement of the coating thickness is desired, scanning electron microscope (SEM) analysis of cross-sections of the boehmite-coated plastic separator can be utilized.

[0069] Using a SEM analysis for the total coating thickness showed a density of about 1.06 gsm per micron of coating thickness. From this, the % porosity for this 12:1 pigment to binder ratio is estimated as 62%.

[0070] One objective for minimizing the cost of the boehmite coating is to apply the total weight of coating that is needed to reduce the heat shrinkage to the target value, such as less than 5% shrinkage at 150°C. Figure 2 shows that this level of heat shrinkage is obtained with an average coating thickness per side of about 1.9 microns or thicker on the 12 micron polyethylene substrate of this example. For coatings on two sides, this number is doubled to give a total coating thickness of about 3.8 microns or thicker. Based on the density of the

boehmite coating determined by SEM analysis, a total coating thickness of about 3.8 microns on both sides equates to about 4.0 gsm of coating weight.

[0071] For better wetting and coating adhesion of the boehmite slurry, a corona treatment of the polyethylene substrate is preferred. Besides wire wound coating rods, a variety of other coating application methods may be used, such as, but not limited to, gravure coating, screen printer coating, reverse roll coating, blade coating, and slot die coating.

[0072] Example 2

[0073] A boehmite slurry was prepared by adding 12 parts of boehmite (DISPERAL D60 supplied by Sasol Inc., Hamburg, Germany) and 1 part of a vinyl pyrrolidone copolymer (Soteras CCS grade supplied by Ashland Inc., Wilmington, DE) to 73 parts of 9:1 water:isopropyl alcohol and stirring the slurry. The primary particle size of the boehmite was about 75 nm. The boehmite slurry was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of coating weights for both sides from about 2.4 grams per square meter (gsm) to about 9.2 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. Then, the boehmite coating was applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0074] The 12:1 by weight ratio of boehmite pigment to vinyl pyrrolidone copolymer binder is calculated to have 18.7% by volume of polymer binder in the solid portion of the porous boehmite coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.1 g/cc for the vinyl pyrrolidone copolymer.

[0075] Figure 3 shows a plot of the average boehmite coating thickness and coating weight per side of about 1.2 gsm to about 4.6 gsm. For two sided coating, these numbers are doubled to give a total coating weight range of about 2.4 gsm to about 9.2 gsm. The total boehmite coating thickness for both sides varies from about 2.8 microns to about 9.0 microns.

[0076] As discussed, one objective for minimizing the cost of the boehmite coating is to apply the total weight of coating that is needed to reduce the heat shrinkage to the target value, such as less than 5% shrinkage at 150°C for 1 hour. Figure 4 shows that this level of heat shrinkage is obtained with an average coating thickness per side of about 2.0 microns or thicker on the 12 micron polyethylene substrate of this example. For the coatings on two sides, this number is doubled to give a total coating thickness of about 4.0 microns or thicker.

Based on Figure 3, a total coating thickness of about 4.0 microns on both sides equates to about 3.9 gsm of coating weight.

[0077] A boehmite slurry of an 8 to 1 ratio by weight of the DISPERAL D60 to vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol at 15% solids was also coated using wire wound rods on the 12 micron polyethylene substrate and dried, as for the 12:1 boehmite D60:polymer coating in this example. It was calculated to have a volume percent of polymer binder of 25.6%. Figure 5 shows a plot of the average boehmite coating weight and coating thickness per side of about 0.8 gsm to about 4.2 gsm. For the coatings on two sides for this boehmite coating with a 8:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a range of about 1.6 gsm to 8.4 gsm. The corresponding total boehmite coating thicknesses for both sides varies from about 1.6 microns to 8.2 microns. Figure 6 shows that a less than 5% shrinkage at 150°C for 1 hour is obtained with an average coating thickness per side of about 2.2 microns or thicker. For the coatings on two sides, this number is doubled to give a total coating thickness on both sides of about 4.4 microns or thicker. Based on Figure 5, a total coating thickness of about 4.4 microns equates to about 4.8 gsm of coating weight.

[0078] Example 3

[0079] Using the procedure of Example 2, a boehmite slurry of a 12:1 by weight of DISPAL 25F4 boehmite and the vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol at 15% solids was prepared. The DISPAL 25F4 is supplied by Sasol, Lake Charles, LA, and has a primary crystallite particle size of 8 nm. This boehmite slurry was coated on the EP12 12 micron polyethylene substrate from ENTEK and dried, similarly to the process used in Examples 1 and 2. It was calculated to have a volume percent of polymer binder of 18.7%. Figure 7 shows a plot of the average boehmite coating thickness and the coating weight per side of about 0.8 gsm to about 3.0 gsm. For the coatings on two sides for this boehmite coating with a 12:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a total coating weight range of about 1.6 gsm to about 6.0 gsm. Figure 8 shows that a less than 5% shrinkage at 150°C for 1 hour is obtained with an average coating thickness per side of about 1.8 microns or thicker. For two sided coatings, this number is doubled to give a total coating thickness on both sides of about 3.6 microns or thicker. Based on Figure 7, a total coating thickness of about 3.6 microns equates to about 3.6 gsm.

[0080] A boehmite slurry of an 8 to 1 ratio by weight of the DISPAL 25F4 to vinyl pyrrolidone copolymer in 9:1 water:isopropyl alcohol was also coated on the 12 micron polyethylene substrate and dried, similarly to the process used in Examples 1 and 2. It was calculated to have a volume percent of polymer binder of 25.6%. Figure 9 shows a plot of the average boehmite coating thickness and coating weight per side of about 0.8 gsm to about 4.7 gsm. For the coatings on two sides for this boehmite coating with a 8:1 boehmite pigment:vinyl pyrrolidone polymer binder ratio by weight, these numbers are doubled to give a total coating weight range of about 1.6 gsm to 9.4 gsm. Figure 10 shows that a less than 5% shrinkage at 150°C for 1 hour is obtained with an average coating thickness per side of about 0.9 microns or thicker. For the coatings on two sides, this number is doubled to give a total coating thickness on both sides of about 1.8 microns or thicker. Based on Figure 9, a total coating thickness of about 1.8 microns equates to about 2.9 gsm of coating weight. For a thickness of about 1.8 microns of the boehmite coating, the coating weight should have been about 2.0 gsm based on the density of the boehmite coating of about 1.1 g/cc shown in Example 1, while factoring in the lower boehmite weight % of the 8:1 boehmite pigment:vinyl pyrrolidone copolymer and the higher vinyl pyrrolidone copolymer weight % and lower density of the vinyl pyrrolidone copolymer compared to chitosan. This shows that about 0.9 gsm of the boehmite coating penetrated into and dried in the pores of the polyethylene substrate to produce a boehmite/polyethylene substrate with a boehmite coating on its surface. At the 8:1 pigment:binder (P:B) ratio of this coating, the about 0.9 gsm of boehmite coating in the pores equates to about 0.8 gsm of boehmite particles in the pores of the boehmite/polyethylene substrate or layer. Since this boehmite coating was applied on both sides of the polyethylene substrate, about one-half, or about 0.4 gsm of boehmite, would have penetrated and dried in the pores on each side of the polyethylene substrate to form the boehmite/polyethylene substrate or layer that resulted in a boehmite-coated boehmite/polyethylene substrate.

[0081] A similar 8:1 boehmite coating with Dispall 25F4, with chitosan substituted for the vinyl pyrrolidone copolymer and a 12 micron thick polyethylene substrate from Shenzhen Senior, Shenzhen, China, used instead of the polyethylene substrate from Entek, was made using the procedure of this Example to give a coat weight of 1.8 gsm. This coating had a heat shrinkage of less than 5% when heated at 150°C for 1 hour. When the procedure of removing the boehmite coating with an adhesive tape was used, even with very careful tape pulls, it was not possible to remove the boehmite coating from the 10 x 10 cm samples without pulling off

some of the polyethylene substrate. This was in contrast to the tape pull results of easy and complete removal of the boehmite coatings from the polyethylene substrate for Examples 1 and 2 and Comparative Examples 1 and 2. This result was consistent with some of the boehmite coating penetrating into the pores of the polyethylene layer and, upon drying, providing increased adhesive and cohesive strength to the boehmite coating such that the point of failure in the tape pull was below the surface of the polyethylene layer at most areas, rather than at the interface of the polyethylene surface and the boehmite coating on the polyethylene surface. After the tape pull, some areas were found where the polyethylene layer had not been disturbed. From these areas, the added thickness of the boehmite layer was found to be about 1.8 microns.

[0082] Since the tape pulls were not able to cleanly remove the boehmite coating from the polyethylene layer, a methanol-saturated soft cloth was used to wash and rub the coating off. The 10 x 10 cm sample after washing had a weight of 0.0757 grams (g). Two more methanol rubs did not change this weight of the washed and rubbed sample. The polyethylene substrate used in this Example contained no inorganic pigment, such as alumina or boehmite. A furnace combustion of the methanol-washed sample in a crucible, at a temperature of 1000 °C for 70 minutes showed 0.0045 g of anhydrous aluminum oxide after dehydrating the boehmite pigment with a weight loss of 15% of the boehmite pigment weight. Using this boehmite to aluminum oxide conversion factor by dividing the aluminum oxide weight by 0.85, this gives a boehmite content of about 0.0053, or at least about 0.53 gsm of boehmite pigment in the pores of the polyethylene layer. Since the wet rubbing to remove the boehmite coating on the surface may have removed a small amount of the boehmite coating in the upper pores of the polyethylene layer, it is possible that the weight of boehmite pigment in the pores was somewhat greater than 0.53 gsm. Since there were two coatings on the polyethylene layer, the amount of boehmite in the pores from a single coating is about one-half of this total, or about 0.27 gsm. As further evidence for penetration of the boehmite coating into the pores of the polyethylene substrate, the dimensional stability of this “washed” polyethylene with about 0.53 gsm of boehmite in the pores after heating at 150°C for 1 hour was 54% and 36.5% in the two directions(--machine direction (MD) and cross direction (CD)). By contrast, the polyethylene substrate before coating showed heat shrinkage of 70% and 75.5%. This increased heat stability with boehmite pigment in the pores of the polyethylene substrate is one explanation for the greater dimensional stability at 150°C with a much lower coat weight and coating thickness added to the surface of the

polyethylene layer. For analyzing the presence and amount of boehmite particles in the pores of the polymeric layer, the removal of the surface coating of the nanoporous layer and subsequent combustion of all of the organic polymer, as described above, in a combustion and/or thermogravimetric method may be used for the case of only boehmite particles in the nanoporous layer and a polymer layer that contained no inorganic or boehmite particles before the application of the boehmite coating. Another method is to use the TEM or SEM method used for measuring the average crystallite size to make images of the presence of boehmite particles in the pores of the polymeric layer and to determine their crystallite size at the same time. Combining this method with the combustion and/or thermogravimetric method above to quantify the weight of the boehmite, or other ceramic particles in the pores, if any, would provide the weight in gsm of boehmite particles in the separator. The known analytical techniques for boehmite particles, such as FTIR, X-ray diffraction, thermogravimetric analysis (TGA), high resolution photoelectron spectroscopy (ESCA), and ^{27}Al nuclear magnetic resonance (nmr) can be utilized to further confirm the presence and quantity of boehmite particles in the pores of the polymeric layer, if needed. An alternative approach for collecting the ceramic particles, such as boehmite particles, in the pores of the polymeric layer, after removing any surface coating from the polymeric layer, would be to do a Soxhlet extraction of the polymeric layer with a hot solvent that dissolves the polymer from the polymeric layer to isolate the insoluble inorganic particles. Besides providing the weight of inorganic particles, this could be combined with the above analytical methods, combustion and/or thermogravimetric analysis, and TEM or SEM method, to confirm the presence and quantity of boehmite particles. One benefit of this penetration of nano-sized boehmite particles would be the ability to make and sell boehmite/polyethylene substrates, with no or minimal surface boehmite or other coating, for various applications, including for manufacturing CCS, where the boehmite/polyethylene substrate would have multiple advantages over polyethylene substrates, including, but not limited to, a much greater dimensional stability at high heats, a modified and smaller pore size distribution, and other benefits, such as ease of wetting by subsequent coatings. One example of a particularly useful application for boehmite/polymeric substrates would be filtration applications, where the very small nano-sized boehmite particles, with and without various surface treatments, in the pores of the polymer could provide the desired nanofiltration and ultrafiltration properties not available from the polymeric substrate without boehmite particles in its pores, and could be adapted for desalination of sea water. This benefit would also be available for other

boehmite/polymeric substrates besides polyethylene, such as for boehmite/polypropylene and boehmite/other polyolefin substrates.

[0083] Example 4

[0084] The procedure of Example 1 was used to make a nanoporous separator comprising a 50:50 blend by weight of boehmite and boron nitride pigments, except that a 12 micron thick porous polyethylene separator from Toray Industries, Tokyo, Japan, was substituted for the Entek polyethylene separator, and the vinyl pyrrolidone polymer, Soteras CCS, was substituted for the chitosan. The boron nitride pigment was NX1, supplied by Momentive Performance Materials. The total coating weight on both sides was 2.5 gsm. Figure 11 shows the reduced water content of this separator compared to the control with only boehmite pigment and no boron nitride. Specifically, as shown in Figure 11, at a 50:50 by weight ratio of boehmite and boron nitride, the water content at 20% RH was reduced by one-third from its value with boehmite pigment only.

[0085] Comparative Example 1

[0086] An alumina mix was prepared in water by adding 20 parts of alumina pigment (HJA-0719 supplied by AnHui Estone Materials Technology Co., Ltd., Anhui Province, China) and 1 part of a hydroxypropyl cellulose polymer (Klucel H grade supplied by Ashland Inc., Wilmington, DE) to 63 parts of water and stirring the slurry. This 20:1 weight ratio of pigment to binder provided a similar Gurley air permeability and similar ionic conductivity in electrolyte as the 8:1 weight ratio of pigment to binder in Example 2. Reducing the 20:1 weight ratio had a progressively negative effect with higher Gurley air permeability and lower ionic conductivity. The primary particle size of the alumina was greater than 400 nm. The alumina mix was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of total coating weights for both sides from about 0.8 grams per square meter (gsm) to about 4.4 gsm. The alumina coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. Then, the alumina coating was applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0087] The 20:1 by weight ratio of alumina pigment to hydroxypropyl cellulose polymer binder is calculated to have 13.6% by volume of polymer binder in the solid portion of the porous alumina coating layer. This calculation is based on a specific gravity of 4.0 g/cc for alumina and of 1.27 g/cc for the hydroxypropyl cellulose polymer. At a total coating

weight of about 4.4 gsm, this alumina coating had a shrinkage of about 60% at 150°C for 1 hour.

[0088] Comparative Example 2

[0089] A boehmite mix was prepared in water by adding 20 parts of boehmite pigment (BG-613 supplied by AnHui Estone Materials Technology Co., Ltd., Anhui Province, China) and 1 part of a hydroxypropyl cellulose polymer (Klucel H grade supplied by Ashland Inc., Wilmington, DE) to 63 parts of water and stirring the slurry. This 20:1 weight ratio of pigment to binder provided a similar Gurley air permeability and similar ionic conductivity in electrolyte as the 8:1 weight ratio of pigment to binder in Example 2. Reducing the 20:1 weight ratio had a progressively negative effect with higher Gurley air permeability and lower ionic conductivity. The primary particle size of the boehmite was greater than 400 nm. The boehmite mix was coated onto both sides of a 12 micron thick porous polyethylene separator (EP12 supplied by ENTEK, Lebanon, Oregon) using wire-wound rods of different wire thicknesses to obtain a range of total coating weights for both sides from about 0.8 grams per square meter (gsm) to about 7.0 gsm. The boehmite coating was first applied to one side of the polyethylene separator and dried at 70°C for 2 minutes in a convection oven. Then, the alumina coating was applied to the reverse uncoated side of the polyethylene separator and dried under the same oven heating conditions as for the first coating.

[0090] The 20:1 by weight ratio of boehmite pigment to hydroxypropyl cellulose polymer binder is calculated to have 10.7% by volume of polymer binder in the solid portion of the porous boehmite coating layer. This calculation is based on a specific gravity of 3.03 g/cc for boehmite and of 1.27 g/cc for the hydroxypropyl cellulose polymer. At a total coating weight of about 7.0 gsm, this boehmite coating had a shrinkage of about 60% at 150°C for 1 hour.

[0091] Table 1 shows a summary of the data from Examples 1 to 3.

Table 1

Example	Boehmite Type	Primary Particle Size (nm)	Pigment:Polymer Binder Weight Ratio	Volume % Polymer	Total Coat Weight for <5% Shrinkage at 150°C for 1 hour
1	10F4	40	12:1	15.3%	≥4.0 gsm
2	D60	75	12:1	18.7%	≥3.9 gsm
2	D60	75	8:1	25.6%	≥4.8 gsm

3	25F4	8	12:1	18.7%	≥ 3.6 gsm
3	25F4	8	8:1	25.6%	≥ 2.9 gsm

[0092] One challenge in using nano-sized ceramic particles, such as nano-sized boehmite particles, is their increased tendency to absorb water because of their much larger surface areas compared to micro-sized ceramic particles, such as micro-sized alumina and micro-sized boehmite particles. For lithium ion cells, lithium metal cells, and other electrochemical cells with sensitivity to water content, the separator needs to have a low water content, such as under 600 ppm of water in the separator, preferably less than 300 ppm of water, and more preferably under 100 ppm of water as typically found in porous polymer only separators with no ceramic materials in them. Although vacuum drying of the separator at up to 80 °C can help to lower this moisture content in separators with nano-sized ceramic particles, and the separators of this invention have increased thermal stability that permits even higher vacuum drying temperatures of up to 90 °C to about 110 °C, some users of separators may prefer not to perform this extra step of vacuum drying. It would be preferable if the greater heat stability, lower cost from the use of a lower amount of ceramic coating materials, and lower ceramic coating thickness, including from the penetration of the ceramic coating and ceramic particles into the porous polymeric separator layer, of the separators of this invention would include a lower moisture content in the separators from the use of more hydrophobic ceramic particles and organic polymers. Having a lower amount of ceramic particles from the lower coating weights of the invention is a significant benefit for lowering the moisture content of the separator, but it would be further useful if the ceramic particles were more hydrophobic and less prone to moisture pickup.

[0093] One approach which provides this lower moisture absorption is to treat the surface of the ceramic particles with a hydrophobic material, such as by using hydrophobically-treated boehmite particles, particularly if the hydrophobic material is chemically reacted with the boehmite particles and thus is permanently bonded to the boehmite particles. Suitable hydrophobic materials for chemical modification of boehmite pigments include, but are not limited to, organic sulfonic acids, organic carbonates, and polymers with hydrophobic groups and with reactive groups, such as carboxylic acid and hydroxyl groups, that can be made to react with hydroxyl groups of the boehmite or other ceramic particles.

[0094] Examples of the organic sulfonic acid-treated boehmite particles include Dispal® 25SR, available from Sasol in Lake Charles, LA, a boehmite pigment chemically reacted with

p-toluene sulfonic acid through the sulfonic acid groups. The reaction leaves the hydrophobic p-toluene group on the outer surface of the boehmite particle to make it less prone to pick up water. An even more hydrophobic example from Sasol is p-dodecylbenzene sulfonic acid surface reacted to boehmite particles through the sulfonic acid group and having the very hydrophobic dodecyl group on the outer surface of the boehmite pigment. This makes the boehmite particles so hydrophobic that this treated boehmite can be dispersed in hydrocarbon solvents. The use of boehmite particles surface treated with organic sulfonic acids in boehmite coatings for separators is disclosed in U.S. Patent Application Publication No. 2013/0171500 by Xu et al.

[0095] Examples of the organic carbonate-treated boehmite particles include, but are not limited to, those with ethylene carbonate and fluoroethylene carbonate as the organic carbonates. Besides the option of choosing an organic carbonate that is very hydrophobic in the part of the molecule that will face away from the surface of the boehmite, the choice can also be made with a view to picking an organic carbonate which will provide increased ionic conductivity to the separator. The use of boehmite particles surface treated with ethylene carbonate to provide separators with increased conductivity is described in U.S. Pat. No. 8,883,354 to Carlson et al.

[0096] The use of boehmite particles surface treated with organic polymers to provide separators is described in U.S. Pat. No. 9,871,239, to Carlson et al. As the primary crystallite size and resulting pore size become smaller in the boehmite separators and the ionic conductivity of the separator becomes lower because of the smaller pore sizes, it is useful to have surface treatments, such as the three types described above, that can be utilized to increase the ionic conductivity of the separator, as well as to decrease the water content of the separator.

[0097] The preparation of a ceramic mix or slurry with the surface-treated hydrophobic materials can be done conveniently by adding a precursor, such as an organic carbonate, to the slurry containing the untreated boehmite particles and heating the slurry to a temperature, such as 80 °C, for a sufficient time, such as 1 hour, to react the surface treatment material, such as an organic carbonate, with the boehmite particles to the desired level of surface treatment. There is no need to isolate the surface-treated boehmite particles by a drying process, but rather the slurry can be used as is, usually after some dilution with water and/or solvents to the desired % solids level for the coating of the porous polymer substrate. If it is desired to further reduce the particle size of the ceramic particles, such as boehmite particles,

and/or to homogenize the slurry for more uniformity and perhaps a lower viscosity at the desired % solids level for coating, the slurry, before and/or after any surface treatment, can be further mixed in an impingement mill or similar milling device. Examples of the use of an impingement mill for reducing and homogenizing coating mixes containing pigments is described in U.S. Pat. Nos. 5,210,114 and 5,292,588, to Katsen.

[0098] While the invention has been described in detail and with reference to specific and general embodiments thereof, it will be apparent to one skilled in the art that various changes and modifications can be made therein without departing from the spirit and scope thereof.

CLAIMS

What is claimed:

1. A separator for a lithium battery comprising:
a porous polymeric layer having a first side and a second side;
a first layer adjacent the first side of the porous polymeric layer, said first layer comprising a porous portion and a solid portion, wherein
 - (a) the solid portion comprises inorganic oxide particles and one or more polymers;
 - (b) a plurality of the inorganic oxide particles in the first layer having a crystallite size of about 5 nm to about 90 nm;
 - (c) the thickness of the first layer is about 3 microns or less and the total coating weight of the first layer is about 0.8 gsm to about 4.7 gsm; and
 - (d) the ratio of the total thickness of the first layer to the total thickness of the separator is 10% to 80%.
2. The separator of claim 1, wherein a plurality of pores in the first layer have a diameter of about 100 nm or less when measured by an SEM or TEM cross section of the porous layer.
3. The separator of claim 1, wherein the crystallite size of the inorganic oxide particle comprises about 5-25 nm, 5-80 nm, 30 nm-50 nm, 5-8 nm, or 5-75 nm.
4. The separator of claim 1, wherein the inorganic oxide particles of the first layer comprise a hydrated Al_2O_3 .

5. The separator of claim 1, wherein the inorganic oxide particles of the first layer comprise one or more of boehmite, SiO_2 , boron nitride pigments, AlN , hydrated aluminum oxide, AlOOH , and $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$.

6. The separator of claim 5, wherein when the inorganic oxide particles of the first layer comprise $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$, x is about 1.0 to 2.0.

7. The separator of claim 1, wherein the at least one polymer in the first layer comprises one or more of the following: cellulosic polymers, hydroxyethyl cellulose, vinyl pyrrolidone polymers and copolymers, polysaccharides, chitosan, polyethylene oxides, polyvinyl alcohols, polyolefin (e.g., polyethylene, polypropylene, homopolymers of 1-butene, 1-pentene, 1-hexene, and 1-octene, as well as copolymers and terpolymers thereof.

8. The separator of claim 1, wherein the volume fraction is about 12% to about 45%, 15% to about 45%, 20% to about 45%, about 25% to about 45%, or about 18% to about 45%.

9. The separator of claim 1, wherein the porous portion of the first layer has porosity of between about 30% to about 70%.

10. The separator of claim 1, further comprising a second layer adjacent to the second side of the porous polymeric layer, said second layer comprising inorganic oxide particles and one or more polymers.

11. The separator of claim 10, wherein the combined total coating weight of the first layer and second layer is less than about 9.4 gsm or less than about 8.4 gsm.

12. The separator of claim 1, wherein the total coating weight of the first layer is about 0.8 gsm to about 3.0 gsm.

13. The separator of Claim 10, wherein the composition of the first layer is different in composition from the composition of the second layer.

14. The separator of Claim 1, wherein the ratio of inorganic oxide to polymer in the first layer is 12:1 or 8:1.

15. The separator of Claim 10, wherein the combined total thickness of the first layer and second layer is less than about 3.8 microns.

16. The separator of Claim 1, wherein the pore volume of the first layer is between 0.8 cc/g and 1.0 cc/g.

17. The separator of Claim 1, wherein the porous polymeric layer comprises a polyolefin.

18. The separator of Claim 1, wherein the polyolefin is selected from the group consisting of polyethylenes, polypropylenes, homopolymers of 1-butene,

homopolymers of 1-pentene, homopolymers of 1-hexene, homopolymers of 1-octene, and copolymers and terpolymers thereof.

19. The separator of claim 9, wherein the porous portion of the first layer has porosity of between about 30% - 55%.

ABSTRACT

A separator for a lithium battery having (a) a porous polymeric layer, such as a polyethylene layer; and (b) a nanoporous inorganic particle/polymer layer on both sides of the polymeric layer, the nanoporous layer having an inorganic oxide and one or more polymers; the volume fraction of the polymers in the nanoporous layer is about 15% to about 50%, and the crystallite size of the inorganic oxide is 5 nm to 90 nm.



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ELECTRONIC PAYMENT RECEIPT

APPLICATION #
19/023,905

RECEIPT DATE / TIME
01/16/2025 12:06:14 PM Z ET

ATTORNEY DOCKET #
00249/0086

Title of Invention

MULTILAYER NANOPOROUS SEPARATOR

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3614

FILED BY Joy Chapman

PATENT CENTER # 68770157

AUTHORIZED BY Brian Amos

CUSTOMER # 1912

FILING DATE -

CORRESPONDENCE ADDRESS -

FIRST NAMED INVENTOR David W. Avison

Payment Information

PAYMENT METHOD
CARD / 1028

PAYMENT TRANSACTION ID
E20251FC08289491

PAYMENT AUTHORIZED BY
Joy Chapman

PRE-AUTHORIZED ACCOUNT
011785

PRE-AUTHORIZED CATEGORY
37 CFR 1.16 (National application filing, search, and examination fees); 37 CFR 1.17 (Patent application and reexamination processing fees); 37 CFR 1.19 (Document supply fees); 37 CFR 1.20 (Post Issuance fees); 37 CFR 1.21 (Miscellaneous fees and charges)

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1830	PROCESSING FEE, EXCEPT IN PROVISIONAL APPLICATIONS	140.00	1	140.00
1817	REQUEST FOR PRIORITIZED EXAMINATION	4200.00	1	4200.00
1111	UTILITY PATENT APPL. SEARCH FEE	700.00	1	700.00
1011	BASIC FILING FEE - UTILITY (PAPER FILING ALSO REQUIRES NON-ELECTRONIC FILING FEE UNDER 1.16(T))	320.00	1	320.00

1311	PATENT APPL. EXAMINATION FEE	800.00	1	800.00
TOTAL AMOUNT:				\$6,160.00

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Application Number: 19023905

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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
19/023,905	01/16/2025		1820	00249/0086	19	1

CONFIRMATION NO. 3614

1912

AMSTER, ROTHSTEIN & EBENSTEIN LLP
405 LEXINGTON AVENUE
FLOOR 48
NEW YORK, NY 10174

FILING RECEIPT



0000000084002371

Date Mailed: 02/05/2025

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

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Inventor(s)

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Applicant(s)

LG Energy Solution, Ltd., Seoul, KOREA, REPUBLIC OF;

Power of Attorney: The patent practitioners associated with Customer Number 1912

Domestic Applications for which benefit is claimed

This application is a CON of 18/631,783 04/10/2024
which is a CON of 17/543,259 12/06/2021 PAT 11,973,181
which is a CON of 17/223,845 04/06/2021 PAT 11,769,903
which is a CON of 16/755,294 04/10/2020 PAT 10,998,536
which is a 371 of PCT/US2018/055862 10/15/2018
which claims benefit of 62/572,083 10/13/2017

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - **None.**
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Projected Publication Date: 05/15/2025

Non-Publication Request: No

Early Publication Request: No

Title

MULTILAYER NANOPOROUS SEPARATOR

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
19/023,905

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	320
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	700
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	800
TOTAL CLAIMS (37 CFR 1.16(i))	19 minus 20 =	*			OR	x =	0
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*				x =	0
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1820

APPLICATION AS AMENDED - PART II

AMENDMENT A	(Column 1)	(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
		CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)	RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=	x =		OR	x =
Independent (37 CFR 1.16(h))	*	Minus	***	=	x =		OR	x =
Application Size Fee (37 CFR 1.16(s))							OR	
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR	
				TOTAL ADD'L FEE			OR	TOTAL ADD'L FEE

AMENDMENT B	(Column 1)	(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
		CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)	RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=	x =		OR	x =
Independent (37 CFR 1.16(h))	*	Minus	***	=	x =		OR	x =
Application Size Fee (37 CFR 1.16(s))							OR	
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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19/023,905

01/16/2025

David W. Avison

00249/0086

3614

1912 7590 02/25/2025

AMSTER, ROTHSTEIN & EBENSTEIN LLP

405 LEXINGTON AVENUE

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NEW YORK, NY 10174

EXAMINER

CENTRAL, DOCKET

ART UNIT

PAPER NUMBER

OPAP

MAIL DATE

DELIVERY MODE

02/25/2025

PAPER

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

<i>Decision Granting Request for Prioritized Examination (Track I)</i>	Application No. 19/023,905	Applicant(s) Avison et al.	
	Examiner FIKIRTE A GEREMEW	Art Unit OMBL	AIA (FITF) Status Yes

1. THE REQUEST FILED 16 January 2025 IS **GRANTED** .

The above-identified application has met the requirements for prioritized examination

A. ☒ for an original nonprovisional application (Track I).

B. ☐ for an application undergoing continued examination (RCE).

2. **The above-identified application will undergo prioritized examination.** The application will be accorded special status throughout its entire course of prosecution until one of the following occurs:

A. filing a **petition for extension of time** to extend the time period for filing a reply;

B. filing an **amendment to amend the application to contain more than four independent claims, more than thirty total claims**, or a multiple dependent claim;

C. filing a **request for continued examination** ;

D. filing a notice of appeal;

E. filing a request for suspension of action;

F. mailing of a notice of allowance;

G. mailing of a final Office action;

H. completion of examination as defined in 37 CFR 41.102; or

I. abandonment of the application.

Telephone inquiries with regard to this decision should be directed to FIKIRTE GEREMEW at (703) 756-1930. In his/her absence, calls may be directed to Petition Help Desk at (571) 272-3282.

/FIKIRTE A GEREMEW/ PROGRAM SUPPORT SPECIALIST, OMBL	
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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19/023,905

01/16/2025

David W. Avison

00249/0086

3614

1912

7590

03/25/2025

AMSTER, ROTHSTEIN & EBENSTEIN LLP

405 LEXINGTON AVENUE

FLOOR 48

NEW YORK, NY 10174

EXAMINER

PARK, LISA S

ART UNIT

PAPER NUMBER

1729

MAIL DATE

DELIVERY MODE

03/25/2025

PAPER

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

DETAILED CORRESPONDENCE

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Priority

2. Applicant's claim for the benefit of a prior-filed application under 35 U.S.C. 119(e) or under 35 U.S.C. 120, 121, or 365(c) is acknowledged.

Information Disclosure Statement

3. Information disclosure statement (IDS), submitted January 16, 2025, has been received and considered by the examiner.

Claim Interpretation

4. All "wherein" clauses are given patentable weight unless otherwise noted. Please see MPEP 2111.04 regarding optional claim language.

Specification

5. The specification is objected to as failing to provide proper antecedent basis for the claimed subject matter. See 37 CFR 1.75(d)(1) and MPEP § 608.01(o). Correction of the following is required: the specification does not provide any discussion of a "first layer" or a "porous portion" in said "first layer". The first layer could reasonably be equated to the nanoporous inorganic oxide/polymer composite layer, but the Office has no suggestion as to how the "porous portion" terminology issue of the first layer would be resolved.

Claim Rejections - 35 USC § 112

The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

(a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

6. Claims 1-19 are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the written description requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for applications subject to pre-AIA 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention. The claims recite “said first layer comprising a porous portion and a solid portion” and the solid portion is discussed in e.g. paras 0005-0007. However, the “porous portion” of the first layer (i.e. the nanoporous inorganic oxide/polymer composite layer) is not described at all, and there are no drawings that show how such a layer is constructed. Claims 9 and 19 recite a porosity of the porous portion, and this range of porosities is discussed in e.g. para 0038 of the disclosure as filed, but said discussion states that the first/nanoporous layer itself has a porosity of 30% to 70% and so this description is not helpful in showing that the Applicant had possession of the claimed invention.

The following is a quotation of 35 U.S.C. 112(b):

(B) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

7. Claims 1-19 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.

Claim 1 recites “said first layer comprising a porous portion” but because the instant disclosure does not recite what the porous portion is, then it is impossible to determine what is meant by “porous portion” in claim 1, and accordingly, impossible to determine the scope of the claim. Claims 2-19 are rejected as being dependent upon a rejected base claim.

Claim 3 recites “wherein the crystallite size of the inorganic oxide particles comprises about 5-15 nm, 5-80 nm, 20 nm-50 nm, 5-8 nm, or 5-75 nm.” A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, as just one example, claim 3 recites the broad recitation “5-75 nm”, and the claim also recites “5-25 nm”, which is a narrower statement of the range/limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by

such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims.

Claim 3 fails to provide units to all of the numerical values, e.g. “5- 25 nm” should be “5 nm – 25 nm” (consistent with other ranges written in the claim).

Claims 9 and 19 recite a porosity of the porous portion, but as explained above, it is unclear what is meant by “porous portion”.

Claim 11 recites “wherein the combined total coating weight of the first layer and second layer is less than about 9.4 gsm or less than about 8.4 gsm...” A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, claim 11 recites the broad recitation “less than about 9.4 gsm”, and the claim also recites “less than about 8.4 gsm”, which is a narrower statement of the range/limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims.

Claim 14 recites “wherein the ratio of inorganic oxide to the polymer in the first layer is 12:1 or 8:1”. A broad range or limitation together with a narrow range or limitation that falls within the broad range or limitation (in the same claim) may be considered indefinite if the resulting claim does not clearly set forth the metes and bounds of the patent protection desired. See MPEP § 2173.05(c). In the present instance, claim 11 recites the broad recitation “12:1”, and the claim also recites “8:1”, which is a narrower statement of the range/limitation. The claim(s) are considered indefinite because there is a question or doubt as to whether the feature

introduced by such narrower language is (a) merely exemplary of the remainder of the claim, and therefore not required, or (b) a required feature of the claims.

Claim 18 recites “wherein the polyolefin is selected from...” but this limitation lacks antecedent basis because there has not been a polyolefin recited. Accordingly, it is unclear how to interpret this limitation.

Claims 1-3, 6, 8-9, 11-12, 15-16 and 19 recite “about” before nearly all numerical values. This use of “about” is unclear because Applicant has not provided any numerical guides pertaining to the term “about” such that the skilled artisan could avoid infringement of the claims, for example. Claims 4-5, 10, 13-14, and 17-18 are rejected as being dependent upon a rejected base claim.

8. No rejections under prior art have been made but this is not an indication of allowable subject matter. A proper search cannot be conducted until the extensive issues under 35 USC 112 can be resolved.

9. Claims 1-19 are provisionally rejected on the ground of nonstatutory obviousness-type double patenting as being unpatentable over claims 1-19 of copending Application No. 19/022365.

Although the conflicting claims are not identical, they are not patentably distinct from each other because:

The combination of copending claims 1-19 read on currently pending claims 1-19.

The only differences in the claims are:

Claim 1 of the instant application recites “(a) the solid portion comprises inorganic oxide particles and one or more polymers” while the co-pending Claim 1 recites “(a) the solid portion comprises inorganic oxide particles and one or more polymers, and about 10% to about 50% of the solid portion comprises one or more polymers”. The scopes are different in that the instant application does not recite any volume fraction of polymer while the co-pending application recites that a certain portion (10% - 50 %) of the solid portion comprises polymer, which is a much broader recitation and therefore encompasses the instant claim.

Instant claim 12 recites one range for the total coating weight by co-pending claim 12 recites a few different ranges that ultimately encompass the instant claimed range.

Claims 2-11 and 13-19 are identical between the applications.

This is a provisional obviousness-type double patenting rejection because the conflicting claims have not in fact been patented.

10. Claims 1-19 are provisionally rejected on the ground of nonstatutory obviousness-type double patenting as being unpatentable over claims 1-19 of copending Application No. 19/023913.

Although the conflicting claims are not identical, they are not patentably distinct from each other because:

The combination of copending claims 1-19 read on currently pending claims 1-19.

The only differences in the claims are: Claim 1 of the instant application recites “(a) the solid portion comprises inorganic oxide particles and at least one polymer” while the co-pending Claim 1 recites “(a) a volume fraction of the one or more polymers in the solid fraction is 10% to about 50%”. The scopes are different in that the instant application recites that a certain portion

of the solid portion comprises polymer while the co-pending application does not recite any particular volume or mass composition of the polymer in the first layer, which is a much broader recitation. However, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to determine the amount of polymer in the solid portion to arrive at a first layer having a solid portion with the desirable properties since this would require only a routine optimization of composition. “[W]here the general conditions of a claim are disclosed in the prior art, it is not inventive to discover the optimum or workable ranges by routine experimentation.” See *In re Aller*, 220 F.2d 454, 456, 105 USPQ 233, 235 (CCPA 1955).

Co-pending Claim 12 recites multiple ranges for the total weight of the coating layer while the instant claim has only one range, but the instant claimed ranges are obvious over the co-pending range because in the case where the claimed ranges "overlap or lie inside ranges disclosed by the prior art" a *prima facie* case of obviousness exists. *In re Wertheim*, 541 F.2d 257, 191 USPQ 90 (CCPA 1976); *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990) (The prior art taught carbon monoxide concentrations of "about 1-5%" while the claim was limited to "more than 5%." The court held that "about 1-5%" allowed for concentrations slightly above 5% thus the ranges overlapped.); *In re Geisler*, 116 F.3d 1465, 1469-71, 43 USPQ2d 1362, 1365-66 (Fed. Cir. 1997) (Claim reciting thickness of a protective layer as falling within a range of "50 to 100 Angstroms" considered *prima facie* obvious in view of prior art reference teaching that "for suitable protection, the thickness of the protective layer should be not less than about 10 nm [i.e., 100 Angstroms]." The court stated that "by stating that 'suitable protection' is provided if the protective layer is 'about' 100 Angstroms thick, [the prior art reference] directly teaches the use of a thickness within [applicant's] claimed range."). Similarly,

a *prima facie* case of obviousness exists where the claimed ranges and prior art ranges do not overlap but are close enough that one skilled in the art would have expected them to have the same properties. *Titanium Metals Corp. of America v. Banner*, 778 F.2d 775, 227 USPQ 773 (Fed. Cir. 1985) (Court held as proper a rejection of a claim directed to an alloy of "having 0.8% nickel, 0.3% molybdenum, up to 0.1% iron, balance titanium" as obvious over a reference disclosing alloys of 0.75% nickel, 0.25% molybdenum, balance titanium and 0.94% nickel, 0.31% molybdenum, balance titanium.).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to LISA S PARK whose telephone number is (571)270-3597. The examiner can normally be reached M-Th 5:30a to 3p Eastern Time.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ula Tavares-Crockett can be reached on 5712721481. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about

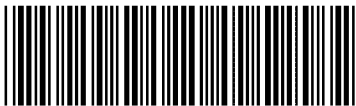
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/LISA S PARK/
Primary Examiner, Art Unit 1729

<i>Index of Claims</i> 	Application/Control No. 19/023,905	Applicant(s)/Patent Under Reexamination Avison et al.
	Examiner LISA S PARK	Art Unit 1729

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

CLAIMS										
☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	03/11/2025								
	1	✓								
	2	✓								
	3	✓								
	4	✓								
	5	✓								
	6	✓								
	7	✓								
	8	✓								
	9	✓								
	10	✓								
	11	✓								
	12	✓								
	13	✓								
	14	✓								
	15	✓								
	16	✓								
	17	✓								
	18	✓								
	19	✓								

<i>Search Notes</i> 	Application/Control No. 19/023,905	Applicant(s)/Patent Under Reexamination Avison et al.
	Examiner LISA S PARK	Art Unit 1729

CPC - Searched*		
Symbol	Date	Examiner
H01M10/0525 or H01M50/417 or H01M50/446 or H01M50/451 or H01M50/457 or H01M50/489 or H01M50/491	03/11/2025	/L.S.P./

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
PE2E keyword search (see attached search notes) (preliminary)	03/11/2025	/L.S.P./
PE2E inventor/assignee search	03/11/2025	/L.S.P./

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

/LISA S PARK/ Primary Examiner, Art Unit 1729	
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Bibliographic Data

Application No: 19/023,905

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /LISA S PARK/

Examiner's Signature

Initials

Title: MULTILAYER NANOPOROUS SEPARATOR

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
01/16/2025	429	1729	00249/0086
RULE			

APPLICANTS

LG Energy Solution, Ltd., Seoul, KOREA, REPUBLIC OF

INVENTORS

David W. Avison, Boxborough, MA, UNITED STATES

Steven A. Carlson, Cambridge, MA, UNITED STATES

Benjamin Sloan, Exeter, NH, UNITED STATES

CONTINUING DATA

This application is a CON of 18631783 04/10/2024

18631783 is a CON of 17543259 12/06/2021 PAT 11973181

17543259 is a CON of 17223845 04/06/2021 PAT 11769903

17223845 is a CON of 16755294 04/10/2020 PAT 10998536

16755294 is a 371 of PCT/US2018/055862 10/15/2018

PCT/US2018/055862 has PRO of 62572083 10/13/2017

FOREIGN APPLICATIONS

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02/05/2025

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\$6,160

Doc code: IDS

PTO/SB/08a (02-18)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 11/30/2020. OMB 0651-0031

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Application Number		
Filing Date		
First Named Inventor	David W. Avison	
Art Unit		
Examiner Name		
Attorney Docket Number	00249/0086	

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	9293752	B2	2016-03-22	Murata et al.	
	2	9847519	B2	2017-12-19	Pekalaet et al.	
	3	7709152	B2	2010-05-04	Kim et al.	
	4	7662517	B2	2010-02-16	Lee et al.	
	5	7638241	B2	2009-12-29	Lee et al.	
	6	6432586	B1	2002-08-13	Zhang	
	7	6723467	B2	2004-04-20	Yoshida et al.	
	8	7883799	B2	2011-02-08	Seo et al.	

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Application Number		
Filing Date		
First Named Inventor	David W. Avison	
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Examiner Name		
Attorney Docket Number	00249/0086	

	9	8318360	B2	2012-11-27	Seo et al.	
	10	9484565	B2	2016-11-01	Miyazaki et al.	
	11	9312527	B2	2016-04-12	Muramatsu et al.	
	12	9543077	B2	2017-01-10	Honda et al.	
	13	9478784	B2	2016-10-25	Iwase	
	14	4024323	A	1977-05-17	Versteegh	

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	1	20160172705	A1	2016-06-16	Hoshiba	
	2	20170294638	A1	2017-10-12	Joo et al.	
	3	20180019457	A1	2018-01-18	Alharizah et al.	(U.S. counterpart of WO 2016/123404 A1 listed below)

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

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Application Number		
Filing Date		
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Art Unit		
Examiner Name		
Attorney Docket Number	00249/0086	

	4	20180248164	A1	2018-08-30	Pekala et al.	
	5	20170301903	A1	2017-10-19	Choi et al.	
	6	20160104876	A1	2016-04-14	Avison et al.	
	7	20110165450	A1	2011-07-07	Nishikawa	
	8	20110003209	A1	2011-01-06	Katayama et al.	
	9	20070190427	A1	2007-08-16	Carlson et al.	
	10	20150155601	A1	2015-06-04	Nakatsutsumi et al.	
	11	20170012264	A1	2017-01-12	Carlson et al.	
	12	20170373295	A1	2017-12-28	Kim et al.	
	13	20160043370	A1	2016-02-11	Hatta et al.	

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Examiner Name		
Attorney Docket Number	00249/0086	

Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1	2016123404	WO	A1	2016-08-04	Hercules Incorporated	(WO counterpart of U.S. 2018/0019457 A1 listed above)	☐
	2	10-2016-0072009	KR	A	2016-06-22	Samsung SDI Co., Ltd.		☒
	3	10-2011-0116489	KR	A	2011-10-26	Korea Institute of Science and Technology		☒

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NON-PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	F. Sola et al., "A new scanning electron microscopy approach to image aerogels at the nanoscale; A new scanning electron microscopy approach to image aerogels at the nanoscale," Nanotechnology, IOP, BRISTOL, GB, Vol. 22, No. 17, March 16, 2011, page 175704, XP020188888, ISSN: 0957-4484, DOI: 10.1088/0957-4484/22/17/175704.	☐
	2	Chongwen Yang et al., "Boehmite particle coating modified microporous polyethylene membrane: A promising separator for lithium ion batteries," Journal of Power Sources, Vol. 348, March 3, 2017, pp. 80-86, XP029949881, ISSN: 0378-7753, DOI: 10.1016/J.POWSOUR.2017.02.078.	☐
	3	International Search Report issued in PCT Intl. Patent Application No. PCT/US2018/055862, mailed on April 11, 2019, 6 pages.	☐
	4	Written Opinion issued in PCT Intl. Patent Application No. PCT/US2018/055862, mailed on April 11, 2019, 13 pages.	☐
	5	Chinese Office Action dated April 12, 2022 issued in Chinese Patent Application No. 201880079213.4.	☐

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First Named Inventor	David W. Avison
Art Unit	
Examiner Name	
Attorney Docket Number	00249/0086

6	Communication Pursuant to Article 94(3) EPC dated September 8, 2023 issued in European Patent Application No. 18 799 922.2.	☐
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Examiner Signature	/LISA S PARK/	Date Considered	03/11/2025
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¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	
First Named Inventor	David W. Avison
Art Unit	
Examiner Name	
Attorney Docket Number	00249/0086

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Brian J. Amos/	Date (YYYY-MM-DD)	2025-01-16
Name/Print	Brian J. Amos	Registration Number	63,679

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	491584	(H01M10/0525 or H01M50/417 or H01M50/446 or H01M50/451 or H01M50/457 or H01M50/489 or H01M50/491).cpc.	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/03/11 08:53 AM
L2	6477	1 AND (multilayer composite hybrid) ADJ separator	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/03/11 08:54 AM
L3	1169	2 AND porous ADJ (base substrate middle central) AND (coat\$4 (first ADJ layer)) WITH (oxide inorganic alumina)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/03/11 08:54 AM
L4	0	2 AND porous ADJ (base substrate middle central) AND (coat\$4 (first ADJ layer)) WITH (oxide inorganic alumina) WITH non\$1porous WITH porous ADJ (section region portion area)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/03/11 08:54 AM
L5	1	2 AND porous ADJ (base substrate middle central) AND (coat\$4 (first ADJ layer)) WITH (oxide inorganic alumina) WITH porous ADJ (section region portion area)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/03/11 08:55 AM
L6	7	2 AND (base substrate middle central) AND (coat\$4 film layer) WITH (oxide inorganic alumina) WITH porous ADJ (section region portion area)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/03/11 08:55 AM
L7	5630	((("LG") near3 ("ENERGY") near3 ("SOLUTION") near3 ("LTD"))).AS,AANM.	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/11 08:56 AM
L8	9	7 AND multilayer ADJ separator	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/11 08:56 AM
L9	16	7 AND multilayer near2 separator	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/11 08:57 AM

L10	0	9 AND first ADJ layer WITH porous near2 polymer\$2	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/11 08:57 AM
L11	477	((("AVISON") near3 ("David")) OR (("CARLSON") near3 ("Steven")) OR (("SLOAN") near3 ("Benjamin"))).INV.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2025/03/11 09:07 AM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number		19023905
	Filing Date		2025-01-16
	First Named Inventor	David W. Avison	
	Art Unit	1729	
	Examiner Name	Lisa S. Park	
	Attorney Docket Number	00249/0086	

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	1	2017010780	WO	A1	2017-01-19	LG Chemical Ltd.		☒

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number		19023905
	Filing Date		2025-01-16
	First Named Inventor	David W. Avison	
	Art Unit		1729
	Examiner Name	Lisa S. Park	
	Attorney Docket Number		00249/0086

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Examiner Signature		Date Considered	
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¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.			

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

Application Number	19023905
Filing Date	2025-01-16
First Named Inventor	David W. Avison
Art Unit	1729
Examiner Name	Lisa S. Park
Attorney Docket Number	00249/0086

TIMING STATEMENT

Please see 37 CFR 1.97(e) to make the appropriate selection(s):

- ☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

- ☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

- ☐ See attached timing statement under 37 CFR 1.97(e).
- ☒ A timing statement under 37 CFR 1.97(e) is not submitted herewith.

TIMING FEE

- ☒ The fee set forth in 37 CFR 1.17(p) has been submitted herewith.

COPIES

- ☐ An identification of an earlier application pursuant to 37 CFR 1.98(d)(1) is attached.

SIZE FEE ASSERTION

Please see 37 CFR 1.17(v) and the [IDS Auto-Load Instructions](#) for completing this form to make the appropriate selection of an assertion under 37 CFR 1.98. For the information disclosure statement (IDS) submitted herewith, the applicant or patent owner certifies the following with respect to the cumulative number of applicant-provided or patent-owner provided items of information submitted to date including those in the accompanying IDS (select only one):

- ☒ No IDS size fee is required under 37 CFR 1.17(v) at this time.
- ☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(1).
- ☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(2).
- ☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(3).

SIGNATURE

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 11.18. Please see 37 CFR 1.4(d) for the form of the signature.

Signature	/Brian A. Comack/	Date (YYYY-MM-DD)	2025-04-07
Name/Print	Brian A. Comack	Registration Number	45,343

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

Application Number	19023905
Filing Date	2025-01-16
First Named Inventor	David W. Avison
Art Unit	1729
Examiner Name	Lisa S. Park
Attorney Docket Number	00249/0086

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
19/023,905

RECEIPT DATE / TIME
04/07/2025 04:37:17 PM Z ET

ATTORNEY DOCKET #
00249/0086

Title of Invention

MULTILAYER NANOPOROUS SEPARATOR

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3614

FILED BY Joy Chapman

PATENT CENTER # 69899759

FILING DATE 01/16/2025

CUSTOMER # 1912

FIRST NAMED
INVENTOR David W. Avison

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY Brian Comack

Documents

TOTAL DOCUMENTS: 7

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
00249-0086_Response-to-Office-Action.pdf	8	-	208 KB
00249-0086_Response-to-Office-Action-A....pdf (1-1)	1	Amendment/Request for Reconsideration-After Non-Final Rejection	131 KB
00249-0086_Response-to-Office-Action-CLM.pdf (2-5)	4	Claims	114 KB
00249-0086_Response-to-Office-Action-REM.pdf (6-8)	3	Applicant Arguments/Remarks Made in an Amendment	176 KB
00249-0086_Terminal-Disclaimer_US-Appln-19-022365_fillable-pdf.pdf	2	Terminal Disclaimer Filed	397 KB
00249-0086_Terminal-	2	Terminal Disclaimer Filed	396 KB

Disclaimer_US-Appln-19-021913_fillable-pdf.pdf

00249-0086_IDS-with-Size-Certification_fillable-pdf.pdf	5	Information Disclosure Statement (IDS) Form (SB08)	1314 KB
WO2017010780_with-Eng-Abstract.pdf	22	Foreign Reference	1465 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
00249-0086_Response-to-Office-Action.pdf	411DD1FB44835F5B4334CDBA126D08D5624F9FD380B9D5A13A69717C8E524DB9963C15466174059C2628741669FCF88C97D6B9CFEBC885F2352928B9F4866B4E
00249-0086_Response-to-Office-Action-A....pdf	E5D9F1E55C95DDFA8E82BD6EA5825B489AA8EC764A8A7BFAA09381E7A4C543B218FC49B16DD5D8CFB2150CD96D1DEB8DB45D96ECD86C75162B1DE2712182C75D
00249-0086_Response-to-Office-Action-CLM.pdf	7FB16E651BE74DA9FB6B4411C056CDCC11DD4879AED5EA00BF8F5FF4D2A827637B783311A5C5250D2AAD8DD5A5FD88D957F4D968FAA03ACC85F9E7D043FBE124
00249-0086_Response-to-Office-Action-REM.pdf	FC65EA58C1DE5A247CCC14001C50E2DA2BB70F6D1E84B1B4634D48C26CBB37E0763F8B0F6DD596BCBB9DF0BE9D9C4B603B0C5C5FD60360F53934513B448371EF
00249-0086_Terminal-Disclaimer_US-Appln-19-022365_fillable-pdf.pdf	549F7A6CD1D767A26CF798AD74CD4F01353B6DCF470BC02B965CE12C1DAD7EFE391C0BAC0B0F10F80DC131F7395840F53F32D2DCCAF7DDB42CCFC52AF4BFBFFA
00249-0086_Terminal-Disclaimer_US-Appln-19-021913_fillable-pdf.pdf	AC29B11852669975DBF0D1887E15E6CAD637FADA6388D3409135F5BBD8E9C2276117C50DE1DC3124B1292FDEBB81EE679EAB989AF368DF619EB5C74A4F77C9C5
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WO2017010780_with-Eng-Abstract.pdf	5CF376F5307115C1EB77B88AF2521D7A60DE560061E1B5550010DE2121F068AA6121EDB9634C8FC148AF8C2A253D17408F4C79A9EB49F45D4214F709A9A95948

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Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

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If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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ELECTRONIC PAYMENT RECEIPT

APPLICATION #
19/023,905

RECEIPT DATE / TIME
04/07/2025 04:37:17 PM Z ET

ATTORNEY DOCKET #
00249/0086

Title of Invention

MULTILAYER NANOPOROUS SEPARATOR

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3614

FILED BY Joy Chapman

PATENT CENTER # 69899759

AUTHORIZED BY Brian Comack

CUSTOMER # 1912

FILING DATE 01/16/2025

CORRESPONDENCE ADDRESS -

FIRST NAMED INVENTOR David W. Avison

Payment Information

PAYMENT METHOD
CARD / 1051

PAYMENT TRANSACTION ID
E202547G37515321

PAYMENT AUTHORIZED BY
Joy Chapman

PRE-AUTHORIZED ACCOUNT
011785

PRE-AUTHORIZED CATEGORY
37 CFR 1.16 (National application filing, search, and examination fees); 37 CFR 1.17 (Patent application and reexamination processing fees); 37 CFR 1.19 (Document supply fees); 37 CFR 1.20 (Post Issuance fees); 37 CFR 1.21 (Miscellaneous fees and charges)

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1806	SUBMISSION OF AN INFORMATION DISCLOSURE STATEMENT	280.00	1	280.00
1202	EACH CLAIM IN EXCESS OF 20	200.00	10	2000.00
1814	STATUTORY DISCLAIMER, INCLUDING TERMINAL DISCLAIMER	183.00	2	366.00
TOTAL AMOUNT:				\$2,646.00

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**TERMINAL DISCLAIMER TO OBVIATE A PROVISIONAL DOUBLE PATENTING
REJECTION OVER A PENDING "REFERENCE" APPLICATION**

Docket Number (Optional)

00249/0086

In re Application of: David W. Avison et al.

Application No.: 19/023,905

Filed: January 16, 2025

For: MULTILAYER NANOPOROUS SEPARATOR

The applicant, LG Energy Solution, Ltd., owner of 100 percent interest in the instant application hereby disclaims, except as provided below, the terminal part of the statutory term of any patent granted on the instant application which would extend beyond the expiration date of the full statutory term of any patent granted on pending **reference** Application Number 19/021,913, filed, January 15, 2025, as the term of any patent granted on said **reference** application may be shortened by any terminal disclaimer filed prior to the grant of any patent on the pending **reference** application. The applicant hereby agrees that any patent so granted on the instant application shall be enforceable only for and during such period that it and any patent granted on the **reference** application are commonly owned. This agreement runs with any patent granted on the instant application and is binding upon the grantee, its successors or assigns.

In making the above disclaimer, the applicant does not disclaim the terminal part of any patent granted on the instant application that would extend to the expiration date of the full statutory term of any patent granted on said **reference** application, "as the term of any patent granted on said **reference** application may be shortened by any terminal disclaimer filed prior to the grant of any patent on the pending **reference** application," in the event that: any such patent granted on the pending **reference** application expires for failure to pay a maintenance fee, is held unenforceable, is found invalid by a court of competent jurisdiction, is statutorily disclaimed in whole or terminally disclaimed under 37 CFR 1.321, has all claims canceled by a reexamination certificate, is reissued, or is in any manner terminated prior to the expiration of its full statutory term as shortened by any terminal disclaimer filed prior to its grant.

Check either box 1 or 2 below, if appropriate.

1. ☐ The undersigned is the applicant. If the applicant is an assignee, the undersigned is authorized to act on behalf of the assignee.

I hereby acknowledge that any willful false statements made are punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

2. ☒ The undersigned is an attorney or agent of record. Reg. No. 45,343

/Brian A. Comack/
Signature

April 7, 2025
Date

Brian A. Comack
Typed or printed name

Attorney for Applicant
Title

(212) 336-8000
Telephone Number

- ☒ Terminal disclaimer fee under 37 CFR 1.20(d) is included.

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant(s) : LG Energy Solution, Ltd.
U.S. Serial No. : 19/023,905 Examiner : Lisa S. Park
Filing Date : January 16, 2025 Art Unit : 1729
Confirmation No. : 3614
For : MULTILAYER NANOPOROUS SEPARATOR

405 Lexington Avenue
48th Floor
New York, NY 10174
April 7, 2025

BY EFS
Commissioner for Patents
Alexandria, VA 22313-1450

RESPONSE TO MARCH 25, 2025 OFFICE ACTION

This Amendment is submitted in response to the March 25, 2025 Office Action issued in connection with the above-identified application. A response to the March 25, 2025 Office Action is due June 25, 2025. Accordingly, this Amendment is being timely filed.

Amendments to the Claims begin on page **2** of this paper.

Remarks begin on page **6** of this paper.

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Serial No. : 19/023,905
Filed : January 16, 2025
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AMENDMENTS TO THE CLAIMS

Please amend the claims by replacing all prior versions of the claims with the listing of claims below pursuant to 37 C.F.R. § 1.121.

1. (Currently Amended) A separator for a lithium battery comprising:
a porous polymeric layer having a first side and a second side;
a first layer adjacent the first side of the porous polymeric layer, said first layer comprising ~~a porous portion~~ pores and ~~[[a]] solid material portion~~, wherein
 - (a) the solid ~~material portion~~ comprises inorganic oxide particles and ~~one or more polymers~~ at least one polymer;
 - (b) a plurality of the inorganic oxide particles in the first layer ~~comprise having a crystallites having a size in the range of about~~ 5 nm to ~~about~~ 90 nm;
 - (c) ~~the a~~ thickness of the first layer is ~~about~~ 3 microns or less and ~~the a~~ total coating weight of the first layer is ~~about~~ 0.8 gsm to ~~about~~ 4.7 gsm; and
 - (d) ~~the a~~ ratio of the ~~total~~ thickness of the first layer to ~~the a~~ total thickness of the separator is 10% to 80%.
2. (Currently Amended) The separator of claim 1, wherein a plurality of the pores in the first layer have a diameter of ~~about~~ 100 nm or less ~~when measured by an SEM or TEM cross section of the porous layer~~.
3. (Currently Amended) The separator of claim 1, wherein the crystallites have a size of the ~~inorganic oxide particle comprises about 5-25 nm, 5-80 nm, 30 nm-50 nm, 5-8 nm, or 5-75 nm~~ 5 nm to 80 nm.
4. (Original) The separator of claim 1, wherein the inorganic oxide particles of the first layer comprise a hydrated Al₂O₃.

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5. (Original) The separator of claim 1, wherein the inorganic oxide particles of the first layer comprise one or more of boehmite, SiO_2 , boron nitride pigments, AlN , hydrated aluminum oxide, AlOOH , and $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$.

6. (Currently Amended) The separator of claim 5, wherein when the inorganic oxide particles of the first layer comprise $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$, x is ~~about~~ 1.0 to 2.0.

7. (Currently Amended) The separator of claim 1, wherein the at least one polymer in the first layer comprises one or more of the following: cellulosic polymers, hydroxyethyl cellulose, vinyl pyrrolidone polymers and copolymers, polysaccharides, chitosan, polyethylene oxides, polyvinyl alcohols, and polyolefin (~~e.g., polyethylene, polypropylene, homopolymers of 1-butene, 1-pentene, 1-hexene, and 1-octene, as well as copolymers and terpolymers thereof.~~

8. (Currently Amended) The separator of claim 1, wherein ~~the~~ a volume fraction of the at least one polymer in the solid material is ~~about 12% to about 45%, 15% to about 45%, 20% to about 45%, about 25% to about 45%, or about 18% to about 45%~~ 10% to 50%.

9. (Currently Amended) The separator of claim 1, wherein ~~the porous portion of the first layer has~~ a porosity of ~~between about 30% to about 70%.~~

10. (Original) The separator of claim 1, further comprising a second layer adjacent to the second side of the porous polymeric layer, said second layer comprising inorganic oxide particles and one or more polymers.

11. (Currently Amended) The separator of claim 10, wherein ~~the~~ a combined total coating weight of the first layer and second layer is less than ~~about 9.4 gsm or less than about 8.4 gsm.~~

12. (Currently Amended) The separator of claim 1, wherein the total coating weight of the first layer is ~~about~~ 0.8 gsm to ~~about~~ 3.0 gsm.

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13. (Currently Amended) The separator of [[C]]claim 10, wherein ~~the composition of the first layer is different in composition from the composition of the second layer.~~

14. (Currently Amended) The separator of [[C]]claim 1, wherein ~~the~~ a ratio by weight of inorganic oxide to polymer in the first layer is about 12:1 ~~or 8:1~~.

15. (Currently Amended) The separator of [[C]]claim 10, wherein ~~the~~ a combined total thickness of the first layer and second layer is less than ~~about~~ 3.8 microns.

16. (Currently Amended) The separator of [[C]]claim 1, wherein ~~the~~ a pore volume of the first layer is between 0.8 cc/g and 1.0 cc/g.

17. (Original) The separator of Claim 1, wherein the porous polymeric layer comprises a polyolefin.

18. (Currently Amended) The separator of [[C]]claim 17, wherein the polyolefin is selected from the group consisting of polyethylenes, polypropylenes, homopolymers of 1-butene, homopolymers of 1-pentene, homopolymers of 1-hexene, homopolymers of 1-octene, and copolymers and terpolymers thereof.

19. (Currently Amended) The separator of claim 9, wherein ~~the porous portion of the first layer~~ has a porosity of ~~between about~~ 30% ~~[-]~~ to 55%.

20. (New) The separator of claim 1, wherein the crystallites have a size of 5 nm to 25 nm.

21. (New) The separator of claim 1, wherein the crystallites have a size of 30 nm to 50 nm.

22. (New) The separator of claim 1, wherein the crystallites have a size of 5 nm to 75 nm.

23. (New) The separator of claim 8, wherein the volume fraction is 12% to 45%.

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24. (New) The separator of claim 8, wherein the volume fraction is 25% to 45%.
25. (New) The separator of claim 8, wherein the volume fraction is 15.3% to 25.6%.
26. (New) The separator of claim 10, wherein a combined total coating weight of the first layer and second layer is 8.4 gsm or less.
27. (New) The separator of claim 1, wherein the total coating weight of the first layer is 5.2 gsm or less.
28. (New) The separator of claim 1, wherein the total coating weight of the first layer is 3.6 gsm or less.
29. (New) The separator of claim 1, wherein a ratio by weight of inorganic oxide to polymer in the first layer is about 8:1.
30. (New) The separator of claim 1, wherein a ratio by weight of inorganic oxide to polymer in the first layer is about 4:1.

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REMARKS

Claims 1-30 are pending in the subject application, with Claims 1-3, 6-9, 11-16, 18, and 19 being amended and new Claims 20-30 added via amendment. Support for the amendments is provided in the discussion below. Applicant maintains that the amendments to the claims raise no issue of new matter and entry of these amendments is respectfully requested.

Claims 1-19 have been rejected under 35 U.S.C. § 112(b) or 35 U.S.C. § 112, second paragraph (*see* Office Action, Par. 7) and on provisional nonstatutory double patenting grounds (*see id.*, Pars. 9 and 10). The specification has also been objected to under 35 U.S.C. § 112(a) or 35 U.S.C. § 112, first paragraph (pre-AIA).

Non-Statutory Double Patenting Rejections

Regarding the provisional nonstatutory double patenting rejections, Applicant submits herewith a terminal disclaimer over the cited co-pending applications, namely, U.S. Patent Application No. 19/021,913 and U.S. Patent Application No. 19/022,365.

Section 112 Rejections and Objections

Regarding the Section 112 rejections of the claims and the specification, Applicant has amended Claims 1-3, 6-9, 11-16, 18, and 19 and has added new Claims 20-30. Applicant respectfully submits that these amendments obviate and overcome the Section 112 rejections of the claims. Nevertheless, Applicant makes the following remarks in support of these amendments.

“First Layer” and “Porous Portion”: The Office Action asserts that “the specification does not provide any discussion of a ‘first layer’ or a ‘porous portion’ in said ‘first layer’”. (*See* Office Action, Pars. 5-7.) Although the specification does not use the words “first layer”, it nevertheless teaches that the polyethylene separator (e.g., porous polymeric layer) can have a single “boehmite coating” on just one side (e.g., the claimed “first layer”) or a second boehmite coating on the opposite side as well (e.g., the “second layer” of Claim 10). *See, e.g.*, Pars. [0008], [0073]. Thus, the specification does indeed disclose the claimed “first layer.”

Regarding the phrase “porous portion”, Applicant has amended Claim 1 to clarify that the first layer comprises “pores” (instead of a “porous portion”) and “solid material” (instead of a “solid portion”).

These amendments seek to clarify that the first layer has both pores and a solid material (e.g., inorganic oxide particles and polymer). This is consistent with and supported by Paragraph

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[0038] of the specification, which teaches that the solid material includes an “inorganic particle/polymer composite layer” and an “air volume (e.g., pores) [that] is not included in the solid portion.”

Dependent Claims 9 and 19 have been amended to delete the term “porous portion” altogether since such language is not necessary.

It is respectfully submitted that these amendments obviate the Section 112 objections and rejections.

Multiple Claimed Ranges: The Office Action objects to Claims 3, 11 and 14 as each reciting multiple numeric ranges in a single claim. To advance prosecution, Applicant has amended these claims to recite a single range and has added new Claims 20-30, which also recite single ranges. To address another objection in the Office Action, all claimed ranges recite units of measure for each of the numbers recited in the claimed ranges.

“About”: The Office Action objects to the term “about” in Claims 1-3, 6, 8-9, 11-12, 15, and 19. To advance prosecution and without making any admissions, Applicant has deleted the term “about” from these claims. Applicant has amended Claim 14 and added new Claim 30 to include the term “about” with the recited ratios. A person of ordinary skill in the art would understand that the claimed ratios allow for some variation when the ingredients of the separator are combined and processed to form a separator.

“Polyolefin”: Claim 18 has been amended to provide proper antecedent basis for the term “polyolefin.”

In view of the foregoing, Applicant respectfully requests the withdrawal of the rejections.

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No fee, other than the fee for adding ten (10) extra total claims in excess of twenty (20) paid herewith, is deemed necessary in connection with the filing of this Amendment. However, if any fee is required, authorization is hereby given to charge the amount of any such fee to Deposit Account No. 01-1785.

Respectfully submitted,

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Dated: April 7, 2025
New York, New York

By /Brian A. Comack/
Brian A. Comack, Reg. No. 45,343



Espacenet

Bibliographic data: WO2017010780 (A1) — 2017-01-19

SEPARATOR AND ELECTROCHEMICAL DEVICE COMPRISING SAME

Inventor(s): SUNG DONG-WOOK [KR]; KIM MIN-HYUNG [KR]; SUNG IN-HYOUK [KR]; SHIN HYUN-KYUNG [KR]; LEE JI-EUN [KR]; LEE JOO-SUNG [KR] ± (성동욱, ; 김민형, ; 성인혁, ; 신현경, ; 이지은, ; 이주성)

Applicant(s): LG CHEMICAL LTD [KR] ± (주식회사 엘지화학)

Classification: - **international:** H01M10/052; H01M50/414; H01M50/434; H01M50/443; H01M50/449; H01M50/417; H01M50/489; H01M50/491
 - **cooperative:** C08K3/22 (KR); H01M10/052 (KR, US); H01M10/0525 (US); H01M50/414 (EP, KR, US); H01M50/417 (EP, KR); H01M50/434 (EP, KR, US); H01M50/443 (EP, KR, US); H01M50/446 (EP, KR, US); H01M50/449 (EP, US); H01M50/451 (KR); H01M50/489 (KR); H01M50/491 (KR); C08K2003/2227 (KR); H01M50/417 (US); H01M50/489 (EP, US); H01M50/491 (EP, US); Y02E60/10 (EP, KR, US); Y02T10/70 (KR, US)

Application number: WO2016KR07532 20160711 Global Dossier

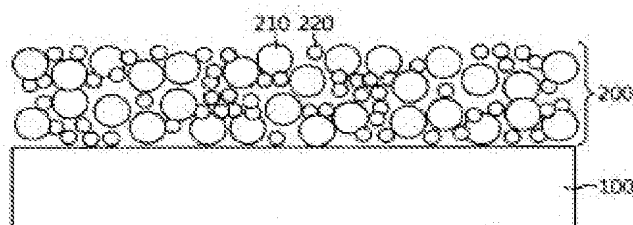
Priority number(s): KR20150098672 20150710

Also published as: CN107851762 (A) CN107851762 (B) EP3316346 (A1) EP3316346 (A4) EP3316346 (B1) more

Abstract of WO2017010780 (A1)

Disclosed are a separator and an electrochemical device comprising the same, the separator comprising: a porous substrate having a plurality of pores; and a porous coating layer formed on at least one surface of the porous substrate or at least one surface of the porous substrate and a portion of the pore, the porous coating layer containing a plurality of inorganic particles and a binder polymer disposed on a part or the entirety of a surface of the inorganic particle to connect and fix the inorganic particles, wherein the inorganic particle comprises an alumina particle and an

[51]



aluminum hydroxide particle having a smaller average particle diameter than the alumina particle.

(12) 특허협력조약에 의하여 공개된 국제출원

(19) 세계지식재산권기구
국제사무국



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(81) 지정국 (별도의 표시가 없는 한, 가능한 모든 종류의 국내 권리의 보호를 위하여): AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CZ, DE, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IR, IS, JP, KE, KG, KN, KP, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, SM, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, ZA, ZM, ZW.

(84) 지정국 (별도의 표시가 없는 한, 가능한 모든 종류의 역내 권리의 보호를 위하여): ARIPO (BW, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ, UG, ZM, ZW), 유라시아 (AM, AZ, BY, KG, KZ, RU, TJ, TM), 유럽 (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

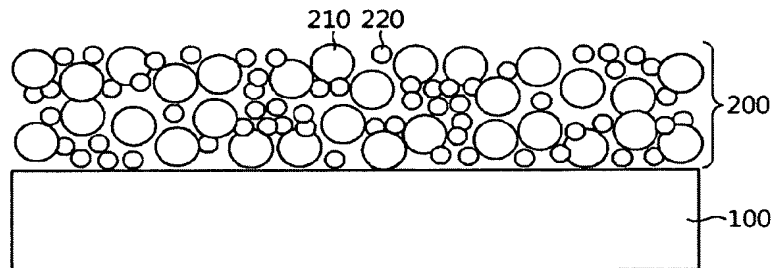
공개:

— 국제조사보고서와 함께 (조약 제 21 조(3))

(54) Title: SEPARATOR AND ELECTROCHEMICAL DEVICE COMPRISING SAME

(54) 발명의 명칭 : 세퍼레이터 및 이를 포함하는 전기화학소자

[도1]



(57) Abstract: Disclosed are a separator and an electrochemical device comprising the same, the separator comprising: a porous substrate having a plurality of pores; and a porous coating layer formed on at least one surface of the porous substrate or at least one surface of the porous substrate and a portion of the pore, the porous coating layer containing a plurality of inorganic particles and a binder polymer disposed on a part or the entirety of a surface of the inorganic particle to connect and fix the inorganic particles, wherein the inorganic particle comprises an alumina particle and an aluminum hydroxide particle having a smaller average particle diameter than the alumina particle.

(57) 요약서: 다수의 기공을 갖는 다공성 기재; 및 상기 다공성 기재의 적어도 일면에 또는 상기 다공성 기재의 적어도 일면 및 기공의 일부에 형성되어 있으며, 다수의 무기물 입자 및 상기 무기물 입자의 표면의 일부 또는 전부에 위치하여 상기 무기물 입자 사이를 연결 및 고정시키는 바인더 고분자를 포함하는 다공성 코팅층을 포함하고, 상기 무기물 입자가 알루미늄 입자 및 상기 알루미늄 입자 보다 평균입경이 작은 수산화알루미늄 입자를 포함하는 세퍼레이터 및 이를 포함하는 전기화학 소자가 제시된다.



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명세서

발명의 명칭: 세퍼레이터 및 이를 포함하는 전기화학소자 기술분야

- [1] 본 발명은 세퍼레이터 및 이를 포함하는 전기화학소자에 관한 것으로서, 열수축 특성이 세퍼레이터 및 이를 포함하는 전기화학소자에 관한 것이다.
- [2] 본 출원은 2015년 7월 10일에 출원된 한국출원 제10-2015-0098672호에 기초한 우선권을 주장하며, 해당 출원의 명세서 및 도면에 개시된 모든 내용은 본 출원에 원용된다.

배경기술

- [3] 최근 에너지 저장 기술에 대한 관심이 갈수록 높아지고 있다. 휴대폰, 캠코더 및 노트북 PC, 나아가서는 전기 자동차의 에너지까지 적용분야가 확대되면서 전기화학소자의 연구와 개발에 대한 노력이 점점 구체화되고 있다. 전기화학소자는 이러한 측면에서 가장 주목 받고 있는 분야이고 그 중에서도 충방전이 가능한 이차전지의 개발은 관심의 초점이 되고 있으며, 최근에는 이러한 전지를 개발함에 있어서 용량 밀도 및 비에너지를 향상시키기 위하여 새로운 전극과 전지의 설계에 대한 연구개발로 진행되고 있다.
- [4] 현재 적용되고 있는 이차전지 중에서 1990년대 초에 개발된 리튬 이차전지는 수용액 전해액을 사용하는 Ni-MH, Ni-Cd, 황산-납 전지 등의 재래식 전지에 비해서 작동 전압이 높고 에너지 밀도가 월등히 크다는 장점으로 각광을 받고 있다. 그러나 이러한 리튬 이온 전지는 유기 전해액을 사용하는 데 따르는 발화 및 폭발 등의 안전 문제가 존재하고, 제조가 까다로운 단점이 있다.
- [5] 최근의 리튬 이온 고분자 전지는 이러한 리튬 이온 전지의 약점을 개선하여 차세대 전지의 하나로 꼽히고 있으나 아직까지 전지의 용량이 리튬 이온 전지와 비교하여 상대적으로 낮고, 특히 저온에서의 방전 용량이 불충분하여 이에 대한 개선이 시급히 요구되고 있다.
- [6] 상기와 같은 전기화학소자는 많은 회사에서 생산되고 있으나 그들의 안전성 특성은 각각 다른 양상을 보인다. 이러한 전기화학소자의 안전성 평가 및 안전성 확보는 매우 중요하다. 가장 중요한 고려사항은 전기화학소자가 오작동시 사용자에게 상해를 입혀서는 안 된다는 것이며, 이러한 목적으로 안전규격은 전기화학소자 내의 발화 및 발연 등을 엄격히 규제하고 있다. 전기화학소자의 안전성 특성에 있어서, 전기화학소자가 과열되어 열폭주가 일어나거나 분리막이 관통될 경우에는 폭발을 일으키게 될 우려가 크다. 특히, 전기화학소자의 분리막으로서 통상적으로 사용되는 폴리올레핀계 다공성 기재는 재료적 특성과 연신을 포함하는 제조공정상의 특성으로 인하여 100°C 이상의 온도에서 극심한 열 수축 거동을 보임으로서, 양극과 음극 사이의 단락을 일으켰다.

- [7] 이와 같은 전기화학소자의 안전성 문제를 해결하기 위하여, 다수의 기공을 갖는 다공성 기재의 적어도 일면에, 과량의 무기물 입자와 바인더 고분자의 혼합물을 코팅하여 다공성 유기-무기 코팅층을 형성한 세퍼레이터가 제안되었다.
- [8] 이러한 다공성 유기-무기 코팅층을 형성한 세퍼레이터에서 입경이 지나치게 큰 큰 입자를 사용하는 경우 열수축 제어가 어려웠고, 입경이 지나치게 작은 입자를 사용하는 경우, 코팅층 형성용 조성물의 안정성이나 공정성이 떨어져 사용이 곤란하다는 문제가 있다.

발명의 상세한 설명

기술적 과제

- [9] 따라서 본 발명이 해결하고자 하는 과제는, 열수축을 최소화하여 안정성을 확보한 세퍼레이터를 제공하는 것이다.
- [10] 본 발명이 해결하고자 하는 또 다른 과제는 상기 세퍼레이터를 구비하는 전기화학소자를 제공하는 것이다.

과제 해결 수단

- [11] 상기 과제를 해결하기 위하여 본 발명의 일 측면에 따르면, 하기 구현예들의 세퍼레이터가 제공된다.
- [12] 제1 구현예는,
- [13] 다수의 기공을 갖는 다공성 기재; 및
- [14] 상기 다공성 기재의 적어도 일면에 또는 상기 다공성 기재의 적어도 일면 및 기공의 일부에 형성되어 있으며, 다수의 무기물 입자 및 상기 무기물 입자의 표면의 일부 또는 전부에 위치하여 상기 무기물 입자 사이를 연결 및 고정시키는 바인더 고분자를 포함하는 다공성 코팅층을 포함하고,
- [15] 상기 무기물 입자가 알루미나 입자 및 상기 알루미나 입자 보다 평균입경이 작은 수산화알루미늄 입자를 포함하는 세퍼레이터에 관한 것이다.
- [16] 제2 구현예는, 제1 구현예에 있어서,
- [17] 상기 수산화알루미늄 입자의 평균입경이 상기 알루미나 입자의 평균입경의 이 0.01 내지 0.9 배인 세퍼레이터에 관한 것이다.
- [18] 제3 구현예는, 제1 구현예 또는 제2 구현예에 있어서,
- [19] 상기 알루미나 입자의 평균입경이 0.5 내지 3 μm 인 세퍼레이터에 관한 것이다.
- [20] 제4 구현예는, 제1 구현예 내지 제3 구현예 중 어느 한 구현예에 있어서,
- [21] 상기 수산화알루미늄 입자의 평균입경이 0.05 내지 0.4 μm 인 세퍼레이터에 관한 것이다.
- [22] 상기 알루미나 입자 대 상기 수산화알루미늄 입자의 함량비가 50:50 내지 97:3인 세퍼레이터에 관한 것이다.
- [23] 제5 구현예는, 제1 구현예 내지 제4 구현예 중 어느 한 구현예에 있어서,
- [24] 상기 수산화알루미늄 입자가 보헤마이트($\gamma\text{-AlO(OH)}$), 유사 보헤마이트(Al_2O_3

· H₂O), 다이어스포(diaspore, α -AlO(OH)), 바이어리트(bayerlite, α -AlO(OH)₃), 깁사이트(gibbsite, γ -AlO(OH)₃), 노드스트란디트(nordstrandite, AlO(OH)₃) 로 이루어진 군으로부터 선택된 1종 이상인 세퍼레이터에 관한 것이다.

- [25] 제6 구현예는, 제1 구현예 내지 제5 구현예 중 어느 한 구현예에 있어서,
- [26] 상기 다공성 기재가 폴리올레핀계 다공성 기재인 세퍼레이터에 관한 것이다.
- [27] 제7 구현예는, 제6 구현예에 있어서,
- [28] 상기 폴리올레핀계 다공성 기재가 폴리에틸렌, 폴리프로필렌, 폴리부틸렌 및 폴리펜텐으로 이루어진 군으로부터 선택된 어느 하나의 고분자 또는 이들의 혼합물로 형성된 세퍼레이터에 관한 것이다.
- [29] 제8 구현예는, 제1 구현예 내지 제7 구현예 중 어느 한 구현예에 있어서,
- [30] 상기 다공성 기재의 두께가 5 내지 50 μ m이고, 기공 크기 및 기공도는 각각 0.01 내지 50 μ m 및 10 내지 95%인 세퍼레이터에 관한 것이다.
- [31] 제9 구현예는, 제1 구현예 내지 제8 구현예 중 어느 한 구현예에 있어서,
- [32] 상기 무기물 입자와 바인더 고분자의 중량비가 50:50 내지 99:1인 세퍼레이터에 관한 것이다.
- [33] 제10 구현예는, 제1 구현예 내지 제9 구현예 중 어느 한 구현예에 있어서,
- [34] 상기 바인더 고분자가 폴리비닐리덴 플루오라이드-헥사플루오로프로필렌 (polyvinylidene fluoride-co-hexafluoropropylene), 폴리비닐리덴 플루오라이드-트리클로로에틸렌 (polyvinylidene fluoride-co-trichloroethylene), 폴리메틸메타크릴레이트 (polymethylmethacrylate), 폴리부틸아크릴레이트 (polybutylacrylate), 폴리아크릴로니트릴 (polyacrylonitrile), 폴리비닐피롤리돈 (polyvinylpyrrolidone), 폴리비닐아세테이트 (polyvinylacetate), 에틸렌 비닐 아세테이트 공중합체 (polyethylene-co-vinyl acetate), 폴리에틸렌옥사이드 (polyethylene oxide), 폴리아릴레이트(polyarylate), 셀룰로오스 아세테이트 (cellulose acetate), 셀룰로오스 아세테이트 부틸레이트 (cellulose acetate butyrate), 셀룰로오스 아세테이트 프로피오네이트 (cellulose acetate propionate), 시아노에틸플루란 (cyanoethylpullulan), 시아노에틸폴리비닐알콜 (cyanoethylpolyvinylalcohol), 시아노에틸셀룰로오스 (cyanoethylcellulose), 시아노에틸수크로오스 (cyanoethylsucrose), 플루란 (pullulan) 및 카르복실 메틸 셀룰로오스 (carboxyl methyl cellulose)로 이루어진 군으로부터 선택된 어느 하나 또는 이들 중 2종 이상의 혼합물인 세퍼레이터에 관한 것이다.
- [35] 본 발명의 다른 측면에 따르면, 하기 구현예들의 전기화학소자가 제공된다.
- [36] 제11 구현예는, 캐소드, 애노드, 상기 캐소드 및 애노드 사이에 개재된 세퍼레이터를 포함하는 전기화학소자에 있어서, 상기 세퍼레이터가 제1 구현예 내지 제10 구현예 중 어느 한 구현예의 세퍼레이터인 전기화학소자에 관한 것이다.
- [37] 제12 구현예는, 제11 구현예에 있어서,
- [38] 상기 전기화학소자가 리튬 이차전지인 전기화학소자에 관한 것이다.

발명의 효과

- [39] 본 발명의 일 실시예에 따르면, 무기물 입자로서 크기와 재질이 상이한 알루미늄 입자 및 상기 알루미늄 입자 보다 입경이 작은 수산화알루미늄 입자를 사용한 다공성 코팅층을 구비함으로써, 무기물 입자의 다공성 코팅층에 의해 다공성 기재의 열수축율 특성이 크게 개선되어, 전기화학소자가 과열되는 경우에도 캐소드와 애노드 사이의 단락을 억제할 수 있으므로 안정성이 향상된 세퍼레이터와 이를 구비한 전기화학소자를 제공할 수 있다.
- [40] 또한, 밀도가 상대적으로 작은 수산화알루미늄 입자를 사용함으로써, 세퍼레이터 및 전기화학소자의 경량화를 도모할 수 있고, 수산화알루미늄 입자의 우수한 흡열 특성에 의해, 이상 작동에 따른 전기화학소자의 내부 온도 급속한 온도 상승을 제어할 수 있다.

도면의 간단한 설명

- [41] 도 1은 본 발명의 일 실시예에 따른 세퍼레이터를 도시한 개략도이다.
도 2 및 도 3은 실시예 1 및 비교예 1에서 얻어진 세퍼레이터의 열수축율 측정시 외관 사진을 각각 나타낸 것이다.

발명의 실시를 위한 형태

- [42] 이하, 본 발명을 상세히 설명하기로 한다. 본 명세서 및 청구범위에 사용된 용어나 단어는 통상적이거나 사전적인 의미로 한정해서 해석되어서는 아니 되며, 발명자는 그 자신의 발명을 가장 최선의 방법으로 설명하기 위해 용어의 개념을 적절하게 정의할 수 있다는 원칙에 입각하여 본 발명의 기술적 사상에 부합하는 의미와 개념으로 해석되어야만 한다.
- [43] 본 발명의 일 측면에 따른 세퍼레이터는, 다수의 기공을 갖는 다공성 기재; 및 상기 다공성 기재의 적어도 일면 및 상기 다공성 기재의 기공 중 1종 이상의 영역에 형성되어 있으며, 다수의 무기물 입자 및 상기 무기물 입자의 표면의 일부 또는 전부에 위치하여 상기 무기물 입자 사이를 연결 및 고정시키는 바인더 고분자를 포함하는 다공성 코팅층;을 포함하고, 상기 무기물 입자가 알루미늄 입자 및 상기 알루미늄 입자 보다 평균입경이 작은 수산화알루미늄 입자를 포함한다.
- [44] 본 발명에서는 무기물 입자로서 평균입경 크기와 그 물질이 상이한 2종의 무기물 입자, 알루미늄 입자 및 상기 알루미늄 입자 보다 평균입경이 작은 수산화알루미늄 입자를 포함한다.
- [45] 상기 수산화알루미늄 입자의 평균입경은 상기 알루미늄 입자의 평균입경의 0.01 내지 0.9 배, 바람직하게는 0.015 내지 0.85 배, 더 바람직하게는 0.02 내지 0.08 배일 수 있다.
- [46] 본 발명의 일 실시예에 따르면, 알루미늄 입자의 평균입경은 0.5 내지 3 μm , 바람직하게는 0.7 내지 2.7 μm , 더 바람직하게는 0.9 내지 2.5 μm 일 수 있고, 수산화알루미늄 입자의 평균입경은 0.05 내지 0.4 μm , 바람직하게는 0.07 내지 0.37 μm , 더 바람직하게는 0.09 내지 0.35 μm 일 수 있다.
- [47] 본 발명의 무기물 입자는 상기와 같이 서로 상이한 평균입경을 갖는 2종을

포함하고 있으므로, 다공성 코팅층 형성용 조성물 내의 무기물 입자의 분산성 및 코팅 공정성이 향상되고, 코팅층의 두께의 조절이 용이하며, 기계적 물성 및 전기적 특성이 저하되는 것을 개선할 수 있다. 더불어, 대입경 입자들 사이의 큰 공극 내에, 소입경 입자가 위치할 수 있어, 생성되는 다공성 코팅층의 기공의 크기를 제어할 수 있으며, 다공성 코팅층의 치밀성이 개선되어 다공성 기체의 열수축 현상을 억제함으로써 전지의 충전 및 방전시 내부 단락 현상을 방지할 수도 있다.

- [48] 특히, 본 발명에서는, 알루미나 입자 및 상기 알루미나 입자 보다 평균입경이 작은 수산화알루미늄 입자를 함께 사용함으로써, 단지 크기가 상이한 동종 무기물 입자 2종을 사용하는 것에 비하여, 보다 유리한 효과를 질 수 있는 바, 이하, 살펴보겠다.
- [49] 상기 알루미나 입자는 Al_2O_3 의 화학식을 갖는 알루미늄과 산소의 화합물로서, 상대적으로 높은 열전도성을 갖는 전기 절연체로서 알려져 있고, 그 밀도는 3.95 내지 4.1g/cm^3 이다.
- [50] 한편, 상기 수산화알루미늄 입자로는 수산화기의 종류에 따라서, 수산화기가 1개인 보헤마이트($\gamma\text{-AlO}(\text{OH})$), 다이어스포(*diaspore*, $\alpha\text{-AlO}(\text{OH})$), 수산화기가 2개인 바이어리트(*bayerlite*, $\alpha\text{-AlO}(\text{OH})_3$), 깁사이트(*gibbsite*, $\gamma\text{-AlO}(\text{OH})_3$), 노드스트란디트(*nordstrandite*, $\text{AlO}(\text{OH})_3$)가 있고, 또한 알루미나의 수화물인 유사 보헤마이트($\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$)를 들 수 있다. 이들 수산화알루미늄 입자로는 이들을 단독 또는 2종 이상의 혼합물로도 사용할 수 있다. 상기 수산화알루미늄 입자는 통상 밀도가 약 2.4 내지 3.1g/cm^3 의 범위를 가진다.
- [51] 또한, 수산화알루미늄 입자는 흡열 특성이 뛰어나고, 원형 이외에, 판 형상(박편 형상), 침 형상(피브릴 형상), 소정의 결정축 방향으로 길게 연장된 육각판 형상, 사각 판 형상을 비롯한 다각 판 형상, 방추 형상 등을 가질 수 있다.
- [52] 따라서, 본 발명의 다공성 코팅층에, 대입경 무기물 입자로서 알루미나 입자와 함께, 알루미나 입자 보다 평균입경이 작고 알루미나에 대비하여 상대적으로 밀도가 작은 수산화알루미늄 입자를 사용하는 경우, 세퍼레이터의 전체 중량을 낮출 수 있어, 전기화학소자의 경량화를 도모할 수 있다. 또한, 수산화알루미늄 입자는 구형 외에 판 형상 등의 다양한 형상으로 구현할 수 있어, 구형의 알루미나 입자가 서로 이웃하여 배열되거나 상당히 이격되어 배열되어 있는 경우에도 알루미나 입자들 사이의 다양한 형태의 공극내에 용이하게 배치될 수 있다. 그 결과, 다공성 코팅층의 치밀성과 기계적 물성을 크게 향상시키고, 다공성 기체의 열수축 현상을 억제하여 전기화학 소자 내부의 단락을 방지하는 역할을 할 수 있다.
- [53] 또한, 평균입경이 상이한 알루미나 입자, 즉 대입경의 알루미나 입자 및 소입경의 알루미나 입자를 사용하는 경우, 대입경의 알루미나 입자에 대비하여 소입경의 알루미나 입자는 그 가격이 현저히 고가이고, 취급이 용이하지 않은 등의 문제가 있으나, 본 발명에서는 알루미나 입자 및 알루미나 입자 보다

평균입경이 작은 수산화알루미늄 입자를 함께 사용함으로써, 이러한 문제가 해소되는 이점도 있다.

[54] 또한, 본 발명의 일 실시예에 따르면, 상기 알루미나 입자 대 수산화알루미늄 입자의 함량비는 50:50 내지 97:3, 바람직하게는 55:45 내지 95:5, 더 바람직하게는 60:40 내지 93:7일 수 있다.

[55] 상기 함량비가 이러한 범위를 만족하는 경우, 다공성 코팅층 형성용 코팅 조성물의 분산 안정성이나 공정성이 크게 개선되고, 균일한 두께의 다공성 코팅층의 형성이 가능하고, 다공성 코팅층의 공극율의 제어가 용이할 수 있다.

[56] 상기 다공성 기재는 다공성 고분자 기재일 수 있고, 구체적으로 다공성 고분자 필름 기재 또는 다공성 고분자 부직포 기재를 들 수 있다.

[57] 상기 다공성 고분자 필름 기재로는 폴리에틸렌, 폴리프로필렌과 같은 폴리올레핀으로 이루어진 다공성 고분자 필름일 수 있으며, 이러한 폴리올레핀 다공성 고분자 필름 기재는 예를 들어 80 내지 130 °C의 온도에서 섰다운 기능을 발현한다.

[58] 이때, 폴리올레핀 다공성 고분자 필름은 고밀도 폴리에틸렌, 선형 저밀도 폴리에틸렌, 저밀도 폴리에틸렌, 초고분자량 폴리에틸렌과 같은 폴리에틸렌, 폴리프로필렌, 폴리부틸렌, 폴리펜텐 등의 폴리올레핀계 고분자를 각각 단독 또는 이들의 2종 이상 혼합하여 고분자로 형성할 수 있다.

[59] 또한, 상기 다공성 고분자 필름 기재는 폴리올레핀 외에 폴리에스테르 등의 다양한 고분자들을 이용하여 필름 형상으로 성형하여 제조될 수도 있다. 또한, 상기 다공성 고분자 필름 기재는 2층 이상의 필름층이 적층된 구조로 형성될 수 있으며, 각 필름층은 전술한 폴리올레핀, 폴리에스테르 등의 고분자 단독으로 또는 이들을 2종 이상 혼합한 고분자로 형성될 수도 있다.

[60] 또한, 상기 다공성 고분자 필름 기재 및 다공성 부직포 기재는 상기와 같은 폴리올레핀계 외에 폴리에틸렌테레프탈레이트(polyethyleneterephthalate), 폴리부틸렌테레프탈레이트(polybutyleneterephthalate), 폴리에스테르(polyester), 폴리아세탈(polyacetal), 폴리아미드(polyamide), 폴리카보네이트(polycarbonate), 폴리이미드(polyimide), 폴리에테르에테르케톤(polyetheretherketone), 폴리에테르설폰(polyethersulfone), 폴리페닐렌옥사이드(polyphenyleneoxide), 폴리페닐렌설파이드(polyphenylenesulfide), 폴리에틸렌나프탈렌(polyethylenenaphthalene) 등을 각각 단독으로 또는 이들을 혼합한 고분자로 형성될 수 있다. 상기 다공성 기재의 두께는 특별히 제한되지 않으나, 바람직하게는 1 내지 100 μm , 더욱 바람직하게는 5 내지 50 μm 이고, 다공성 기재에 존재하는 기공 크기 및 기공도 역시 특별히 제한되지 않으나 각각 0.01 내지 50 μm 및 10 내지 95%인 것이 바람직하다.

[61] 본 발명의 일 측면에 따른 세퍼레이터에 있어서, 다공성 코팅층 형성에 사용되는 바인더 고분자로는 당 업계에서 다공성 코팅층 형성에 통상적으로 사용되는 고분자를 사용할 수 있다. 특히, 유리 전이 온도(glass transition

temperature, T_g)가 -200 내지 200°C인 고분자를 사용할 수 있는데, 이는 최종적으로 형성되는 다공성 코팅층의 유연성 및 탄성 등과 같은 기계적 물성을 향상시킬 수 있기 때문이다. 이러한 바인더 고분자는 무기물 입자들 사이를 연결 및 안정하게 고정시켜주는 바인더 역할을 충실히 수행함으로써, 다공성 코팅층이 도입된 세퍼레이터의 기계적 물성 저하 방지에 기여한다.

[62] 또한, 상기 바인더 고분자는 이온 전도 능력을 반드시 가질 필요는 없으나, 이온 전도 능력을 갖는 고분자를 사용할 경우 전기화학소자의 성능을 더욱 향상시킬 수 있다. 따라서, 상기 바인더 고분자는 가능한 유전율 상수가 높은 것을 사용할 수 있다. 실제로 전해액에서 염의 해리도는 전해액 용매의 유전율 상수에 의존하기 때문에, 상기 바인더 고분자의 유전율 상수가 높을수록 전해질에서의 염 해리도를 향상시킬 수 있다. 이러한 바인더 고분자의 유전율 상수는 1.0 내지 100 (측정 주파수 = 1 kHz) 범위가 사용 가능하며, 특히 10 이상일 수 있다.

[63] 전술한 기능 이외에, 상기 바인더 고분자는 액체 전해액 함침시 겔화됨으로써 높은 전해액 팽윤도(degree of swelling)를 나타낼 수 있는 특징을 가질 수 있다. 이에 따라, 상기 바인더 고분자의 용해도 지수, 즉 힐더브랜드 용해도 지수(Hildebrand solubility parameter)는 15 내지 45 $\text{MPa}^{1/2}$ 또는 15 내지 25 $\text{MPa}^{1/2}$ 및 30 내지 45 $\text{MPa}^{1/2}$ 범위이다. 따라서, 폴리올레핀류와 같은 소수성 고분자들보다는 극성기를 많이 갖는 친수성 고분자들이 더 사용될 수 있다. 상기 용해도 지수가 15 $\text{MPa}^{1/2}$ 미만 및 45 $\text{MPa}^{1/2}$ 를 초과할 경우, 통상적인 전지용 액체 전해액에 의해 팽윤(swelling)되기 어려울 수 있기 때문이다.

[64] 이러한 바인더 고분자의 비제한적인 예로는 폴리비닐리덴 플루오라이드-헥사플루오로프로필렌 (polyvinylidene fluoride-co-hexafluoropropylene), 폴리비닐리덴 플루오라이드-트리클로로에틸렌 (polyvinylidene fluoride-co-trichloroethylene), 폴리메틸메타크릴레이트 (polymethylmethacrylate), 폴리부틸아크릴레이트 (polybutylacrylate), 폴리아크릴로니트릴 (polyacrylonitrile), 폴리비닐피롤리돈 (polyvinylpyrrolidone), 폴리비닐아세테이트 (polyvinylacetate), 에틸렌 비닐 아세테이트 공중합체 (polyethylene-co-vinyl acetate), 폴리에틸렌옥사이드 (polyethylene oxide), 폴리아릴레이트(polyarylate), 셀룰로오스 아세테이트 (cellulose acetate), 셀룰로오스 아세테이트 부틸레이트 (cellulose acetate butyrate), 셀룰로오스 아세테이트 프로피오네이트 (cellulose acetate propionate), 시아노에틸플루란 (cyanoethylpullulan), 시아노에틸폴리비닐알코올 (cyanoethylpolyvinylalcohol), 시아노에틸셀룰로오스 (cyanoethylcellulose), 시아노에틸수크로오스 (cyanoethylsucrose), 플루란 (pullulan) 및 카르복실 메틸 셀룰로오스 (carboxyl methyl cellulose)등을 들 수 있으며, 이에 한정되는 것은 아니다.

[65] 상기 무기물 입자와 바인더 고분자의 중량비는 예를 들어 50:50 내지 99:1 범위가 바람직하며, 더욱 바람직하게는 70:30 내지 95:5이다. 바인더 고분자에 대한 무기물 입자의 함량비가 상기 범위를 만족하는 경우, 바인더 고분자의

함량이 많아지게 되어 형성되는 코팅층의 기공 크기 및 기공도가 감소되는 문제가 방지될 수 있고, 바인더 고분자 함량이 적기 때문에 형성되는 코팅층의 내필링성이 악화되는 문제도 해소될 수 있다.

[66] 본 발명의 일 측면에 따른 세퍼레이터는 다공성 코팅층 성분으로 전술한 무기물 입자 및 고분자 이외에, 기타 첨가제를 더 포함할 수 있다.

[67] 본 발명의 일 실시예에 따른 세퍼레이터는, 알루미늄 입자 및 상기 알루미늄 입자 보다 평균입경이 작은 수산화알루미늄 입자를 포함하는 무기물 입자 및 바인더 고분자를 구비하는, 다공성 코팅층 형성용 조성물을 준비하고, 이러한 조성물을 다공성 기재의 적어도 일면 상에 도포하고, 이를 건조함으로써 제조될 수 있다.

[68] 먼저, 다공성 코팅층 형성용 조성물은, 바인더 고분자를 용매에 용해시킨 다음 무기물 입자를 첨가하고 이를 분산시켜 제조할 수 있다. 무기물 입자들은 미리 소정의 평균입경을 갖도록 파쇄된 상태에서 첨가할 수 있으며, 또는 바인더 고분자의 용액에 무기물 입자를 첨가한 후 무기물 입자를 볼밀법 등을 이용하여 소정의 평균입경을 갖도록 제어하면서 파쇄하여 분산시킬 수도 있다.

[69] 상기 다공성 코팅층 형성용 조성물을 상기 다공성 기재에 코팅하는 방법은 특별히 한정하지는 않지만, 슬랏 코팅이나 딥 코팅 방법을 사용하는 것이 바람직하다. 슬랏 코팅은 슬랏 다이를 통해 공급된 조성물이 기재의 전면에 도포되는 방식으로 정량 펌프에서 공급되는 유량에 따라 코팅층 두께의 조절이 가능하다. 또한 딥 코팅은 조성물이 들어있는 탱크에 기재를 담그어 코팅하는 방법으로, 조성물의 농도 및 조성물 탱크에서 기재를 꺼내는 속도에 따라 코팅층 두께의 조절이 가능하며 보다 정확한 코팅 두께 제어를 위해 침지 후 메이어바 등을 통해 후계량할 수 있다.

[70] 이렇게 다공성 코팅층 형성용 조성물이 코팅된 다공성 기재를 오븐에서 건조하여 다공성 기재의 적어도 일면에 코팅층을 형성하게 된다.

[71] 상기 다공성 코팅층에서는 무기물 입자들은 충전되어 서로 접촉된 상태에서 상기 바인더 고분자에 의해 서로 결합되고, 이로 인해 무기물 입자들 사이에 인터스티셜 볼륨(interstitial volume)이 형성되고, 상기 무기물 입자 사이의 인터스티셜 볼륨(Interstitial Volume)은 빈 공간이 되어 기공을 형성한다.

[72] 즉, 바인더 고분자는 무기물 입자들이 서로 결합된 상태를 유지할 수 있도록 이들을 서로 부착, 예를 들어, 바인더 고분자가 무기물 입자 사이를 연결 및 고정시키고 있다. 또한, 상기 다공성 코팅층의 기공은 무기물 입자들 간의 인터스티셜 볼륨(interstitial volume)이 빈 공간이 되어 형성된 기공이고, 이는 무기물 입자들에 의한 충전 구조(closed packed or densely packed)에서 실질적으로 면접하는 무기물 입자들에 의해 한정되는 공간이다.

[73] 본 발명의 일 실시예에 따른 세퍼레이터를 도 1에 도시하였다. 도 1을 참조하면, 상기 세퍼레이터는 다공성 기재(100); 및 상기 다공성 기재(100)의 일 면 상에 형성되고, 알루미늄 입자(210) 및 상기 알루미늄 입자 보다 평균입경이 작은

수산화알루미늄 입자(220)를 포함하는 다공성 코팅층(200) (바인더 고분자는 미도시)을 구비하고 있다.

[74] 본 발명의 일 측면에 따른 전기화학소자는 캐소드, 애노드, 상기 캐소드 및 애노드 사이에 개재된 세퍼레이터를 포함하고, 상기 세퍼레이터가 전술한 본 발명의 일 실시예에 따른 세퍼레이터이다.

[75] 이러한 전기화학소자는 전기 화학 반응을 하는 모든 소자를 포함하며, 구체적인 예를 들면, 모든 종류의 1차, 이차 전지, 연료 전지, 태양 전지 또는 수퍼 캐패시터 소자와 같은 캐퍼시터(capacitor) 등이 있다. 특히, 상기 2차 전지 중 리튬 금속 이차 전지, 리튬 이온 이차 전지, 리튬 폴리머 이차 전지 또는 리튬 이온 폴리머 이차 전지 등을 포함하는 리튬 이차전지가 바람직하다.

[76] 본 발명의 세퍼레이터와 함께 적용될 양극과 음극의 양 전극으로는 특별히 제한되지 않으며, 당업계에 알려진 통상적인 방법에 따라 전극활물질을 전극 전류집전체에 결합된 형태로 제조할 수 있다. 상기 전극활물질 중 양극활물질의 비제한적인 예로는 종래 전기화학소자의 양극에 사용될 수 있는 통상적인 양극활물질이 사용 가능하며, 특히 리튬망간산화물, 리튬코발트산화물, 리튬니켈산화물, 리튬철산화물 또는 이들을 조합한 리튬복합산화물을 사용하는 것이 바람직하다. 음극활물질의 비제한적인 예로는 종래 전기화학소자의 음극에 사용될 수 있는 통상적인 음극활물질이 사용 가능하며, 특히 리튬 금속 또는 리튬 합금, 탄소, 석유코크(petroleum coke), 활성화 탄소(activated carbon), 그래파이트(graphite) 또는 기타 탄소류 등과 같은 리튬 흡착물질 등이 바람직하다. 양극 전류집전체의 비제한적인 예로는 알루미늄, 니켈 또는 이들의 조합에 의하여 제조되는 호일 등이 있으며, 음극 전류집전체의 비제한적인 예로는 구리, 금, 니켈 또는 구리 합금 또는 이들의 조합에 의하여 제조되는 호일 등이 있다.

[77] 본 발명의 전기화학소자에서 사용될 수 있는 전해액은 A+B와 같은 구조의 염으로서, A⁺는 Li⁺, Na⁺, K⁺와 같은 알칼리 금속 양이온 또는 이들의 조합으로 이루어진 이온을 포함하고 B⁻는 PF₆⁻, BF₄⁻, Cl⁻, Br⁻, I⁻, ClO₄⁻, AsF₆⁻, CH₃CO₂⁻, CF₃SO₃⁻, N(CF₃SO₂)₂⁻, C(CF₂SO₂)₃⁻와 같은 음이온 또는 이들의 조합으로 이루어진 이온을 포함하는 염이 프로필렌 카보네이트(PC), 에틸렌 카보네이트(EC), 디에틸카보네이트(DEC), 디메틸카보네이트(DMC), 디프로필카보네이트(DPC), 디메틸설폭사이드, 아세토니트릴, 디메톡시에탄, 디에톡시에탄, 테트라하이드로퓨란, N-메틸-2-피롤리돈(NMP), 에틸메틸카보네이트(EMC), 감마 부티로락톤 (γ-부티로락톤) 또는 이들의 혼합물로 이루어진 유기 용매에 용해 또는 해리된 것이 있으나, 이에만 한정되는 것은 아니다.

[78] 상기 전해액 주입은 최종 제품의 제조 공정 및 요구 물성에 따라, 전지 제조 공정 중 적절한 단계에서 행해질 수 있다. 즉, 전지 조립 전 또는 전지 조립 최종 단계 등에서 적용될 수 있다.

[79] 이하, 본 발명을 구체적으로 설명하기 위해 실시예를 들어 상세하게

설명하기로 한다. 그러나, 본 발명에 따른 실시예는 여러 가지 다른 형태로 변형될 수 있으며, 본 발명의 범위가 아래에서 상술하는 실시예에 한정되는 것으로 해석되어서는 안 된다. 본 발명의 실시예는 당업계에서 평균적인 지식을 가진 자에게 본 발명을 보다 완전하게 설명하기 위해서 제공되는 것이다.

[80]

[81] 실시예 1

[82] PVdF-CTFE (폴리비닐리덴플로라이드-클로로트리플로로에틸렌 공중합체) 10 중량부를 아세톤 95 중량부에 첨가하여 50°C에서 약 12시간 이상 용해시켜 바인더 고분자 용액 10 중량부를 준비하였다. 제조된 바인더 고분자 용액에 알루미나 입자로서 평균입경 0.5 μm 의 알루미나(Al_2O_3)입자와, 수산화알루미늄 입자로서 평균입경 0.2 μm 의 보헤마이트($\gamma\text{-AlO}(\text{OH})$) 입자를 90:10의 중량비이면서, 바인더 고분자/무기물 입자 총합 = 10/90의 중량비가 되도록 첨가하고, 분산하여 슬러리를 제조하였다.

[83] 이와 같이 제조된 슬러리를 딥(dip) 코팅법으로, 두께 9 μm 인 폴리에틸렌 다공성 막(기공도 40%)에 코팅하고, 코팅 두께는 약 5 내지 6 μm 정도로 조절하여, 세퍼레이터를 제조하였다.

[84]

[85] 비교예 1

[86] 무기물 입자로 평균 입경 0.5 μm 의 Al_2O_3 입자만을 사용한 점을 제외하고는 실시예 1과 동일한 방법으로 세퍼레이터를 제조하였다.

[87]

[88] 실시예 2

[89] PVdF-CTFE (폴리비닐리덴플로라이드-클로로트리플로로에틸렌 공중합체) 5 중량부를 아세톤 95 중량부에 첨가하여 50°C에서 약 12시간 이상 용해시켜 바인더 고분자 용액 5 중량부를 준비하였다. 제조된 바인더 고분자 용액에 알루미나 입자로서 평균입경 2 μm 의 알루미나(Al_2O_3)입자와, 수산화알루미늄 입자로서 평균입경 0.2 μm 의 보헤마이트($\gamma\text{-AlO}(\text{OH})$) 입자를 90:10의 중량비이면서, 바인더 고분자/무기물 입자 총합 = 20/80의 중량비가 되도록 첨가하고, 분산하여 슬러리를 제조하였다.

[90] 이와 같이 제조된 슬러리를 딥(dip) 코팅법으로, 두께 20 μm 인 폴리에틸렌 다공성 막(기공도 40%)에 코팅하고, 코팅 두께는 약 10 μm 정도로 조절하여, 세퍼레이터를 제조하였다.

[91]

[92] 비교예 2

[93] 무기물 입자로 평균 입경 500 nm의 Al_2O_3 입자와 평균 입경 20 nm인 Al_2O_3 입자를 90:10의 중량비로 사용한 점을 제외하고는 실시예 2와 동일한 방법으로 세퍼레이터를 제조하였다.

[94]

[95] 비교예 3

[96] 무기물 입자로 평균 입경 500 nm의 보헤마이트(γ -AlO(OH)) 입자와 평균 입경 20 nm인 Al₂O₃ 입자를 90:10의 중량비로 사용한 점을 제외하고는 실시예 2와 동일한 방법으로 세퍼레이터를 제조하였다.

[97]

[98] 실시예 1 및 비교예 1에서 얻어진 세퍼레이터의 두께, 걸리(Gurley) 공기투과도, 및 기계방향(MD)과 횡방향(TD)의 열수축율을 측정하고, 그 결과를 하기 표 1에 기재하였다.

또한, 실시예 1 및 비교예 1에서 얻어진 세퍼레이터의 열수축율 측정시 외관 사진을 도 2 및 도 3에 각각 나타내었다.

[99]

[100] 걸리(Gurley) 공기투과도(통기시간)

[101] 걸리 공기투과도 값은 ASTM D726-94 방법에 의해 측정하였다. 여기서 사용된 걸리는, 공기의 흐름에 대한 저항으로서, 걸리 덴소미터(densometer)에 의해 측정된다. 여기서 설명된 걸리 공기투과도 값은 100 cc의 공기가 12.2 inH₂O의 압력하에서 1 in²의 단면을 통과하는 데 걸리는 시간(초), 즉 통기시간으로 나타낸다.

[102]

[103] 열수축율

[104] 실시예 1 및 비교예 1에서 제조된 세퍼레이터를 50mm * 50mm 크기로 재단하여 A4 용지 사이에 놓고 150℃의 컨벡션 오븐(convection oven)에 30분 동안 넣은 후 기계방향(MD)과 횡방향(TD)의 열수축율을 측정하였다. 이때, 열수축율은 [(최초 길이 - 열처리 후 길이)/(최초 길이)×100]으로 계산하였다.

[105]

[106] [표1]

	실시예 1	비교예 1
두께(μ m)	13.25	14.7
통기시간(Gurley) (s/100cc)	417	572
열수축율(MD/TD) (%) 150℃, 30분	9/9	47/41

[107]

[108] 상기 표 1을 참조하면, 알루미나 입자 및 상기 알루미나 입자보다 입경이 작은 수산화알루미늄 입자를 사용한 다공성 코팅층을 구비한 실시예 1의 세퍼레이터가 알루미나 입자만을 사용한 비교예 1과 비교하여, 통기도와, MD 방향/TD 방향의 열수축율 면에서 우수한 효과를 나타냄을 알 수 있었다.

[109] 특히, 실시예 1의 다공성 코팅층의 두께는 $4.25\mu\text{m}$ 이고, 비교예 1의 다공성 코팅층의 두께는 $5.7\mu\text{m}$ 로서, 실시예 1은 비교예 1에 비하여 더 작은 다공성 코팅층의 두께를 가짐에도 대입경 입자들 사이의 큰 공극 내에, 소입경 입자가 위치할 수 있어, 다공성 코팅층의 치밀성이 개선되어 다공성 기재의 열수축 현상을 더 탁월하게 억제한 것으로 볼 수 있다. 이를 통하여, 밀도가 상대적으로 작은 수산화알루미늄 입자를 사용함으로써, 세퍼레이터 및 전기화학소자의 경량화를 도모하면서도, 전기화학소자가 과열되는 경우에도 캐소드와 애노드 사이의 단락을 억제할 수 있으므로 안정성이 향상된 세퍼레이터와 이를 구비한 전기화학소자를 제공할 수 있을 것이다.

[110] [부호의 설명]

[111] 100: 다공성 기재, 200: 다공성 코팅층

[112] 210: 알루미나 입자, 220: 수산화알루미늄 입자

청구범위

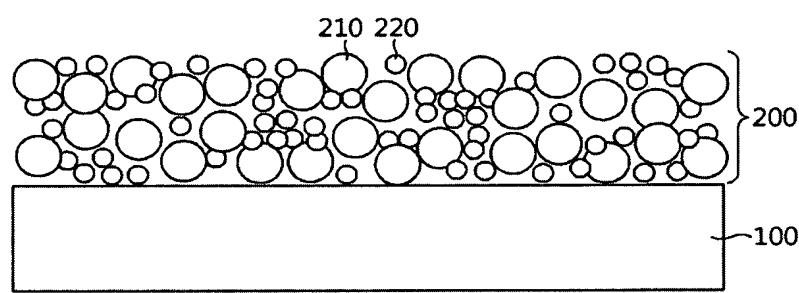
- [청구항 1] 다수의 기공을 갖는 다공성 기재; 및
상기 다공성 기재의 적어도 일면에 또는 상기 다공성 기재의 적어도 일면 및 기공의 일부에 형성되어 있으며, 다수의 무기물 입자 및 상기 무기물 입자의 표면의 일부 또는 전부에 위치하여 상기 무기물 입자 사이를 연결 및 고정시키는 바인더 고분자를 포함하는 다공성 코팅층을 포함하고, 상기 무기물 입자가 알루미나 입자 및 상기 알루미나 입자 보다 평균입경이 작은 수산화알루미늄 입자를 포함하는 세퍼레이터.
- [청구항 2] 제1항에 있어서,
상기 수산화알루미늄 입자의 평균입경이 상기 알루미나 입자의 평균입경의 0.01 내지 0.9 배인 세퍼레이터.
- [청구항 3] 제1항에 있어서,
상기 알루미나 입자의 평균입경이 0.5 내지 3 μm 인 세퍼레이터.
- [청구항 4] 제1항에 있어서,
상기 수산화알루미늄 입자의 평균입경이 0.05 내지 0.4 μm 인 세퍼레이터.
- [청구항 5] 제1항에 있어서,
상기 알루미나 입자 대 상기 수산화알루미늄 입자의 함량비가 50:50 내지 97:3인 세퍼레이터.
- [청구항 6] 제1항에 있어서,
상기 수산화알루미늄 입자가 보헤마이트($\gamma\text{-AlO}(\text{OH})$), 유사 보헤마이트($\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$), 다이어스포(diaspore, $\alpha\text{-AlO}(\text{OH})$), 바이어리트(bayerlite, $\alpha\text{-AlO}(\text{OH})_3$), 깁사이트(gibbsite, $\gamma\text{-AlO}(\text{OH})_3$), 노드스트란디트(nordstrandite, $\text{AlO}(\text{OH})_3$) 로 이루어진 군으로부터 선택된 1종 이상인 세퍼레이터.
- [청구항 7] 제1항에 있어서,
상기 다공성 기재가 폴리올레핀계 다공성 기재인 세퍼레이터.
- [청구항 8] 제7항에 있어서,
상기 폴리올레핀계 다공성 기재가 폴리에틸렌, 폴리프로필렌, 폴리부틸렌 및 폴리펜텐으로 이루어진 군으로부터 선택된 어느 하나의 고분자 또는 이들의 혼합물로 형성된 세퍼레이터.
- [청구항 9] 제1항에 있어서,
상기 다공성 기재의 두께가 5 내지 50 μm 이고, 기공 크기 및 기공도는 각각 0.01 내지 50 μm 및 10 내지 95%인 세퍼레이터.
- [청구항 10] 제1항에 있어서,
상기 무기물 입자와 바인더 고분자의 중량비가 50:50 내지 99:1인 세퍼레이터.
- [청구항 11] 제1항에 있어서,

상기 바인더 고분자가 폴리비닐리덴
 플루오라이드-헥사플루오로프로필렌 (polyvinylidene
 fluoride-co-hexafluoropropylene), 폴리비닐리덴
 플루오라이드-트리클로로에틸렌 (polyvinylidene
 fluoride-co-trichloroethylene), 폴리메틸메타크릴레이트
 (polymethylmethacrylate), 폴리부틸아크릴레이트 (polybutylacrylate),
 폴리아크릴로니트릴 (polyacrylonitrile), 폴리비닐피롤리돈
 (polyvinylpyrrolidone), 폴리비닐아세테이트 (polyvinylacetate), 에틸렌
 비닐 아세테이트 공중합체 (polyethylene-co-vinyl acetate),
 폴리에틸렌옥사이드 (polyethylene oxide), 폴리아릴레이트(polyarylate),
 셀룰로오스 아세테이트 (cellulose acetate), 셀룰로오스 아세테이트
 부틸레이트 (cellulose acetate butyrate), 셀룰로오스 아세테이트
 프로피오네이트 (cellulose acetate propionate), 시아노에틸플루란
 (cyanoethylpullulan), 시아노에틸폴리비닐알콜
 (cyanoethylpolyvinylalcohol), 시아노에틸셀룰로오스 (cyanoethylcellulose),
 시아노에틸수크로오스 (cyanoethylsucrose), 플루란 (pullulan) 및 카르복실
 메틸 셀룰로오스 (carboxyl methyl cellulose)로 이루어진 군으로부터
 선택된 어느 하나 또는 이들 중 2종 이상의 혼합물인 세퍼레이터.

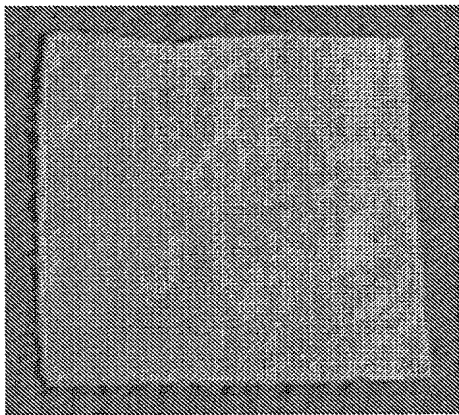
[청구항 12] 캐소드, 애노드, 상기 캐소드 및 애노드 사이에 개재된 세퍼레이터를
 포함하는 전기화학소자에 있어서, 상기 세퍼레이터가 제1항 내지 제11항
 중 어느 한 항의 세퍼레이터인 전기화학소자.

[청구항 13] 제12항에 있어서,
 상기 전기화학소자가 리튬 이차전지인 전기화학소자.

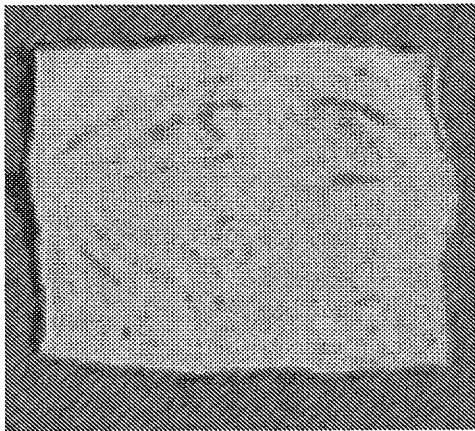
[도1]



[도2]



[도3]



INTERNATIONAL SEARCH REPORT

International application No.

PCT/KR2016/007532**A. CLASSIFICATION OF SUBJECT MATTER*****H01M 2/16(2006.01)i, H01M 10/052(2010.01)i***

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

H01M 2/16; H01M 10/05; H01M 10/0525; H05B 3/10; C08J 5/24; B32B 27/36

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Korean Utility models and applications for Utility models: IPC as above

Japanese Utility models and applications for Utility models: IPC as above

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

eKOMPASS (KIPO internal) & Keywords: separator, alumina, aluminum hydroxide, binder, polyolefin, electrochemical device

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	KR 10-2012-0079515 A (EIG LTD.) 13 July 2012 See abstract; paragraphs [0005]-[0044]; claims 1-9; and figures 1-3.	1-13
A	KR 10-2014-0116415 A (SIHL GMBH.) 02 October 2014 See abstract; paragraphs [0021]-[0052]; and claims 1-11.	1-13
A	KR 10-2013-0007740 A (GUMI ELECTRONICS & INFORMATION TECHNOLOGY RESEARCH INSTITUTE et al.) 21 January 2013 See abstract; paragraphs [0024]-[0033]; and claims 1-5.	1-13
A	JP 4311208 B2 (TORAY IND. INC.) 12 August 2009 See paragraphs [0052]-[0072]; and claims 1-3.	1-13
A	KR 10-2012-0124239 A (LG CHEM. LTD.) 13 November 2012 See abstract; paragraphs [0022]-[0058]; and claims 1-14.	1-13



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents:

“A” document defining the general state of the art which is not considered to be of particular relevance

“E” earlier application or patent but published on or after the international filing date

“L” document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

“O” document referring to an oral disclosure, use, exhibition or other means

“P” document published prior to the international filing date but later than the priority date claimed

“T” later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

“X” document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

“Y” document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

“&” document member of the same patent family

Date of the actual completion of the international search

21 OCTOBER 2016 (21.10.2016)

Date of mailing of the international search report

25 OCTOBER 2016 (25.10.2016)

Name and mailing address of the ISA/KR


 Korean Intellectual Property Office
 Government Complex-Daejeon, 189 Seonsa-ro, Daejeon 302-701,
 Republic of Korea

Facsimile No. 82-42-472-7140

Authorized officer

Telephone No.

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.

PCT/KR2016/007532

Patent document cited in search report	Publication date	Patent family member	Publication date
KR 10-2012-0079515 A	13/07/2012	EP 2662920 A2 KR 10-1282067 B1 WO 2012-093864 A2	13/11/2013 05/07/2013 12/07/2012
KR 10-2014-0116415 A	02/10/2014	CN 104115306 A EP 2805366 A1 EP 2805366 B1 JP 2015-504234 A US 2015-0030933 A1 WO 2013-107911 A1	22/10/2014 26/11/2014 24/08/2016 05/02/2015 29/01/2015 25/07/2013
KR 10-2013-0007740 A	21/01/2013	NONE	
JP 4311208 B2	12/08/2009	CN 1517402 A EP 1441001 A2 EP 1441001 B1 JP 2004-243760 A KR 10-0996022 B1 US 2004-0161620 A1 US 7459217 B2	04/08/2004 28/07/2004 28/11/2007 02/09/2004 22/11/2010 19/08/2004 02/12/2008
KR 10-2012-0124239 A	13/11/2012	CN 103733380 A EP 2706591 A2 JP 2014-510388 A JP 2016-076492 A JP 5875673 B2 KR 10-1358761 B1 US 2013-0149587 A1 US 2015-0086714 A1 US 8927134 B2 WO 2012-150838 A2	16/04/2014 12/03/2014 24/04/2014 12/05/2016 02/03/2016 07/02/2014 13/06/2013 26/03/2015 06/01/2015 08/11/2012

A. 발명이 속하는 기술분류(국제특허분류(IPC))

H01M 2/16(2006.01)i, H01M 10/052(2010.01)i

B. 조사된 분야

조사된 최소문헌(국제특허분류를 기재)

H01M 2/16; H01M 10/05; H01M 10/0525; H05B 3/10; C08J 5/24; B32B 27/36

조사된 기술분야에 속하는 최소문헌 이외의 문헌

한국등록실용신안공보 및 한국공개실용신안공보: 조사된 최소문헌란에 기재된 IPC
일본등록실용신안공보 및 일본공개실용신안공보: 조사된 최소문헌란에 기재된 IPC

국제조사에 이용된 전산 데이터베이스(데이터베이스의 명칭 및 검색어(해당하는 경우))

eKOMPASS(특허청 내부 검색시스템) & 키워드: 세퍼레이터, 알루미늄, 수산화알루미늄, 바인더, 폴리올레핀, 전기화학소자

C. 관련 문헌

카테고리*	인용문헌명 및 관련 구절(해당하는 경우)의 기재	관련 청구항
X	KR 10-2012-0079515 A (주식회사 이아이지) 2012.07.13 요약; 단락 [0005]-[0044]; 청구항 1-9; 및 도면 1-3 참조.	1-13
A	KR 10-2014-0116415 A (실 개엠베하) 2014.10.02 요약; 단락 [0021]-[0052]; 및 청구항 1-11 참조.	1-13
A	KR 10-2013-0007740 A (재단법인 구미전자정보기술원 등) 2013.01.21 요약; 단락 [0024]-[0033]; 및 청구항 1-5 참조.	1-13
A	JP 4311208 B2 (TORAY IND INC.) 2009.08.12 단락 [0052]-[0072]; 및 청구항 1-3 참조.	1-13
A	KR 10-2012-0124239 A (주식회사 엘지화학) 2012.11.13 요약; 단락 [0022]-[0058]; 및 청구항 1-14 참조.	1-13

☐ 추가 문헌이 C(계속)에 기재되어 있습니다.

☒ 대응특허에 관한 별지를 참조하십시오.

* 인용된 문헌의 특별 카테고리:

“A” 특별히 관련이 없는 것으로 보이는 일반적인 기술수준을 정의한 문헌

“E” 국제출원일보다 빠른 출원일 또는 우선일을 가지나 국제출원일 이후에 공개된 선출원 또는 특허 문헌

“L” 우선권 주장에 의문을 제기하는 문헌 또는 다른 인용문헌의 공개일 또는 다른 특별한 이유(이유를 명시)를 밝히기 위하여 인용된 문헌

“O” 구두 개시, 사용, 전시 또는 기타 수단을 언급하고 있는 문헌

“P” 우선일 이후에 공개되었으나 국제출원일 이전에 공개된 문헌

“T” 국제출원일 또는 우선일 후에 공개된 문헌으로, 출원과 상충하지 않으며 발명의 기초가 되는 원리나 이론을 이해하기 위해 인용된 문헌

“X” 특별한 관련이 있는 문헌. 해당 문헌 하나만으로 청구된 발명의 신규성 또는 진보성이 없는 것으로 본다.

“Y” 특별한 관련이 있는 문헌. 해당 문헌이 하나 이상의 다른 문헌과 조합하는 경우로 그 조합이 당업자에게 자명한 경우 청구된 발명은 진보성이 없는 것으로 본다.

“&” 동일한 대응특허문헌에 속하는 문헌

국제조사의 실제 완료일

2016년 10월 21일 (21.10.2016)

국제조사보고서 발송일

2016년 10월 25일 (25.10.2016)

ISA/KR의 명칭 및 우편주소

 대한민국 특허청
(35208) 대전광역시 서구 청사로 189,
4동 (둔산동, 정부대전청사)

팩스 번호 +82-42-481-8578

심사관

민인규

전화번호 +82-42-481-3326



국제조사보고서에서 인용된 특허문헌	공개일	대응특허문헌	공개일
KR 10-2012-0079515 A	2012/07/13	EP 2662920 A2 KR 10-1282067 B1 WO 2012-093864 A2	2013/11/13 2013/07/05 2012/07/12
KR 10-2014-0116415 A	2014/10/02	CN 104115306 A EP 2805366 A1 EP 2805366 B1 JP 2015-504234 A US 2015-0030933 A1 WO 2013-107911 A1	2014/10/22 2014/11/26 2016/08/24 2015/02/05 2015/01/29 2013/07/25
KR 10-2013-0007740 A	2013/01/21	없음	
JP 4311208 B2	2009/08/12	CN 1517402 A EP 1441001 A2 EP 1441001 B1 JP 2004-243760 A KR 10-0996022 B1 US 2004-0161620 A1 US 7459217 B2	2004/08/04 2004/07/28 2007/11/28 2004/09/02 2010/11/22 2004/08/19 2008/12/02
KR 10-2012-0124239 A	2012/11/13	CN 103733380 A EP 2706591 A2 JP 2014-510388 A JP 2016-076492 A JP 5875673 B2 KR 10-1358761 B1 US 2013-0149587 A1 US 2015-0086714 A1 US 8927134 B2 WO 2012-150838 A2	2014/04/16 2014/03/12 2014/04/24 2016/05/12 2016/03/02 2014/02/07 2013/06/13 2015/03/26 2015/01/06 2012/11/08

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**TERMINAL DISCLAIMER TO OBVIATE A PROVISIONAL DOUBLE PATENTING
REJECTION OVER A PENDING "REFERENCE" APPLICATION**

Docket Number (Optional)

00249/0086

In re Application of: David W. Avison et al.

Application No.: 19/023,905

Filed: January 16, 2025

For: MULTILAYER NANOPOROUS SEPARATOR

The applicant, LG Energy Solution, Ltd., owner of 100 percent interest in the instant application hereby disclaims, except as provided below, the terminal part of the statutory term of any patent granted on the instant application which would extend beyond the expiration date of the full statutory term of any patent granted on pending **reference** Application Number 19/022,365, filed, January 15, 2025, as the term of any patent granted on said **reference** application may be shortened by any terminal disclaimer filed prior to the grant of any patent on the pending **reference** application. The applicant hereby agrees that any patent so granted on the instant application shall be enforceable only for and during such period that it and any patent granted on the **reference** application are commonly owned. This agreement runs with any patent granted on the instant application and is binding upon the grantee, its successors or assigns.

In making the above disclaimer, the applicant does not disclaim the terminal part of any patent granted on the instant application that would extend to the expiration date of the full statutory term of any patent granted on said **reference** application, "as the term of any patent granted on said **reference** application may be shortened by any terminal disclaimer filed prior to the grant of any patent on the pending **reference** application," in the event that: any such patent granted on the pending **reference** application expires for failure to pay a maintenance fee, is held unenforceable, is found invalid by a court of competent jurisdiction, is statutorily disclaimed in whole or terminally disclaimed under 37 CFR 1.321, has all claims canceled by a reexamination certificate, is reissued, or is in any manner terminated prior to the expiration of its full statutory term as shortened by any terminal disclaimer filed prior to its grant.

Check either box 1 or 2 below, if appropriate.

1. ☐ The undersigned is the applicant. If the applicant is an assignee, the undersigned is authorized to act on behalf of the assignee.

I hereby acknowledge that any willful false statements made are punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

2. ☒ The undersigned is an attorney or agent of record. Reg. No. 45,343

/Brian A. Comack/
Signature

April 7, 2025
Date

Brian A. Comack
Typed or printed name

Attorney for Applicant
Title

(212) 336-8000
Telephone Number

- ☒ Terminal disclaimer fee under 37 CFR 1.20(d) is included.

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<i>Application Number</i> * 19/023,905 *	Application/Control No. 19/023,905	Applicant(s)/Patent under Reexamination Avison et al.	
	Examiner PARK, LISA S	Art Unit 1729	
Document Code - DISQ		Internal Document - DO NOT MAIL	

TERMINAL DISCLAIMER	☒ APPROVED	☐ DISAPPROVED
Date Filed: <u>07 April 2025</u>	This patent is subject to a Terminal Disclaimer	

Approved/Disapproved by:
/TRINA STEPTOE/ Technology Center: <u>OPLC</u> Telephone: <u>(571)272-2577</u> <u>2 TD's</u>

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 19/023,905	Filing Date 01/16/2025	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 = *		x \$100 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 = *		x \$480 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

		(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	04/07/2025	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		
	Total (37 CFR 1.16(i))	* 30	Minus	** 20	= 10	x \$200 =	2000
	Independent (37 CFR 1.16(h))	* 1	Minus	*** 3	= 0	x \$600 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	2000
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =	
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE	
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/NICOLE LOVE HENSLEY/	
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".							
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.							

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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19/023,905

01/16/2025

David W. Avison

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04/15/2025

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EXAMINER

PARK, LISA S

ART UNIT

PAPER NUMBER

1729

MAIL DATE

DELIVERY MODE

04/15/2025

PAPER

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

DETAILED CORRESPONDENCE

1. The text of those sections of Title 35, U.S. Code not included in this action can be found in a prior Office action.

Notice of Pre-AIA or AIA Status

2. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Amendment

3. In response to the amendment received on 4/3/2025:
 - a. Claims 1-30 are pending in the current application. Claims 1-3, 6-9, 11-16, 18-19 have been amended, and Claims 20-30 are newly added.
 - b. The previous objection to the specification is overcome in light of the claim amendments.
 - c. The previous rejections under 35 USC 112 is overcome in light of the amendment. New rejections under 35 USC 112 are raised by the amendment, as explained below.
 - d. The double patenting rejections are withdrawn. The terminal disclaimers filed on 4/3/2025 disclaiming the terminal portion of any patent granted on this application which would extend beyond the expiration date of Application numbers 19/021913 and 19/022365 have been reviewed and are accepted. The terminal disclaimers have been recorded.

Claim Interpretation

4. All “wherein” clauses are given patentable weight unless otherwise noted. Please see MPEP 2111.04 regarding optional claim language.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(d):

(d) REFERENCE IN DEPENDENT FORMS.—Subject to subsection (e), a claim in dependent form shall contain a reference to a claim previously set forth and then specify a further limitation of the subject matter claimed. A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.

The following is a quotation of pre-AIA 35 U.S.C. 112, fourth paragraph:

Subject to the following paragraph [i.e., the fifth paragraph of pre-AIA 35 U.S.C. 112], a claim in dependent form shall contain a reference to a claim previously set forth and then specify a further limitation of the subject matter claimed. A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.

5. Claims 27 and 28 are rejected under 35 U.S.C. 112(d) or pre-AIA 35 U.S.C. 112, 4th paragraph, as being of improper dependent form for failing to further limit the subject matter of the claim upon which it depends, or for failing to include all the limitations of the claim upon which it depends. Claim 27 attempts to further limit Claim 1’s limitation “a total coating weight of the first layer is 0.8 gsm to 4.7 gsm” by reciting “the total coating weight of the first layer is 5.2 gsm or less” but the presence of “or less” broadens the upper and lower limits of the range of claim 1. Claim 28’s recitation of “the total coating weight of the first layer is 3.6 gsm or less” broadens the lower limit of Claim 1. Applicant may cancel the claim(s), amend the claim(s) to place the claim(s) in proper dependent form, rewrite the claim(s) in independent form, or present a sufficient showing that the dependent claim(s) complies with the statutory requirements.

Claim Rejections - 35 USC § 103

6. Claims 1-15 and 17-30 are rejected under 35 U.S.C. 103 as being unpatentable over Iwase US PG Publication 2014/0170464 in view of Nishikawa US PG Publication 2011/0143185.

Regarding Claims 1, 3-8, 12, 17-18, and 20-25, 27-28, Iwase discloses a separator for a lithium battery comprising a porous polymeric layer (separator sheet) **40** having a first side and a second side (para 0042-0047) and formed of a polyolefin such as polyethylene or polypropylene (para 0042, meeting **Claims 17-18**), a first layer (porous heat resistance layer) **30** adjacent the first side of the porous polymeric layer **40**, the first layer comprising pores and solid material, wherein (a) the solid material comprises inorganic oxide particles (e.g. alumina hydrate/boehmite, $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$, para 0053, where x is 1, given the shown formula, meeting **Claims 4-6** and which is inherently crystalline and therefore has crystallites) and at least one polymer (para 0056), and a fraction of the at least one polymer (including carboxymethyl cellulose, which is a cellulosic polymer, meeting **Claim 7**) in the solid material portion is up to 50% by mass (para 0057), (b) a plurality of the inorganic oxide particles in the first layer have an average size (e.g. crystallite size due to the inherency of the presence of crystallites) in the range of 5 nm to 90 nm (0.01 μm or more, which is 10 nm or more, up to about 2 μm , para 0054, also meeting **Claim 3**'s range of 5 nm to 80 nm, and the ranges of 5 nm to 25 nm, 30 nm to 50 nm, and 5 nm to 75 nm as required by **Claims 20-22**), (c) a thickness of the first layer is 3 microns or less (1 μm or more, para 0053, overlapping the claimed range) (see entire disclosure and particularly Fig 5 and paras 0042-0058). Iwase fails to specifically disclose wherein (d) the ratio of the thickness of the first layer **30** to a total thickness of the separator (thickness of **30** + thickness of **40**) is 10% to 80%. However, the porous polymeric layer of Iwase has a typical thickness of 10 μm or more (para 0045) and because thicknesses are provided by Iwase for both the porous polymer layer **40** and

the first layer **30**, the skilled artisan would be capable of selecting any thickness of **30** and **40** in the ranges provided. Therefore, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to design the first layer of the separator of Iwase such that a ratio of the thickness of the first layer **30** to a total thickness of the separator **40** is between 10% and 80% since Iwase teaches ranges of values for the two thicknesses such that the skilled artisan could arrive at a ratio in the claimed range as a matter of routine experimentation. For example, given the parameters laid out by Iwase, the skilled artisan could choose a 2 micron thickness of the first layer and a 10 micron thickness of the porous polymer layer, resulting in a total thickness of 12 microns, and so the ratio of the thickness of the first layer **30** to a total thickness of the separator (thickness of **30** + thickness of **40**) would be 17%, which is between 10% and 80%.

Iwase fails to specifically disclose wherein the solid material comprises the at least one polymer in a volume fraction of 10% to 50% (as opposed to *mass fraction* taught by Iwase) and the specific ranges claimed in **Claims 8 and 23-25**. However, in the same field of endeavor of hybrid separator design, Nishikawa discloses wherein an effective porous heat resistant layer using e.g. alumina or boehmite inorganic filler should use an amount of 0.4 to 4 times the volume of inorganic filler relative to the volume of heat-resistant resin which equates to using 0.25 to 2.5 times the volume of resin relative to filler (i.e. 25% to 250%), and teaches that this quantity is optimized to improve heat resistance effect while maintaining good ion permeation and handling properties (paras 0060-0061).

Therefore, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to design the first layer of the separator of Iwase such that the at least one polymer (and specifically a cellulosic polymer such as

carboxymethyl cellulose) in the first layer solid material is present in a volume fraction of 10% to 50% or specifically 12% to 45% (meeting **Claim 8**) or 15% to 45% (meeting **Claim 23**) or 25% to 45% (meeting **Claim 24**) or 15.3% to 25.6% (meeting **Claim 25**) because Nishikawa teaches that this range improves physical and ion conduction properties of the separator and battery.

Iwase fails to specifically disclose a total coating weight of the first layer such that it is 0.8 gsm to 4.7 gsm, or is 0.8 gsm to 3.0 gsm, or 5.2 gsm or less or 3.6 gsm or less. However, Nishikawa discloses wherein a heat resistant porous layer comprising the same materials and other properties should have a weight per unit (which is the coating weight) of 2 gsm to 10 gsm (para 0059) and so it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to design the first layer of Iwase such that the total coating weight is 0.8 gsm to 4.7 gsm, 0.8 gsm to 3.0 gsm (meeting **Claim 12**), 5.2 gsm or less (meeting **Claim 27**), or 3.6 gsm or less (meeting **Claim 28**) because Nishikawa teaches that the ranges are beneficial for a similar design of battery having excellent heat resistance.

Regarding Claim 2, Iwase does not specifically disclose a pore diameter for the plurality of pores in the first (heat resistance) layer **30**, and in particular does not disclose wherein the pores have a diameter of 100 nm or less. However, Nishikawa discusses both particle size of inorganic filler in the heat resistant layer and the pore size in the heat resistant layer. Nishikawa states in para 0062 that the preferable particle size of inorganic filler in the heat-resistant porous layer is from 100 nm to 1 μ m and states in para 0040 that the heat-resistant porous layer has an average pore size of 100 nm - 200 nm, and so it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to design the separator of Iwase and Nishikawa such that the pores in the first layer have a size that is in the same range of

values as the particle sizes of the filler since Nishikawa teaches that these values used together provide for a battery having good shutdown properties, ion permeation, and heat resistance.

Regarding Claims 9 and 19, Iwase discloses wherein the first layer **30** has a porosity of 20% to 50% (para 0050), a range that largely overlaps the range of **Claim 9** of 20%-70% and the range of **Claim 19** of 30-55%. In the case where the claimed ranges "overlap or lie inside ranges disclosed by the prior art" a *prima facie* case of obviousness exists. *In re Wertheim*, 541 F.2d 257, 191 USPQ 90 (CCPA 1976); *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990) (The prior art taught carbon monoxide concentrations of "about 1-5%" while the claim was limited to "more than 5%." The court held that "about 1-5%" allowed for concentrations slightly above 5% thus the ranges overlapped.); *In re Geisler*, 116 F.3d 1465, 1469-71, 43 USPQ2d 1362, 1365-66 (Fed. Cir. 1997) (Claim reciting thickness of a protective layer as falling within a range of "50 to 100 Angstroms" considered *prima facie* obvious in view of prior art reference teaching that "for suitable protection, the thickness of the protective layer should be not less than about 10 nm [i.e., 100 Angstroms]." The court stated that "by stating that 'suitable protection' is provided if the protective layer is 'about' 100 Angstroms thick, [the prior art reference] directly teaches the use of a thickness within [applicant's] claimed range."). Similarly, a *prima facie* case of obviousness exists where the claimed ranges and prior art ranges do not overlap but are close enough that one skilled in the art would have expected them to have the same properties. *Titanium Metals Corp. of America v. Banner*, 778 F.2d 775, 227 USPQ 773 (Fed. Cir. 1985) (Court held as proper a rejection of a claim directed to an alloy of "having 0.8% nickel, 0.3% molybdenum, up to 0.1% iron, balance titanium" as obvious over a reference disclosing alloys of 0.75% nickel, 0.25% molybdenum, balance titanium and 0.94% nickel, 0.31% molybdenum, balance titanium.).

Regarding Claims 10 and 13, Iwase teaches that the first layer **30** can be formed on both opposing sides of a porous polymer layer **40** (resulting in a first layer **30** and a second layer **30**) and teaches that the kind of inorganic filler used in the two layers **30** can be different (para 0058) which would result in a different composition (meeting **Claim 13**). Although Iwase does not specifically discuss that a polymer is also used in the second layer **30**, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to design the second layer **30** of Iwase modified by Nishikawa such that it has an inorganic filler and a polymer because Iwase teaches in para 0057 that the layer **30** comprises less than 100% of the inorganic filler and at least a polymeric binder is also present.

Regarding Claims 11 and 26, Iwase modified by Nishikawa fails to specifically disclose wherein a total coating weight of the first layer and second layer is less than 9.4 gsm or 8.4 gsm or less. However, Nishikawa discloses wherein a heat resistant porous layer comprising the same materials and other properties should have a weight per unit (which is the coating weight) of 2 gsm to 10 gsm (para 0059) and Iwase teaches the use of the first and second layers, and so it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to design the total weight of the first and second weight as 9.4 gsm or less than 8.4 because Nishikawa teaches that this range is beneficial for a similar design of battery having excellent heat resistance.

The Office notes that while Nishikawa does not specifically teach the coating weight of *two* layers, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to apply the same reasoning to both the first and second layers of Iwase and given the range of values taught by Nishikawa, the skilled artisan would be capable of selecting values for both layers such that the total falls in the claimed range.

Regarding Claims 14 and 30, Iwase teaches that a ratio by weight of inorganic oxide to polymer in the first layer is about 12:1 or about 8:1 since Iwase teaches that a ratio of inorganic filler is set to 85% by mass to 99% by mass, which encompasses or overlaps the claimed values of “about 12:1” or 92% of the mixture or “about 4:1” or 80% of the mixture. In the case where the claimed ranges “overlap or lie inside ranges disclosed by the prior art” a *prima facie* case of obviousness exists. *In re Wertheim*, 541 F.2d 257, 191 USPQ 90 (CCPA 1976); *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990) (The prior art taught carbon monoxide concentrations of “about 1-5%” while the claim was limited to “more than 5%.” The court held that “about 1-5%” allowed for concentrations slightly above 5% thus the ranges overlapped.); *In re Geisler*, 116 F.3d 1465, 1469-71, 43 USPQ2d 1362, 1365-66 (Fed. Cir. 1997) (Claim reciting thickness of a protective layer as falling within a range of “50 to 100 Angstroms” considered *prima facie* obvious in view of prior art reference teaching that “for suitable protection, the thickness of the protective layer should be not less than about 10 nm [i.e., 100 Angstroms].” The court stated that “by stating that ‘suitable protection’ is provided if the protective layer is ‘about’ 100 Angstroms thick, [the prior art reference] directly teaches the use of a thickness within [applicant's] claimed range.”).

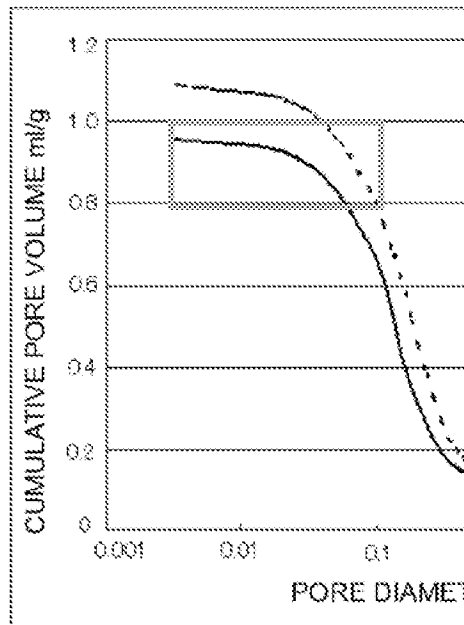
Regarding Claim 15, Iwase discloses that the thickness of the first (heat resistance) layer **30** is usually 1 micron or more (para 0052) and discloses that a first layer **30** can be formed on each side of the porous substrate (separator) **40** (para 0058), but Iwase modified by Nishikawa fails to specifically disclose wherein a combined thickness of the first and second layer of Claim 10 is less than 3.8 microns. However, since Iwase teaches that this thickness is a result-effective variable that is optimized to maintain protection against short circuit while ensuring resistance against foreign matter contained in the battery, and that battery characteristics and reliability are

maintained (para 0052), it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to design the separator of Iwase and Nishikawa such that a combined thickness of the first and second layer of is less than 3.8 microns because Iwase teaches that this would ensure good battery characteristics and reliability. “[W]here the general conditions of a claim are disclosed in the prior art, it is not inventive to discover the optimum or workable ranges by routine experimentation.” See *In re Aller*, 220 F.2d 454, 456, 105 USPQ 233, 235 (CCPA 1955). The discovery of an optimum value of a known result effective variable, without producing any new or unexpected results, is within the ambit of a person of ordinary skill in the art. See *In re Boesch*, 205 USPQ 215 (CCPA 1980) (see MPEP § 2144.05, II.).

7. Claim 16 is rejected under 35 U.S.C. 103 as being unpatentable over Iwase US PG Publication 2014/0170464 in view of Nishikawa US PG Publication 2011/0143185, as applied to Claim 1, and further in view of Morigaki US PG Publication 2001/0041290.

Regarding Claim 16, Iwase modified by Nishikawa discloses the claimed separator as described in the rejection of Claim 1, which is incorporated herein in its entirety. Nishikawa acknowledges the importance of pore volume (Fig. 1 and para 0109 and measures pore volume for the total separator and the substrate, and discloses that the pore volumes of both components are within the claimed range while having the same pore diameters as claimed). See the grey box in the annotated Fig. 1 of Nishikawa, below.

[Fig. 1]



It would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to design the first layer of Iwase modified by Nishikawa to have a similar pore volume as the total separator and the substrate because Nishikawa provides these values (between 0.8 cc/g and 1.0 cc/g for pores of the same size as claimed) as part of a well-functioning separator and the combination of familiar elements is likely to be obvious when it does no more than yield predictable results. See *KSR International Co. v. Teleflex Inc.*, 550 U.S. ___, ___, 82 USPQ2d 1385, 1395 – 97 (2007) (see MPEP § 2143, A.).

Further, although Iwase modified by Nishikawa does not specifically disclose the pore volume of just the *first layer*, such that it is between 0.8 cc/g and 1.0 cc/g, in the same field of endeavor of battery separator design, Morigaki teaches that the void volume and void size/diameter in all components of a separator and electrodes all have an effect on absorption of sufficient electrolyte for improved battery capacity (see entire disclosure and especially e.g. paras 0061, 0074-0077).

Therefore, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to design the first layer of Iwase modified by Nishikawa to optimize the pore volume of the first layer of Iwase and Nishikawa such that it is between 0.8 cc/g and 1.0 cc/g because Morigaki teaches that this property is part of an important consideration in improving battery capacity because it relates to absorption of electrolyte.

Response to Arguments

8. Applicant's arguments with respect to the claims are based on the claims as amended. The amended claims have been addressed in the new rejection above.

Conclusion

9. Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to LISA S PARK whose telephone number is (571)270-3597. The examiner can normally be reached M-Th 5:30a to 3p Eastern Time.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ula Tavares-Crockett can be reached on 5712721481. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/LISA S PARK/
Primary Examiner, Art Unit 1729

<i>Notice of References Cited</i>	Application/Control No. 19/023,905		Applicant(s)/Patent Under Reexamination Avison et al.	
	Examiner LISA S PARK		Art Unit 1729	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Name	CPC Classification	US Classification
*	A	US-20140170464-A1	2014-06-19	IWASE; Kosuke	H01M50/434	429/246
*	B	US-20110143185-A1	2011-06-16	Nishikawa; Satoshi	H01M50/423	429/145
*	C	US-20010041290-A1	2001-11-15	Morigaki, Kenichi	H01M10/0525	429/231.95
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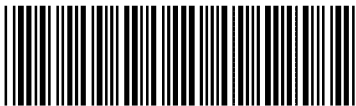
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*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in YYYY-MM-DD format are publication dates. Classifications may be US or foreign.

<i>Search Notes</i> 	Application/Control No. 19/023,905	Applicant(s)/Patent Under Reexamination Avison et al.
	Examiner LISA S PARK	Art Unit 1729

CPC - Searched*		
Symbol	Date	Examiner
H01M10/0525 or H01M50/417 or H01M50/446 or H01M50/451 or H01M50/457 or H01M50/489 or H01M50/491	03/11/2025, 04/14/2025	/L.S.P./

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
PE2E keyword search (see attached search notes) (preliminary)	03/11/2025, 04/14/2025	/L.S.P./
PE2E inventor/assignee search	03/11/2025	/L.S.P./
Analysis of prior art in family and related applications	04/14/2025	/L.S.P./

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

/LISA S PARK/ Primary Examiner, Art Unit 1729	
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<i>Index of Claims</i> 	Application/Control No. 19/023,905	Applicant(s)/Patent Under Reexamination Avison et al.
	Examiner LISA S PARK	Art Unit 1729

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

CLAIMS										
☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	03/11/2025	04/14/2025							
	1	✓	✓							
	2	✓	✓							
	3	✓	✓							
	4	✓	✓							
	5	✓	✓							
	6	✓	✓							
	7	✓	✓							
	8	✓	✓							
	9	✓	✓							
	10	✓	✓							
	11	✓	✓							
	12	✓	✓							
	13	✓	✓							
	14	✓	✓							
	15	✓	✓							
	16	✓	✓							
	17	✓	✓							
	18	✓	✓							
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	26		✓							
	27		✓							
	28		✓							
	29		✓							
	30		✓							

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L12	16	1 AND (multilayer composite hybrid) ADJ separator AND inorganic WITH crystallite	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/14 12:07 PM
L13	48	1 AND separator AND inorganic WITH crystallite	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/14 12:07 PM
L14	62	1 AND separator WITH (alumina boehmite silica oxide) WITH crystallite	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/14 12:10 PM
L15	3	"20200266414"	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/14 12:13 PM
L16	2	("20200266414").pn.	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/14 12:13 PM
L17	1	("20200266414").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/14 12:13 PM
L18	21	("20160172705" "20170294638" "20110003209" "20110165450" "20160104876" "20160172705" "20170301903" "20180019457" "20180248164" "6432586" "6723467" "7638241" "7662517" "7709152" "7883799" "8318360" "9293752" "9312527" "9478784" "9484565" "9543077" "9847519").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/14 12:16 PM
L19	2	18 AND crystallite	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/14 12:16 PM
L20	39	1 AND separator AND filler WITH crystallite	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/14 12:24 PM
L21	2759	1 AND (alumina boehmite silica oxide)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR,	OR	ON	ON	2025/04/14 12:27 PM

L22	214	WITH crystallite 1 AND (alumina boehmite silica oxide) near2 crystallite	JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) (US-PGPUB; USPAT)	OR	ON	ON	2025/04/14 12:27 PM
L23	26	1 AND (alumina boehmite silica oxide) near2 crystallite AND heat ADJ resistan\$4	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/14 12:27 PM
L24	683	21 AND separat\$4	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/14 12:29 PM
L25	15	1 AND (alumina boehmite silica oxide) near2 crystallite SAME separat\$4	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/14 12:29 PM
L26	44	1 AND (alumina boehmite silica oxide) near2 crystallite SAME separat\$4	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/14 12:29 PM
L27	3	("20140170464" "20110143185" "20010041290").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/14 12:49 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

Doc code: IDS

PTO/SB/08 (01-25)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 11/30/2027. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number		19023905
	Filing Date		2025-01-16
	First Named Inventor	David W. Avison	
	Art Unit	1729	
	Examiner Name	Lisa S. Park	
	Attorney Docket Number	00249/0086	

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	1	2017010780	WO	A1	2017-01-19	LG Chemical Ltd.		☒

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number		19023905
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	First Named Inventor	David W. Avison	
	Art Unit		1729
	Examiner Name	Lisa S. Park	
	Attorney Docket Number		00249/0086

	1		☐
If you wish to add additional non-patent literature document citation information, please click the Add button Add			
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Examiner Signature	/LISA S PARK/		Date Considered 04/14/2025
*EXAMINER: Initial if reference considered and whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.			
¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.			

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**

Application Number	19023905
Filing Date	2025-01-16
First Named Inventor	David W. Avison
Art Unit	1729
Examiner Name	Lisa S. Park
Attorney Docket Number	00249/0086

TIMING STATEMENT

Please see 37 CFR 1.97(e) to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached timing statement under 37 CFR 1.97(e).

☒ A timing statement under 37 CFR 1.97(e) is not submitted herewith.

TIMING FEE

☒ The fee set forth in 37 CFR 1.17(p) has been submitted herewith.

COPIES

☐ An identification of an earlier application pursuant to 37 CFR 1.98(d)(1) is attached.

SIZE FEE ASSERTION

Please see 37 CFR 1.17(v) and the [IDS Auto-Load Instructions](#) for completing this form to make the appropriate selection of an assertion under 37 CFR 1.98. For the information disclosure statement (IDS) submitted herewith, the applicant or patent owner certifies the following with respect to the cumulative number of applicant-provided or patent-owner provided items of information submitted to date including those in the accompanying IDS (select only one):

☒ No IDS size fee is required under 37 CFR 1.17(v) at this time.

☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(1).

☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(2).

☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(3).

SIGNATURE

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 11.18. Please see 37 CFR 1.4(d) for the form of the signature.

Signature	/Brian A. Comack/	Date (YYYY-MM-DD)	2025-04-07
Name/Print	Brian A. Comack	Registration Number	45,343

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**

Application Number	19023905
Filing Date	2025-01-16
First Named Inventor	David W. Avison
Art Unit	1729
Examiner Name	Lisa S. Park
Attorney Docket Number	00249/0086

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	491584	(H01M10/0525 or H01M50/417 or H01M50/446 or H01M50/451 or H01M50/457 or H01M50/489 or H01M50/491).cpc.	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/03/11 08:16 AM
L2	27	("20070190427" OR "20110003209" OR "20110165450" OR "20150155601" OR "20160043370" OR "20160104876" OR "20160172705" OR "20170012264" OR "20170294638" OR "20170301903" OR "20170373295" OR "20180019457" OR "20180248164" OR "4024323" OR "6432586" OR "6723467" OR "7638241" OR "7662517" OR "7709152" OR "7883799" OR "8318360" OR "9293752" OR "9312527" OR "9478784" OR "9484565" OR "9543077" OR "9847519").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/11 08:35 AM
L4	6477	1 AND (multilayer composite hybrid) ADJ separator	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/03/11 08:37 AM
L5	1169	4 AND porous ADJ (base substrate middle central) AND (coat\$4 (first ADJ layer)) WITH (oxide inorganic alumina)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/03/11 08:37 AM
L6	0	4 AND porous ADJ (base substrate middle central) AND (coat\$4 (first ADJ layer)) WITH (oxide inorganic alumina) WITH	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/03/11 08:37 AM

L7	5630	non\$1porous WITH porous ADJ (section region portion area) (((("LG") near3 ("ENERGY") near3 ("SOLUTION") near3 ("LTD")))).AS,AANM.	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/11 08:44 AM
L8	9	7 AND multilayer ADJ separator	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/11 08:44 AM
L9	16	7 AND multilayer near2 separator	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/11 08:45 AM
L10	0	9 AND first ADJ layer WITH porous near2 polymer\$2	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/11 08:45 AM
L11	477	(((("AVISON") near3 ("David")) OR ((("CARLSON") near3 ("Steven")) OR ((("SLOAN") near3 ("Benjamin")))).INV.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2025/03/11 08:46 AM
L12	18	11 AND multilayer near2 separator	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2025/03/11 08:47 AM
L13	6	2 AND (quantity amount ratio content percent\$4) WITH (polymer\$2 resin) near2 (coat\$4 layer film)	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/11 08:47 AM
L14	7	4 AND (base substrate middle central) AND (coat\$4 film layer) near3 porous WITH (polymer\$2 resin) near2 volume ADJ fraction	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/11 08:47 AM
L15	944	4 AND porous ADJ (base substrate middle central) AND (coat\$4 (first ADJ layer)) WITH (oxide inorganic alumina) WITH (pore porous)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/09 03:14 PM
L16	20	2 AND alumina	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/09 03:15 PM
L17	2	2 AND alumina AND crystallite	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/10 05:35 AM
L18	50	4 AND porous ADJ (base substrate middle central) AND (coat\$4 (first ADJ layer)) WITH (oxide inorganic alumina) AND crystallite	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/10 05:44 AM
L19	1	("20120145468").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/10 05:46 AM

L20	1	("20080292968").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/10 05:51 AM
L21	19	18 AND (vol volume) ADJ (percent\$4 "%" fraction)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/10 06:12 AM
L22	18	14 AND (vol volume) ADJ (percent\$4 "%" fraction)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/10 06:15 AM
L23	1	("20160172705").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/10 06:21 AM
L24	1627	(h01m10/0525.ipc. h01m50/417.ipc. h01m50/446.ipc. h01m50/451.ipc.)	(JPO)	OR	ON	ON	2025/04/10 06:51 AM
L25	16	24 AND porous AND (layer substrate) AND filler	(JPO)	OR	ON	ON	2025/04/10 06:56 AM
L26	1	("20140170464").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/10 07:10 AM
L27	27	("20070190427" OR "20110003209" OR "20110165450" OR "20150155601" OR "20160043370" OR "20160104876" OR "20160172705" OR "20170012264" OR "20170294638" OR "20170301903" OR "20170373295" OR "20180019457" OR "20180248164" OR "4024323" OR "6432586" OR "6723467" OR "7638241" OR "7662517" OR "7709152" OR "7883799" OR "8318360" OR "9293752" OR "9312527" OR "9478784" OR "9484565" OR "9543077" OR "9847519").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/10 07:16 AM
L28	0	("2009006119").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/10 09:45 AM
L29	0	4 AND porous ADJ (base substrate middle central) AND (coat\$4	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS;	OR	ON	ON	2025/04/10 12:59 PM

L30	42	(first ADJ layer)) WITH (oxide inorganic alumina) WITH crystallite separator AND h01m\$.cpc. AND porous ADJ (base substrate middle central separator) AND (coat\$4 (first ADJ layer)) WITH (oxide inorganic alumina) WITH crystallite	EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/10 12:59 PM
L31	24	separator AND h01m\$.cpc. AND porous ADJ (base substrate middle central separator) AND (coat\$4 (first ADJ layer)) WITH (oxide inorganic alumina) near3 crystallite	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/10 01:00 PM
L32	21	h01m\$.cpc. AND separator WITH porous ADJ (base substrate middle central separator) AND (coat\$4 (first ADJ layer)) WITH (oxide inorganic alumina) near3 crystallite	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/10 01:03 PM
L33	26	h01m\$.cpc. AND porous ADJ (base substrate middle central separator) AND (coat\$4 (first ADJ layer)) WITH (oxide inorganic alumina) near3 crystallite	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/10 01:16 PM
L34	26	h01m\$.cpc. AND porous ADJ (base substrate middle central separator) AND (coat\$4 (first ADJ layer)) WITH (oxide inorganic alumina filler) near3 crystallite	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/10 01:19 PM
L35	15	h01m\$.cpc. AND separator WITH (coat\$4 (first ADJ layer)) WITH (oxide inorganic alumina filler) WITH crystallite	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/10 01:20 PM
L36	4938	h01m\$.cpc. AND (oxide inorganic alumina filler)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR,	OR	ON	ON	2025/04/10 01:20 PM

L37	95	WITH crystallite h01m\$.cpc. AND (alumina beohmite) WITH crystallite	JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/10 01:21 PM
L38	201	h01m\$.cpc. AND (alumina beohmite (inorganic ADJ oxide) silica) WITH crystallite	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/10 01:34 PM
L39	24	h01m\$.cpc. AND (alumina beohmite (inorganic ADJ oxide) silica) WITH crystallite AND separator	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/10 01:34 PM
L40	250	(alumina beohmite (inorganic ADJ oxide) silica) WITH crystallite AND separator	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/10 01:38 PM
L41	13	(alumina beohmite (inorganic ADJ oxide) silica) WITH crystallite SAME separator	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/10 01:38 PM
L42	33	(alumina beohmite (inorganic ADJ oxide) silica) WITH crystallite SAME separator	(USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/10 01:42 PM
L43	499	1 AND (multilayer composite hybrid) ADJ separator AND (inorganic filler heat- resistant) WITH pore ADJ (size diameter) WITH layer	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/12 06:56 AM
L44	1	("20170309915").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/12 06:59 AM
L45	0	1 AND (multilayer composite hybrid) ADJ separator AND (inorganic filler heat- resistant) WITH pore ADJ (size diameter) WITH layer WITH gsm	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/12 07:09 AM
L46	0	1 AND (multilayer composite hybrid) ADJ separator AND (inorganic filler heat- resistant) WITH pore ADJ (size diameter) WITH layer SAME gsm	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/12 07:09 AM
L47	0	1 AND (multilayer	(US-PGPUB; USPAT;	OR	ON	ON	2025/04/12

		composite hybrid) ADJ separator AND (inorganic filler heat-resistant) WITH layer WITH gsm	USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)				07:10 AM
L48	19	1 AND (multilayer composite hybrid) ADJ separator AND (inorganic filler heat-resistant) WITH gsm	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/12 07:14 AM
L49	19	h01m\$.cpc. AND (multilayer composite hybrid) ADJ separator AND (inorganic filler heat-resistant) WITH gsm	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/12 07:14 AM
L50	17	h01m\$.cpc. AND (multilayer composite hybrid) ADJ separator AND (alumina) WITH gsm	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/12 07:15 AM
L51	27	h01m\$.cpc. AND (multilayer composite hybrid) ADJ separator AND (alumina) WITH coating ADJ weight	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/12 07:15 AM
L52	4	1 AND (multilayer composite hybrid) ADJ separator AND (inorganic filler heat-resistant) WITH pore ADJ (size diameter) WITH layer WITH g/m.sup.2	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/12 07:15 AM
L53	0	1 AND (multilayer composite hybrid) ADJ separator AND (inorganic filler heat-resistant) WITH (void pore) near2 (ml/g cc/g)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/14 07:09 AM
L54	6	1 AND (multilayer composite hybrid) ADJ separator AND (inorganic filler heat-resistant) WITH (void pore) WITH (ml/g cc/g)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/14 07:09 AM
L55	26	1 AND (multilayer composite hybrid) ADJ separator WITH (void pore) WITH (ml/g cc/g)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/14 07:29 AM
L56	26	(multilayer composite hybrid) ADJ separator WITH (void pore) WITH (ml/g cc/g)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO;	OR	ON	ON	2025/04/14 07:30 AM

L57	141	separator WITH (void pore) WITH (ml/g cc/g)	DERWENT; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (CN, FR, JP, KR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2025/04/14 07:30 AM
L58	42	separator WITH (void pore) WITH (ml/g cc/g)	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/14 07:30 AM
L59	1020	separator WITH (void pore) ADJ volume	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/14 08:20 AM
L60	637	1 AND 59	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/14 08:20 AM
L61	79	60 AND inorganic ADJ filler	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/14 08:21 AM
L62	7	60 AND inorganic ADJ filler AND (cc/g ml/g)	(US-PGPUB; USPAT)	OR	ON	ON	2025/04/14 08:22 AM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
19/023,905	01/16/2025	David W. Avison	00249/0086

CONFIRMATION NO. 3614

1912
AMSTER, ROTHSTEIN & EBENSTEIN LLP
405 LEXINGTON AVENUE
FLOOR 48
NEW YORK, NY 10174

PUBLICATION NOTICE



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Date Mailed: 05/15/2025

Title:MULTILAYER NANOPOROUS SEPARATOR

Publication No.US-2025-0158107-A1

Publication Date:05/15/2025

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/023,905	01/16/2025	David W. Avison	00249/0086	3614
1912 7590 07/14/2025 AMSTER, ROTHSTEIN & EBENSTEIN LLP 405 LEXINGTON AVENUE FLOOR 48 NEW YORK, NY 10174			EXAMINER PARK, LISA S	
			ART UNIT	PAPER NUMBER
			1729	
			MAIL DATE	DELIVERY MODE
			07/14/2025	PAPER

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

<i>Applicant-Initiated Interview Summary</i>	Application No. 19/023,905	Applicant(s) Avison et al.		
	Examiner LISA S PARK	Art Unit 1729	AIA (First Inventor to File) Status Yes	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
LISA S PARK	Primary Examiner	Telephonic
Brian Comack	Attorney of Record	
Andrew Sutton	Attorney	

Date of Interview: 11 July 2025

Issues Discussed:

35 U.S.C. 103

Arguments directed to the rejections over the Nishikawa reference were discussed. No specific agreement was reached. Applicant is encouraged to submit arguments for consideration after-final.

☒ Attachment

/LISA S PARK/ Primary Examiner, Art Unit 1729	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

From: Brian Amos
To: Park, Lisa
Cc: Brian Cornack
Subject: Interview Agenda U.S. Patent Application Serial No. 19/021,913 (Our file: 00249/84)
Date: Thursday, July 10, 2025 11:50:24 AM

CAUTION: This email has originated from a source outside of USPTO. **PLEASE CONSIDER THE SOURCE** before responding, clicking on links, or opening attachments.

Dear Examiner Park:

Thank you for agreeing to a telephonic interview tomorrow at 10 am regarding the above referenced case. Below is an agenda regarding the topics that we plan to discuss:

- The cited Iwase and Nishikawa references do not disclose, alone or in combination, the following claim limitation: a "volume fraction of the at least one polymer in the solid material is 10%-50%."
- There is no motivation to combine Nishikawa with Iwase to arrive that the above claimed volume fraction limitation.
- The cited Iwase and Nishikawa references do not disclose, alone or in combination, the combination of the claimed thickness and coating weights of Claims 12, 27, 28 and 29.

We look forward to talking to you tomorrow.

Best regards,

Brian Amos, PhD
AMSTER ROTHSTEIN & EBENSTEIN LLP
Chrysler Building
405 Lexington Avenue
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APPLICATION #
19/023,905

RECEIPT DATE / TIME
07/15/2025 05:31:28 PM Z ET

ATTORNEY DOCKET #
00249/0086

Title of Invention

MULTILAYER NANOPOROUS SEPARATOR

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	3614	FILED BY	Joy Chapman
PATENT CENTER #	71383439	FILING DATE	01/16/2025
CUSTOMER #	1912	FIRST NAMED INVENTOR	David W. Avison
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Brian Comack

Documents

TOTAL DOCUMENTS: 4

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
00249-0086_Response-to-Office-Action.pdf	14	-	239 KB
00249-0086_Response-to-Office-Action-A.NE.pdf (1-1)	1	Response After Final Action	138 KB
00249-0086_Response-to-Office-Action-CLM.pdf (2-5)	4	Claims	117 KB
00249-0086_Response-to-Office-Action-REM.pdf (6-14)	9	Applicant Arguments/Remarks Made in an Amendment	208 KB
00249-0086_Executed-Declaration_1-132_with-Appendix-A-and-B.pdf	62	Affidavit-traversing rejections or objections rule 132	1050 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
00249-0086_Response-to-Office-Action.pdf	658409C7F2B8DF4639EA4FFB9ADE9DE67F8E1F48840BE6F739E5CBBE054DE048C33FD8396A41D26DAC4CFAFF01EFE5774726DDE66FBF5E5E21E96D402BABDB47
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00249-0086_Executed-Declaration_1-132_with-Appendix-A-and-B.pdf	FEAA840FE9C974C6189A1DC1A3383212456EC069674B6676A265901FD1ECD90C4658D53DB1A38C1FD5FE7D7A3EB2CD9C3F64B25918AEBF8DF6E3400281ADB5D8

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If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant(s) : LG Energy Solution, Ltd.
U.S. Serial No. : 19/023,905 Examiner : Lisa S. Park
Filing Date : January 16, 2025 Art Unit : 1729
Confirmation No. : 3614
For : MULTILAYER NANOPOROUS SEPARATOR

DECLARATION OF ANDREW R. BARRON, PH.D., F.R.S.C.
PURSUANT TO 37 CFR § 1.132

I, Andrew R. Barron, Ph.D., F.R.S.C., declare as follows:

1. I have been asked by Applicant to provide my expert opinions regarding certain statements made in an April 15, 2025 Office Action (hereinafter, "Office Action") issued by the U.S. Patent and Trademark Office in connection with the above pending patent application.
2. A copy of my *curriculum vitae*, which summarizes my experience, education and publications is attached hereto as Appendix A. A more fulsome list of my publications is attached hereto as Appendix B. My experience is briefly summarized below.
3. I hold a Bachelor of Science degree in Chemistry from Imperial College, as well as a Ph.D. in Chemistry from Imperial College. For most of my 35+ year academic career, I have taught inorganic chemistry, material science, materials and characterization, analytical chemistry, and nanoscience at various universities, including Harvard University, Rice University, and Swansea University. I have written an eight-volume series of textbooks on analytical chemistry and materials characterization. See Appendix 1, pp. 3-4. As a result, I have extensive experience and expertise testing and analyzing inorganic oxide particles, polymers, and their related porosities.
4. I am currently Emeritus at Rice University and Swansea University and retain a visiting professorship at the University of Arizona.
5. Among my 553 peer reviewed publications, I have written extensively about porous polymers and inorganic oxides. See, e.g., Appendix 2, Publication Nos. 132, 184, 284, 408, 440,
6. In forming my opinions set forth in this Declaration, I have reviewed the following materials:

Applicant(s) : LG Energy Solution, Ltd.
Serial No. : 19/023,905
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PURSUANT TO 37 CFR § 1.132

- The specification filed in connection with the above referenced patent application;
- The currently pending claims as filed by the Applicant in its “RESPONSE TO APRIL 15, 2025 FINAL OFFICE ACTION”, which I understand will be filed concurrently with this Declaration;
- The Office Action issued by the U.S. Patent and Trademark Office in connection with the above referenced patent application; and
- U.S. Patent Publication No. 2011/0143185 (“Nishikawa”).

7. Based on my review of the specification for the above referenced application, it is my opinion that, as of the filing date of the above referenced application, a person of ordinary skill in the art (“POSITA”) of the analysis and characterization of materials would have a bachelor of science degree in chemistry or material science with at least four years of experience of methods of analysis of polymers and inorganic materials, including, for example, analysis of porosity, density, and thermal stability of such materials. In the course of my academic career, I have taught many students who would qualify as POSITAs.

8. By way of background, Claim 1 of the present application requires a separator comprising a “first layer” that has “*pores and a solid* material”, wherein the solid material comprises “inorganic oxide particles and at least one polymer.” The “*volume fraction of the at least one polymer* in the *solid* material is 10%-50%.” Dependent Claims 8 and 23-25 respectively recite additional volume fraction ranges of 12%-45%, 15%-45%, 25%-45% and 15.3%-25.6%.

9. The specification explains that the “term ‘volume fraction of the one or more polymers in the solid portion’ is a “fraction expressed as a percentage *by volume of the polymer content of the solid portion* of the inorganic particle/polymer composite layer.” In this regard, where the layer “has a % porosity of between about 30% to about 70%”, “*much of the volume* of the nanoporous layer *is air*, and not a solid material. The *air volume (e.g., pores)* is *not included* in the solid portion *for purposes of determining the volume fraction of the one or more polymers* portion in the solid portion of the nanoporous layer.” See Par. [0038]. Thus, the air volume (pores) in the first layer is excluded from the volume fraction calculation.

10. Because the separator of the claimed invention is a formed and finished product, in order to calculate the volume fraction of the at least one polymer, it is necessary to determine volume of

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Serial No. : 19/023,905
Filed : January 16, 2025
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PURSUANT TO 37 CFR § 1.132

the *solid* polymer *in the solid portion of the first layer*. Since the first layer includes both a solid portion and pores, it is necessary to exclude the pores from the volume fraction calculation. To do this, the POSITA needs to independently determine the volume of solid polymer and the solid inorganic oxide particles in the solid portion of the first layer. The specification teaches an example of a method to make this calculation.

11. Specifically, the specification explains that such exemplary (see Par. [0038]) as involves the following steps:

- (a) removing the first layer from the separator (e.g., by scraping),
- (b) weighing a sample of the first layer,
- (c) combusting the sample in a crucible and weighing the resulting residue such that the weight of the residue is equal to the weight of the inorganic oxide particles,
- (d) subtracting the weight of inorganic oxide particles from the total weight of the sample of the first layer, which gives the weight of the solid polymer,
- (e) calculating the volume of the solid polymer using the known density of the polymer,
- (f) in order to calculate the appropriate density of the inorganic oxide particles, heating the sample to 900°C to cause dehydration, whereby the decrease in weight is due to the level of hydration in the particles,
- (g) using the calculated hydration, the appropriate density is used to calculate the volume of inorganic oxide particles,¹
- (h) calculating the volume fraction of the at least one polymer in the solid material in the first layer as follows: volume of the polymer ÷ (volume of the polymer + volume of inorganic oxide particles).

12. The Office Action asserts that Paragraph [0061] of Nishikawa discloses a volume fraction of 25% to 250% because “Nishikawa discloses an effective porous heat-resistant layer using . . . 0.4 to 4 times the volume of inorganic filler relative to the volume of the heat-resistant resin which equates to using 0.25 to 2.5 times the volume of resin relative to filler (i.e. 25% to 250%).” Office

¹ The density of an inorganic oxide particle is dependent upon the level of hydration. For example, the density for $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ is dependent on the value of x ; for example, when $x=0$, the density is 3.99 g/cm³ (Al_2O_3); when $x=1$, the density is 3.07 g/cm³ (AlOOH); and when $x=3$, the density is 2.42 g/cm³ ($\text{Al}(\text{OH})_3$).

Action at p. 4.

13. As discussed below, at the time of the filing date of the present application, it is my opinion that a POSITA would not have understood Nishikawa as teaching the volume fraction limitation, as recited in the pending claims of the present application.

14. Nishikawa teaches that its “heat resistant *porous* layer” is made of inorganic oxide *filler* and a heat resistant *resin*. Such layer has pores throughout its *entire* thickness. Indeed, Nishikawa explains that “heat resistant porous layer has . . . a structure such that the *micropores* are connected with one another, and a gas or liquid can pass through the heat-resistant porous layer from one side to the other.” Par. 0056. Nishikawa’s heat-resistant porous layer “preferably has a porosity in the range of from 20% to 80%.”; *id.*; *see also id.*, pars. 0057-0059, 0063.

15. At the time of the filing of the present application, a POSITA would have understood that the pores in the *finished* heat-resistant porous layer of Nishikawa are formed throughout the resin. This is because the inorganic materials are not porous materials and it is well known that when forming a separator, pores are formed in the polymer that binds the inorganic materials.

16. Indeed, in Example 1-1, the polymer is dissolved in a solvent (e.g., DMAc and TPG); when dried, the pores are formed due to the evaporation of the solvents. Nishikawa, Par. [0167]. Thus, much of the *formed* heat-resistant porous layer of Nishikawa is air (20% to 80% is porous), and not a solid material.

17. Against this background, the Office Action relies on the following passage in Nishikawa as teaching that its heat-resistant resin has a volume fraction of 25% to 250%:

It is preferred that the amount of the inorganic filler contained in the heat-resistant *porous layer* is 0.4 to 4 times the volume of the heat-resistant resin.

See Nishikawa, ¶0061.

18. This passage refers to a *formed* “heat-resistant *porous* layer” that includes “inorganic filler” and “heat-resistant resin.” As discussed above, a POSITA would understand that much of heat resistant-resin in the formed layer is porous (e.g., it has a porosity of 20%-80%).

19. In order to determine the volume fraction of the heat-resistant resin under the pending claims of the present application, the pores (i.e., air) in the resin of Nishikawa must be excluded. However, Nishikawa does not discuss excluding the 20% to 80% pores in the heat-resistant resin

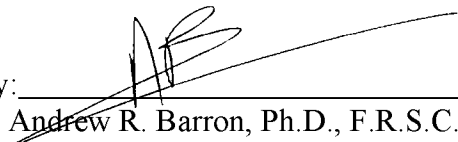
Applicant(s) : LG Energy Solution, Ltd.
Serial No. : 19/023,905
Filed : January 16, 2025
Page 5 of 5 : DECLARATION OF ANDREW R. BARRON, Ph.D., F.R.S.C.
PURSUANT TO 37 CFR § 1.132

such as by the methodology described in Paragraph 11 above or by any other methodology.

20. Moreover, I note that Paragraph [0061] in Nishikawa compares the *amount* of inorganic filler to the *volume* of the heat-resistant resin. An amount and a volume are not the same thing. Thus, it is my opinion a POSITA would find this statement uninformative. It may be that Nishikawa was comparing volume to volume, amount to amount, or amount to volume. But there is no way to determine this from the text and working examples of Nishikawa. Thus, Paragraph [0061] does not teach the volume fraction limitation of the pending claims for this reason alone.

21. I declare under penalty of perjury under the laws of the United States that the foregoing is true and correct to the best of my knowledge and belief. I understand that willfully false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code.

Dated: July 15, 2025
Swansea, UK

By: 
Andrew R. Barron, Ph.D., F.R.S.C.

CURRICULUM VITAE

ANDREW R. BARRON

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ORCHID (0000-0002-2018-8288)
DoB: 20 May 1962
Nationality: UK (USA Green Card)

PRESENT POSITIONS:

- 1) Professor Emeritus and Sêr Cymru Chair of Low Carbon Energy and Environment Swansea University, Wales, UK (2024-).
- 2) Professor Emeritus, Chemistry, Rice University (2019-).
- 3) Professor Emeritus, Material Science and Nanoengineering, Rice University (2019-).
- 4) Visiting Professor, Arizona Institute for Resilient Environments and Societies (AIRES), University of Arizona (2021-).

PREVIOUS POSITIONS:

- 1) Sêr Cymru Chair of Low Carbon Energy and Environment, Swansea University, Wales, UK (2013-2024).
- 2) Founder & Director, Energy Safety Research Institute, Swansea University, Wales, UK (2013-2024).
- 3) Charles W. Duncan, Jr. - Welch Chair of Chemistry, Rice University (1995-2019).
- 4) Professor of Nanoengineering and Materials Science, Rice University (1995-2019).
- 5) Associate Dean of Industry Interactions and Technology Transfer, Rice University, (2006-2009).
- 6) Visiting Professor, Universiti Teknologi Brunei (2020-2022).
- 7) Director, National Corrosion Center, Rice University (2011-2012).
- 8) Director, Texas Center for Crystallography at Rice University (1997-2008).
- 9) Visiting Professor, Swansea University (2009-2013).
- 10) Visiting Professor, University of Wales (2009-2010).
- 11) Associate Professor of Chemistry, Harvard University (1991-1995).
- 12) Assistant Professor of Chemistry, Harvard University (1987-1991).
- 13) Post-doctoral Research Associate, University of Texas at Austin (1986-1987).

Non-academic

- 1) Team Principle Ross Racing (Chiefie Motorsport) American LeMans Series GTS (1999-2000).
- 2) Manager, Freemantle Hotel, London (1983).

EDUCATION:

- 1) B.Sc. (1st Class, Hons), Imperial College of Science and Technology, University of London (1983).
- 2) Associate of the Royal College of Science (ARCS), Imperial College of Science and Technology, University of London (1983).
- 3) Ph.D. Imperial College of Science and Technology, University of London. Thesis title "Transition Metal Aluminohydrides". Research under the supervision of Professor Sir Geoffrey Wilkinson F.R.S. (1986).
- 4) Diploma of Imperial College (D.I.C.) Imperial College of Science and Technology, University of London (1986).

AWARDS:

- 1) Top 100 Influential People - 2025
- 2) The Hydrogen Awards - 2024

- 3) Albert Nelson Marquis Lifetime Achievement Award (Marquis Who's Who) – 2021
- 4) Star of Asia International Award – 2019.
- 5) Erasmus+ Vilnius University - 2016.
- 6) Applied Inorganic Chemistry Award (Royal Society of Chemistry) – 2013.
- 7) World Technology Award (Materials) – 2011.
- 8) Lifetime Achievement Award in Nanotechnology – 2011.
- 9) Prince of Wales Visiting Innovator – 2009.
- 10) Welch Foundation Norman Hackerman Award in Chemical Research – 2002.
- 11) Houston Police Bomb Squad Challenge Medal presented by the Houston Police Department and the International Association of Bomb Technicians and Investigators (IABTI) – 2001.
- 12) Fellow, Royal Society of Chemistry - 1995.
- 13) Humboldt Senior Scientist Research Award – 1997.
- 14) Corday Morgan Medal and Prize (Royal Society of Chemistry) – 1995.
- 15) Meldola Medal and Prize (Royal Society of Chemistry) – 1991.
- 16) Alcoa Directors Fellowship – 1992.
- 17) Alcoa Directors Fellowship – 1993.
- 18) Alcoa Directors Fellowship – 1994.
- 19) Du Pont Young Faculty Fellow – 1987.
- 20) HVA Briscoe Prize – 1983.

ORGANIZATIONS:

- 1) Royal Society of Chemistry
- 2) American Chemical Society
- 3) Windsor Energy Group.
- 4) UK Carbon Capture and Research Centre (UKCCSRC).
- 5) Sigma Xi, The Scientific Research Honor Society

RESEARCH INTERESTS:

- 1) Technology for industrial decarbonisation.
- 2) Inorganic nanomaterials for green energy production.
- 3) Improved environmental efficacy of energy processes.

FINANCIAL SUPPORT: 1987 – Present (funding details upon request)

Government Agencies

- 1) Advanced Research Projects Agency (ARPA).
- 2) Center for Biological and Environmental Nanotechnology (NSF).
- 3) Defense Advanced Research Projects Agency (DARPA).
- 4) Department of Defense (DOD).
- 5) Department of Defense: Navy Surface Warfare Center Panama City.
- 6) Department of Energy (DOE).
- 7) Environmental Protection Agency (EPA).
- 8) EPSRC.
- 9) European Regional Development Fund (ERDF).
- 10) Innovate UK.
- 11) National Academies Keck Futures Initiatives.
- 12) National Aeronautics and Space Administration (NASA).
- 13) National Research Network (NRN) Wales.
- 14) National Science Foundation (NSF).
- 15) Office of Naval Research (ONR).
- 16) Technology Strategy Board (TSB).
- 17) US Air Force Research Laboratory.
- 18) Welsh Government Sêr Cymru Programme.
- 19) Welsh European Funding Office (WEFO).
- 20) UK Foreign, Commonwealth and Development Office (FCDO).

Foundations and Professional Societies

- 1) ACS Petroleum Research Fund (PRF)
- 2) Advanced Energy Consortium (AEC).
- 3) American Chemical Society, SEED Fund
- 4) Coleman Foundation
- 5) Dreyfus Foundation
- 6) Elsa U. Pardee Foundation.
- 7) Fleming and Davenport Awards.
- 8) Ford Foundation.
- 9) Richard E. Smalley Institute for Nanoscale Science and Technology
- 10) Robert A. Welch Foundation
- 11) Texas Medical Center Fleming and Davenport Awards
- 12) Virginia and L. E. Simmons Family Foundation

Industrial

- 1) Akzo Nobel
- 2) Albemarle Corporation
- 3) Apache Corporation
- 4) BP
- 5) Colgate-Palmolive Company
- 6) Condea Vista Company - SASOL North America
- 7) eCORP International, Inc.
- 8) Gentex Optics Inc.
- 9) Halliburton Energy Services
- 10) ICI
- 11) Isotron Corp.
- 12) Idemitsu Chemicals Corp.
- 13) Lithion Corp.
- 14) Natcore Technology, Inc.
- 15) Malakoff Industries (Reynolds International)
- 16) Lockheed Martin Corp.
- 17) Oman Oil.
- 18) SALTS Healthcare
- 19) TDA Research Inc.
- 20) Yardney Technical Products.
- 21) YTL.
- 22) Worldwide Eco-Corporation
- 23) Middle East Water Solutions
- 24) NIQ Filter Works

PUBLICATIONS:

Peer review Journals

- 1) 551 publications to date
- 2) h-factor (impact) = 76
- 3) Citations = 22,433
- 4) i10-index = 400
- 5) Full list attached.

Books

- 1) M. Barron and A. R. Barron, *Project Management*. ISBN:978-1-83800-852-9.
- 2) M. Barron and A. R. Barron, *Project Management*. e-book, ISBN:978-1-45835-779-3.
- 3) A. R. Barron, *Organometallic Transition Metal Catalysis: A Holistic Approach to Understanding and Predicting their Mechanisms*, ISBN:978-1-6780-0353-1.
- 4) A. R. Barron, *Chemistry of the Main Group Elements*. ISBN:978-1-83800-851-2.

- 5) A. R. Barron, *Portland Cement in the Energy Industry*. ISBN:978-1-8380085-3-6.
- 6) A. R. Barron, *Chemistry of Electronic Materials*. ISBN: 978-1-8384167-1-3.
- 7) A. R. Barron, *Physical Methods in Chemistry and Nano Science. Volume 1: Elemental Analysis*, ISBN:978-1-8380085-4-3.
- 8) A. R. Barron, *Physical Methods in Chemistry and Nano Science. Volume 2: Physical and Thermal Methods*, MiDAS Green Innovations, ISBN:978-1-8380085-5-0.
- 9) A. R. Barron, *Physical Methods in Chemistry and Nano Science. Volume 3: Chromatography*, MiDAS Green Innovations, ISBN:978-1-8380085-6-7.
- 10) A. R. Barron, *Physical Methods in Chemistry and Nano Science. Volume 4: Chemical Speciation*, MiDAS Green Innovations, ISBN:978-1-8380085-7-4.
- 11) A. R. Barron, *Physical Methods in Chemistry and Nano Science. Volume 5: Molecular and Solid State Structure*, MiDAS Green Innovations, ISBN:978-1-8380085-8-1.
- 12) A. R. Barron, *Physical Methods in Chemistry and Nano Science. Volume 6: Surface Morphology and Structure at the Nanoscale*, MiDAS Green Innovations, ISBN:978-1-8380085-9-8.
- 13) A. R. Barron, *Physical Methods in Chemistry and Nano Science. Volume 7: Device Performance*, MiDAS Green Innovations, ISBN:978-1-8380803-0-3.
- 14) A. R. Barron, *Physical Methods in Chemistry and Nano Science. Volume 7: Applications of Analytical Methods*, MiDAS Green Innovations, ISBN:978-1-8380803-1-0.
- 15) A. Apblett, A. R. Barron, and A. Hepp, *Nanomaterials via Single-Source Precursors: Synthesis, Processing and Applications*, Elsevier, ISBN:978-0-1282034-0-8.
- 16) V. Griffiths and C. W. Dunnill, *An Examination of the Feasibility of Producing Green Hydrogen from Curtailed, Onshore Wind Power using a North Wales Case Study*, Ed. A. R. Barron, MiDAS Green Innovations, ISBN:978-1-6780-1790-3.
- 17) W. Al-Shatty, S. Alexander, and A. R. Barron, *Micromodel Evaluation of Nanoparticles for Enhanced Oil Recovery*, MiDAS Green Innovations, ISBN:978-1-4717-3392-5.
- 18) S. Kiani, S. Alexander, and A. R. Barron, *Emerging Platforms in Fluid Flow Transport: Surfactants, Polymers, and Nanoparticles*. MiDAS Green Innovations, ISBN: 978-1-8384167-0-6.
- 19) F. Al-Sulaiti and A. R. Barron, *A Case Study for the Potential of Carbon Capture and Storage in Qatar*. MiDAS Green Innovations, ISBN:978-1-4716-9043-3.
- 20) E. Mkaratigwa and A. R. Barron, *Hydrogen as an Energy Vector: Production, Storage and Distribution*. MiDAS Green Innovations, ISBN:978-1-4716-9517-9.
- 21) J. C. Fernandes Serodio and A. R. Barron, *Water-in-Diesel Emulsions as a Route to Lower Transport Emissions*, MiDAS Green Innovations, ISBN 978-1-4709-5934-0.
- 22) M. Barron and A. R. Barron, *Project Management*. OpenStax CNX. May 15, 2019 <http://cnx.org/contents/5e9177d7-9998-43d0-9b98-91a369c6a371@11.6>.
- 23) A. R. Barron, *Hydrogen*. OpenStax CNX. Jan 27, 2010 <http://cnx.org/contents/fe64dc8c-59d1-4e69-acdf-22381522a60f@4.6>.
- 24) D. J. Flood and A. R. Barron, *Carbon Nanotubes*. OpenStax CNX. Sep 30, 2013 <http://cnx.org/contents/a0309d0a-3759-4a07-8ef7-f6eb29220eb6@1.1>.
- 25) A. R. Barron, *Nanomaterials and Nanotechnology*. OpenStax CNX. Mar 24, 2015 <http://cnx.org/contents/7b4dfcf8-b28e-4141-8f2f-1ff39052669f@13.7>.

Patents

- 1) US Patent 5,139,999, Gordon et al., Chemical vapor deposition process where an alkaline earth metal organic precursor material is volatilized in the presence of an amine or ammonia and deposited onto a substrate
- 2) US Patent 5,159,983, Barron et al., Apparatus and method for capping oil or gas wells
- 3) US Patent 5,238,711, Barron et al., Method of coating carbon fibers with a carbide

- 4) US Patent 5,300,320, Barron et al., Chemical vapor deposition from single organometallic precursors
- 5) US Patent 5,527,851, Barron et al., Stabilised olefin carbon monoxide copolymer compositions
- 6) US Patent 5,738,721, Barron et al., Liquid precursor and method for forming a cubic-phase passivating/buffer film
- 7) US Patent 5,760,462, Barron et al., Metal, passivating layer, semiconductor, field-effect transistor
- 8) US Patent 6,008,525, Barron et al., Minority carrier device comprising a passivating layer including a Group 13 element and a chalcogenide component
- 9) US Patent 6,207,130, Kareiva et al., Metal-exchanged carboxylato-alumoxanes and process of making metal-doped alumina
- 10) US Patent 6,322,890, Barron et al., Supra-molecular alkylalumoxanes
- 11) US Patent 6,369,183, Cook et al., Methods and materials for fabrication of alumoxane polymers
- 12) US Patent 6,770,773, Rose et al., Organic acid-Fe-OOH (ferroxane) particles and ferroxane-derived ceramics and ceramic membranes
- 13) US Patent 6,936,306, Barron et al., Chemical control over ceramic porosity using carboxylate-alumoxanes
- 14) US Patent 7,115,764, Barron et al., Mechanical shear based synthesis of alumoxane nanoparticles
- 15) US Patent 7,220,454, Barron et al., Production method of high strength polycrystalline ceramic spheres
- 16) US Patent 7,253,014, Barron et al., Fabrication of light emitting film coated fullerenes and their application for in-vivo light emission
- 17) US Patent 7,459,209, Smith et al., Composition and method for making a proppant
- 18) US Patent 7,491,444, Smith et al., Composition and method for making a proppant
- 19) US Patent 7,569,199, Barron et al., Method to remove sulfur or sulfur-containing species from a source
- 20) US Patent 7,669,658, Barron et al., High strength polycrystalline ceramic spheres
- 21) US Patent 7,682,527, Barron et al., Fabrication of light emitting film coated fullerenes and their application for in-vivo light emission
- 22) US Patent 7,692,218, Barron et al., Method for creating a functional interface between a nanoparticle, nanotube or nanowire, and a biological molecule or system
- 23) US Patent 7,718,550, Barron et al., Method for low temperature growth of inorganic materials from solution using catalyzed growth and re-growth
- 24) US Patent 7,736,430, Barron et al., Compositions and methods for controlling the setting behavior of cement slurries using carbonated fly ash
- 25) US Patent 7,867,613, Smith et al., Composition and method for making a proppant
- 26) US Patent 7,883,773, Smith et al., Composition and method for making a proppant
- 27) US Patent 7,887,918, Smith et al., Composition and method for making a proppant
- 28) US Patent 7,914,892, Smith et al., Composition and method for making a proppant
- 29) US Patent 8,003,212, Smith et al., Composition and method for making a proppant
- 30) US Patent 8,012,533, Smith et al., Composition and method for making a proppant
- 31) US Patent 8,062,702, Barron et al., Coated fullerenes, composites and dielectrics made therefrom
- 32) US Patent 8,075,997, Smith et al., Composition and method for making a proppant
- 33) US Patent 8,168,570, Barron et al., Method of manufacture and the use of a functional proppant for determination of subterranean fracture geometries
- 34) US Patent 8,201,517, Barron et al., Method for low temperature growth of inorganic materials from solution using catalyzed growth and re-growth
- 35) US Patent 8,217,137, Barron et al., Fullerene-based amino acids
- 36) US Patent 8,298,667, Smith et al., Composition and method for making a proppant
- 37) US Patent 8,361,349, Barron et al., Fabrication of light emitting film coated fullerenes and their application for in-vivo light emission

- 38) US Patent 8,562,935, Smalley et al., Amplification of carbon nanotubes via seeded-growth methods
- 39) US Patent 8,575,548, Barron et al., Analyzing the transport of plasmonic particles through mineral formations
- 40) US Patent 8,603,578, Smith et al., Composition and method for making a proppant
- 41) US Patent 8,636,830, Barron et al., Aliphatic amine based nanocarbons for the absorption of carbon dioxide
- 42) US Patent 9,034,085, Barron et al., Aliphatic amine based nanocarbons for the absorption of carbon dioxide
- 43) US Patent 9,242,876, Barron et al., Methods, systems and membranes for separation of organic compounds from liquid samples
- 44) US Patent 9,290,665, Barron et al., Coated fullerenes, compositions and dielectrics made therefrom
- 45) US Patent 10,232,342, Ghosh et al., Method, synthesis, activation procedure and characterization of an oxygen rich activated porous carbon sorbent for selective removal of carbon dioxide with ultra high capacity
- 46) US Patent 10,376,861, Ghosh et al., Method, synthesis, activation procedure and characterization of an oxygen rich activated porous carbon sorbent for selective removal of carbon dioxide with ultra high capacity
- 47) US Patent D668,213, Barron, Horizontal solar energy collector
- 48) US Patent D668,212, Barron, Vertical solar energy collector
- 49) European Patent 1,070,029, Bailey et al., Chemical control over ceramic porosity using carboxylate-alumoxanes
- 50) European Patent 1,200,498, Barron et al., Method and materials for fabrication of alumoxane polymers
- 51) European Patent 1,319,639, Barron et al., Chemical control over ceramic porosity using carboxylate-alumoxanes
- 52) European Patent 1,456,124, Barron et al., Coated fullerenes, composites and dielectrics made therefrom
- 53) European Patent 1,476,399, Rose et al., Organic acid-FeOOH (ferroxane) particles and ferroxane-derived ceramics and ceramic membranes
- 54) European Patent 1,563,530, Barron et al., Fabrication of light emitting film coated fullerenes and their application for in-vivo light emission
- 55) European Patent 1,563,545, Barron et al., Field effect transistor with functionalised nanotube and corresponding manufacturing method.
- 56) European Patent 1,579,490, Barron et al., Method for low temperature growth of inorganic materials from solution using catalyzed growth and re-growth
- 57) European Patent 1,713,723, Barron et al., Fullerene-based amino acids

COLLABORATIONS (past and present):

UK

- 1) Dr. Joseph C. Bear, Kingston University.
- 2) Dr. Richard Cobley, Swansea University.
- 3) Dr. Shareen Doak, Swansea University.
- 4) Dr. Julian Eastoe, University of Bristol.
- 5) Prof. Chenfeng Li, Swansea University.
- 6) Prof. Alex M. Lord, Swansea University.
- 7) Prof. Iseult Lynch, University of Birmingham.
- 8) Prof. Darren L Oatley-Radcliffe, Swansea University.
- 9) Prof. Paul O'Brien, Manchester University.
- 10) Prof. D. Roger J. Owen, Swansea University.
- 11) Prof. David R. Rankin, University of Edinburgh, Scotland.
- 12) Dr. Martin B. Ward, Leeds University.

USA

- 1) Dr. Atta Arif, University of Utah, UT.
- 2) Prof. Jerry L. Atwood, University of Missouri-Columbia, MO
- 3) Dr. Manoop Bhutani, University of Texas MD Anderson Cancer Center, Houston, TX.
- 4) Prof. Simon Bott, University of Houston, TX.
- 5) Dr. M. Bowes Hamill, Baylor College of Medicine, Houston, TX.
- 6) Prof. W. Edward Billups, Rice University, TX.
- 7) Dr. S. A. Curley, University of Texas, MD Anderson Cancer Center, TX.
- 8) Prof. Seamus A. Curran, University of Houston, TX.
- 9) Prof. Geoffrey Davies, North Eastern University, MA
- 10) Prof. Kerwin D. Dobbs, University of Delaware, DE
- 11) Prof. Mark E. Eberhart, Colorado School of Mines, CO
- 12) Prof. Michelle Franci, Bryn Mawr College, PA
- 13) Dr. Jason B. Fleming, Department of Surgical Oncology, The University of Texas, M. D. Anderson Cancer Center, TX.
- 14) Dr. Dennis J. Flood, NASA Glenn, Cleveland, OH.
- 15) Prof. Thomas R. Gilbert, North Eastern University, MA
- 16) Prof. R. G. Gordon, Harvard University, MA
- 17) Prof. Norman Hackerman, Rice University, TX.
- 18) Dr. Al Hepp, NASA Glenn, Cleveland, OH.
- 19) Dr. Philip P. Jenkins, NASA Glenn, Cleveland, OH.
- 20) Dr. Tom R. Jervis, Los Alamos National Laboratory, NM.
- 21) Prof. Robert Kren, University of Central Michigan, MI
- 22) Prof. Paul E. Laibinis, Vanderbilt University, TN
- 23) Prof. Quilin Li, Rice University, TX.
- 24) Prof. Robert W. Lenz, University of Massachusetts, MA
- 25) Prof. Dennis L. Lichtenberger, The University of Arizona, AZ
- 26) Prof. Andreas Luttmann, Rice University, TX.
- 27) Prof. Antonios G. Mikos, Rice University, TX.
- 28) Dr. Nancy Monteiro-Riviere, North Carolina State University, NC.
- 29) Dr. Benji Maruyama, Wright Patterson Air Force Base, OH.
- 30) Prof. Michael Nastasi, University of Nebraska-Lincoln, NE
- 31) Dr. Lewis Norman, Halliburton Energy Services, Duncan, OK.
- 32) Dr. Anthony J. Perrotta, ALCO, PA
- 33) Prof. W. S. Rees, Jr. Georgia Institute of Technology, GA.
- 34) Prof. Massood Tabib-Azar, University of Utah, UT.
- 35) Prof. David J. Singel, Montana State University, MT.
- 36) Prof. Richard Smalley, Rice University, TX.
- 37) Dr. Rei Suzuki, Department of Gastroenterology, Hepatology, and Nutrition, The University of Texas M. D. Anderson Cancer Center, Houston, USA.
- 38) Prof. Mark Wiesner, Duke University, NC.
- 39) Prof. Lon J. Wilson, Rice University, TX
- 40) Dr. Joseph Ziller, University of California, Irvine, CA.
- 41) Prof. Maria M. Fidalgo de Cortalezzi, University of Missouri, MO.

Europe

- 1) Dr. Jérôme Rose, Institut national des sciences de l'univers (INSU), France.
- 2) Prof. Jean-Yves Bottero, Centre Européen de Recherche et d'Enseignement des Géosciences de l'Environnement, France.
- 3) Dr. Frédéric Guittard, University of Nice Sophia Antipolis and CNRS, France.
- 4) Prof. Aivaras Kareiva, Vilnius University, Lithuania.
- 5) Dr. Marta Sevilla, Instituto Nacional del Carbón (CSIC), Spain.
- 6) Prof. Antonio B. Fuertes, Instituto Nacional del Carbón (CSIC), Spain.
- 7) Prof. J. M. Tsangaris, University of Ioannina, Greece.

- 8) Prof. Manthos G. Papadopoulos, The National Hellenic Research Foundation, Greece.
- 9) Dr. Ionel Haiduc, Babeş-Bolyai University, Turkey.
- 10) Dr. Cristian Silvestru, Babeş-Bolyai University, Turkey.
- 11) Prof. Claudiu T. Supurm, University of Florence, Italy.
- 12) Prof. Michael Stuke, Max Planck Institute, Germany.
- 13) Prof. Herbert W. Roesky, University of Gottingen, Germany.
- 14) Dr. Stephan Schulz, University of Duisburg-Essen, Germany.
- 15) Prof. Janusz Lewinski, Warsaw University of Technology, Poland.
- 16) Prof. Janusz Lipkowski, Cardinal Stefan Wyszyński University, Poland.

Rest of World

- 1) Prof. Masanobu Sagisaka, Hirosaki University, Japan.
- 2) Prof. Zeyad Almutairi, King Saud University, Saudi Arabia
- 3) Dr. Sattam Fahad Al-Mojil, King Saud University, Saudi Arabia.
- 4) Dr. Abdullah A. Alabdulkarem, King Saud University, Saudi Arabia.
- 5) Dr. Rajan Jose, Universiti Malaysia Pahang, Malaysia.
- 6) Dr. Goshtasp Cheraghian, Islamic Azad University, Iran.
- 7) Prof. David Potter, University of Calgary, Canada.
- 8) Prof. Serdar Durdagi, University of Calgary, Canada.
- 9) Dr. Nashaat N. Nassar, University of Calgary, Canada.
- 10) Prof. Felipe Feijoo, Pontificia Universidad Católica de Valparaíso (PUCV), Chile.

SERVICES TO ACADEMIC DISCIPLINE:

Editorial Board (Past and Present):

- 1) North American Regional Editor: Advanced Materials for Optics and Electronics, Wiley (1992- 2000)
- 2) Editorial Board: Advanced Materials, VCH. (1998-2005)
- 3) Editorial Board: Chemistry of Materials, ACS. (1998-2002)
- 4) Editorial Board: Polyhedron, Pergamon (1998 - 2002).
- 5) Editor/Symposium Organizer: Covalent Ceramics II, Materials Research Society Symposium Proceedings, Boston Meeting 1993.
- 6) Editor: Polyhedron Symposium-in-print number 10, Aluminium, Gallium and Indium, 1990, 9, (2,3), 149-453.
- 7) Editorial Board: Main Group Metal Chemistry (1995-1999).
- 8) Editorial Board, Dalton (2000-2008).
- 9) Editorial Board, Recent Patents in Nanotechnology (2006-2009).
- 10) Editorial Board: Main Group Chemistry (2006-present).
- 11) Editorial Board: Materials Science in Semiconductor Processing (2011-present).
- 12) Editor: Journal of Nanomaterials (2013-present).
- 13) Editor: Scientific Reports (Nature) (2014-present).
- 14) Editorial Board: Journal of Materials & Research (2016-present).
- 15) Editorial Board: Journal of Nanostructure in Chemistry (2018-present).

Organisation of Scientific Meetings

- 1) Conference Organizer, Transcending Incrementalism, ESRI, Swansea, UK, 2019.
- 2) Conference Committee, International Conference on Applied Energy (ICAE2017), Cardiff, 2019.
- 3) Co-chair, 1st European Workshop on Metal Phosphonates Chemistry – Materials for Energy Applications and Beyond, ESRI, Swansea, UK, 2018.
- 4) Co-chair, MSA/NASA Technical Workshop, Houston, 2000.
- 5) Chair, Inorganic Chemistry Gordon Conference, Newport, USA, 1997.
- 6) Symposium Chair for special symposium in honor of Sir. Geoffrey Wilkinson, ACS National Meeting, San Francisco, USA, 1997.

- 7) Symposium Organizer, Covalent Ceramics II, MRS Fall Meeting, Boston, USA, 1993
- 8) Co-chair, Inorganic Chemistry Gordon Conference, Newport, USA, 1996.

TEACHING:

- 1) Integrated Topics in Chemistry, CH-349 (2022).
- 2) Engineering Management, EG 386 (2021-2022).
- 3) Project Management, EGSM 31 (2020-2022).
- 4) Transition Metal Chemistry, CHEM 495 (1996-2013).
- 5) Chemistry of Electronic Materials, CHEM 496 (1995, 1997).
- 6) Honors Lab, CHEM 106 (1998).
- 7) Inorganic Chemistry, CHEM 360 (1997, 2001, 2003, 2004).
- 8) Special Topics in Inorganic Chemistry, CHEM 595 (1996).
- 9) Physical Methods in Inorganic Chemistry, CHEM 575 (1996-Present).
- 10) Physical Methods in Inorganic and Nano Chemistry, CHEM 575 (2009, 2010).
- 11) Effective Presentations in Chemistry, CHEM 606 (1997-2003).
- 12) Chemistry of Electronic Materials, CHEM 596 (1997, 1999).
- 13) Inorganic Seminar. CHEM 600 (1996, 1997, 1998, 1999, 2003)
- 14) Technology Management for Scientist and Engineers. CHEM/MSCI/MGMT 603 (1998, 1999).
- 15) Entrepreneurial Management for Science and Engineering, CHEM/MSCI/MGMT 750 (2000 - 2009).
- 16) New Venture Creation for Science and Engineering, CHEM/MSCI/MGMT 751 (2001-2009).
- 17) Advanced Module in Catalysis, CHEM 391 (2000, 2001).
- 18) Advanced Module in Experimental Chemistry, CHEM 351 (2000).
- 19) Management for Science and Engineering, NSCI/ENGI 610 (2006-2015).
- 20) Automotive Engineering: Materials and Dynamics, MSCI 615-616, 2005-2015).
- 21) Organometallic Chemistry, CHEM 105 (1987-1995).
- 22) Inorganic Chemistry, CHEM 40 (1993-1994).
- 23) General Chemistry, CHEM 5 (1988-1994).
- 24) Chemical Safety Seminar Series for the Faculty of Arts and Science, "Handling Hazardous Chemicals" (1988-1995).

EDUCATION DEVELOPMENT

- 1) Scientific Advisor to the National Science Resource Center (jointly operated by the Smithsonian Institution and the National Academy of Science) aimed at the development of a national hands-on school science curriculum for K through 12th grade. Review and advisory role of all new science based programs.
- 2) Development of Open Access Education programs through Connexions (cnx.org) to allow students (85 students to date) to create user defined textbook for University Courses to pass on their learning to the next generation.
- 3) Managerial Chemistry, Jones School of Business courses for senior petroleum, energy and chemical industry executives (1998, 1999).
- 4) Management for Science and Engineering, Rice University, joint education programme between Colleges of Science, Engineering and Management to provide Management skills for Science and Engineering students (1998-2019).
- 5) Co-founder of College of Science, Rice University, Professional Masters Program (<https://profms.rice.edu/>).
- 6) Development training program for Middle School Teachers, Houston, TX, through research experiences (2004-2001).
- 7) Developed Automotive Engineering programme at Rice University in partnership with Society of Automotive Engineering (SAE).
- 8) Promotion of the Athena SWAN Charter for excellence in Science, Technology, Engineering, Medicine and Mathematics (STEMM) employment in Higher Education at Swansea University.

REGIONAL AND MEDIA DEVELOPMENT

- 1) Science Advisor to the Library of Congress. Review and advisory role for mass deacidification program (1992 - 1993).
- 2) Scientific advisor appointed to the Cambridge School Committee (1990 – 1995).
- 3) Scientific advisor and lecturer to the Cambridge Fire Department, "Hazardous Metals", "Acids and Corrosives" and "Pyrophorics", (1990 - 1995).
- 4) Scientific advisor and lecturer to the Houston Fire Department (1995 - 2015).
- 5) Scientific advisor and lecturer to the Houston Police Department Bomb Squad (2006-2015).
- 6) Technical consultant Harris County District Attorney (1998-2019).
- 7) Scientific programs for CBS News (1997 - 2009).
- 8) Scientific programs for Fox News (1998 - 2010).
- 9) Scientific programs for NBC News (1998 - 2010).
- 10) Scientific programs for BBC (2018 - 2019).
- 11) Scientific programs for Discovery Channel (2000).

INTERNATIONAL DEVELOPMENT

- 1) Special advisor to the President Vilnius University, Lithuania (2017-present).
- 2) Scientific Advisory Board Member for Scientific Fellowship Program, Republic of Ireland (2008-2009).
- 3) Founder and Manager, Texas/UK Collaborative (2000).
- 4) PI for BP 21CPD Education Program, which involved a 4-party collaboration between Rice University, Baylor College of Medicine, Manchester University, and Heriot Watt University to deliver education to BP staff world-wide.
- 5) Founder King Saud-Swansea University K(SU)² Collaborative (2017).
- 6) Advisory Board, King Abdullah University of Science and Technology (KAUST), Saudi Arabia (2009-2012).
- 7) Advisor Yellow River Delta Efficient Eco-economic Development, Binzhou, China (2010-2011).
- 8) Board of Directors, Zhu Zhou International Research Institute China (ZIRIC) (2011-2012).
- 9) Member, Innovation and Leadership Committee of Greater Houston Energy Collaborative, Greater Houston Partnership (2006).
- 10) Developed & Negotiated Swansea University's partnership with King Khalid University (KKU) Saudi Arabia.
- 11) Developed & Negotiated Swansea University's partnership with Princess Nourah University (PNU) Saudi Arabia.
- 12) Developed & Negotiated Swansea University's partnership with Universiti Brunei Darussalam (UBD).
- 13) Developed & Negotiated Swansea University's partnership with Universiti Technology Brunei (UTB).
- 14) Developed & Negotiated Swansea University's partnership with University Technology Petronas (UTP).
- 15) Developed & Negotiated Swansea University's partnership with Universiti Malaysia Pahang (UMP).
- 16) Developed & Negotiated Swansea University's partnership with University of Malaya Power and Energy System (UMPES).

UNIVERSITY SERVICE:

- 1) Graduate Admissions Committee (1996 - 2000)
- 2) Chemistry Department Curriculum Committee (1996 -1998)
- 3) Department of Mechanical Engineering and Materials Science Chair Search Committee (1996)
- 4) School of Sciences Steering Committee (1996)

- 5) Jones School Dean Search Committee (1997)
- 6) Technology Transfer Committee (1996, 1997)
- 7) School of Continuing Studies: Lecture on Topics in Contemporary Science (1996)
- 8) Curriculum Innovations: Development of Weiss School/Jones School joint programs (1996)
- 9) Committee on Scholarships and Awards (1997, 1998)
- 10) Promotion and Tenure Committee (1998)
- 11) Search Committee for Vice Provost for Development (1999)
- 12) Materials Science Steering Committee (1998)
- 13) Departmental Seminar Committee (1998)
- 14) Provost Search Committee (1998)
- 15) Executive Education - Managerial Chemistry (1997 - 1999)
- 16) Academic Advisor, Baker College (1996 - Present)
- 17) Graduate Admissions Committee (2001)
- 18) Curriculum Committee (2001, 2002)
- 19) Graduate Recruiting Committee (2002)
- 20) Division Contacts Committee (2002)
- 21) Industrial Enterprise Committee (2002)
- 22) Steering Committee, Rice Alliance for Technology and Entrepreneurship (1999 - present).
- 23) Advisory Board for the Energy and Environmental Systems Institute, Rice University.
- 24) Board Member, National Corrosion Center, Rice University.
- 25) Founder and Director of the Energy Safety Research Institute (ESRI), Swansea University (2013-present).
- 26) College of Engineering Executive Committee, Swansea University (2018-present).

CORPORATE SERVICES:

- 1) Founder and Chairman Scientific Advisory Board, Gallia Inc. (1992-1997).
- 2) President: Aluminum Research Board (1991-1997).
- 3) Board Member, Velocita, Inc. (2008-2011).
- 4) Board Member, Houston Clean Energy Park (2009-2011).
- 5) Science Advisory Board, Nanotech Innovations (2008-2009).
- 6) Science Advisory Board, Yellowstone Energy Ventures (2008-2009).
- 7) Scientific Advisory Board, Tego Biosciences (2007-2009).
- 8) Scientific Advisory Board, and Board Member, Oxane Materials, Inc. (2002-2014).
- 9) Scientific Advisory Board, NatCore Technology, Inc. (2004-present).
- 10) Scientific Advisory Board, Vanguard Solar (2007-2009 acquired by Natcore Technology, Inc.).
- 11) Scientific Advisory Board, Molecular Filtration, Inc. (2009-2010).
- 12) Scientific Advisory Board, Lance Energy Services (2010-2014).
- 13) Chief Technology Officer, C-Bond Systems (2017-present).
- 14) Board Member, Trimtabs Ltd (2019-present).
- 15) Technical Advisor, Worldwide Eco-Corporation (2019 – present).
- 16) Board Member, Apex Water Systems (2020-present).
- 17) Founder, MiDAS Green Innovations, Ltd (2019-present).
- 18) Advisory Board, Macrocaps (2020-2021).
- 19) Board Member, FXI, Inc. (2020-present).
- 20) Board Member, H2-NRG (2022-present).
- 21) Board Member, Welsh Racing Drivers Association (2022-present).
- 22) Technical Advisor, Clean Air Tech, Ltd. (2023-present).

ENTREPRENEURSHIP:

- 1) Founder, Gallia, Inc. Materials technology for GaAs chip devices for communications systems.

- 2) Founder, Natcore Technology, Inc. Solar cell materials technology, founded from Rice University IP.
- 3) Founder, Oxane Materials, Inc. Technology for reducing waste in oil and gas production through enabling higher yields per well, from Rice University IP.
- 4) Founder, Vanguard Solar, Inc. Nanotechnology systems for compound solar cells.
- 5) Founder, C-Bond Systems, Inc. developed technology for increasing glass strength to allow thinner glass with reduced energy consumption in manufacture.
- 6) Co-Founder, VLS-Decarbonisation Partners, developing technology for large scale Carbon Capture and Storage (CCS) with zero carbon energy.
- 7) Founder, CTO, Apex Water Systems, developed technology for water treatment of industrial waste-water and pre-treatment of water for desalination.
- 8) Co-founder, Board member, FXI, Inc., developed technology anti-viral PPE.
- 9) Co-Founder Alliance for Technology and Entrepreneurship (<https://alliance.rice.edu/>) internationally-recognized initiative devoted to the support of technology commercialization, entrepreneurship education, and the launch of technology companies. It is a catalyst for building successful ventures through education, guidance and connections. Its mission is to support the creation of technology-based companies and the commercialization of new technologies. Since its inception in 2000, more than 2300 companies have participated in over 190 Rice Alliance programs and have raised more than \$5.9 billion in early-stage capital. More than 48,000 individuals have attended Rice Alliance events and over 36,000 individuals subscribe to the Rice Alliance Digest newsletter. The Rice University Business Plan Competition is the World's Richest and Largest, awarding more than \$10 million in prizes since 2001. Over 201 past competitors are in business today or successfully exited having raised nearly \$1.9 billion in capital and exit dollars.
- 10) Advisor to the Celtic Alliance for Nanohealth (2012). A partnership of between Swansea University, Trinity College Dublin, Dublin City University and University College Dublin to replicate the Rice Alliance for Technology and Entrepreneurship model between Ireland and Wales.

RESEARCH GROUP: Present (Total since 1987 in parentheses) (42% Female)

- 1) Graduate Students 6 (53)
- 2) Postdoctoral Associates 5 (33)
- 3) Undergraduate Assistants 6 (30)
- 4) Summer Research interns 0 (24)
- 5) Sabbatical and Visiting Scholars 0 (9)

THESES SUPERVISION:

- 1) Sterically Crowded Aryloxides of Aluminum, M. D. Healy, Harvard University (1992).
- 2) 1,3-Diphenyltriazene Compounds of Aluminum and Indium, J. T. Leman, Ph.D., Harvard University (1993).
- 3) *Tert*-butyl Compounds of Gallium, W. M. Cleaver, Ph.D., Harvard University (1994).
- 4) Chemical Routes to Group 13-16 Materials, C. C. Landry, Ph.D., Harvard University (1994).
- 5) Intramolecular Coordination in Compounds of Aluminum, J. A. Francis, Ph.D., Rice University (1999).
- 6) Surface and Coordination Chemistry Related to GaAs, A. Keys, Ph.D., Rice University (1999).
- 7) Advanced Ceramic Composites and Coatings via Alumoxane Nanoparticles, R. L. Callender, Ph.D., Rice University (1999).
- 8) Chemical Vapor Deposition of Alumina-Based Thin Films, B. Fahlman, Ph.D., Rice University (2000).
- 9) Inorganic-Organic Materials Incorporating Alumoxane Nano-Particles, C. Vogelson, Ph.D., Rice University (2000).

- 10) Controlled Ceramic Porosity and Membrane Fabrication via Alumoxane Nanoparticles, C. Jones, Ph.D., Rice University (2000).
- 11) Cement Hydration Inhibition and Crosslinking in the Guar-Borate System, M. Bishop, Ph.D., Rice University (2001).
- 12) Aluminum and Gallium Chloride Stabilized Arene-Mercury Complexes, A. Borovik, Ph.D., Rice University (2001).
- 13) Development of Homogeneous and Heterogeneous Alkylalumoxane Catalysts, S. Obrey, Ph.D., Rice University (2001).
- 14) Controlling Ceramic Porosity Using Carboxylate-Alumoxane Nanoparticles, K. DeFriend, Ph.D., Rice University (2002).
- 15) Reversible Binding of Lewis Bases to Aluminum and Gallium Aryloxides, L. van Poppel, Ph.D., Rice University (2002).
- 16) Chemistry of Group 13 Lewis Acids, C. S. Branch, Ph.D., Rice University (2002).
- 17) Liquid Phase Deposition of Silica: Thin Films, Colloids and Fullerenes. E. Whitsitt, Ph.D., Rice University (2004).
- 18) Nanoparticles as crosslinking agents in polymer systems. N. Shahid, Ph.D., Rice University (2004).
- 19) Formation of Alumina Features & Cadmium Chalcogenide Coatings of Single-Walled Carbon Nanotubes. J. R. Loscutova, Ph.D., Rice University (2004).
- 20) Metal catalyzed reactions of fullerenes and single walled carbon nanotubes. D. Ogrin, Ph.D., Rice University (2005).
- 21) Transition Metal Catalyzed Reactions of Fullerenes and Carbon Nanotubes, R. Anderson, Ph.D., Rice University (2005).
- 22) From Fullerene Amino Acids to Fullerene Peptides. J. Yang, Ph.D., Rice University (2006).
- 23) Functionalization of Single Walled Carbon Nanotubes. L. Zhang, Ph.D., Rice University (2006).
- 24) Characterization via Nuclear Magnetic Resonance of Portland Cement and Related Materials. C. Edwards, Ph.D., Rice University (2006).
- 25) Nucleation and Growth of Carbon Nanotubes as a Function of Catalyst Composition. Christopher Crouse, Ph.D., Rice University (2008).
- 26) Single-Walled Carbon Nanotubes: Functionalization, Characterization and Application. L. Zeng, Ph.D., Rice University (2008).
- 27) Metal Ion Interactions with Single Walled Carbon Nanotubes. Jonathan Brege, Ph.D., Rice University (2009).
- 28) Functionalization, Characterization, and Coordination of Carbon Nanomaterials. C. Hamilton, Ph.D., Rice University (2009).
- 29) Various Coatings of Carbon fibers and Single Wall Carbon Nanotubes: Synthesis and Applications. H. Jafry, Ph.D., Rice University (2010).
- 30) In-vitro Model System for Calcific Band Keratopathy and Inhibitory Effects of C60 Fullerene Derivatives. N. Doostdar, Ph.D., Rice University (2010).
- 31) Fullerene Amino Acids and Peptides: Synthesis and Applications. T. A. Strom, Ph.D., Rice University (2010).
- 32) Steric Considerations in Copper(II)-Olefin Complexes Incorporating Substituted *Bis*-2-pyrrolamines. J. Allen, Ph.D., Rice University (2011).
- 33) Regular Arrays of QDs by Solution Processing, B. Oliva, MS, Rice University (2011).
- 34) Polyethylenimine functionalized nano-carbons for the absorption of carbon dioxide. E. Dillon, Ph.D., Rice University (2011).
- 35) Fabrication of Petrochemical and Viral Resistant Membranes, S. Maguire Boyle, M.S., Rice University (2012).
- 36) Synthesis of black silicon anti-reflection layers for silicon solar cells, Y.-T. Lu, Ph.D., Rice University (2015).
- 37) Coating and doping of Ge QDs, B. Oliva-Chatelain, Ph.D., Rice University (2016).
- 38) Branched aluminum oxide nanoparticles for enhanced oil recovery applications, W. A. A. D. Al-Shatty, MSc, Swansea University (2016).

- 39) Functionalization of carbon materials with metals, K. D. Wright, PhD, Rice University (2017).
- 40) Catalytic growth of carbon nanotubes using a single molecule catalyst precursor, Gibran L. Esquenazi, PhD, Rice University (2018).
- 41) A Design Approach to the Synthesis and Characterization of Metal Phosphonate MOFs, Derek B. Barbee, PhD, Rice University (2019).
- 42) The Use of Steelmaking Slags in Marine Applications, Lucy Fisher, MSc, Swansea University (2019).
- 43) Surfactants and Nanoparticles as Emerging Platforms in Fluid Flow Transport, Sajad Kiani, PhD, Swansea University (2020).
- 44) A Case Study for the Potential of Carbon Capture and Storage in Qatar, Fahad Al Sulaiti, PhD, Swansea University (2021).
- 45) Value generation by recovering by-products from steelmaking processes: Dezincification of basic oxygen steelmaking dust, Daniel J. C. Stewart, PhD, Swansea University (2022).

PRESENTATIONS: 1995 - present

- 1) "A chemical approach for structural, composite, and coating materials for automotive applications", invited lecture, World Car Conference '96, Riverside, CA, January 1996
- 2) "Alkyl alumoxanes: destroying the myth", invited lecture, University of Texas at Arlington, Arlington, TX, January 1996
- 3) "Alkyl alumoxanes: destroying the myth", invited lecture, University of North Texas, Denton, TX, February 1996
- 4) "Molecular control over materials synthesis", NREL Photochemical Sciences Workshop, Estes Park, CO, February 1996
- 5) "Environmentally benign processing of aluminum-based ceramic materials: water soluble alumoxanes", ACS National Meeting; New Orleans, LA, March 1996
- 6) "Gallium sulfide: molecules materials and transistors", invited lecture, University of Delaware, DE, May 1996
- 7) "Alkyl alumoxanes: destroying the myth", invited lecture, DuPont Central Research, Delaware, DE, May 1996
- 8) "Alkyl alumoxanes: destroying the myth", invited lecture, GE; Schenectady, NY, July 1996
- 9) "Dialkylaluminum carboxylates: models for carboxylate alumoxanes", International Coordination Chemistry Conference, Vancouver, BC, August 1996
- 10) "The future of inorganic chemistry: nano-science and technology", Monsanto, St. Louis, MO, October 1996
- 11) "Alkyl alumoxanes: destroying the myth", invited lecture, ACS Southwestern Regional Meeting; Houston, TX, October 1996
- 12) "Gallium sulfide: molecules materials and transistors", invited lecture, University of Southern Illinois, IL, October 1996
- 13) "A new route to alumina-based ceramics via a novel transmetalation reaction", American Ceramic Society, San Antonio, TX; October 1996
- 14) "Alumoxane precursors to designer catalysts and catalyst supports: catalytic oxidation of dichloromethane", invited lecture, MRS Fall Meeting, Boston, MA, December, 1996
- 15) Inorganic Gordon Conference, Salve Regina University, Newport, RI, July 1997
- 16) Symposium Chair for special symposium in honor of Sir. Geoffrey Wilkinson, ACS National Meeting, San Francisco, CA. April 1997
- 17) Organometallic Gordon Conference, Salve Regina University, Newport, RI, July 1997
- 18) MRS Fall Meeting, Boston, MA, December 1997.
- 19) 81st Canadian Society for Chemistry Conference & Exhibition, Vancouver, Canada, May 31 - June 4, 1998
- 20) 2nd Fargo Conference on Main Group Chemistry, Fargo, ND, June 4 - June 6, 1998
- 21) Hunter College Symposium, New York, NY, April 28 - April 30, 1998

- 22) "A new route to hexaluminate ceramics via a novel transmetalation reaction", American Ceramic Society, Coco Beach, FL, Jan. 25-Jan. 28, 1999
- 23) "Carboxylate Alumoxanes Environmentally Benign Routes to Ceramics", Alcoa Chemical Co., Alcoa, PA, March 9, 1999
- 24) "Enhancement of Intermolecular Hydrogen Bonding Through Coordination", American Chemical Society, Anaheim, CA, March 21, 1999
- 25) "Carboxylate Alumoxane Nanoparticles", University of Oldenburg, Germany, May 17, 1999
- 26) "Carboxylate Alumoxane Nanoparticles", University of Munchen, Munich, Germany, May 19, 1999
- 27) "Analytical Facilities at Rice University", Baker Hughes, Sugar Land, TX, June 4, 1999
- 28) "CVD of Conformal Alumina Films", Gordon Conference, Chemistry of Electronic Materials, New England College, New Hampshire, July 4 - July 9, 1999
- 29) "Aluminate Interphase Coatings for FRCMC", NASA Glenn Research Center, Cleveland, OH, July 13, 1999
- 30) "Gas Phase Structure of $\text{Al}(\text{t-Bu})_3$ and $\text{Ga}(\text{t-Bu})_3$ ", Gordon Conference, Inorganic Chemistry, Newport, RI, July 19 - July 23, 1999
- 31) "Reaction of Tert-Butylalumoxane with Trimethylaluminum", American Chemical Society, New Orleans, LA, Aug. 23 - Aug. 26, 1999
- 32) "Cleavage of Organosiloxanes by Aluminum Hydrides", American Chemical Society Regional Meeting, El Paso, TX, Oct. 20 - Oct. 22, 1999
- 33) "Activation of Small Molecules by Group 12/13 Complexes", International Conference on Lewis Acidity, Nagoya, Japan, Nov. 1 - Nov. 3, 1999
- 34) "Alumoxane Nanoparticles", Toyota Motor Company, Toyota City, Japan, Nov. 4, 1999
- 35) "Understanding Methyl Alumoxane", Sumitomo Chemical, Chiba, Japan, Nov. 5, 1999
- 36) "Understanding MAO", Albemarle Corp., Baton Rouge, LA, Dec. 7, 1999
- 37) "Understanding Alumoxanes", Albemarle Corp., Baton Rouge, LA, April 2000
- 38) "Aluminum Oxides: How a New Look at an Old Material Can Provide Unexpected Results", Society of Minerals, Metals, and Materials, Rice University, Spring, 2000
- 39) "An Investigation into the Mechanism of Cement Hydration Inhibition", Halliburton Energy, Duncan, OK, Spring, 2000
- 40) "A ^{11}B NMR Investigation into the Mechanism of Crosslinking in the Guar-Boron System", Halliburton Energy, Duncan, OK, Spring, 2000
- 41) "Fiber-Reinforced Ceramic Matrix Composites Using Alumina Nanoparticles", American Ceramic Society, St. Louis, MO, May 2000
- 42) "Simple Solutions are Often the Best: Cross-Disciplinary Research from the Barron Group", presentation to Rice University Alumni, Rice University, March 2000
- 43) "Carboxylate Alumoxanes: Inorganic-Organic Composite Materials for Automotive Applications", Rice University School of Continuing Studies, Spring, 2000
- 44) "Activation of Aromatics Using Group 12/13 Lewis Acids", Gordon Conference, Inorganic Chemistry, Salve Regina College, Newport, RI, July 2000
- 45) "Aluminum and gallium chloride stabilized arene-mercury complexes", American Chemical Society National Meeting, San Diego, CA, April 2001
- 46) "Cement hydration inhibition: In situ creation of composite structures", American Chemical Society National Meeting, San Diego, CA, April 2001
- 47) "Carboxylate-Alumoxanes: Environmentally Benign Precursors for Developing Aluminum Based Ceramic Membranes and Filters", Halliburton Energy Services, Duncan, OK, June 2001
- 48) "Arene-Mercury Complexes Stabilized by Aluminum and Gallium Chloride: Catalysis for H/D Exchange of Aromatic Compounds." University of Arizona, Tucson, Arizona. January 18, 2002

- 49) "Arene-Mercury Complexes Stabilized by Aluminum and Gallium Chloride: Catalysis for H/D Exchange of Aromatic Compounds." Los Alamos National Laboratory, Los Alamos, New Mexico. January 24, 2002
- 50) "Carboxylate-Alumoxanes: A Journey from Understanding Sol-Gels to Catalysts, Composites, and Membranes." Sandia National Laboratory, Albuquerque, New Mexico. January 23, 2002
- 51) "Carboxylic Acid Functionalized Alumina Nanoparticles: A Flexible Class of Pre-Ceramics for Structural Composites and Ultrafiltration Membranes." Georgia Institute of Technology, Atlanta, GA. 9/16/2002
- 52) "Carboxylic Acid Functionalized Alumina Nanoparticles: A Flexible Class of Pre-Ceramics for Structural Composites and Ultrafiltration Membranes." Central Michigan University, Mount Pleasant, MI. 9/18/2002
- 53) "Carboxylic Acid Functionalized Alumina Nanoparticles: A Flexible Class of Pre-Ceramics for Structural Composites and Ultrafiltration Membranes." University of Houston, Houston, TX. 10/8/2002
- 54) "Nanotechnology Presentation." Cleantech Venture Network, Toronto, Canada. Workshop, November 13, 2002
- 55) "A Structural and Reactivity Model for MAO: Does Anyone Believe Me Now?." American Chemical Society, New Orleans, LA, March 22, 2003
- 56) "Controlling Surface Chemistry of Oxide Membranes." CBEN NanoDays 2003, Rice University. October 14, 2003
- 57) "Coating and Exposing SWNTs." CBEN NanoDays 2003, Rice University. October 14, 2003
- 58) "Coating and Exposing SWNTs." Welch Foundation, Houston, Texas. October 28, 2003
- 59) "Coatings, Spontaneous Interconnects, Regioselective Functionalization, and Interactions with Biological Systems of SWNTs." National Science Foundation, Jackson Hole, WY. June 1, 2003
- 60) "Interface of Wet and Dry Nanotechnology." Affymetrix, Inc., San Francisco, CA, San Francisco, CA. September 9, 2003
- 61) "Interface of Wet and Dry Nanotechnology." Puretech, Inc., Boston, MA. October 23, 2003
- 62) "Nanostructure Coating Technology." Valve Technologies Technical Conference International Seminar 2003, Houston, TX. October 24, 2003
- 63) "Use of Carboxylate Alumoxanes in Bone Replacement Materials." IRIS X 10th International Symposium on Inorganic Ring Systems, Burlington, VT. August 22, 2003
- 64) "Fabrication of Nanostructured Ceramic Membrane", France/Texas Water Treatment Research Congress: Emerging Technologies and Challenges. January 12, 2004
- 65) Lecture to CBEN for high school teachers, February 17th, 2004
- 66) Teaching at Lee High School for CBEN, February 17th, 2004
- 67) "Biological and Environmental Applications of Nanotechnology", Case Fellowship 'Societal Implications of Nanotechnology' March 15, 2004
- 68) "Reinforcement of Poly Propylene Fumarate -Based Networks with Surface Modified Alumoxane Nanoparticles for Bone Tissue Engineering". Alliance for NanoHealth Workshop, Texas Heart Institute, Houston, TX. May 14, 2004
- 69) "New High Density Fuels: A Chemical Approach" DARPA/TTO Seedling Study, Rice University, Houston, TX. (August 31, 2005)
- 70) "Alumoxanes: A Journey from Research to Commercialization." Nanotechnology Colloquium, Houston, Texas. (October 31, 2005)
- 71) "Alumoxanes: A Journey from Ceramics and Catalysts to Bone Replacement and Fuel Cells" Baylor College of Medicine, Houston, TX. (January 14, 2005)
- 72) "Coating and Exposing SWNTs" University of Notre Dame, Notre Dame, IN. (February 17, 2005)

- 73) "Coating and Exposing SWNTs" United States Air Force, Wright Patterson Air Force Base, Ohio. (February 22, 2005)
- 74) "SWNTs and Fullerenes: From Composites and Catalysis to Biology and Remediation" Strategic Partnership for Research in Nanotechnology, Houston, Texas. (October 11-12, 2005)
- 75) "Membranes" Nanotechnologies for a Sustainable Environment, Rice University, Houston, TX. (December 15, 2005)
- 76) "Nano-Applications in Energy: Fuel Cells/Catalysis" Rice Energy Symposium, Rice University, Houston, TX. (January 10, 2006)
- 77) "Technology Transfer at Rice University" Nanotechnology Venture Forum 5, Rice University. (January 20, 2006)
- 78) "Evaluation and Accessing University Based Technology" Sino-US Small Business Development Forum & Expo, Houston, TX. (April 26-29, 2006)
- 79) "Water Purification with Nanostructured Membranes" VIII International Physics Symposium, Monterrey, Mexico. (February 16-18, 2006)
- 80) "How the Presence of Nanoparticles Control the Reactivity/Mobility of Biological Materials" Nano Tox Conference, Boston, MA. (April 24, 2006)
- 81) "The Chemistry Behind Explosives and the Application of Nanotechnology" Tactical Operations Division, Houston Police Department, Houston, TX. (September 11, 2006)
- 82) "Big Energy Solutions with Complex Licensing Issues" Licensing Executives Society, New York, NY. (September 10, 2006)
- 83) "Catalysts and SWNT Modification: Towards a Vision of SWNT Amplification" ACS Conference, San Francisco, CA. (September 12, 2006)
- 84) "Growth of Single Walled Carbon Nanotubes from Seeds" Trinity College, Dublin, Dublin, Ireland. (December 7, 2006)
- 85) "Fullerene Amino Acids as a Passport for Peptides Through Cell Membranes" University College Dublin, Dublin, Ireland. (December 8, 2006)
- 86) "Intellectual Property: Issues and Solutions", Baker Institute and Texas/UK Collaborative, Rice University, Houston, TX. (January 22, 2007)
- 87) "Energetic Materials", Strategic Partnership for Research in Nanotechnology, University of Houston, Houston, TX. (February 6, 2007)
- 88) "The Barron Research Group: Tackling Problems in Health and Energy Using Nanotechnology", Rice University, Houston, TX. (February 22, 2007)
- 89) "Rice's View of an Energy Future", Energy Innovation Academic Roundtable, San Francisco, CA. (March 8, 2007)
- 90) "Nanotechnology Commercialization and IP Issues", Nanoforum, Milan, Italy. (September 18, 2007)
- 91) "An Overview of Energy-Related Programs at Rice", Rice Alliance for Technology and Entrepreneurship, Houston, TX. (September 27, 2007)
- 92) "Presentation of Research and Q&A for the Beverage Institute for Health and Wellness", Rice University, Houston, TX. (September 28, 2007)
- 93) "A Vision of SWNT Amplification", NanoTX'07 Conference and Expo, Dallas, TX. (October 4, 2007)
- 94) "Nano Applications for Lockheed Martin", Rice University, Houston, TX. (October 12, 2007)
- 95) "Rice University Energy Vision", Next Generation Biofuels, Greater Houston Partnership/UK Trade & Investments, Houston, TX. (October 15, 2007)
- 96) "Recent Developments in Energy Research at Rice University", Texas/UK Research Collaborative Meeting, Glasgow, Scotland. (October 22, 2007)
- 97) "Academic and/or Company Research", Windsor Energy Group: Global Energy-Policy Needs and Priorities, Houston, TX. (November 6, 2007)
- 98) "Down Hole Nano: Investigating Nanotechnology in the Petroleum Reservoir", Baker Institute, Rice University, Houston, TX (January 8-9, 2008)

- 99) "Amplification of Carbon Nanotubes: A Problem in Understanding Catalysis", Texas Christian University, Dallas, TX. (January 15, 2008)
- 100) "The Unique Properties of Nano Materials Enable Alternative Approaches to Therapeutic Agents", Alliance for Nanohealth, Baker Institute, Rice University, Houston, TX. (March 17, 2008)
- 101) "NanoEnabled Intracellular and Trans-dermal Drug Delivery", DARPA Meeting on Nanohealth, M. D. Anderson Cancer Center, Houston, TX. (March 24, 2008)
- 102) "Down Hole Nano", BP-Rice University Meeting, Rice University, Houston, TX. (March 31, 2008)
- 103) "Single Walled Carbon Nanotubes: Metal Ion Interactions and Composite Applications", 2008 NanoMaterials for Defense Conference, Arlington, VA. (April 21-23, 2008)
- 104) "Revolutionizing Solar Energy", New Orleans Investment Conference, New Orleans, LA. (November 15, 2008)
- 105) "Carboxylate Alumoxanes - Modeling Catalysts, Surfaces, and Nanoparticles", Afton Chemicals, Richmond, VA. (November 21, 2008)
- 106) "Future of Energy at Rice University", Texas-United Kingdom Collaborative Workshop, Austin, TX. (January 25, 2009)
- 107) "Small Things, Big Changes", Windsor Energy Group, London, England. (March 8, 2009)
- 108) "Proving Covalent Attachment to Carbon Nanotubes", PittCon Conference and Expo, Chicago, IL. (March 12, 2009)
- 109) "Fabrication Approaches", American Physical Society, Pittsburgh, OH. (March 14, 2009)
- 110) "Nanotechnology for the Oil & Gas Industry." Houston Technology Center, Houston, TX. (September 18)
- 111) "Reactive Nanoparticle Materials for Enhanced Warfighter Operational Capabilities" Naval Surface Warfare Center, Panama City, FL. (October 8, 2009)
- 112) "Nanotechnology Investing: Where Is It Going?" New Orleans 2009 Investment Conference, New Orleans, LA. (October 10, 2009)
- 113) "An Academic Perspective of Collaboration: Bringing the Human Element to the Front." King Abdullah University of Science and Technology, Riyadh, Saudi Arabia. (November 1, 2009)
- 114) "Environmental Catalysis and Nanotechnology" to Nankai University Delegation Meeting, Rice University. (January 21, 2010)
- 115) "Patent Pooling in Energy and Nanotechnology". Energy R&D and Intellectual Property in the New "Green Economy", Baker Institute, Rice University. (January 26, 2010)
- 116) "Applications of Connexions in Teaching", Connexions Conference, Rice University. (February 2, 2010)
- 117) "Applications in Energy", Rice Alliance for Technology and Entrepreneurship, Rice University. (February 18, 2010)
- 118) "Challenges for the Future of the Automobile". Galveston College, Galveston, TX. (March 24, 2010).
- 119) "One Man's Inhibition is a Blind Man's Cure", South Texas Section of the American Institute of Chemical Engineers, Rice University. (April 8, 2010)
- 120) "Innovation in Carbon Capture Through Nanotechnology", Saudi Aramco Technology Symposium, Houston, TX. (October 11, 2010)
- 121) "Nanotechnology Applications for CO₂ Extraction From Gas Streams", Saudi Aramco Technology Symposium, Houston, TX. (October 12, 2010)
- 122) "The Evolution of a Serial Entrepreneur in Nanotechnology", The Rice Alliance Nanotechnology and Sustainability Venture Forum, Duncan Hall Atrium, Rice University, February 17, 2011
- 123) "How We Take Nanotechnology to Market", The MIT Enterprise Forum, McMurtry Hall, Rice University, February 16, 2011

- 124) "SWNT Amplification: Concepts and Results" Guadalupe Workshop on Nucleation and Growth Mechanisms of Single Wall Carbon Nanotubes (SWCNT), Boerne, TX, (April 12, 2011)
- 125) "Can Nanotechnology Provide a New Approach to Oil and Gas Shale Production?", Ryder Scott Reserves Conference, Houston, TX, (September 16, 2011)
- 126) "How Can Nanotechnology Make Your Frac More Productive, Environmentally Friendly, and Less Open to Litigation?", American Association of Mineral Owners, Houston, TX (September 15, 2011)
- 127) "Can Nanotechnology Provide a New Approach to Oil and Gas Shale Production?", Platts 6th Annual Oil & Gas Shale Developer Conference, Houston, TX (June 21, 2011)
- 128) "Nano Technology & Shale Gas Production", Korean-American Energy E&P Society, Houston, TX (November 4, 2011)
- 129) "Nanotechnology Has the Potential to Provide Power to the World", Pioneer Oil Producers Society, Houston, TX (November 21, 2011)
- 130) "How Can Nanotechnology Make Your Play More Productive, Environmentally Friendly, and Less Open to Litigation?", The Energy Forum Frontier Plays 2011, Farmers Branch, TX (September 14, 2011).
- 131) "Saving the E & P Industry, Nano Style", EPNano Net Summit, Houston, TX (June 22, 2011).
- 132) "More Oil, More Gas, Less Water", Presentation to National Science Foundation workshop on hydraulic fracturing, Washington, DC (May 15, 2012)
- 133) "Nanotechnology: More Oil, More Gas, Less Water, Less Pollution", Invited speaker at the Roughneck Camp Conference, Schlumberger-WesternGeco Office, Houston, TX (June 28, 2012)
- 134) "How nanotechnology can create more value and less risk in the energy industry", Presentation at the Houston ASA Energy Valuation Seminar, Houston Chapter of the American Society of Appraisers, Houston, TX (September 13, 2012)
- 135) "Creating Successful Technology Spin-Out Through Different Worlds", Celtic Alliance for Nanohealth, Venture Forum Swansea 2012, Swansea, UK (September 26, 2012).
- 136) "Turning Innovation into a Commercial Reality", BioWales 2013: Unveiled Connectivity Delivering Tomorrow's Health Solution, UK (March 19, 2013)
- 137) "Nanoparticle Chemistry for Water Treatment - Health and Energy Applications", Center for the Environmental Implications of Nanotechnology, Duke University, Durham, NC. (April 10, 2013)
- 138) "New Technology for a Better Oil and Gas Future", 11th Annual Forum of the Korean-American Offshore Engineers Association (May 9, 2013)
- 139) "Produced Water from Shale Oil", BlueTech Webinar Series (June 27, 2013)
- 140) "What Can Nanotechnology Do for the Oil and Gas Industry?", The Society of Petroleum Engineers Research and Development, Houston, TX (October 2013)
- 141) "Analysis, treatment and tracing of frac and produced water.", American Chemical Society Symposium, 44th American Chemical Society, Western Regional Meeting, Santa Clara, CA (October 2013)
- 142) "Metal coordination to oxidized fullerene and carbon nanotubes as routes to water purification/metal reclamation", American Chemical Society, Baylor University, Waco, TX (November 2013)
- 143) "From macroscopic coatings to molecular treatments of surfaces: how can an understanding of nano help with corrosion?", CORROSION 2014 Conference, San Antonio, TX. (March 8, 2014)
- 144) "Evolution of Serial Entrepreneurs", Friends of the Fondren Library, Houston, TX (October 2, 2014).
- 145) "From Macroscopic Coatings to Molecular Treatments of Surfaces: How Can an Understanding of Nano Help With Corrosion?" CORROSION 2014 Conference, held in San Antonio, TX. (March 8, 2014)

- 146) "Imaging a Frac and Tracing the water Using Nanotechnology", Korean-American Energy E&P Society Meeting, Houston, TX, (November 2014)
- 147) "Nanotechnology for Enhanced Oil Recovery". Moderator and Presentation, Maximizing Oil Recovery: Boosting Production, UK Trade and Investment Breakfast Briefing, Houston, Texas. (May 5, 2014)
- 148) "The Global Skilled Workforce Shortage in the Energy Industry", Swansea University, Wales, UK (October 27, 2014)
- 149) "Design of Super Hydrophilic Membranes for Non-Fouling Treatment of Frac and Produced Water" Shell Technology Center Produced Water Symposium in Galveston, TX (May 20, 2015).
- 150) "What's Up? (Science United for Philanthropy)", Advanced Materials Workshop: Transcending Incrementalism, Texas A&M University (June 23, 2015).
- 151) "Fullerene Based Amino Acid and peptides: Designing Drug Delivery and Interactions", 3rd International BAU Drug Design Congress, Istanbul (October 2015).
- 152) "Energy Safety Research Institute (ESRI): Building Bridges to Sustainable Energy", University of Brunei Darussalam (May 2016).
- 153) "Copper/CNT Composites", Rice University (July 2016).
- 154) "SWNT Amplification: Concepts and Results" Guadalupe Workshop on Nucleation and Growth Mechanisms of Single Wall Carbon Nanotubes (SWCNT)", University of Vilnius, Lithuania (September 2016).
- 155) "Flexible and Integrated Energy Systems: a smart opportunity" Welsh Government's Horizon 2020 Annual Event, Cardiff (March 2017).
- 156) "Fracking: The Good the Bad and the Ugly" Pennard Mens Club (March 2017).
- 157) "Creating Research Opportunities through Ser Cymru II", SER Cymru II: Celebrating and Growing" Cardiff, (February 2017).
- 158) "Fracking: The Good the Bad and the Ugly" Swansea U3A (March 2017).
- 159) "Electronic Measurements of Single- and Multi-Walled Carbon nanotubes: Evidence for Unusual Behaviour at Nanotube-Nanotube Junctions" 231st ECE Meeting, New Orleans (May 2017).
- 160) "Energy Safety Research Institute", EPSRC UK Nanotube Collaborative Inaugural Workshop, ESRI (March 2017)
- 161) "Next Generation of Energy Distribution: Electronic measurements of CNTs and copper-CNT composite fabrication", NCEM-5.4, Wolfson College, Cambridge (July 2017).
- 162) "Fracking: The Good the Bad and the Ugly" Bishopston Men's Society (September 2017).
- 163) Keynote speaker EcoBalt 2018 "Flexible approaches to water treatment and metal remediation: from a village to the oil patch" Vilnius, Lithuania (October 2018).
- 164) "The need for water re-use", British Council's Science Collaboration Symposium, Kuwait (March 2018).
- 165) "Copper/Carbon Nanotube Ultraconductive Wire: Conduction, Processing, and Stability", Knoxville, USA (December 2018)
- 166) "Possible game-changers – making carbon capture and storage commercial" Windsor Energy Group, Windsor Castle (March 2019).
- 167) "Helping welsh industries to reduce carbon dioxide emissions and drive a stronger, greener economy", Storytelling and the Environment Symposium, The George Ewart Evans Centre for Storytelling, Cardiff (April 2019).
- 168) "Large scale carbon sequestration – making carbon capture and storage commercial", Transcending Incrementalism, ESRI, Swansea, UK (April 2019).
- 169) "Large scale carbon sequestration – making carbon capture and storage commercial", The 10th Trondheim Conference on CO₂ Capture, Transport and Storage, Trondheim, Norway (June 2019).
- 170) "Angular and overlap dependence of conduction between carbon nanotubes of identical chirality and diameter: towards increased CNT fiber conduction", UltraWire 2019, Cambridge, UK (July 2019).

- 171) Large scale carbon sequestration - could shale make carbon storage commercially viable?", Achieving Net Zero International Conference, Oxford, UK (September 2019).
- 172) "Nanoscale technologies in medicine, climate mitigation, water conservation", Transcending Incrementalism-3, Rabat, Morocco (November 2019).
- 173) "Summary of field trial results of the treatment of contaminated water using nonfouling superhydrophilic functionalized ceramic membranes", QEERI ICSEWEN - Qatar 2019 (December 2019).
- 174) "Planning a successful grant application", Universiti Teknologi Brunei (January 2020).
- 175) "Fracking: The Good the Bad and the Ugly" Mumbles Probus Club (February 2020).
- 176) "Possible game-changers – making carbon capture and storage commercially viable", International Conference on Fossil and Renewable Energy, Houston, USA (February 2020).
- 177) "The hydrogen option: Brunei experience", 20th Annual Windsor Energy Group Conference, Windsor Castle (March 2020).
- 178) "Finding green energy technologies: likely options", 20th Annual Windsor Energy Group Conference, Windsor Castle (March 2020).
- 179) "Future green energy", Ambassadorial Briefing, 20th Annual Windsor Energy Group Conference, Windsor Castle (March 2020).
- 180) "Copper-carbon nanotubes hybrid wire for electrical conduction", UltraWire 2019, Virtual (July 2020).
- 181) "Reducing Industrial carbon emissions", Oystermouth Probus Club, Virtual (July 2020)
- 182) "How KTN helped me move from saving the seals to the next generation PPE", KTN Open Forum, Virtual (November 2020).
- 183) "Reducing industrial carbon emissions and very large scale decarbonization (VLSD): a new approach to economic CCS", Enabling CO₂ Geological Storage within a Low-Carbon Economy, KAUST, virtual (February 2021).
- 184) Reducing Industrial Carbon Emissions: Creating and Operating an Industrial Demonstrator for Multiple Decarbonization Technologies", F&R Energy Conference, Houston, virtual (March 2021).
- 185) "Reducing industrial carbon emissions: helping industries to reduce carbon dioxide emissions and drive a stronger, greener economy", BEIS, virtual (March 2021).
- 186) "Planning a successful grant application", Princess Nourah Bint Abdulrahman University, virtual (March 2021).
- 187) "Reducing industrial carbon emissions: an overview with the impact of COVID-19", Oystermouth Probus Club, virtual (June 2021).
- 188) "Reducing industrial carbon emissions: helping industries to reduce carbon dioxide emissions and drive a stronger, greener economy", 2nd UK CCUS and Hydrogen Decarbonisation Summit, Birmingham (September 2021).
- 189) "Reducing industrial carbon emissions: helping industries to reduce carbon dioxide emissions and drive a stronger, greener economy", Endress+Hauser, virtual (September 2021).
- 190) "Cluster for decarbonization", Decarb Connect Europe, virtual (October 2021).
- 191) "From saving the seals to the next generation PPE: True anti-viral masks and HVAC filters with unique COVID trapping and inactivation performance", Pumps and Pipes, virtual (December 2021).
- 192) "Identifying pathways and establishing market opportunities to scale resilience through technological solutions", University of Arizona (January 2022).
- 193) "True Anti-Viral masks and HVAC filters with Enhanced Trapping and Inactivation", United Conference of COVID-19 and Post COVID Sequelae, virtual (February 2022).

- 194) "Simultaneous hydrogen generation and carbon sequestration from recycling steel manufacturing waste", 6th International Conference on Fossil & Renewable Energy, virtual (February 2022).
- 195) "Getting to net zero through industrial decarbonisation", Teachout event, virtual (March 2022).
- 196) "Carbon Negative Hydrogen Generation as Part of a Holistic Approach to Industrial Decarbonization", 3rd UK CCUS and Hydrogen Decarbonisation Summit, Birmingham (May 2022).
- 197) "Reducing Industrial Carbon Emissions: Helping Industries to Reduce Carbon Dioxide Emissions and Drive a Stronger, Greener Economy", Salters Institute Festival of Chemistry Discovery Days, Swansea University (May 2022).
- 198) "Reducing Industrial Carbon Emissions - A Case Study in the Energy Transition", Go-to Market Conference Endress+Hauser, Geneva, Switzerland (June 2022)
- 199) "Reducing Industrial Carbon Emissions - A Case Study in Making Real Impact Towards Net Zero", GSESRE2022, Copenhagen, Denmark (June 2022).
- 200) "Creating a symbiotic relationship between regional industry as a holistic approach to industrial decarbonization" F&R Energy Conference, Boston (March 2023).
- 201) "TRL jumping and a new approach for industry to de-risk technology adoption", 4th UK CCUS and Hydrogen Decarbonisation Summit, Leeds (March 2023).
- 202) "From Purifying Water to Protection from COVID", Mumbles Probus Club (April 2023).

List of Publications of Andrew R. Barron

1. Synthesis of rhodium (II) pyrazolate complexes. Crystal structure of tetra- μ -3,5-dimethylpyrazolato dirhodium (II) bis-acetonitrile, (Rh-Rh). A. R. Barron, G. Wilkinson, M. Motevalli, and M. B. Hursthouse, *Polyhedron*, 1985, **4**, 1131-1134.
2. The synthesis and structure of a new type of bridged hydrido-aluminate complex: 1,2;1,2;2,3;2,3-tetra- μ -hydrido-1,1,1,2,3,3,3,-heptahydrido-1,1,1,3,3,3,-hexakis(trimethylphosphine)-1,3,-ditungsten-(IV)-2-aluminum. A. R. Barron, M. Motevalli, M. B. Hursthouse, and G. Wilkinson, *J. Chem. Soc., Chem. Commun.*, 1985, 664-665.
3. Synthesis and characterization of tungsten and rhenium aluminopolyhydrides: X-ray crystal structure of $(\text{Me}_3\text{P})_3\text{H}_3\text{W}(\mu\text{-H})_2\text{AlH}(\mu\text{-O}^n\text{Bu})_2$ and $[(\text{Me}_3\text{P})_3\text{H}_3\text{W}(\mu\text{-H})]_2\text{AlH}_3$. A. R. Barron, D. Lyons, G. Wilkinson, M. Motevalli, A. J. Howes, and M. B. Hursthouse, *J. Chem. Soc., Dalton Trans.*, 1986, 279-285.
4. Tertiary phosphine aluminohydride complexes of ruthenium and osmium. A. R. Barron and G. Wilkinson, *J. Chem. Soc., Dalton Trans.*, 1986, 287-289.
5. Tertiary phosphine aluminohydride complexes of chromium, molybdenum, and tungsten. A. R. Barron, J. E. Salt, and G. Wilkinson, *J. Chem. Soc., Dalton Trans.*, 1986, 1329-1332.
6. Transition metal polyhydride anions: a new synthetic route. X-ray crystal structure of $\{[(\text{Me}_3\text{P})_3\text{WH}_5]\text{Li}\}_4$. A. R. Barron, M. B. Hursthouse, M. Motevalli, and G. Wilkinson, *J. Chem. Soc., Chem. Commun.*, 1986, 81-82.
7. Transition metal aluminohydride complexes. A. R. Barron and G. Wilkinson, *Polyhedron*, 1986, **5**, 1897-1915.
8. Tertiary phosphine borohydride complexes of chromium, tungsten and rhenium. Crystal structure of trans-hydrido-(η^2 -tetrahydridoborato)-bis [1,2-bis(dimethylphosphino) ethane]chromium (II). A. R. Barron, J. E. Salt, G. Wilkinson, M. Motevalli, and M. B. Hursthouse, *Polyhedron*, 1986, **5**, 1833-1837.
9. Reactions of aluminopolyhydride complexes of tungsten. X-ray crystal structures of $[(\text{Me}_3\text{P})_3\text{H}_3\text{W}(\mu\text{-H})_2\text{AlCl}(\mu\text{-N}=\text{CHEt})]_2$, $\{[(\text{Me}_3\text{P})_3\text{WH}_5]\text{Li}\}_4$, $(\text{Me}_3\text{P})_3\text{WH}_2\text{I}(\text{SiMe}_3)$ and $(\text{Me}_3\text{P})_3\text{H}_2\text{W}(\mu\text{-H})_3\text{AlCl}_2\text{NMe}_3$. A. R. Barron, G. Wilkinson, M. Motevalli, and M. B. Hursthouse, *J. Chem. Soc., Dalton Trans.*, 1987, 837-846.
10. Crystal structure of hydridochloro(carbonyl)tris-(diphenylmethylphosphine)ruthenium (II) $[\text{RuHCl}(\text{CO})(\text{PPh}_2\text{Me})_3]$. M. Motevalli, M. B. Hursthouse, A. R. Barron, and G. Wilkinson, *Acta Crystallogr. C.*, 1987, **43**, 214-216.
11. The chemistry of chromium nitrile complexes of 1,2-bis(dimethylphosphino)ethane. X-ray crystal structure of trans- $[\text{Cr}^{\text{IV}}\text{Cl}(\text{NEt})(\text{dmpe})_2]\text{CF}_3\text{SO}_3$, trans- $[\text{Cr}^{\text{IV}}(\text{N}=\text{CHMe})_2(\text{dmpe})_2](\text{BPh}_4)_2$ and trans- $[\text{Cr}(\text{H.NCMe})_2(\text{dmpe})_2](\text{BPh}_4)_2$. A. R. Barron, J. E. Salt, G. Wilkinson, M. Motevalli, and M. B. Hursthouse, *J. Chem. Soc., Dalton Trans.*, 1987, 2947-2854.
12. Organonitrile complexes of iron and ruthenium. X-ray crystal structure of trans- $[\text{Fe}(\text{NCMe})_2(\text{dmpe})_2](\text{BPh}_4)_2$. A. R. Barron, G. Wilkinson, M. Motevalli, and M. B. Hursthouse, *Polyhedron*, 1987, **6**, 1089-1095.
13. Remarkable differences in the reactivities of the E and Z isomers of a phosphoalkene. A. R. Barron, S. W. Hall, and A. H. Cowley, *J. Chem. Soc., Chem. Commun.*, 1987, 980-981.
14. Synthesis of a metal-free tri-coordinate phosphorus(V) hydride and its conversion to a phosphoalkene via a reductive hydride shift. A.R. Barron and A.H. Cowley, *J. Chem. Soc., Chem. Commun.*, 1987, 1092-1093.

15. $(\eta^5\text{-C}_5\text{H}_5)\text{Fe}(\text{CO})_2\text{-}\lambda^1\text{-P}[\text{C}(\text{SiMe}_3)_2]_2$, the first three coordinate metallo-bis(methylene)phosphorane. A. R. Barron and A. H. Cowley, *J. Chem. Soc., Chem. Commun.*, 1987, 1272-1273.
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17. Cyclotrimerization of a phospho-alkyne and the formation of a coordinated 1,3,5-triphospha benzene. A. R. Barron and A. H. Cowley, *Angew. Chem., Int. Ed. Engl.*, 1987, **26**, 907-908.
18. Cyclic carboxylic monophosphides: A new class of phosphorus heterocycles. A. R. Barron, S. W. Hall and A. H. Cowley, *J. Chem. Soc., Chem. Commun.*, 1987, 1753-1754.
19. Preparation and structural characterization of a stibido-indium dimer. A. R. Barron, A. H. Cowley, R. A. Jones, C. M. Nunn, and D. L. Westmoreland, *Polyhedron*, 1988, **7**, 77-78.
20. Reaction of the phospho-alkyne $\text{ArC}\equiv\text{P}$ ($\text{Ar} = 2,4,6\text{-tBu}_3\text{C}_4\text{H}_3$) with nucleophiles: A new approach to 1,3-diphospha butadiene synthesis. A. R. Barron, A. H. Cowley, and S. W. Hall, *J. Chem. Soc., Chem. Commun.*, 1988, 171-172.
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22. Structure of pentacarbonyl tris(trimethylsilyl) phosphine iron (0). A. R. Barron, A. H. Cowley, and C. M. Nunn, *Acta. Crystallogr. C.*, 1988, **44**, 750.
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28. Molecular structure of α -trans-cinnamic acid. D. A. Wierda, T. Feng, and A. R. Barron, *Acta. Crystallogr. C.*, 1989, **45**, 338.
29. Adducts of trimethylaluminum with phosphine ligands: X-ray structure of $\text{Me}_3\text{AlPPh}_3$ and $\text{Me}_3\text{AlP(o-tolyl)}_3$. D. A. Wierda and A. R. Barron, *Polyhedron*, 1989, **8**, 831-834.
30. The reaction of indium(III)chloride with tris(trimethyl-silyl)phosphine: A novel route to indium-phosphide. M. D. Healy, P. E. Laibinis, P. D. Stupik and A. R. Barron, *J. Chem. Soc., Chem. Commun.*, 1989, 359-360.
31. Synthesis and molecular structure of $\{[\text{N}(\text{CH}_2\text{CH}_2\text{O})_3\text{Al}_2(\text{CH}_3)_3]\}_2$ The first six-coordinate aluminum alkyl. M. D. Healy and A. R. Barron, *J. Am. Chem. Soc.*, 1989, **111**, 398.

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34. Synthesis of 1,3-diphenyltriazene complexes of aluminum, gallium and indium: Crystal structure of tris(1,3-diphenyl-triazenido)aluminum(III). J. T. Leman, A. R. Barron, J. W. Ziller, and R. M. Kren, *Polyhedron*, 1989, **8**, 1909-1912.
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41. Sterically crowded aryloxide compounds of aluminium: Reactivity of coordinated benzaldehyde. M. B. Power and A. R. Barron, *Polyhedron*, 1990, **9**, 233-237.
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44. Complexes of aluminium (III) with picolinic and pipercolinic acids: an ^{27}Al NMR investigation. T. L. Feng, J. M. Tsangaris, and A. R. Barron, *Montash. Chemie*, 1990, **121**, 113.
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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

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U.S. Serial No. : 19/023,905 Examiner : Lisa S. Park
Filing Date : January 16, 2025 Art Unit : 1729
Confirmation No. : 3614
For : MULTILAYER NANOPOROUS SEPARATOR

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Alexandria, VA 22313-1450

RESPONSE TO APRIL 15, 2025 FINAL OFFICE ACTION

This Communication is submitted in response to the April 15, 2025 Office Action issued in connection with the above-identified application. A response to the April 15, 2025 Office Action is due July 15, 2025. Accordingly, this Amendment is being timely filed.

LISTING OF Claims begins on page **2** of this paper.

Remarks begin on page **6** of this paper.

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LISTING OF CLAIMS

1. (Previously Presented) A separator for a lithium battery comprising:
a porous polymeric layer having a first side and a second side;
a first layer adjacent the first side of the porous polymeric layer, said first layer comprising pores and solid material, wherein
 - (a) the solid material comprises inorganic oxide particles and at least one polymer;
 - (b) a plurality of the inorganic oxide particles in the first layer comprise crystallites having a size in the range of 5 nm to 90 nm;
 - (c) a thickness of the first layer is 3 microns or less and a total coating weight of the first layer is 0.8 gsm to 4.7 gsm; and
 - (d) a ratio of the thickness of the first layer to a total thickness of the separator is 10% to 80%.
2. (Previously Presented) The separator of claim 1, wherein a plurality of the pores in the first layer have a diameter of 100 nm or less.
3. (Previously Presented) The separator of claim 1, wherein the crystallites have a size of 5 nm to 80 nm.
4. (Original) The separator of claim 1, wherein the inorganic oxide particles of the first layer comprise a hydrated Al_2O_3 .
5. (Original) The separator of claim 1, wherein the inorganic oxide particles of the first layer comprise one or more of boehmite, SiO_2 , boron nitride pigments, AlN , hydrated aluminum oxide, AlOOH , and $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$.
6. (Previously Presented) The separator of claim 5, wherein when the inorganic oxide particles of the first layer comprise $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$, x is 1.0 to 2.0.

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7. (Previously Presented) The separator of claim 1, wherein the at least one polymer in the first layer comprises one or more of the following: cellulosic polymers, hydroxyethyl cellulose, vinyl pyrrolidone polymers and copolymers, polysaccharides, chitosan, polyethylene oxides, polyvinyl alcohols, and polyolefin.

8. (Previously Presented) The separator of claim 1, wherein a volume fraction of the at least one polymer in the solid material is 10% to 50%.

9. (Previously Presented) The separator of claim 1, wherein the first layer has a porosity of 30% to 70%.

10. (Original) The separator of claim 1, further comprising a second layer adjacent to the second side of the porous polymeric layer, said second layer comprising inorganic oxide particles and one or more polymers.

11. (Previously Presented) The separator of claim 10, wherein a combined total coating weight of the first layer and second layer is less than 9.4 gsm.

12. (Previously Presented) The separator of claim 1, wherein the total coating weight of the first layer is 0.8 gsm to 3.0 gsm.

13. (Previously Presented) The separator of claim 10, wherein the first layer is different in composition from the second layer.

14. (Previously Presented) The separator of claim 1, wherein a ratio by weight of inorganic oxide to polymer in the first layer is about 12:1.

15. (Previously Presented) The separator of claim 10, wherein a combined total thickness of the first layer and second layer is less than 3.8 microns.

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16. (Previously Presented) The separator of claim 1, wherein a pore volume of the first layer is between 0.8 cc/g and 1.0 cc/g.

17. (Original) The separator of Claim 1, wherein the porous polymeric layer comprises a polyolefin.

18. (Previously Presented) The separator of claim 17, wherein the polyolefin is selected from the group consisting of polyethylenes, polypropylenes, homopolymers of 1-butene, homopolymers of 1-pentene, homopolymers of 1-hexene, homopolymers of 1-octene, and copolymers and terpolymers thereof.

19. (Previously Presented) The separator of claim 9, wherein the first layer has a porosity of 30% to 55%.

20. (Previously Presented) The separator of claim 1, wherein the crystallites have a size of 5 nm to 25 nm.

21. (Previously Presented) The separator of claim 1, wherein the crystallites have a size of 30 nm to 50 nm.

22. (Previously Presented) The separator of claim 1, wherein the crystallites have a size of 5 nm to 75 nm.

23. (Previously Presented) The separator of claim 8, wherein the volume fraction is 12% to 45%.

24. (Previously Presented) The separator of claim 8, wherein the volume fraction is 25% to 45%.

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25. (Previously Presented) The separator of claim 8, wherein the volume fraction is 15.3% to 25.6%.

26. (Previously Presented) The separator of claim 10, wherein a combined total coating weight of the first layer and second layer is 8.4 gsm or less.

27. (Previously Presented) The separator of claim 1, wherein the total coating weight of the first layer is 5.2 gsm or less.

28. (Previously Presented) The separator of claim 1, wherein the total coating weight of the first layer is 3.6 gsm or less.

29. (Previously Presented) The separator of claim 1, wherein a ratio by weight of inorganic oxide to polymer in the first layer is about 8:1.

30. (Previously Presented) The separator of claim 1, wherein a ratio by weight of inorganic oxide to polymer in the first layer is about 4:1.

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REMARKS

Claims 1-30 are pending in the subject application. No claim amendments are made herein.

The Office Action has rejected Claims 1-15 and 17-30 as obvious over U.S. Patent Publication No. 2014/0170464 (“Iwase”) in view of U.S. Patent Publication No. 2011/0143185 (“Nishikawa”). Remaining Claim 16 has been rejected as obvious over Iwase in view of Nishikawa and further in view of U.S. Patent Publication No. 2001/0041290 (“Morigaki”). Applicant respectfully submits that Iwase and Nishikawa (and Morigaki) fail to disclose, alone and in combination, all of the limitations of the pending claims. Applicant also respectfully submits that there was no motivation to combine these references. Thus, Applicant traverses both rejections on these grounds. Applicant also submits the Expert Declaration of Andrew R. Barron, Ph.D., F.R.S.C. pursuant to 37 C.F.R. § 1.132 (hereinafter, “Barron Decl.” or “Barron Declaration”) in support of this response.

Applicant would like to thank the Examiner for participating in a telephonic interview on July 11, 2025. Applicant acknowledges and agrees with the statement in the interview summary issued on July 14, 2025 that “Arguments directed to the rejections over the Nishikawa reference were discussed. No specific agreement was reached.” Applicant greatly appreciates the statement in the interview summary that “Applicant is encouraged to submit arguments for consideration after-final” and accordingly submits the following arguments for consideration after-final.

NEITHER IWASE NOR NISHIKAWA TEACH THE CLAIMED COMBINATION OF THICKNESS AND COATING WEIGHT

Claim 1 specifies that the “thickness of the first layer is about 3 microns or less” and that the first layer has a coating weight of 0.8 to 4.7 gsm. Various dependent claims further specify that this 3 micron or less first layer has the following coating weights: 0.8-3.0 gsm (Claim 12); 5.2 gsm or less” (Claim 27); 3.6 gsm or less (Claim 28). Thus, for such claims, the first layer is **both** thin (3 microns or less) **and** has a low coating weight (0.8 gsm – 5.2 gsm). As explained in the specification, the “separators of the present invention provide superior safety, **lower cost**, lower thickness, and other key performance properties for use in lithium batteries and in other

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electric current producing cells.” (See Specification, Par. 0024; *id.*, Pars. 0044-0046).¹ Using a thin layer with a low coating weight helps to achieve such cost savings.

The Office Action cites paragraph 0053 of Iwase as teaching a first layer of 1 micron or more, but acknowledges that Iwase does not teach the claimed coating weight. (See Office Action, pp. 3, 7). However, the Office Action contends that Nishikawa discloses a “heat resistant porous layer comprising the same materials and other properties [as Iwase] should have a weight per unit (which is the coating weight) of 2 gsm to 10 gsm (**para 0059**) and so it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to design the first layer of Iwase” to have the coating weights of dependent Claims 11, 12, and 26-29. (Office Action, p. 7.)

Applicant respectfully disagrees that Iwase and Nishikawa combine to disclose a first layer that has **both** a thickness of 3 microns or less **and** a coating weight of 5.2 gsm or less. Paragraph 0059 of Nishikawa explains the “weight per unit of the heat-resistant porous layer **varies** depending on the material used and hence **cannot be constant**, but, generally, it is preferably 2 to 10 g/m².” Because of this variation, Nishikawa’s inventors were **unable** to make a heat-resistant porous layer that had **both** a thickness of 3 microns or less **and** a coating weight of 0.8 – 3.0 gsm, 5.2 gsm, and 3.6 as respectively recited in dependent Claims 12, 27 and 28.

Indeed, Nishikawa discloses 13 working examples of its heat-resistant porous layer (see Paras. 0166-0192 and Table 1) where the thickness and weight per unit area of the heat-resistant porous layer is provided. None of the 13 working examples of such layer (i.e., Examples 1-1, 1-2, 1-3, 1-4, 1-5, 2-1, 2-2, 2-3, 3-1, 3-2, 3-3, 3-4, and 3-5) have the **combined** thickness **and** coating weight as claimed.

For example, the heat-resistant porous layer of Example 1-4 of Nishikawa had a thickness of 7 microns (i.e., more than **double the claimed thickness** of 3 microns or less) and a weight per unit area of 4.77 g/m². See Pars. 0172-1073. Similarly, the heat-resistant porous layer in Example 3-3 had a thickness of 5.6 microns and weight per unit area of 5.74 g/m². See Table 1. In all other examples, the thickness of the heat-resistant porous layer was 7.0 – 8.5 microns and the weight per unit area for such layer fell in the range of 5 g/m² to 8.09 g/m². (See Pars. 0166-0192, Table 1).

¹ All emphasis added herein unless otherwise indicated.

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In view of Nishikawa's difficulty in controlling the variability of the coating weight of the heat-resistant porous layer, and further in view of the fact that Nishikawa was only able to achieve a coating weight of 4.77 g/m² or more in a much thicker (5.6– 8.4 microns) heat-resistant porous layer, Nishikawa does not teach the much thinner first layer (3.0 microns or less, Claim 1) that **also** has a low coating weight (0.8 gsm – 5.2 gsm, Claims 1, 12 and 27 and 28).

As such, there is no teaching in Nishikawa as to how to make Iwase's heat resistance layer so that it has **both** a thickness of 3 microns **and** a coating weight of 5.2 gsm or less.

THE CITED ART DOES NOT DISCLOSE THE VOLUME FRACTION LIMITATION

The Office Action Admits That Iwase Does not Disclose This Limitation

Claim 1 requires a separator that comprises a first layer having “pores and solid material” wherein the solid material comprises inorganic oxide particles and at least one polymer.

Dependent Claim 8 further specifies that the “volume fraction of the at least one polymer in the solid material is 10% to 50%.” Dependent Claims 23-25 respectively recite additional volume fraction ranges of 12%-45%, 25%-45%, and 15.3%-25.6%.

Dr. Barron discusses the volume fraction limitation in these claims in his declaration and is well qualified to do so. Indeed, through his 35+ years teaching at academic institutions such as Harvard University and Rice University, he has expertise in testing and analyzing inorganic oxide particles, polymers, and their related porosities. He has published more than 550 peer reviewed articles, many of which relate to this very subject. He has also written textbooks on this subject matter. See Barron Decl., Pars. 2-5, Appendices A, B.

As explained by Dr. Barron and as set forth in the specification, the volume fraction is a volume ratio of the polymer content of the solid portion of the first layer. The first layer is made of pores and the solid portion. The solid portion includes polymer and inorganic particles. See Specification, Par. 0038. However, because much of the volume of the first layer “is air, and not a solid material”, the “**air volume (e.g., pores) is not included** in the solid portion **for purposes of determining the volume fraction** of the one or more polymers in the **solid portion**.” *Id.* Thus, the air volume (pores) in the first layer is excluded from the volume fraction calculation. See Barron Decl., Pars. 7-11.

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To exclude the pores from the volume fraction calculation the POSITA needs to independently determine the volume of solid polymer and the solid inorganic oxide particles in the solid portion of the first layer. The specification teaches an example of a multistep method to make this calculation, see Par. [0038]. Such method involves, *inter alia*, removing the first layer from the separator, weighing a sample of the first layer, combusting the sample, weighing the residue that is left after combustion, calculating the difference in weight of the sample before and after combustion, calculating the volume of the solid polymer using the known density of the polymer, heating and dehydrating the samples, calculating the hydration of the sample, and using the appropriate density to calculate the volume of inorganic oxide particles. Once this is all done, the volume fraction of the at least one polymer in the solid material is calculated. This methodology is described Par. [0038] of the specification and discussed in detail in the Barron Declaration at Pars. 10-11.

The Office Action states that the primary reference Iwase discloses a first layer that includes pores and a solid material, wherein the solid material comprises inorganic oxide particles and at least one polymer. Office Action, p. 3. According to the Office Action, in Iwase, “the solid material portion is up to 50% by **mass** (para 0057)”. Office Action, p. 3. The Office Action acknowledges, however, that “**Iwase fails to specifically disclose wherein the solid material comprises the at least one polymer in a volume fraction of 10% to 50%** (as opposed to *mass fraction* taught by Iwase and the specific ranges claimed in Claims 8 and 23-25.” *Id.*, p. 4 (italicized emphasis in original).

The Secondary Reference, Nishikawa, Does Not Disclose This Limitation

According to the Office Action, “Nishikawa discloses an effective porous heat-resistant layer using . . . 0.4 to 4 times the volume of inorganic filler relative to the volume of the heat-resistant resin which equates to using 0.25 to 2.5 times the volume of resin relative to filler (i.e. 25% to 250%).” *Id.* The Office Action contends that based on this teaching in Nishikawa, it would have been obvious to a person of ordinary skill in the art (“POSITA”) to “design the first layer of the separator of Iwase such that the at least one polymer . . . in the first layer solid material is present in a volume fraction” of the ranges recited in Claims 1, 8 and 23-25.

Applicant respectfully disagrees that Nishikawa discloses the volume fraction limitation. As discussed above, for the claimed volume fraction limitation to be met, Nishikawa must teach

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excluding pores in any calculation of the volume fraction. There is no such teaching in Nishikawa.

Nishikawa teaches a “heat resistant **porous** layer” formed from an inorganic oxide filler and heat resistant resin. Such layer has pores throughout its **entire** thickness. Indeed, Nishikawa explains that “heat resistant porous layer has . . . a structure such that the **micropores** are connected with one another, and a gas or liquid can pass through the heat-resistant porous layer from one side to the other.” Nishikawa’s heat-resistant porous layer “preferably has a porosity in the range of from 20% to 80%.” Par. 0056; *id.*, pars. 0057-0059, 0063; Barron Decl., pars. 13-16. As Dr. Barron explains, a POSITA would understand from the teachings of Nishikawa that when the heat-resistant porous layer is included in a **finished and formed** separator, the **pores are located in the heat-resistant resin**, and not the inorganic oxide filler. Barron Decl., Pars. 7, 15, 16.

Against this background, the portion of Nishikawa that the Office Action bases its volume fraction calculation of 25% to 250% on is as follows:

It is preferred that the amount of the inorganic filler contained in the heat-resistant **porous** layer is 0.4 to 4 times the volume of the heat-resistant resin.

See Nishikawa, ¶0061.

But this passage **does not teach to exclude the pores** from any volume calculation. As Dr. Barron explains, this passage refers to a formed “heat-resistant porous layer” that includes “inorganic filler” and “heat-resistant resin.” As discussed above, a POSITA would understand that much of heat resistant-resin in Nishikawa’s formed layer **is porous** (e.g., it has a porosity of 20%-80%). Barron Decl., par. 18. In order to determine the volume fraction of the heat-resistant resin under the pending claims of the present application, the pores (i.e., air) in the resin of Nishikawa must be excluded. *Id.*, Par. 19. However, Nishikawa does not discuss excluding the 20% to 80% pores in the heat-resistant resin such as by the multistep methodology discussed above or by any other methodology. *Id.* As such, Nishikawa does not teach the volume fraction limitation.

Moreover, Paragraph [0061] in Nishikawa compares the **amount** of inorganic filler to the **volume** of the heat-resistant resin. An amount and a volume are not the same thing. Thus, as Dr. Barron explains, a POSITA would find the statement in Paragraph [0061] of Nishikawa

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“uninformative.” Barron Decl., Par. 20. Indeed, it is not possible to know whether Nishikawa was comparing volume to volume, amount to amount, volume to amount, or some other property in Paragraph [0061]. *Id.* Thus, Paragraph [0061] does not teach the volume fraction limitation of the pending claims for this additional reason.

Since there is no teaching of a volume fraction in Iwase, and further since there is no teaching or suggestion in Nishikawa to exclude pores from any volume fraction calculation, the prior art does not combine to teach the claimed volume fraction limitation. Thus, it is respectfully submitted that all pending claims are allowable.

Nishikawa Teaches Away From Using The Claimed Volume Fraction In Iwase

As discussed above, Claim 8 recites a volume fraction of 10% to 50%, and Claims 23 and 25 respectively recite volume fractions of 12%-45% and 15.3%-25.6%. As discussed below, because Nishikawa teaches that the volume of resin cannot fall below 25%, it teaches away from the **entire** claimed range of volume fractions.

Specifically, Nishikawa explains that when the amount of filler is **more than 4 times** the volume of the resin, this would **negatively affect** the physical and ion conduction properties of Nishikawa's separator:

when the amount of the inorganic filler contained is **more than 4 times** the volume of the heat-resistant resin, the heat-resistant porous layer is likely to be so dense that the **ion permeation is lowered**. Further, in this case, the heat-resistant porous layer is likely to be **brittle**, so that the handling properties become **markedly poor**.

Nishikawa, Par. 0061.

According to the Office Action's methodology for calculating the polymer volume in Nishikawa, when the amount of inorganic filler is **more than 4 times** the volume of the resin, the volume of polymer would fall **below 25%**. For example, if the amount of filler were 5 times the volume of the resin, then according to the Office Action's methodology, the alleged volume of polymer would be 20% (i.e., 1 divided by 5).

Thus, a POSITA would have understood that in Nishikawa, the volume of polymer in the heat-resistant layer **cannot** fall below 25% since this would not achieve improved ion conductivity and the separator would become too brittle to handle and manufacture. Thus, Nishikawa teaches

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away from the lower end of the claimed ranges of Claims 8, 23 and 25 which ranges respectively start at 10%, 12%, and 15.3%.

Indeed, the Federal Circuit addressed similar facts in *Chemours Co. FC, LLC v. Daikin Indus., Ltd.*, 4 F.4th 1370 (Fed. Cir. 2021), and found the challenged claims nonobvious. In *Chemours*, the claims recited “a melt flow rate of within the range of about 30 ± 3 g/10 min” (i.e., 27 – 33 g/10 min). The prior art Kaulbach reference taught melt flow rates of 15 g/10 min **or greater** and the example relied on by the Board had a melt flow rate of “24 g/10 min.” However, the Federal Circuit reversed the Board because the “Board appears to have ignored the express disclosure in Kaulbach that teaches away from the claimed invention.”

Iwase is Incompatible With Nishikawa

Applicant respectfully contends that the Office Action’s reading of Iwase is incorrect and that when correctly read, the separator of Iwase would be incompatible with the separator of Nishikawa.

The Office Action asserts that Iwase’s porous heat resistance layer 30 is the “first layer” of Claim 1. Iwase teaches that the heat resistance layer 30 includes an inorganic filler (e.g., aluminum oxide) and also “**may** contain, in addition to the organic filler . . . , **one kind or two or more** kinds of materials” that can be used as constituent components. Iwase, Pars. 0053, 0056. “[E]xamples of such materials” that can be used in addition to the inorganic filler include “binders and various kinds of additives.” *Id.* The binder can be a polymer (e.g., PE) and the additive can be “a thickener.” *Id.* As such, a POSITA would understand Par. 0056 of Iwase as teaching that the porous heat resistance layer 30 can include an inorganic filler combined with **either** a polymer binder **or** an additive (**or both**).

Against this background, the Office Action asserts that Par. 0057 of Iwase teaches that “a fraction of the at least one **polymer** . . . in the solid portion [of the first layer] is **up to 50% by mass.**” *Id.* Applicant respectfully disagrees. Iwase states that the “ratio of the **inorganic filler** contained in the entire porous heat resistance layer 30 is properly set to about **50% by mass or more, usually** preferably set to **85% by mass to 99.8% by mass** (90% by mass to 99% by mass, for example).”

The Office Action appears to assume that when the inorganic filler takes up 50% by mass of layer 30, that must mean that the polymer binder takes up the other 50%. But this is not

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correct. As explained above, layer 30 does not have to use a polymer binder at all and instead can use a thickener. And the very next sentence of Par. 0057 of Iwase teaches that when the binder is used in layer 30, the **maximum** amount of polymer binder is 10% by mass:

When the binder is used, a ratio of the binder in the entire porous heat resistance layer 30 may be set to about **1% by mass to 10% by mass**.

(Iwase, Par. 0057). Thus, contrary to the Office Action's statement, Iwase does not teach at least one polymer in the solid portion that is up to 50% by mass. The polymer is at most 10% by mass. This is consistent with Iwase's working example, which include only 2% or less by mass of polymer binder in the heat resistance layer. (See Iwase, Par. 0069) (teaching heat resistance layer made according to the following "weight ratio of . . . alumina:SBR:CMC-97:**2:1**").

In view of Iwase's teaching to use a small amount of polymer in its heat resistance layer, and further in view of Nishikawa's teaching away from using a low volume of polymer in its heat resistant porous layer, a POSITA would not have been motivated to apply Nishikawa's heat resistant layer to Iwase.

Conclusion

Applicant disagrees that the cited prior art combines to disclose all of the other limitations recited in the pending claims. However, since the prior art does not disclose the foregoing claim limitations, Applicant does not address such other claim limitations. In all events, in view of the foregoing remarks, Applicant respectfully requests withdrawal of the rejections.

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No fee is deemed necessary in connection with the filing of this Amendment. However, if any fee is required, authorization is hereby given to charge the amount of any such fee to Deposit Account No. 01-1785.

Respectfully submitted,

AMSTER, ROTHSTEIN & EBENSTEIN LLP
Attorneys for Applicant
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Dated: July 15, 2025
New York, New York

By /Brian A. Comack/
Brian A. Comack, Reg. No. 45,343

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 19/023,905	Filing Date 01/16/2025	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	x \$100 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	x \$480 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

		(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	07/15/2025	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		
	Total (37 CFR 1.16(i))	* 30	Minus	** 30	= 0	x \$200 =	0
	Independent (37 CFR 1.16(h))	* 1	Minus	*** 3	= 0	x \$600 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	0
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =	
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE	
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/DIANE FLOYD/	
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".							
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.							

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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NOTICE OF ALLOWANCE AND FEE(S) DUE

1912 7590 08/04/2025
AMSTER, ROTHSTEIN & EBENSTEIN LLP
405 LEXINGTON AVENUE
FLOOR 48
NEW YORK, NY 10174

EXAMINER

PARK, LISA S

ART UNIT

PAPER NUMBER

1729

DATE MAILED: 08/04/2025

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/023,905	01/16/2025	David W. Avison	00249/0086	3614

TITLE OF INVENTION: MULTILAYER NANOPOROUS SEPARATOR

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	11/04/2025

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 40% the amount of undiscounted fees, and micro entity fees are 20% the amount of undiscounted fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

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INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

1912 7590 08/04/2025
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Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/023,905	01/16/2025	David W. Avison	00249/0086	3614

TITLE OF INVENTION: MULTILAYER NANOPOROUS SEPARATOR

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	11/04/2025

EXAMINER	ART UNIT	CLASS-SUBCLASS
PARK, LISA S	1729	429-145000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

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3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

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Date _____

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/023,905	01/16/2025	David W. Avison	00249/0086	3614
1912	7590	08/04/2025	EXAMINER	
AMSTER, ROTHSTEIN & EBENSTEIN LLP			PARK, LISA S	
405 LEXINGTON AVENUE			ART UNIT	
FLOOR 48			PAPER NUMBER	
NEW YORK, NY 10174			1729	

DATE MAILED: 08/04/2025

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013).

<https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>

Routine uses of the information in this record may include disclosure to:

- 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law;
- 2) a federal, state, local, or international agency, in response to its request;
- 3) a contractor of the USPTO having need for the information in order to perform a contract;
- 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
- 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
- 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
- 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
- 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
- 9) the Office of Personnel Management (OPM) for personnel research purposes; and
- 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

Notice of Allowability	Application No. 19/023,905	Applicant(s) Avison et al.	
	Examiner LISA S PARK	Art Unit 1729	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to response filed 7/15/2025.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-30. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov**.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some* c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: ____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____.
- Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| 1. ☐ Notice of References Cited (PTO-892) | 5. ☒ Examiner's Amendment/Comment |
| 2. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date ____. | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material ____. | 7. ☐ Other ____. |
| 4. ☒ Interview Summary (PTO-413),
Paper No./Mail Date. <u>20250722</u> . | |

/LISA S PARK/
Primary Examiner, Art Unit 1729

DETAILED CORRESPONDENCE

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Amendment

2. In response to the amendment received on 7/15/2025:
 - a. Claims 1-30 are pending in the current application.
 - b. The previous prior art rejections are overcome in light of arguments, submitted affidavit, and Examiner's amendment, laid out below.

Claim Interpretation

3. All "wherein" clauses are given patentable weight unless otherwise noted. Please see MPEP 2111.04 regarding optional claim language.

Examiner's Amendment

4. An Examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee. This examiner's amendment was approved by Applicant's representative by telephone on 7/21/2025.
 - 1. (Currently Amended) A separator for a lithium battery comprising:
a porous polymeric layer having a first side and a second side;

a first layer adjacent the first side of the porous polymeric layer, said first layer comprising pores and solid material, wherein

- a) the solid material comprises inorganic oxide particles and at least one polymer;
- (b) a plurality of the inorganic oxide particles in the first layer comprise crystallites having a size in the range of 5 nm to 90 nm;
- (c) a thickness of the first layer is 3 microns or less and a total coating weight of the first layer is less than 6.0 gsm ~~0.8 gsm to 4.7 gsm~~; and
- (d) a ratio of the thickness of the first layer to a total thickness of the separator is 10% to 80%.

- 27. (Currently Amended) The separator of claim 1, wherein the total coating weight of the first layer is 0.8 gsm to 4.7 gsm ~~5.2 gsm or less~~.

Allowable Subject Matter

5. Claims 1-30 are allowed.

The following is an examiner's statement of reasons for allowance: none of the prior art of record, alone or in combination, appears to teach, suggest, or render obvious the invention of Claims 1-30.

Independent Claim 1 recites a separator comprising a porous polymeric layer having a first layer adjacent to a first side of the porous polymeric layer, the first layer comprising pores and solid material, where the solid material comprises at least one polymer and inorganic oxide particles (having a specified size), the first layer having a specified thickness and total coating weight, and a specified ratio of the thickness of the first layer to a total thickness of the separator. As Applicant correctly and convincingly points out in the remarks filed 7/15/2025, the closest

prior art does not teach the layer thickness and loading as claimed, and it's not obvious to modify these teachings to give the instant claimed invention. Thus none of the prior art of the record, alone or in combination, appears to teach, suggest, or render obvious the invention of independent Claim 1. Since Claims 2-30 depend on Claim 1, they are allowable for the same reason.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to LISA S PARK whose telephone number is (571)270-3597. The examiner can normally be reached M-Th 5:30a to 3p Eastern Time.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ula Tavares-Crockett can be reached on 5712721481. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit:

<https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/LISA S PARK/
Primary Examiner, Art Unit 1729

<i>Examiner-Initiated Interview Summary</i>	Application No. 19/023,905	Applicant(s) Avison et al.		
	Examiner LISA S PARK	Art Unit 1729	AIA (First Inventor to File) Status Yes	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
LISA S PARK	Primary Examiner	Telephonic
Brian Comack	Attorney of Record	

Date of Interview: 21 July 2025

Issues Discussed:

Proposed Amendment(s)

The Examiner will amend Claim 1 to recite "a total coating weight of the first layer is less than 6.0 gsm" and move the existing language regarding coating weight (0.8 gsm to 4.7 gsm) to Claim 27 to put the application in condition for allowance. This amendment was approved by Applicant.

/LISA S PARK/ Primary Examiner, Art Unit 1729	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: It is not necessary for applicant to provide a separate record of the substance of interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

<i>Index of Claims</i> 	Application/Control No. 19/023,905	Applicant(s)/Patent Under Reexamination Avison et al.
	Examiner LISA S PARK	Art Unit 1729

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

CLAIMS										
☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	03/11/2025	04/14/2025	07/22/2025						
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	2	✓	✓	=						
	3	✓	✓	=						
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	28		✓	=						
	29		✓	=						
	30		✓	=						

Issue Classification 	Application/Control No. 19/023,905	Applicant(s)/Patent Under Reexamination Avison et al.
	Examiner LISA S PARK	Art Unit 1729

CPC						
Symbol					Type	Version
H01M	/	10	/	0525	F	2013-01-01
H01M	/	50	/	446	I	2021-01-01
H01M	/	50	/	491	I	2021-01-01
H01M	/	50	/	489	I	2021-01-01
H01M	/	50	/	417	I	2021-01-01
H01M	/	50	/	451	I	2021-01-01
H01M	/	50	/	457	I	2021-01-01

CPC Combination Sets				
Symbol	Type	Set	Ranking	Version
/				

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	30	
/LISA S PARK/ Primary Examiner, Art Unit 1729	22 July 2025	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	7

<i>Issue Classification</i> 	Application/Control No. 19/023,905	Applicant(s)/Patent Under Reexamination Avison et al.
	Examiner LISA S PARK	Art Unit 1729

INTERNATIONAL CLASSIFICATION				
CLAIMED				
H01M	/	10	/	0525
H01M	/	50	/	417
H01M	/	50	/	446
H01M	/	50	/	451
NON-CLAIMED				
	/		/	

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

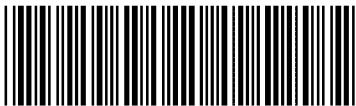
CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	30	
/LISA S PARK/ Primary Examiner, Art Unit 1729 (Primary Examiner)	22 July 2025 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 7

Issue Classification 	Application/Control No. 19/023,905	Applicant(s)/Patent Under Reexamination Avison et al.
	Examiner LISA S PARK	Art Unit 1729

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☒ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
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2	2	11	11	20	20	29	29								
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8	8	17	17	26	26										
9	9	18	18	27	27										

NONE	Total Claims Allowed:	
(Assistant Examiner)	(Date)	30
/LISA S PARK/ Primary Examiner, Art Unit 1729	22 July 2025	O.G. Print Claim(s)
(Primary Examiner)	(Date)	1
		O.G. Print Figure
		7

<i>Search Notes</i> 	Application/Control No. 19/023,905	Applicant(s)/Patent Under Reexamination Avison et al.
	Examiner LISA S PARK	Art Unit 1729

CPC - Searched*		
Symbol	Date	Examiner
H01M10/0525 or H01M50/417 or H01M50/446 or H01M50/451 or H01M50/457 or H01M50/489 or H01M50/491	03/11/2025, 04/14/2025, 07/22/2025	/L.S.P./

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

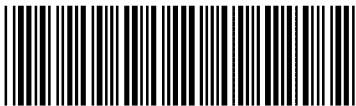
US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
PE2E keyword search (see attached search notes) (preliminary)	03/11/2025, 04/14/2025, 07/22/2025	/L.S.P./
PE2E inventor/assignee search	03/11/2025	/L.S.P./
Analysis of prior art in family and related applications	04/14/2025	/L.S.P./
PE2E similarity search	07/22/2025	/L.S.P./

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
H01M10	0525	07/22/2025	/L.S.P./
H01M50	417, 446, 451, 457, 489, 491	07/22/2025	/L.S.P./

/LISA S PARK/ Primary Examiner, Art Unit 1729	
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<i>Search Notes</i> 	Application/Control No. 19/023,905	Applicant(s)/Patent Under Reexamination Avison et al.
	Examiner LISA S PARK	Art Unit 1729

/LISA S PARK/ Primary Examiner, Art Unit 1729	
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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
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PE2E SEARCH - Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
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UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
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ELECTRONIC PAYMENT RECEIPT

APPLICATION #
19/023,905

RECEIPT DATE / TIME
08/14/2025 04:36:32 PM Z ET

ATTORNEY DOCKET #
00249/0086

Title of Invention

MULTILAYER NANOPOROUS SEPARATOR

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 3614

FILED BY Joy Chapman

PATENT CENTER # 71862288

AUTHORIZED BY Brian Amos

CUSTOMER # 1912

FILING DATE 01/16/2025

CORRESPONDENCE ADDRESS -

FIRST NAMED INVENTOR David W. Avison

Payment Information

PAYMENT METHOD
CARD / 1010

PAYMENT TRANSACTION ID
E20258DG37007573

PAYMENT AUTHORIZED BY
Joy Chapman

PRE-AUTHORIZED ACCOUNT
011785

PRE-AUTHORIZED CATEGORY
37 CFR 1.16 (National application filing, search, and examination fees); 37 CFR 1.17 (Patent application and reexamination processing fees); 37 CFR 1.19 (Document supply fees); 37 CFR 1.20 (Post Issuance fees); 37 CFR 1.21 (Miscellaneous fees and charges)

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1501	UTILITY ISSUE FEE	1290.00	1	1290.00
TOTAL AMOUNT:				\$1,290.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement

Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

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MULTILAYER NANOPOROUS SEPARATOR

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	3614	FILED BY	Joy Chapman
PATENT CENTER #	71862288	FILING DATE	01/16/2025
CUSTOMER #	1912	FIRST NAMED INVENTOR	David W. Avison
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Brian Amos

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
00249-0086_Issue-Fee-Transmittal.pdf	1	Issue Fee Payment (PTO-85B)	135 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
00249-0086_Issue-Fee-Transmittal.pdf	0C1307933B6C59B18A645A4EE5B7184B1075CC82E4ECA18A3 16A27991C89359B4E4A3DF0C212D0CC2EB4E20E6BBF47CD6 8C87B34F8A59413E45F271D35E5D41D

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PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

Brian J. Amos

(Typed or printed name)

/Brian J. Amos/

(Signature)

August 14, 2025

(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/023,905	01/16/2025	David W. Avison	00249/0086	3614

TITLE OF INVENTION: MULTILAYER NANOPOROUS SEPARATOR

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	11/04/2025

EXAMINER	ART UNIT	CLASS-SUBCLASS
PARK, LISA S	1729	429-145000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

Amster, Rothstein & Ebenstein LLP

1 _____

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

LG Energy Solution, Ltd.

Seoul, Korea

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☒ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☒ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 01-1785

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Brian J. Amos/

Date August 14, 2025

Typed or printed name Brian J. Amos

Registration No. 63,679



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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/023,905	09/09/2025	12412924	00249/0086	3614

1912 7590 08/27/2025
AMSTER, ROTHSTEIN & EBENSTEIN LLP
405 LEXINGTON AVENUE
FLOOR 48
NEW YORK, NY 10174

ISSUE NOTIFICATION

The projected patent number and issue date are specified above. The patent will issue electronically. The electronically issued patent is the official patent grant pursuant to 35 U.S.C. § 153. The patent may be accessed on or after the issue date through Patent Center at <https://patentcenter.uspto.gov/>. The patent will be available in both the public and the private sides of Patent Center. Further assistance in electronically accessing the patent, or about Patent Center, is available by calling the Patent Electronic Business Center at 1-888-217-9197.

The USPTO is implementing electronic patent issuance with a transition period, during which period the USPTO will mail a ceremonial paper copy of the electronic patent grant to the correspondence address of record. Additional copies of the patent (i.e., certified and presentation copies) may be ordered for a fee from the USPTO's Certified Copy Center at <https://certifiedcopycenter.uspto.gov/index.html>. The Certified Copy Center may be reached at (800)972-6382.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Center (<https://patentcenter.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Patents Stakeholder Experience (OPSE), Stakeholder Support Division (SSD) at (571)-272-4200.

INVENTOR(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional inventors):

David W. Avison, Boxborough, MA;
Steven A. Carlson, Cambridge, MA;
Benjamin Sloan, Exeter, NH;

APPLICANT(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional applicants):

LG Energy Solution, Ltd., Seoul, KOREA, REPUBLIC OF;

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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19/023,905

01/16/2025

David W. Avison

00249/0086

3614

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7590

09/09/2025

AMSTER, ROTHSTEIN & EBENSTEIN LLP

405 LEXINGTON AVENUE

FLOOR 48

NEW YORK, NY 10174

EXAMINER

PARK, LISA S

ART UNIT

PAPER NUMBER

1729

MAIL DATE

DELIVERY MODE

09/09/2025

PAPER

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.



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APPLICATION NO.	ISSUE DATE	PATENT NO.
19/023,905	09-Sep-2025	12412924

AMSTER, ROTHSTEIN & EBENSTEIN LLP
405 LEXINGTON AVENUE
NEW YORK, NY 10174

EGRANT NOTIFICATION

Your electronic patent grant (eGrant) is now available, which can be accessed via Patent Center at <https://patentcenter.uspto.gov>

The electronic patent grant is the official patent grant under 35 U.S.C. 153. For more information, please visit <https://www.uspto.gov/electronicgrants>